



Theses and Dissertations

2013-07-05

Aqueous Henry's Law Constants, Infinite Dilution Activity Coefficients, and Water Solubility: Critically Evaluated Database, Experimental Analysis, and Prediction Methods

Sarah Ann Brockbank
Brigham Young University - Provo

Follow this and additional works at: <https://scholarsarchive.byu.edu/etd>

 Part of the [Chemical Engineering Commons](#)

BYU ScholarsArchive Citation

Brockbank, Sarah Ann, "Aqueous Henry's Law Constants, Infinite Dilution Activity Coefficients, and Water Solubility: Critically Evaluated Database, Experimental Analysis, and Prediction Methods" (2013). *Theses and Dissertations*. 3691.

<https://scholarsarchive.byu.edu/etd/3691>

This Dissertation is brought to you for free and open access by BYU ScholarsArchive. It has been accepted for inclusion in Theses and Dissertations by an authorized administrator of BYU ScholarsArchive. For more information, please contact scholarsarchive@byu.edu, ellen_amatangelo@byu.edu.

Aqueous Henry's Law Constants, Infinite Dilution Activity Coefficients,
and Water Solubility: Critically Evaluated Database,
Experimental Analysis, and Prediction Methods

Sarah A. Brockbank

A dissertation submitted to the faculty of
Brigham Young University
in partial fulfillment of the requirements for the degree of
Doctor of Philosophy

W. Vincent Wilding, Chair
Richard L. Rowley
Thomas A. Knotts
Randy S. Lewis
Dean R. Wheeler

Department of Chemical Engineering
Brigham Young University

July 2013

Copyright © 2013 Sarah A. Brockbank

All Rights Reserved

ABSTRACT

Aqueous Henry's Law Constants, Infinite Dilution Activity Coefficients, and Water Solubility: Critically Evaluated Database, Experimental Analysis, and Prediction Methods

Sarah A. Brockbank
Department of Chemical Engineering, BYU
Doctor of Philosophy

A database containing Henry's law constants, infinite dilution activity coefficients and solubility data of industrially important chemicals in aqueous systems has been compiled. These properties are important in predicting the fate and transport of chemicals in the environment. The structure of this database is compatible with the existing DIPPR[®] 801 database and DIADDEM interface, and the compounds included are a subset of the compounds found in the DIPPR[®] 801 database. Thermodynamic relationships, chemical family trends, and predicted values were carefully considered when designating recommended values.

Henry's law constants and infinite dilution activity coefficients were measured for toluene, 1-butanol, anisole, 1,2-difluorobenzene, 4-bromotoluene, 1,2,3-trichlorobenzene, and 2,4-dichlorotoluene in water using the inert gas stripping method at ambient pressure (approximately 12.5 psia) and at temperatures between 8°C and 50°C. Fugacity ratios, required to determine infinite dilution activity coefficients for the solid solutes, were calculated from literature values for the heat of fusion and the liquid and solid heat capacities. Chemicals were chosen based on missing or conflicting data from the literature.

A first-order temperature-dependent group contribution method was developed to predict Henry's law constants of hydrocarbons, alcohols, ketones, and formats where none of the functional groups are attached directly to a benzene ring. Efforts to expand this method to include ester and ether groups were unsuccessful. Second-order groups were developed at a reference condition of 298.15 K and 100 kPa. A second-order temperature-dependent group contribution method was then developed for hydrocarbons, ketones, esters, ethers, and alcohols. These methods were compared to existing literature prediction methods.

Keywords: DIPPR, database, Henry's law constant, water solubility, infinite dilution activity coefficient

ACKNOWLEDGEMENTS

I would like to thank my advisor, Dr. Vincent Wilding, as well as Dr. Neil Giles for helping me troubleshoot in the lab and for providing me with feedback on presentations, reports, and papers. I thank Dr. Richard Rowley for his helpful, detailed comments on my writing. AICHE and DIPPR[®] 801 are thanked for project funding. I thank Jenna Russon, Doug Nevers, and Coleman Vaclaw for database, programming, and lab help. I thank the many DIPPR[®] 801 programmers for all of their help programming and debugging.

I want to thank my family members for their love and support throughout this journey. I thank my husband, Bryan Brockbank, for being willing to go to the lab with me at strange hours. I thank his patience as I have missed birthdays and an anniversary on top of having to live in different states for a period of time in order for me to finish this degree. I also thank Brad Brockbank, Jennie Linford, and Quinn Linford for being willing to lend a hand in the lab at strange hours. I thank my Purcell and Brockbank parents as well as grandparents for all of their prayers, love, and support. Most importantly, I want to express gratitude for my Savior, Jesus Christ, who has given me strength and understanding beyond my own abilities.

TABLE OF CONTENTS

LIST OF TABLES	vii
LIST OF FIGURES	xi
1 Introduction	1
2 Overview of Properties	5
2.1 Definitions and Uses	5
2.1.1 Solubility of a Compound in Water	5
2.1.2 Solubility of Water in a Compound	6
2.1.3 Henry's Law Constant	6
2.1.4 Infinite Dilution Activity Coefficients: Chemical in Water	10
2.1.5 Infinite Dilution Activity Coefficients: Water in Chemical	10
2.1.6 Octanol/Water Partition Coefficient	11
2.2 Experimental Techniques	11
2.2.1 Solubilities	11
2.2.2 Henry's Law Constants	12
2.2.3 Infinite Dilution Activity Coefficients	12
2.2.4 Octanol/Water Partition Coefficients	13
3 Database Overview	15
3.1 Background	15
3.2 Database Structure and Data Entry	16
3.3 Unit Conversions	20
4 Data Evaluation	23
4.1 Property Relationships	23
4.2 Experimental Factors	30
4.3 Temperature-Dependent Evaluation	32
4.3.1 Error Calculations	36
4.3.2 Critical Solution Temperatures	40
4.3.3 Chemical Family Plots	43
4.3.4 Other Data Analysis Notes	48
4.4 Constant Values	48
4.4.1 Chemical Family Plot Regressions	49
4.4.2 Averaging	62
4.5 American Petroleum Institute Recommendations	63
4.6 Comparison with Literature Recommendations and Predicted Values	64
4.7 Summary	72
5 Experimental Methods	75
5.1 Materials	75
5.2 Inert Gas Stripping Theory	76
5.3 Apparatus and Procedure	83
6 Experimental Results and Discussion	89
7 Prediction Methods	101
7.1 Overview of Prediction Method Types	101
7.2 Prediction Method Selection	103
7.2.1 Method of Sedlbauer et al. [22, 191, 283]	106

7.2.2	Method of Plyasunov and Shock [284].....	107
7.2.3	Methods of Lau et al. [39].....	108
7.2.4	Method of Kühne, Ebert, and Schüürmann [285].....	109
7.2.5	Existing Method Comparison	109
7.3	Prediction Method Development	110
7.3.1	Data Selection	113
7.3.2	Validation Techniques	114
7.3.3	First-Order Group Contribution Method	116
7.3.4	Second-Order Group Contribution Method	121
7.3.5	Method Application Summary.....	131
7.3.6	Method Comparison and Recommendations	131
8	Summary.....	135
8.1	Recommendations for Future Work.....	137
9	References	139
Appendix A.	Experimental Method Details.....	163
Appendix B.	Fugacity Error Propagation	179
Appendix C.	Summary of Recommended Values.....	181
Appendix D.	Experimental Data.....	273
Appendix E.	Prediction Method Sample Calculations	289
E.1	First-Order Group Contribution Method.....	289
E.1.1.	1,2,4,5-Tetramethylbenzene.....	289
E.1.2.	2,2-Dimethylhexane.....	290
E.1.3.	Methyl ethyl ketone	290
E.2	Second-Order Group Contribution Method	291
E.2.1.	3-Methyl-1-butanol	291
E.2.2.	Ethylbenzene	292
E.2.3.	<i>n</i> -Butyl ethyl ether	292

LIST OF TABLES

Table 3.1: Abbreviations used in 801E database	15
Table 3.2: Summary of 801E entries	17
Table 3.3: Summary of the number of evaluated compounds (#) per chemical family.....	18
Table 3.4: Unit conversion assumptions.....	21
Table 4.1: Number of compounds with recommended values	28
Table 4.2: Summary of property relationships used in this study.....	28
Table 4.3: DIPPR [®] 801 equation numbers	34
Table 4.4: Equation comparisons used in data analysis.....	34
Table 4.5: Summary of uncertainty determinations for temperature-dependent regressions	37
Table 4.6: Chemical families with evaluated compounds exhibiting critical solution behavior ..	40
Table 5.1: Compounds and purities	75
Table 5.2: Constant GC conditions.....	85
Table 5.3: GC conditions for each compound	85
Table 6.1: Toluene and 1-butanol results at ambient pressure (approximately 12.5 psia)	90
Table 6.2: Flow rates tested and used in this study.....	91
Table 6.3: Maximum headspace concentrations possible in this study	92
Table 6.4: k_H and γ_i^∞ values at ambient pressure (approximately 12.5 psia)	96
Table 7.1: k_H prediction methods at 298.15 K	104
Table 7.2: k_H temperature-dependent prediction methods excluded from this study	105
Table 7.3: k_H temperature-dependent prediction methods chosen for evaluation.....	105
Table 7.4 Temperature-dependent k_H prediction method comparison for select compounds	111
Table 7.5: Summary of performance for first-order group contribution temperature-dependent method for hydrocarbons.....	116

Table 7.6: Summary of performance for first-order group contribution temperature-dependent method for alcohols, ketones, and formates	118
Table 7.7: Summary of parameter estimates for the first-order method	120
Table 7.8: Summary of hydrocarbon second-order group contributions for reference values at 298.15 K and 0.1 MPa	123
Table 7.9: Performance summary of the second-order temperature-dependent group contribution method for alkylbenzenes	124
Table 7.10 Second-order alcohol groups	125
Table 7.11: Summary of performance for second-order temperature-dependent group contribution method for esters, ethers, ketones, and alcohols	128
Table 7.12: Summary of second-order temperature-dependent model values	130
Table 7.13: Second-order method correction factor	131
Table 7.14: Comparison of methods from this study with existing methods	133
Table A.1: x_i^{aq} experimental methods	163
Table A.2: k_H experimental methods	165
Table A.3: γ_i^∞ experimental methods	169
Table A.4: KOW experimental methods	172
Table C.1: Abbreviations used in regression tables	181
Table C.2: k_H regression summary	182
Table C.3: γ_i^∞ regression summary	201
Table C.4: x_i^{aq} regression summary	220
Table C.5: Summary of constant value recommendations (298.15 K and 1 atm)	249
Table D.1: Abbreviations used in tables with experimental run information	273
Table D.2: Cell data for each experimental run	274
Table D.3: Experimental run information—additional parameters	279

Table D.4: Experimental run calculated values	284
---	-----

LIST OF FIGURES

Figure 4.1: 2-Methyl-1-butanol k_H , x_i^{aq} , and γ_i^∞ data plotted as $\ln(k_H)$ versus inverse temperature. References: [44-48].....	29
Figure 4.2: 2-Methyl-1-butanol k_H , x_i^{aq} , and γ_i^∞ data plotted as $\ln(x_i^{aq})$ versus inverse temperature References: [44-48].....	30
Figure 4.3: Literature x_i^{aq} data for vanillin. References: a [50], b [51], c [52], d [53], e [54].....	31
Figure 4.4: Uncertainty determination of experimental datasets	39
Figure 4.5: 2-Butanol $\ln(x_i^{aq})$ data and regression. References: [55-62]	41
Figure 4.6: $\ln(x_i^{aq})$ data and regression for p-chlorophenol. References: [63-67]	42
Figure 4.7: Residual plot from p-chlorophenol $\ln(x_i^{aq})$ regression	42
Figure 4.8: n -Alcohol $\ln(x_i^{aq})$ regressions based on experimental data versus inverse temperature. Initial C8 regression based on experimental data (C8) and updated C8 regression based on experimental data and predicted values from the family plot (C8-Exp&Pred) are both shown.	44
Figure 4.9: 1-Heptanol values plotted as $\ln(k_H)$ versus inverse temperature. 1-Octanol (C8) and 1-hexanol (C6) regressions obtained from experimental data shown along with the average of these two regressions (Avg C6 & C8). References: a [47], b [69], c [68] (*reference reports values using two different experimental methods), d [70], e [71], x_i^{aq} data [44, 72-80].....	45
Figure 4.10: n -Alcohol $\ln(\gamma_i^\infty)$ regressions based on experimental data versus inverse temperature	47
Figure 4.11: n -Alcohol $\ln(k_H)$ regressions based on experimental data versus inverse temperature	47
Figure 4.12: 1-Alkene k_H and x_i^{aq} data at 298.15 K plotted as $\ln(x_i^{aq})$ versus carbon number (#C). References: [47, 72, 84-87]	51
Figure 4.13: 1-Alkene $\ln(k_H)$ values at 298.15 K from this study (“Smoothed” and “Predicted”) and HENRYWIN bond-contribution prediction method [83] plotted as a function of carbon number (#C)	51
Figure 4.14: 1-Iodoalkane $\ln(k_H)$ values at 298.15 K from temperature-dependent regressions (Temp. Dep. Smoothed), constant value family plot (Const. Val. Smoothed), the literature (data) [47], and HENRYWIN bond-contribution method [83] plotted as a function of carbon number (#C)	52

Figure 4.15: <i>n</i> -Alcohol “Smoothed” values from temperature-dependent evaluations and literature values plotted as $\ln(x_i^{aq})$ versus carbon number (#C). Solid symbols represent values at 298.15 K. Other values are at temperatures ranging from 277.15 K to 307.15 K. References: a [70], b [91], c [88], d [90], e [92], f [89], g [93]	53
Figure 4.16: <i>n</i> -Alkane “Smoothed” values from temperature-dependent evaluations and literature (Data) x_i^{aq} values at 298.15 K. References: [85, 95-104]	55
Figure 4.17: <i>n</i> -Alkylbenzene $\ln(x_i^{aq})$ values at 298.15 K from temperature-dependent regressions (Temp. Dep. Smoothed), constant value family plot (Const. Val. Smoothed), and the literature (Data) plotted as a function of carbon number (#C). References: [105, 106] ..	55
Figure 4.18: di- <i>n</i> -Alkyl phthalate $\ln(x_i^{aq})$ values at 298.15 K from temperature-dependent regressions (Temp. Dep. Smoothed), constant value family plot (Const. Val. Smoothed), and the literature (Data and Rejected Data) plotted as a function of the number of carbons on the alkyl chains (#C). References: [107-115]	56
Figure 4.19: di- <i>n</i> -Alkyl phthalate “Smoothed” $\ln(k_H)$ values at 298.15 K from this study and from HENRYWIN[83] bond-contribution method.....	57
Figure 4.20: Literature data of diethyl phthalate at 298.15 K plotted as $\ln(k_H)$ compared with HENRYWIN bond-contribution value and reported experimental value [83]. References: a [111], b [116], c [109], d [112], e [110], f [107]	58
Figure 4.21: Smoothed $\ln(x_i^{aq})$ and $\ln(P_i^{sat})$ [117] of di- <i>n</i> -alkyl phthalates at 298.15 K	59
Figure 4.22: Cycloalkane and cycloalkene $\ln(x_i^{aq})$ and $\ln(P_i^{sat})$ values at 298.15 K plotted as a function of carbon number (#C). References: [86, 117]	61
Figure 4.23: Cycloalkane and cycloalkene $\ln(k_H)$ values at 298.15 K compared to HENRYWIN bond-contribution values [83]	61
Figure 4.24: Triethylamine x_i^{aq} , γ_i^∞ , and k_H literature data. References: [118-127]	62
Figure 4.25: Methanol γ_i^∞ regressions from this study and Dohnal et al. [21]. References: [45, 48, 69, 70, 77, 167-182].....	66
Figure 4.26: <i>n</i> -Propyl formate x_i^{aq} data, regression from this study, and IUPAC-NIST [159] regression compared to ethyl formate x_i^{aq} data. References: [183, 184]	67
Figure 4.27: di- <i>n</i> -Propyl ether x_i^{aq} regressions from this study and from IUPAC-NIST [158]. References: [47, 172, 186-189].....	68
Figure 4.28: 3-Methyl-3-pentanol x_i^{aq} regressions from this study and from IUPAC-NIST [150]. References: [44, 190]	69

Figure 4.29: Cumene x_i^{aq} data compared with Sedlbauer et. al [191] predicted values and API [128] and IUPAC-NIST [136] recommendations. References: a [194], b [192], c [193], d [105]	71
Figure 4.30: 2,2,5-Trimethylhexane x_i^{aq} data plotted as $\ln(k_H)$ compared with Sedlbauer et. al [191] predicted values. References: a [195], b [86]	72
Figure 5.1: Inlets and outlets of saturation and stripping cells used in mass balances	78
Figure 5.2: IGS apparatus	85
Figure 6.1: GC data for two separate 1,2-difluorobenzene experiments at 323.15 K and a N_2 flow rate of 40 mL/min that resulted in different slopes (a)	94
Figure 6.2: Cumulative slopes (a) from GC data shown in Figure 6.1 as a function of time	94
Figure 6.3: 1,2,3-Trichlorobenzene k_H values from this study and from literature. Values from references marked with an asterisk (*) are x_i^{aq} data converted to k_H values using Equation 4.1. References: a [209], b [71], c [206], d* [210], e* [211], f* [212], g* [213], h* [205], i* [207], j* [214]	97
Figure 6.4: Anisole k_H values from this study and from the literature. Values from references marked with an asterisk (*) are x_i^{aq} data converted to k_H values using Equation 4.1. References: a [172], b [206], c* [215], d* [118], e* [216]	98
Figure 6.5: 2,4-Dichlorotoluene k_H values from this study and from the literature. Values from x_i^{aq} data were converted to k_H values using Equation 4.1. Reference: [8]	99
Figure 6.6: 1,2-Difluorobenzene k_H values from this study and from the literature. Values from x_i^{aq} data were converted to k_H values using Equation 4.1. Reference: [207]	99
Figure 6.7: 4-Bromotoluene k_H values from this study and from the literature. Values from x_i^{aq} data were converted to k_H values using Equation 4.1. References: [47, 208]	100
Figure 7.1: Plot of experimental versus model values for the first-order temperature-dependent group contribution method for hydrocarbons	117
Figure 7.2: Plot of experimental versus model values for the first-order temperature-dependent group contribution method for alcohols, ketones, and formates	119
Figure 7.3: Plot of experimental versus model values for the second-order temperature-dependent group contribution method for alkylbenzenes and alkanes	125
Figure 7.4: Initial second-order group contribution model performance for functional groups	126

Figure 7.5: Corrected second-order group contribution model performance for functional groups.....	128
Figure 7.6: Plot of experimental versus model values for the second-order temperature-dependent group contribution method for esters, ethers, ketones, and alcohols.....	129

1 Introduction

A growing global concern is the fate and transport of chemicals in the environment in addition to health and safety risks associated with chemical exposure. The Toxic Substances Control Act (TSCA) was passed in 1976 in the United States, and it requires the U.S. Environmental Protection Agency (EPA) to estimate risks of chemicals prior to commercialization [1]. The EPA has developed a process to estimate risks associated with a chemical to ensure it will not pose an unreasonable risk if commercialized. Properties used in these risk assessments include water solubility, octanol/water partition coefficients, and Henry's law constants. The Registration, Evaluation and Authorization of Chemicals (REACH) is a new policy concerning chemical regulations in Europe that became effective June 2007 throughout the European Union [2]. REACH requires industry to provide health, safety, and environmental information that will aid in the management of risks. Properties required by REACH include water solubility and the octanol/water partition coefficient [3, 4].

It is estimated that experimental values are available for less than 1% of the approximately 84,000 compounds included in TSCA and the 100,000 compounds included in REACH [5]. This means that environmental regulations rely heavily on prediction methods. A prediction method is only as good as the data used to develop the method. Prediction methods developed using evaluated data allow for superior prediction method development performance.

DIPPR[®] Projects 911 and 912 were initiated in 1991 by AIChE/DIPPR[®] with the goal of compiling data and prediction methods for important environmental, safety, and health properties of chemicals that are regulated and important in industry [6]. The information was compiled as the Environmental and Safety Properties (ESP) database. In November 2009, the DIPPR[®] 801 steering committee decided that a thorough review should be conducted on select compounds in order to evaluate the status of the DIPPR[®] ESP database. As a result of the review, it was decided to build a new database (801E) focusing on select properties. The properties of primary interest are aqueous Henry's law constant (k_H), infinite dilution activity coefficient of a chemical in water (γ_i^∞), and the solubility of a compound in water (x_i^{aq}). The infinite dilution activity coefficient of water in the chemical, the solubility of water in a chemical, and the octanol/water partition coefficients were included when available. However, these properties were given low priority meaning they were not reviewed as thoroughly due to time constraints and lower interest from the DIPPR[®] steering committee. An overview of the properties included in the 801E database is given in Chapter 2.

The first objective of this project was to compile a database containing primary, experimental data for k_H , γ_i^∞ , and x_i^{aq} for 500 compounds. An overview of the database is given in Chapter 3. Accepted values were designated for k_H , γ_i^∞ , and x_i^{aq} by evaluating the available literature data. Recommendations were given for values at a single temperature or over a temperature range depending on the amount of available literature data. The data evaluation process is described in Chapter 4. The second objective was to perform temperature-dependent measurements of k_H and γ_i^∞ for seven compounds to fill in data gaps and provide clarification for inconsistencies with reported experimental values. The experimental methods and results are

given in Chapters 5 and 6. The third objective was to analyze and refine k_H prediction methods using the evaluated database which is described in Chapter 7.

2 Overview of Properties

2.1 Definitions and Uses

2.1.1 Solubility of a Compound in Water

Water solubility (x_i^{aq}) is the maximum amount of a substance (gas, liquid, or solid) that will dissolve in a specified amount of water at a defined temperature and pressure [1]. This is an important property for predicting a chemical's environmental fate and its effects on biological systems. Highly water-soluble chemicals absorb easily into biological systems and typically degrade readily in the environment. However, highly water-soluble chemicals do not easily adsorb on soils and sediments or bioaccumulate. Solubility is a function of temperature, but it is often treated as a constant over environmental temperature ranges [7]. Generally, solubility increases with increasing temperature [8]. Other factors that affect solubility include pressure, pH, and the presence of additional chemical species [9-11]. In this project, the focus is x_i^{aq} data of a single solute in pure water measured at temperatures below 373 K at or near atmospheric pressure. However, high-temperature data have been included for some compounds.

Some systems exhibit upper and/or lower critical solution temperatures. At temperatures below or above the lower or upper critical solution temperatures, respectively, the compound and water are completely soluble in each other resulting in one liquid phase. Compounds that have

critical solution temperatures within the temperature limits of this study are noted in the database. In some studies reporting solubility data for solid compounds, compositions are reported for multiple portions of the phase diagrams (i.e. ice compositions). These other phase data are input (or mentioned) in the database. However, only data that meet the definition above of x_i^{aq} for liquid water were analyzed in this study.

2.1.2 Solubility of Water in a Compound

The solubility of water in a compound is defined as the maximum amount of water that will dissolve in a given amount of compound at a specified temperature and pressure. This property is often measured in conjunction with x_i^{aq} (x_i^{aq} and the solubility of water in a compound are reported as mutual solubility values). Recommended values were not provided in this study because this is a low priority property, so a thorough literature review was not conducted for this property. This property is not as important as other properties when determining the environmental fate of hydrophobic compounds. However, data were briefly examined during the data analysis portion of this project. This is discussed in more detail in Section 4.1.

2.1.3 Henry's Law Constant

In environmental impact studies, Henry's law constants (k_{HS}) are used to determine the fate and transport of chemicals in air and water by determining volatilization or absorption tendencies [12, 13]. Chemicals with higher k_{HS} partition towards air while chemicals with lower k_{HS} partition towards water [1]. Additional uses of k_H values include the design and optimization

of air-stripping columns that remove contaminants from groundwater as well as applications in the pharmaceutical and food science industries [14].

The definition of k_H is given by

$$k_H \equiv \lim_{x_i \rightarrow 0} \frac{\hat{f}_i^L}{x_i}, \quad 2.1$$

where $\hat{f}_i^L$ is the partial fugacity of compound i in solution, and x_i is the mole fraction in the liquid phase [15-17]. This definition is applicable at all temperatures and pressures [16].

Application of l'Hôpital's rule equates k_H to the slope of the tangent line of the $\hat{f}_i^L$ versus x_i curve at $x_i = 0$. The equation of this tangent line produces Henry's law:

$$\hat{f}_i^L = x_i k_H. \quad 2.2$$

Most systems are represented by Henry's law for small values of x_i . The concentration at which system behavior deviates from Henry's law varies with every system, although an upper concentration of 0.01 mole fraction has been recommended [18, 19]. An activity coefficient can be added to correct for non-idealities caused by higher concentrations. Corrections can also be introduced for systems at high pressures. If ideal gas behavior is assumed, Equation 2.2 simplifies to

$$y_i P = x_i k_H, \quad 2.3$$

where y_i is the mole fraction in the vapor phase and P is the total pressure. In this study, k_H refers to aqueous systems.

The Gibbs energy of hydration, ΔG_{hyd}^{∞} , is defined as the process of transferring a molecule from an ideal gas at a reference pressure to a hypothetical infinitely dilute solution with a solute mole fraction of unity[20-22]. A relationship between k_H and ΔG_{hyd}^{∞} is

$$RT \ln \left(\frac{k_H}{P_{ref}} \right) = \Delta G_{hyd}^{\infty}, \quad 2.4$$

where P_{ref} is 100 kPa. Unit conversions may need to be included in Equation 2.4 depending on the units of the reference state. Temperature and pressure derivatives link ΔG_{hyd}^{∞} to additional properties often referred to as derivative properties. These derivative properties are the enthalpy of hydration (ΔH_{hyd}^{∞}), heat capacity of hydration ($\Delta C_{p_{hyd}}^{\infty}$), and partial molar volume (V^{∞}) at infinite dilution:

$$RT^2 \left(\frac{\partial \ln k_H}{\partial T} \right)_P = -\Delta H_{hyd}^{\infty}, \quad 2.5$$

$$\left(\frac{\partial}{\partial T} \left[RT^2 \frac{\partial \ln k_H}{\partial T} \right] \right)_P = -\Delta C_{p_{hyd}}^{\infty}, \quad 2.6$$

$$RT \left(\frac{\partial \ln k_H}{\partial P} \right)_T = V^{\infty}. \quad 2.7$$

Data for derivative properties were not included in this study. These relationships are mentioned because they are used in some prediction methods (discussed in Chapter 7).

Values of k_H first increase with temperature, reach a maximum typically between 373 and 473 K, and then decrease with temperature [22]. Most compounds do not have enough reported temperature-dependent data to see this whole trend. There are several different reported units for

k_H . Units include the ratio of partial pressure (P_i) and solute concentration in a dilute solution, P_i/x_i or P_i/C_i^L , where C_i^L is the concentration, in units such as moles per liter, in the liquid phase. The aqueous k_H is also reported as a ratio of the air and water concentrations making it unitless—either y_i/x_i or C_i^{air}/C_i^L where C_i^{air} is the concentration in the vapor phase. While both of these ratios are unitless, their numerical values can differ significantly [23]. A universal conversion between the various ratios is done assuming the vapor phase behaves as an ideal gas and that the solution is pure water. These assumptions are valid under environmental conditions.

The air-water partition coefficient is the mole fraction ratio of the solute in air to that in water, and by definition it is not limited by the requirement of infinite dilution. In the environmental field, it is often used synonymously with k_H despite different thermodynamic definitions [17, 24-26]. The conversion between the air-water partition coefficient and k_H is straightforward for dilute solutions at environmental conditions which is why the terms are often used synonymously. The conversion becomes more complex at high temperatures and pressures. In the database, any reported k_{HS} or air-water partition coefficients that are measured for environmental purposes were treated as experimental k_{HS} .

The k_H values measured for compounds that hydrate or dissociate in water are typically apparent k_H values. Hydration or dissociation can be accounted for in order to calculate an intrinsic k_H [23, 27]. No efforts were made to account for dissociation or hydration in this study. It is assumed that literature k_H values are apparent k_H values unless otherwise noted. In this study, k_H data for strong electrolytes were typically left unevaluated due to the spread in data partially caused by the variety of pH conditions reported in the literature.

2.1.4 Infinite Dilution Activity Coefficients: Chemical in Water

Activity coefficients are an indicator of the nonideal behavior of a mixture. The activity coefficient of species i (γ_i) is defined as

$$\gamma_i \equiv \frac{\hat{f}_i}{x_i f_i^L} \quad 2.8$$

where $\hat{f}_i$ is the partial fugacity of compound i in solution, f_i^L is the pure liquid fugacity of species i , and x_i is the mole fraction in the liquid phase. The infinite dilution activity coefficient (γ_i^∞) is the limiting value as x_i goes to zero. In this discussion, γ_i^∞ specifically refers to systems in which the solvent is water.

Thermodynamic relationships exist between γ_i^∞ and x_i^{aq} , k_H and the octanol-water partition coefficient [28]. Relationships between γ_i^∞ , x_i^{aq} , and k_H are given in Section 4.1. Environmental applications of γ_i^∞ 's include assessments of aquatic toxicity, leaching, soil and sediment transport properties, sorption, bioconcentration, and volatilization [29]. Other applications include modeling evaporation rates [12], choosing the most selective solvent for extraction [30], and developing constants in empirical equations for excess functions which are then used to predict fluid phase equilibrium.

2.1.5 Infinite Dilution Activity Coefficients: Water in Chemical

The discussion for γ_i^∞ applies here for the infinite dilution activity coefficient of water in a chemical, but the difference is that water is the solute and the chemical is the solvent. This property has low priority in this study because it is uncommon in environmental applications. Thorough literature and data analysis reviews have not been conducted in this study.

2.1.6 Octanol/Water Partition Coefficient

The octanol/water partition coefficient (KOW) is the equilibrium ratio of the concentrations of a chemical substance in *n*-octanol to that in water [1]. It is dimensionless and is often expressed as the base 10 logarithm. It can be used to predict the distribution of a chemical in living organisms (because octanol has a similar carbon to oxygen ratio as is found in lipid materials in animals [31]) or in the environment. For almost all non-ionic organic substances, KOW can be used to estimate other properties such as x_i^{aq} , soil/sediment absorption, biological absorption, bioaccumulation, and toxicity [1]. This property is temperature dependent [32]. KOW is a low priority property in this study since there are existing databases of KOW values [33, 34]. Data have been input when included in articles containing high priority properties. However, thorough literature and data analysis reviews have not been conducted.

2.2 Experimental Techniques

There are a variety of experimental methods that have been reported to measure the properties included in this study. Many of the methods can be used to obtain more than one property due to the relationships between properties (discussed in Section 4.1).

2.2.1 Solubilities

The methods to measure solubilities in which the solvent is a liquid are similar for all solute-solvent combinations. The solubility of water in a compound and x_i^{aq} are often measured in the same study and reported as mutual solubility data. Many solubility methods consist of equilibrating water with excess compound and analyzing the composition of one or both of the resulting phases. There are a variety of equilibration and analytical techniques employed in the

literature to accomplish this. The experimental methods are applicable over limited solubility ranges due to adsorption and emulsion problems and detection limitations, so the solubility needs to be predicted in order to choose the correct testing method. The methods are discussed in further detail in Appendix A.

2.2.2 Henry's Law Constants

Static or dynamic equilibration techniques can be used to measure k_H . Static equilibration involves the direct measurement of the air and/or water concentrations in a closed, equilibrated system. Methods include single equilibration, multiple equilibration, equilibrium partitioning in closed systems (EPICS), and variable headspace techniques. Static equilibration techniques are usually only done for higher concentrations due to the difficulty in measuring low concentrations in both phases [19]. Dynamic equilibration involves some type of air/water exchange. Two methods of this type are inert gas stripping and concurrent flow. Experimental methods are discussed in more detail along with limitations in Appendix A.

2.2.3 Infinite Dilution Activity Coefficients

There is overlap between the methods used to measure k_H and γ_i^∞ due to the thermodynamic relationship between these two properties (see Section 4.1). Many papers report both properties from the same experimental data. Often multiple techniques are required to measure values over a range of temperatures due to measurement limitations. It is ideal to apply more than one measurement technique to ensure consistency in results [35]. Experimental methods are discussed in more detail along with limitations in Appendix A.

2.2.4 Octanol/Water Partition Coefficients

The methods used to measure KOWs are similar to some of the methods used to measure x_i^{aq} . These methods include equilibrating a compound with 1-octanol and water and determining the concentration using an analytical technique. Another common method involves estimating the KOW based on gas chromatography (GC) retention data of the compound and reference compounds. Each method covers a range of values, so KOW should be predicted in order to determine the proper testing technique. Experimental methods are discussed in more detail along with limitations in Appendix A.

3 Database Overview

3.1 Background

It was decided at the May 2010 DIPPR[®] meeting by BYU staff and DIPPR[®] sponsors to build a new database containing the properties outlined in Chapter 2 because more work would be involved in fixing the ESP database than in starting fresh. In the original ESP database, data for the properties outlined in Chapter 2 were only included at 298.15 K, and the properties were included in the constant values tables. It was decided to include temperature-dependent data in addition to the constant values (defined at 298.15 K) in the new database. Due to the notation allowed in Microsoft Access[®], the properties had to be represented by letter abbreviations instead of symbols. Separate abbreviations were required for the constant value and temperature-dependent properties. The abbreviations are given in Table 3.1. The symbols k_H , x_i^{aq} , and γ_i^∞ will continue to be used throughout this document in the text and plots because they are more intuitive.

Table 3.1: Abbreviations used in 801E database

Property	Constant Value	Temperature-Dependent
Infinite dilution activity coefficient of compound in water	ACCW	ACCWT
Infinite dilution activity coefficient of water in compound	ACWC	ACWCT
Aqueous Henry's law constant	HLC	HLCT
Octanol/water partition coefficient	KOW	KOWT
Solubility of compound in water	SOLC	SOLCT
Solubility of water in a compound	SOLW	SOLWT

3.2 Database Structure and Data Entry

This new database, DIPPR[®] 801E, has a similar format to the 801 database so that it is compatible with DIADEM, the DIPPR[®] 801 software interface. In order to make the process of data entry easier while reducing the opportunities for entry error, DIADEM was modified slightly. Units were added so that they can be selected from dropdown menus. Due to assumptions in the unit conversions, two columns were added to the database so that it stores the original values and units in the database after the data for each property are converted to the default unit. There is not a way to view these original values or perform more complex unit conversions using DIADEM, but a future user of the database could look up the original value and perform a more exact unit conversion if desired.

Data were considered constant values or temperature-dependent. The constant values properties are defined at 298.15 K and 1 atm. Experimental values within ± 5 degrees of 298.15 K were included in the constant values table. The actual temperature and pressure were included in a note if they differed from 298.15 K or 1 atm, respectively. All values at 298.15 K were also input as temperature-dependent data in addition to data at other temperatures. There were instances when articles included measurements for some chemicals only at temperatures outside of the constant value range. In these instances, the data were only entered in the temperature-dependent table. If data of non-dissociating compounds were reported as a function of pH, only the values at $\text{pH} \approx 7$ were analyzed. When data were input, they were labeled as “Unevaluated”. As datasets were evaluated, the label was changed to “Accepted”, “Not Used”, or “Rejected” based on the quality of the dataset. These categories allow a future user to know which data were considered during the evaluation process.

At the conclusion of this project, recommended (“Accepted”) values have been determined for at least one property for 509 compounds. There are entries for over 800 compounds due to the data entry process; however, not all of the data have been evaluated. Data were entered by reference, and all data for any DIPPR[®] 801 chemical mentioned in a reference were included. Table 3.2 summarizes the number of data points that have been considered during evaluation and the number of compounds with recommended values for each property. Due to the lack of reliable data, not all of the high priority properties were analyzed for each compound which is why each property has a different number of compounds with recommendations. Based on the availability of temperature-dependent data, some compounds have recommendations for a temperature range while other compounds only have recommendations for a single temperature (typically 298.15 K). Table 3.3 summarizes the total number of evaluated compounds per chemical family. The entry numbers given in Table 3.2 reflect the actual entries retrieved directly from the literature. They do not include the derived entries that were added during the data evaluation process of this study (i.e. k_H values derived from x_i^{aq} data, etc.). The compound totals include the American Petroleum Institute (API) recommendations that were accepted in this study (discussed in Section 4.5). The data evaluation process is discussed in detail in Chapter 4.

Table 3.2: Summary of 801E entries

Property	Evaluated Data Points	Compounds with Recommendations
γ_i^∞	1350	407
k_H	3950	445
x_i^{aq}	7617	481
Overall	12917	509

Table 3.3: Summary of the number of evaluated compounds (#) per chemical family

Family	#	Family	#	Family	#
1-Alkenes	14	Dialkenes	2	Other Alkanes	2
2,3,4-Alkenes	4	Dicarboxylic Acids	3	Other Alkylbenzenes	10
Acetates	17	Dimethylalkanes	7	Other Amines, Imines	7
Aldehydes	19	Diphenyl/Polyaromatics	1	Other Condensed Rings	11
Aliphatic Ethers	13	Elements	6	Other Ethers/Diethers	7
Alkylcyclohexanes	4	Epoxides	6	Other Inorganics	1
Alkylcyclopentanes	1	Formates	7	Other Monoaromatics	1
Alkynes	8	Inorganic Bases	1	Other Polyfunctional C, H, O	6
Aromatic Alcohols	14	Inorganic Gases	5	Other Polyfunctional Organics	2
Aromatic Amines	16	Ketones	26	Other Saturated Aliphatic Esters	6
Aromatic Carboxylic Acids	4	Mercaptans	5	Peroxides	2
Aromatic Chlorides	11	Methylalkanes	8	Polyfunctional Acids	1
Aromatic Esters	9	Methylalkenes	2	Polyfunctional Amides/Amines	7
C, H, Br Compounds	14	<i>n</i> -Alcohols	18	Polyfunctional C, H, N, Halide, (O)	2
C, H, F Compounds	8	<i>n</i> -Aliphatic Primary Amines	8	Polyfunctional C, H, O, Halide	6
C, H, I Compounds	7	<i>n</i> -Alkanes	14	Polyfunctional C, H, O, N	3
C, H, Multihalogen Compounds	3	<i>n</i> -Alkylbenzenes	11	Polyfunctional C, H, O, S	1
C, H, NO ₂ Compounds	17	Naphthalenes	4	Polyfunctional Esters	4
C1/C2 Aliphatic Chlorides	19	Nitriles	12	Polyols	3
C3 & Higher Aliphatic Chlorides	11	Nitroamines	3	Propionates and Butyrates	13
Cycloaliphatic Alcohols	1	Organic Salts	2	Sulfides/Thiophenes	6
Cycloalkanes	5	Other Aliphatic Alcohols	21	Unsaturated Aliphatic Esters	4
Cycloalkenes	4	Other Aliphatic Amines	9		

Data were entered with the original values and units to avoid any errors from external conversions. The only exceptions were when articles only report data as plots or regressions of experimental values. *DataThief* [36] was used to extract values from plots of experimental data. When authors only reported temperature-dependent correlations of their experimental data, the

correlations were used to calculate values at the measurement temperatures reported by the authors. These values were classified as “Smoothed” in the database. Some authors report multiple values at the same temperature. The average of these values was input in order to not give extra weight to the dataset during analysis.

The focus of this database was on primary, experimental data. Due to time constraints, an additional focus was on post 1950 English references. It was assumed that more recent data are generally of higher quality than older data due to improvements in analytical techniques and higher material purities. Many of the older and/or foreign articles were not readily available, and the foreign articles required translating. Due to time constraints, it was not realistic for all of these types of articles to be included. Data from these articles were selectively input based on the availability of data from more recent English references. Data found in handbooks would be interesting to include in the database to compare with primary data, but these references were also excluded due to ambiguities and the time required for data entry. The original sources of handbook data (experimental versus predicted) are often excluded or unclear making it difficult to determine the reliability of the reported values.

Guidelines were developed that aided in data classification. These guidelines explain how to label data (i.e., classifying data as “experimental” or “predicted”). Issues that were encountered during the data entry process were clarified to establish consistency in the database. The guidelines also include assumptions and limitations for the various measurement techniques which was useful during the data analysis process.

3.3 Unit Conversions

In order to compare values from different references, all literature values needed to be converted to the same units. The unit conversions for the temperature-dependent properties accounted for the temperature-dependence of the conversion. For example, some x_i^{aq} and k_H conversions required liquid densities. When the temperature-dependent data were converted, the liquid densities at the correct temperature were used. There were additional assumptions with some of the unit conversions. These assumptions in addition to the default units are listed in Table 3.4. The assumptions for the k_H unit conversions are valid since the data included in the database were typically measured at atmospheric pressure using dilute solutions. Many of the solubility data were reported in units that did not require any assumptions in order to perform the conversion to the default unit. When assumptions were required, it was assumed that volumes were additive. This assumption was generally only needed for compounds with small solubilities. Any volume changes due to non-ideality would be negligible. Initially only liquid or theoretical liquid densities were used for the solubility conversions. In some cases extrapolating the liquid density curve to temperatures where the compound is a solid led to obviously erroneous values (gave negative solubility values). Therefore, the volume of the liquid or solid solute was calculated using the liquid or solid densities, respectively, when using additive volumes for solubility conversions. If the solute was a gas, it was assumed that the added volume was negligible.

Table 3.4: Unit conversion assumptions

Property	Unit Conversion Assumptions	Default Units
HLC	• Ideal gas with pressure of 1 atm • Solution treated as pure water (since infinitely dilute)	kPa·mol/mol
SOLW/SOLC	• Additive volumes if solute is a solid or liquid • Solution considered pure solvent if solute is gas • Solution considered pure solvent if solute is a solid that does not have a solid density equation (i.e. 1-heptadecanol) • Conversions treated as per amount of solution (i.e. g/100g solution) unless authors state otherwise (i.e. g/100g H₂O)	mol frac.
ACCW/ACWC	None	unitless
KOW	None	Log KOW

4 Data Evaluation

Because of the thermodynamic relationships between k_H , x_i^{aq} , and γ_i^∞ , reported values of these different properties can be conveniently compared. In this study, the combined data from the three properties were often used to obtain temperature regressions and recommended values for each property. For compounds where the assumptions of the thermodynamic relationships were not met (meaning the data from all the properties could not be combined), these relationships still helped identify outliers in reported data making it easier to designate recommended values. Notes in the database indicate the data that were combined to obtain regressions for each compound and property. In addition, experimental factors, temperature-dependent trends, chemical family plots, and literature recommendations were also used to evaluate data. All of these are discussed in further detail.

4.1 Property Relationships

A relationship between k_H and x_i^{aq} is given by evaluating equation 2.3 at the solubility limit

$$k_H = \frac{P_i^{sat}}{x_i^{aq}} \quad 4.1$$

where P_i^{sat} is the vapor pressure of solute i . This relationship is valid for hydrophobic chemicals with x_i^{aq} values below approximately 0.001 mole fraction. It assumes that the solubility of water

in the compound (SOLC) is negligible so that the partial vapor pressure of solute is unaffected by the presence of water allowing it to be treated as P_i^{sat} [12, 19, 23]. The assumptions break down when SOLC exceeds approximately 0.1 mole fraction. In addition, x_i^{aq} and P_i^{sat} must reference the same physical state [37]. In this study, a pressure of 101.325 kPa was assumed for gases in order to convert k_H to x_i^{aq} values and vice versa. For gases, k_H and x_i^{aq} are used interchangeably. In the literature, k_H values calculated from Equation 4.1 are usually reported as experimental.

A relationship between k_H and γ_i^∞ is obtained by substituting the definition of γ_i (Equation 2.8) into that of k_H (Equation 2.1),

$$k_H = \gamma_i^\infty f_i^L, \quad 4.2$$

where f_i^L is the fugacity of pure species i in the liquid phase, and x_i is the mole fraction in the liquid phase. At atmospheric conditions for liquid solutes, this simplifies to:

$$k_H = \gamma_i^\infty P_i^{sat}. \quad 4.3$$

This relationship is more thermodynamically rigorous than Equation 4.1. Combining Equations 4.1 and 4.3 yields a relationship between γ_i^∞ and x_i^{aq} for hydrophobic liquids:

$$\gamma_i^\infty = \frac{1}{x_i^{aq}}. \quad 4.4$$

The same assumptions that make Equation 4.1 valid also apply for Equation 4.4. Equation 4.4 is valid for γ_i^∞ values greater than 1000 [38]. When available, SOLC data were examined during the data analysis procedure to check the validity of Equations 4.1 and 4.4. It is possible to use both SOLC and x_i^{aq} to obtain γ_i^∞ when the assumptions of Equation 4.4 are violated [23, 39]. However, since SOLC data were not critically evaluated, γ_i^∞ and k_H values were not determined from x_i^{aq} when SOLC was above approximately 0.1 mole fraction. All of the valid ranges of the

thermodynamic relationships discussed in this section are approximate. The thermodynamic relationships were applied outside of these ranges if chemical and/or chemical family data indicated a different range of applicability.

When the solute is a solid or a gas, it is necessary to determine a hypothetical f_i^L in order to relate k_H and γ_i^∞ . For compounds that are solids at the specified temperature (so only the solid P_i^{sat} is known), a fugacity ratio must be used to account for the difference in states. The fugacity ratio, Ψ , of species i is defined as the ratio of the solid and liquid fugacities at the system temperature, T , and pressure, P :

$$\Psi_i \equiv \frac{f_i^S(T, P)}{f_i^L(T, P)}. \quad 4.5$$

Substituting Equation 4.5 into 4.2 yields

$$k_H = \gamma_i^\infty \frac{P_i^{sat}}{\Psi}, \quad 4.6$$

which assumes that f_i^S is equal to the solid P_i^{sat} . Combining Equations 4.1 and 4.6 provides a relationship between γ_i^∞ and x_i^{aq} for solids:

$$\gamma_i^\infty = \frac{\Psi}{x_i^{aq}}. \quad 4.7$$

At the melting point (Tm_i), the solid and liquid fugacities are equal in both phases:

$$\frac{f_i^S(Tm_i, P)}{f_i^L(Tm_i, P)} = 1. \quad 4.8$$

Multiplying Equation 4.5 by 4.8 leads to

$$\Psi_i = \frac{f_i^S(T, P)}{f_i^S(Tm_i, P)} \frac{f_i^L(Tm_i, P)}{f_i^L(T, P)}. \quad 4.9$$

The residual Gibbs energy, G_i^R , is related to f and P by

$$G_i^R = RT \ln \left(\frac{f_i}{P} \right). \quad 4.10$$

Rearranging Equation 4.10 and using partial derivatives leads to

$$\left(\frac{\partial \ln(f_i)}{\partial T} \right)_P = \left[\frac{\partial \left(\frac{G_i^R}{RT} \right)}{\partial T} \right]_P = -\frac{H_i^R}{RT^2} \quad 4.11$$

where H_i^R is the residual enthalpy. To integrate for a phase from Tm to T at constant P gives

$$\frac{f_i(T, P)}{f_i(Tm_i, P)} = \exp \int_{Tm_i}^T -\frac{H_i^R}{RT^2} dT. \quad 4.12$$

This is applied to both the solid and liquid ratios in Equation 4.9:

$$\Psi_i = \frac{\exp \int_{Tm_i}^T -\frac{H_i^{R,S}}{RT^2} dT}{\exp \int_{Tm_i}^T -\frac{H_i^{R,L}}{RT^2} dT} = \exp \int_{Tm_i}^T -\frac{(H_i^{R,S} - H_i^{R,L})}{RT^2} dT. \quad 4.13$$

Using the definition of residuals yields

$$-(H_i^{R,S} - H_i^{R,L}) = -[(H_i^S - H_i^{IG}) - (H_i^L - H_i^{IG})] = H_i^L - H_i^S \quad 4.14$$

where IG refers to the ideal gas property. The exact expression for Ψ is

$$\Psi_i = \exp \int_{Tm_i}^T \frac{H_i^L - H_i^S}{RT^2} dT . \quad 4.15$$

The temperature dependence of H can be described using

$$H_i^L(T) - H_i^S(T) = \Delta H_{f,Tm} + \int_{Tm_i}^T Cp_i^L dT - \int_{Tm_i}^T Cp_i^S dT \quad 4.16$$

where $\Delta H_{f,Tm}$ is the heat of fusion at the melting point and Cp_i is the constant pressure heat capacity.

In the literature, the fugacity ratio is often approximated as

$$\Psi \cong \exp \left(\frac{\Delta H_{f,Tm}}{RT} \left(\frac{T}{Tm_i} - 1 \right) \right) = \exp \left(\frac{\Delta S_{f,Tm}}{R} \left(1 - \frac{T}{Tm_i} \right) \right) \quad 4.17$$

where $\Delta S_{f,Tm}$ is the entropy of fusion at the melting point [40, 41]. In many instances $\Delta S_{f,Tm}$ is estimated to be a constant value, 56.5 J/(mol·K), for all solid compounds. Van Noort found that using this estimated value for $\Delta S_{f,Tm}$ led to substantial random errors for some aromatic compounds [40].

In this study, Equations 4.15 and 4.16 were used with DIPPR[®] 801 property values to calculate Ψ s. When Cp_i data were not available, Equation 4.17 was used with DIPPR[®] 801 $\Delta H_{f,Tm}$ values. The Cp_i values in DIPPR[®] 801 are at saturation pressure above the normal boiling point (NBP). Below the NBP, saturated heat capacity and constant pressure heat capacity are used interchangeably. For most of the solid compounds included in this study, the temperature range of data did not extend beyond the NBP. Equation 4.16 requires Cp_i^L values at temperatures below Tm_i which cannot be measured. In order to use this equation, the DIPPR[®] 801 Cp_i^L equations had to be extrapolated below Tm_i which resulted in hypothetical values. The equation

used for the error estimation of Ψ was determined using propagation of error, and the derivation is included in Appendix B. Fugacity Error Propagation. Many of the recommended solid γ_i^∞ values from this study have high uncertainties due to the uncertainty in Ψ .

For compounds that are gases, estimating a hypothetical f_i^L can be difficult for temperatures above the compound critical temperature [42]. Due to this difficulty, γ_i^∞ values were not computed for any compound that is a gas at 1 atm and 298.15 K. This limit was chosen in order to simplify the data analysis process even though 298.15 K is not above the critical temperature for every gaseous compound. The total number of gaseous compounds analyzed in the database is small compared to compounds that are solids or liquids. Table 4.1 summarizes the number of compounds with recommended values (determined in this study) that are gases, liquids, or solids at 298.15 K and 1 atm. A summary of the property relationships used in this study is shown in Table 4.2 by the physical state of the solute.

Table 4.1: Number of compounds with recommended values per physical state at 298.15 K

Phase at 298.15 K	Number of Compounds with Recommended Values
Gas	39
Liquid	393
Solid	77

Table 4.2: Summary of property relationships used in this study

Solid	$k_H = \frac{P_i^{sat}}{x_i^{aq}}$	$k_H = \gamma_i^\infty \frac{P_i^{sat}}{\Psi}$	$\gamma_i^\infty = \frac{\Psi}{x_i^{aq}}$
Liquid	$k_H = \frac{P_i^{sat}}{x_i^{aq}}$	$k_H = \gamma_i^\infty P_i^{sat}$	$\gamma_i^\infty = \frac{1}{x_i^{aq}}$
Gas	$k_H = \frac{101.325 \text{ kPa}}{x_i^{aq}}$	-	-

As was mentioned, k_H and γ_i^∞ were analyzed together and x_i^{aq} analyzed separately for compounds where Equations 4.1 and 4.4 are not valid. As an example, Figure 4.1 is a plot of k_H , x_i^{aq} , and γ_i^∞ data for 2-methyl-1-butanol plotted as $\ln(k_H)$ versus inverse temperature. Literature x_i^{aq} and γ_i^∞ data were converted to k_H using Equations 4.1 and 4.3. Although SOLC is significant (reported SOLC values range between 0.325-0.553 mole fraction over the temperature range of 293.15-303.15 K [43]), it is difficult to see a difference between the curves obtained from using only k_H and γ_i^∞ values and the curve that includes x_i^{aq} data. The difference is more pronounced when all of the data are converted to $\ln(x_i^{aq})$ (Figure 4.2). Although all of the properties cannot be combined to obtain a regression, plotting the properties together aids in the data evaluation process because the data still follow similar temperature trends. This is especially useful when data for a property are only available from a single reference.

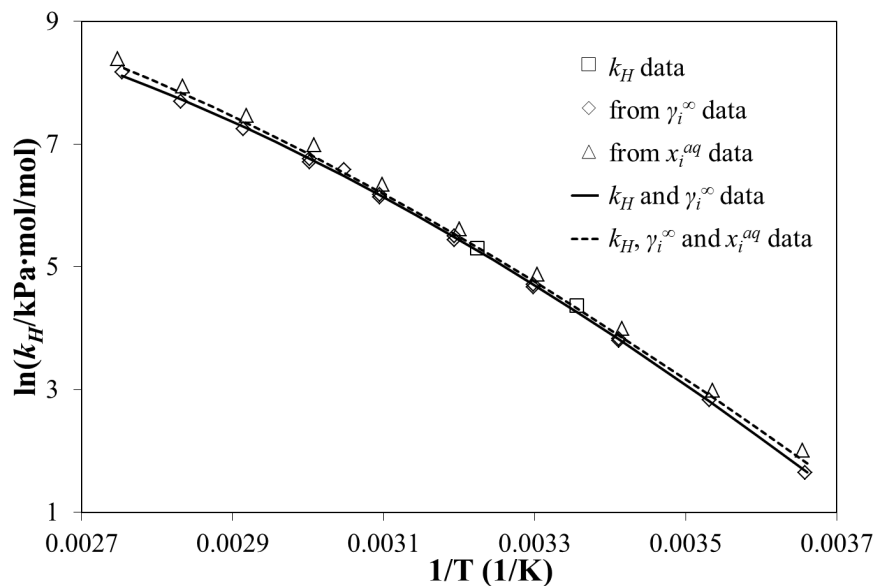


Figure 4.1: 2-Methyl-1-butanol k_H , x_i^{aq} , and γ_i^∞ data plotted as $\ln(k_H)$ versus inverse temperature. References: [44-48]

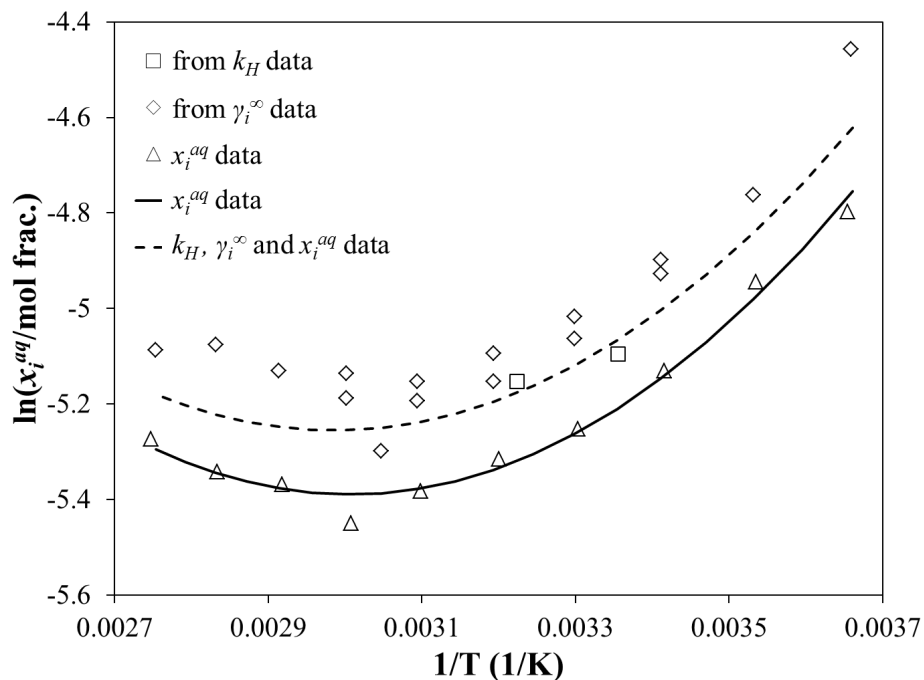


Figure 4.2: 2-Methyl-1-butanol k_H , x_i^{aq} , and γ_i^∞ data plotted as $\ln(x_i^{aq})$ versus inverse temperature
References: [44-48]

As was mentioned in Section 3.2, not all of the properties were analyzed for every compound. Some compounds did not have data for every property, and the assumptions used to relate the properties were not valid. For example, temperature-dependent x_i^{aq} data were available for many aromatic alcohols but no literature k_H or γ_i^∞ data were found. Due to the high x_i^{aq} and high SOLC, Equations 4.1 and 4.4 are not valid, so k_H and γ_i^∞ recommendations were not determined.

4.2 Experimental Factors

Experimental factors that were considered when choosing between conflicting datasets include accuracy of experimental method, reported chemical purity, reported experimental error, and agreement with other references. For example, adsorption problems have been noted for

aqueous systems using the gas-liquid chromatography (GLC) method [49]. If GLC values are inconsistent with other values obtained using different methods, then the GLC dataset was rejected. In addition, if an author reported a dataset as having a large uncertainty and the dataset does not match up with other data, then the dataset with the high reported experimental error was rejected. Experimental limitations are included with experimental descriptions in Appendix B.

As an example, Figure 4.3 gives a plot of experimental x_i^{aq} data for vanillin reported in the literature. The data from one of the references, d, does not match up with the data from the other references even though a reasonable experimental method was used with a low reported experimental uncertainty. Because the other references are consistent with each other, the data from reference d were rejected.

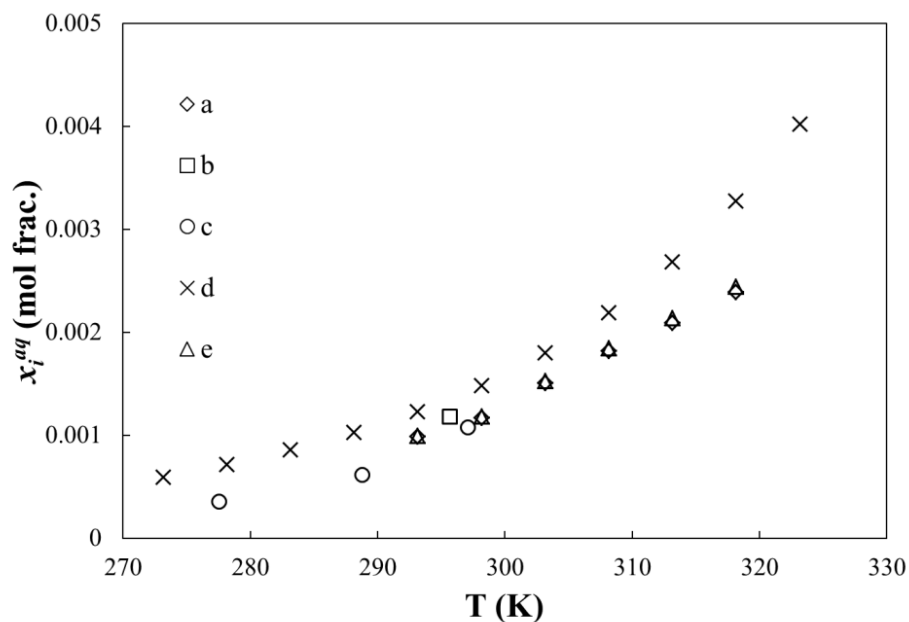


Figure 4.3: Literature x_i^{aq} data for vanillin. References: a [50], b [51], c [52], d [53], e [54]

4.3 Temperature-Dependent Evaluation

When temperature-dependent data were available, correlations modeling the temperature-dependence of k_H , x_i^{aq} , and γ_i^∞ were regressed. This section outlines a general procedure that was used to select the data that were suitable for the regressions. A Microsoft Excel[®] workbook was made to aid with data evaluation of these high priority properties. Temperature-dependent data were extracted from the database for an individual compound. A macro converted each of the values to k_H and γ_i^∞ using P_i^{sat} from the DIPPR[®] 801 database. Plots were generated with all of the data to easily identify outliers. Any outliers were checked to ensure that the value was input correctly and that there was not a mistake with the unit conversion. At this stage, data were classified as “Accepted”, “Rejected”, or “Not Used”. Criteria discussed in Sections 4.1 and 4.2 were used to determine this data classification.

The data were then checked for “doubles”. Some published papers report values for more than one property from the same experiment by using the property relationships discussed in Section 4.1. For example, k_H and γ_i^∞ values are both reported in some papers from the same experimental data. For a paper with both of these properties, converting k_H values to γ_i^∞ (or vice versa) would have given double weight to these values during data analysis since they were already reported as the other property. In addition, there are some instances where data have been doubly published. It was assumed that identical datasets reported by the same author in different papers are duplicates. Only one dataset of duplicate sets was included when determining regression coefficients, but both datasets were labeled as “Accepted”. Once the doubles were identified, data were sorted to be input into a Mathcad[®] file. One last check was performed to ensure that all of the data points from the same dataset had the same classification. Datasets cannot have some points that are “Accepted” and others that are “Rejected”. If there

was a special case where there was a valid reason for rejecting some points from a dataset while accepting others (for example, the authors reported large errors at low temperatures), then the dataset had to be split into two separate datasets.

The classified data were then put into a template that was made using Mathcad[®]. Other input values required were pure-component properties available in DIPPR[®] 801. The user designated which data were to be used to determine recommended temperature-dependent regressions of all of the properties (i.e. x_i^{aq} , k_H , and γ_i^∞ data were all included in the k_H regression; only k_H and γ_i^∞ data were included in the k_H regression; etc.).

The equations used to model temperature-dependence are given in Table 4.3 along with the corresponding DIPPR[®] equation numbers. DIPPR[®] Equations 5000 and 6000 were added specifically for this project for compounds with x_i^{aq} data that do not conform to DIPPR[®] Equation 101. Various Mathcad[®] files were available that compared different regressions for compounds over the temperature range of the data. The equation comparisons available in the files are summarized in Table 4.4. The analyzer used F-test p-values to determine if regressed parameters were statistically significant (p-values < 0.05 indicated that an additional term was significant) when comparing equations that only differed by a single term. When comparing DIPPR[®] Equations 5000 and 6000 (case 3 for $\ln(x_i^{aq})$ in Table 4.4), the best equation was selected visually or based on R^2 values. Most of the compounds that deviate from DIPPR[®] Equation 101 have upper or lower critical solution temperatures within the temperature range of data in the database. These new equations are simple and generic resulting in a wide range of applicability. With all of the regressions, the best equation was used even if it did not fit the data perfectly. The equations are meant to be generic, so terms were not modified on a compound basis.

Table 4.3: DIPPR[®] 801 equation numbers

DIPPR [®] Equation Number	Equation ^a
101	$\ln(Y) = A + \frac{B}{T} + C \ln(T) + DT^E$
5000	$\ln(Y) = A + \frac{B}{T} + C \ln(T) + D \cdot \exp\left(1 - \frac{T}{E}\right)$
6000	$\ln(Y) = A + \frac{B}{T} + C \cdot T + D \left(1 - \frac{T}{E}\right)^{\frac{1}{3}}$

^a Y represents the property and A, B, C, D, and E are compound-specific coefficients

Table 4.4: Equation comparisons used in data analysis

Property	Equation Comparisons
$\ln(k_H)$ & $\ln(\gamma_i^\infty)$	1) $A + \frac{B}{T} + C \ln(T) + DT$ and $A + \frac{B}{T} + C \ln(T)$ 2) $A + \frac{B}{T} + C \ln(T)$ and $A + \frac{B}{T}$
$\ln(x_i^{aq})$	1) $A + \frac{B}{T} + C \ln(T) + \frac{D}{T^2}$ and $A + \frac{B}{T} + \frac{D}{T^2}$ 2) $A + \frac{B}{T} + \frac{D}{T^2}$ and $A + \frac{B}{T}$ 3) $A + \frac{B}{T} + C \ln(T) + D \cdot \exp\left(1 - \frac{T}{E}\right)$ and $A + \frac{B}{T} + C \cdot T + D \left(1 - \frac{T}{E}\right)^{\frac{1}{3}}$

The γ_i^∞ regressions were derived from the k_H regressions and not directly from the γ_i^∞ data due to the scatter in γ_i^∞ data. If the regressions were obtained directly from the γ_i^∞ data, curvature often would not be significant due to the scatter in the data. In addition, compounds generally have more k_H data than γ_i^∞ data. The γ_i^∞ regressions are derived from k_H instead of x_i^{aq} because the relationship between γ_i^∞ and k_H (Equation 4.3) is more thermodynamically rigorous than the relationship between γ_i^∞ and x_i^{aq} (Equation 4.4). For solid compounds, Ψ s were used to convert the k_H regressions to the γ_i^∞ regressions. Since the γ_i^∞ curves were derived from the k_H

regressions, there were really two γ_i^∞ curves for every compound that had data above and below the melting point. For these compounds, a single curve was fit to the γ_i^∞ values so that one equation covered the entire temperature range.

A temperature-dependent regression was classified as “Experimental” if experimental data from any of the properties were used and the property relationship assumptions discussed in Section 4.1 were valid. As was mentioned in Section 4.1, experimental SOLC values were checked when available. If SOLC values were significant at higher temperatures ($> \sim 0.1$ mole fraction), then only lower temperature x_i^{aq} data were used to obtain the other properties. For example, with 3,5-xylene, only k_H and γ_i^∞ data were available at high temperatures. Six x_i^{aq} values at temperatures of 313 K and below were combined with the k_H and γ_i^∞ data in order to extend the temperature ranges of the k_H and γ_i^∞ regressions. At these low temperatures for this compound, x_i^{aq} and SOLC are small so the assumptions required to use Equations 4.1 and 4.4 are valid. When SOLC data were not available for some compounds, family trends were used to estimate the validity of the assumptions required to use Equations 4.1 and 4.4 in order to determine the classifications of the regressions. If data were available that fell within reasonable limits of the assumptions used for the thermodynamic relationships, then the temperature dependent regressions were labeled as “Smoothed”. If there was any doubt concerning the validity of the assumptions, the regressions were labeled as “Predicted” or “Exp&Pred” (Experimental and Predicted) depending on the amount of data available for the different properties.

Recommended constant values were determined for the three properties from the temperature-dependent regressions evaluated at 298.15 K. These values were entered as new

constant values and classified as “Smoothed” or “Predicted” depending on if the temperature-dependent regression was classified as “Smoothed” or “Predicted”, respectively.

At the end of the Mathcad[®] sheets, there were multiple output files linked to Microsoft Excel[®]. Information determined from the data analysis process (i.e. acceptance, temperature-dependent coefficients, recommended constant values, assigned errors for literature datasets, etc.) were exported to Microsoft Excel[®]. The Microsoft Excel[®] file was then used to directly update the Microsoft Access[®] database. This procedure was different than the update procedure used in DIPPR[®] 801. With the DIPPR[®] 801 procedure, values are overwritten in the database with the updated information. However, with the 801E database, the information was updated and not overwritten in order to save the original values and units.

All of the converted data used to obtain the regressions were added to the database and labeled as “Derived”. The notes contain the references from which they were derived. This allows a user to view all of the values that were used to obtain the regression including values that were calculated from other properties.

4.3.1 Error Calculations

Linear regression statistics and propagation of error were used to determine the uncertainty of the temperature-dependent regressions. Table 4.5 summarizes the error analyses used in this study. The γ_i^∞ regressions and resulting errors were derived from the k_H regressions. The confidence intervals were initially determined on a natural log scale, but they were converted to a normal scale in order to determine uncertainty assignments. In some instances, the uncertainty assignments were adjusted based on additional factors. Examples of this are given later in this section. All of the regressions assumed constant variance. In reality, some

data are more reliable than others. Efforts were made to only include the best data in each regression so that the assumption of constant variance was valid.

Table 4.5: Summary of uncertainty determinations for temperature-dependent regressions

Property	Regression	Error Analysis	Assumptions
k_H	Least squares	95% confidence intervals of the mean (determined by regression)	Constant variance
γ_i^∞	Thermodynamically derived from k_H , P_i^{sat} , and Ψ	Propagation of error	Constant variance
x_i^{aq}	Least squares	95% confidence intervals of the mean (determined by regression)	Constant variance

Confidence intervals determined in this manner are not constant over the entire temperature range. Confidence intervals are larger at the minimum and maximum temperatures. In addition, for compounds that cover solid and liquid regions, the uncertainty in γ_i^∞ tends to be larger over the solid region versus the liquid region due to uncertainty in Ψ . In the DIPPR[®] 801 database, confidence intervals are assigned to the entire regression. To be consistent with the 801 database format, single uncertainties were assigned. This was accomplished by a weighted average. First, the average absolute upper and lower percent uncertainties, s , were calculated at every temperature for each experimental data point, j ,

$$s_j = \frac{|\%Error_{up}(T_j)| + |\%Error_{down}(T_j)|}{2}, \quad 4.18$$

where $\%Error$ is the percent difference between the 95% confidence interval of the mean and the regression value. If there were N points at a single temperature, then the uncertainty was calculated N times. These resulting uncertainties were then averaged,

$$s = \frac{\sum_j s_j}{N_{total}}, \quad 4.19$$

where N_{total} is the total number of experimental points used to obtain the regression. This resulted in greater weight for the temperatures where there were more experimental values. These weighted absolute uncertainties were then converted to DIPPR[®] 801 uncertainty ranges. The DIPPR[®] uncertainty options are < 0.2%, < 1%, < 3%, < 5%, < 10%, < 25%, < 50%, < 100%, and > 100%. As an example, if $s = 2.7\%$, the uncertainty for the regression was labeled as < 3%.

When data were only available for one property, uncertainties were initially calculated as outlined in Table 4.5. The uncertainties of the derived property regressions were then compared to the error assigned to the regression of the original data. The derived property uncertainty was kept if it was greater than or equal to the error of the original property data. If it was smaller, it was changed to match the uncertainty of the original property. For example, if k_H and γ_i^∞ regressions were determined from x_i^{aq} data and the uncertainty assigned to the x_i^{aq} regression was < 25%, the smallest uncertainty assigned to the k_H and γ_i^∞ regressions was < 25%.

When data were only available from one reference, an uncertainty was calculated from the regression, and this was compared to the reported author uncertainty. The largest uncertainty of the two was reported in the database. For example, with ethyl mercaptan the uncertainty obtained from the k_H regression was < 5%. However, the uncertainty reported by the author for the dataset was < 10%, so the uncertainty of the regression was changed to < 10%.

Uncertainties were assigned to the individual datasets based on the regression uncertainty. Figure 4.4 shows an example of the error calculation for an experimental data point. Each data point is at the same temperature. The points are spread along the x-axis for viewing

purposes. The correlation value and uncertainty are shown along with an experimental value (shown twice). The uncertainty of the experimental value can be calculated in two different ways as shown in the figure. The uncertainty can be taken as the difference between the experimental and correlated values or the difference between the experimental value and uncertainty limit of the correlated value. The latter calculation was used in this study. The absolute uncertainties of all of the data points within a dataset were averaged to assign an overall uncertainty to the dataset. The lowest possible uncertainty for a dataset was defined as being the uncertainty assigned to the regression. For example, if a regression was assigned an uncertainty of $< 10\%$ and the difference between the experimental data and regression values is 3% , the uncertainty of the dataset was labeled as $< 10\%$.

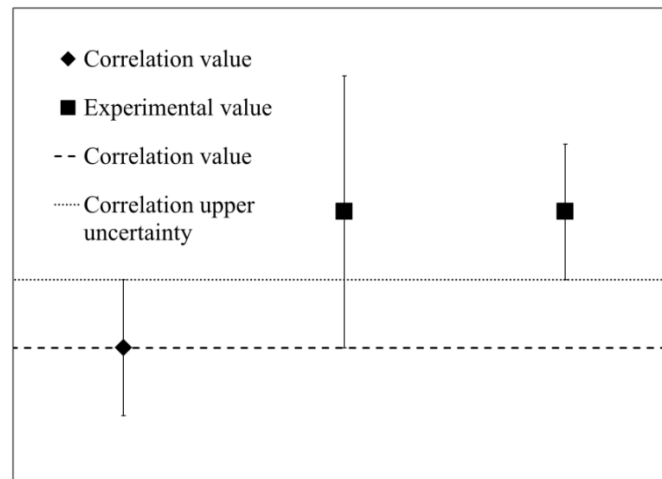


Figure 4.4: Uncertainty determination of experimental datasets

4.3.2 Critical Solution Temperatures

This section discusses compounds with upper or lower critical solution temperatures. Table 4.6 gives the total number of compounds per chemical family with critical solution behavior that were evaluated in this study. For the compounds with critical solution behavior, the actual critical temperatures were not determined. The determination of the actual critical point would involve the evaluation of x_i^{aq} and SOLC data which was beyond the scope of this study. Term “E” in DIPPR[®] Equations 5000 and 6000 (see Table 4.3) was assigned a value (an approximate critical temperature) based on the available x_i^{aq} data. This allowed the remaining coefficients to be found using linear regression. Values should be extrapolated from temperature-dependent equations with caution for all properties discussed in this paper. Errors can result from extrapolation, especially for x_i^{aq} . Some of the more interesting behaviors encountered in this study are discussed below.

Table 4.6: Chemical families with evaluated compounds exhibiting critical solution behavior

Chemical Family: Number Evaluated Compounds/Type of Critical Solution Behavior ^a	
Aldehydes: 1/UCST	Nitriles: 2/UCST
Aromatic Alcohols: 12/UCST	Nitroamines: 1/UCST
Aromatic Amines: 2/LCST	Other Aliphatic Alcohols: 1/UCST
Aromatic Carboxylic Acids: 1/UCST	Other Aliphatic Amines: 5/LCST
Cycloaliphatic Alcohols: 1/UCST	Other Amines, Imines: 3/LCST
Epoxides: 1/UCST	Other Ethers/Diethers: 1/LCST
Ketones: 1/UCST	Other Polyfunctional C, H, O: 1/UCST
n-Alcohols: 1/UCST	Polyfunctional C, H, O, Halide: 3/UCST

^aUCST = Upper Critical Solution Temperature; LCST = Lower Critical Solution Temperature

Figure 4.5 shows $\ln(x_i^{aq})$ data plotted versus inverse temperature for 2-butanol along with the regression obtained in this study. Below 273.15 K (0.00366 K^{-1}), the $\ln(x_i^{aq})$ data plateau. The data below 273.15 K were not included in the regression due to the inability of any of the

equations in Table 4.4 to model the behavior with these data included. Extrapolation of the regression below 273.15 K would obviously give erroneous x_i^{aq} values.

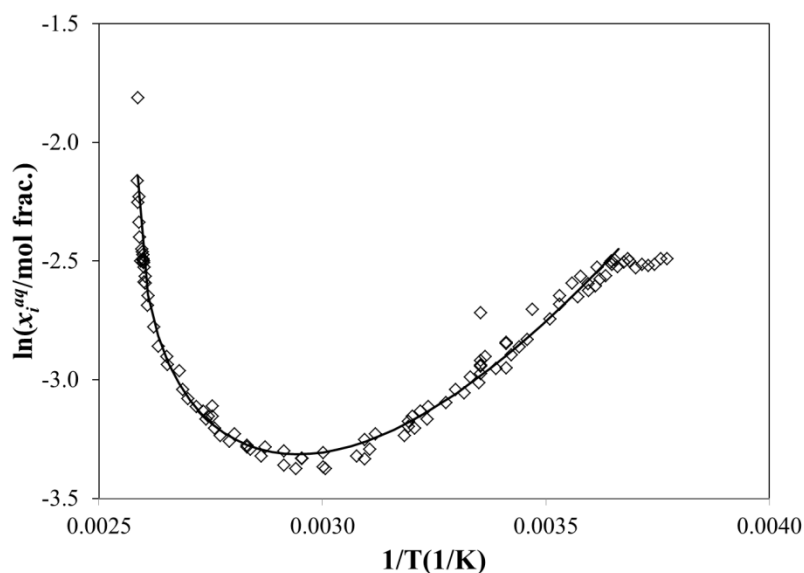


Figure 4.5: 2-Butanol $\ln(x_i^{aq})$ data and regression. References: [55-62]

Figure 4.6 shows $\ln(x_i^{aq})$ data of *p*-chlorophenol plotted as a function of inverse temperature along with the regression obtained by using DIPPR[®] Equation 6000. The regression was purposefully extrapolated beyond the maximum temperature in the figure to show the dangers of extrapolation. DIPPR[®] Equation 6000 is not quite adequate for the x_i^{aq} behavior of *p*-chlorophenol as can be seen by the residual plot in Figure 4.7. The behavior of the residuals suggests that additional or different terms are required to correctly model the temperature-dependence since there is a pattern in the residuals. However, since compounds like this were the exception, DIPPR[®] Equation 6000 was used anyway. As already mentioned, it was desired to have a generic equation that could apply to multiple compounds. Equations were not

developed for individual compounds. The error of the regression for *p*-chlorophenol was taken into account with the regression statistics, and this error is reflected in the assigned uncertainty.

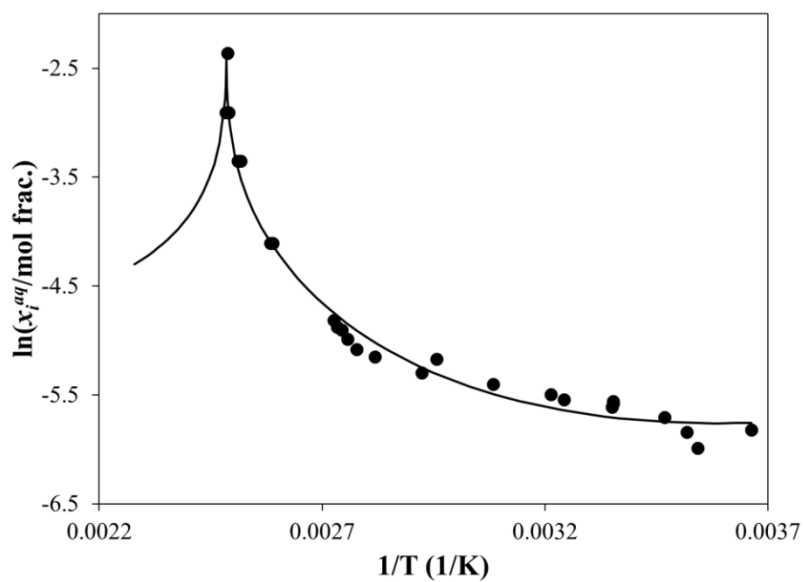


Figure 4.6: $\ln(x_i^{aq})$ data and regression for *p*-chlorophenol. References: [63-67]

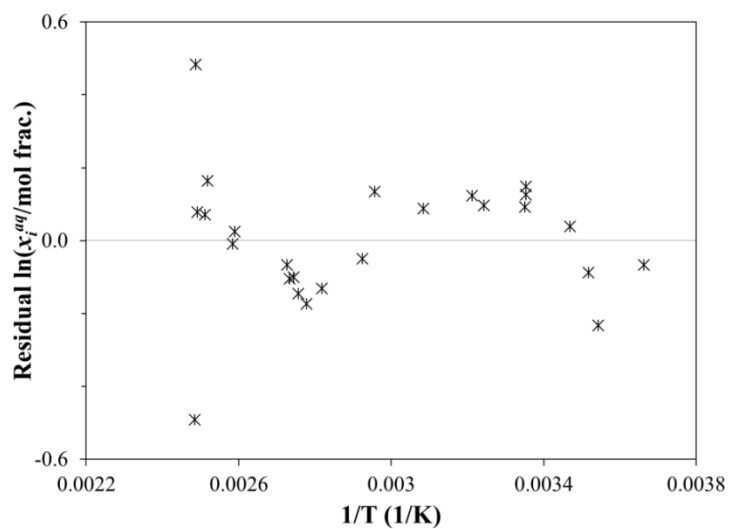


Figure 4.7: Residual plot from *p*-chlorophenol $\ln(x_i^{aq})$ regression

4.3.3 Chemical Family Plots

Chemical family plots of temperature-dependent regressions obtained from experimental data were used to select the best experimental datasets for a compound within the chemical family and qualitatively check the reasonableness of the temperature trends. Temperature-dependent chemical family plots were used throughout this project when enough data were available for similar compounds. Most regressions within a chemical family have varying applicable temperature ranges due to the data available for the individual compounds. Extrapolated values based on experimental data do not always follow the trends of other compounds within the family. When enough chemical family data were available, temperature regressions were modified to change extrapolated behavior to better match the chemical family trends. However, the main concern was ensuring that trends were reasonable over the recommended temperature ranges of the regressions.

Figure 4.8 is a plot of the *n*-alcohol $\ln(x_i^{aq})$ regressions (determined in this study from experimental data) versus inverse temperature starting with 1-butanol. Smaller alcohols were not included because of their miscibility with water. As the number of carbons (#C) increases, x_i^{aq} decreases which is what is expected. 1-Butanol exhibited an upper critical solution temperature in the temperature range of available data. Two regressions are shown for 1-octanol. The C8 regression is based solely on experimental data. As can be seen in the figure, the C8 regression does not follow the same trend as the other *n*-alcohols. A second regression was developed (C8-Exp&Pred) using the experimental data and predicted points based on the family trend. This second regression was accepted.

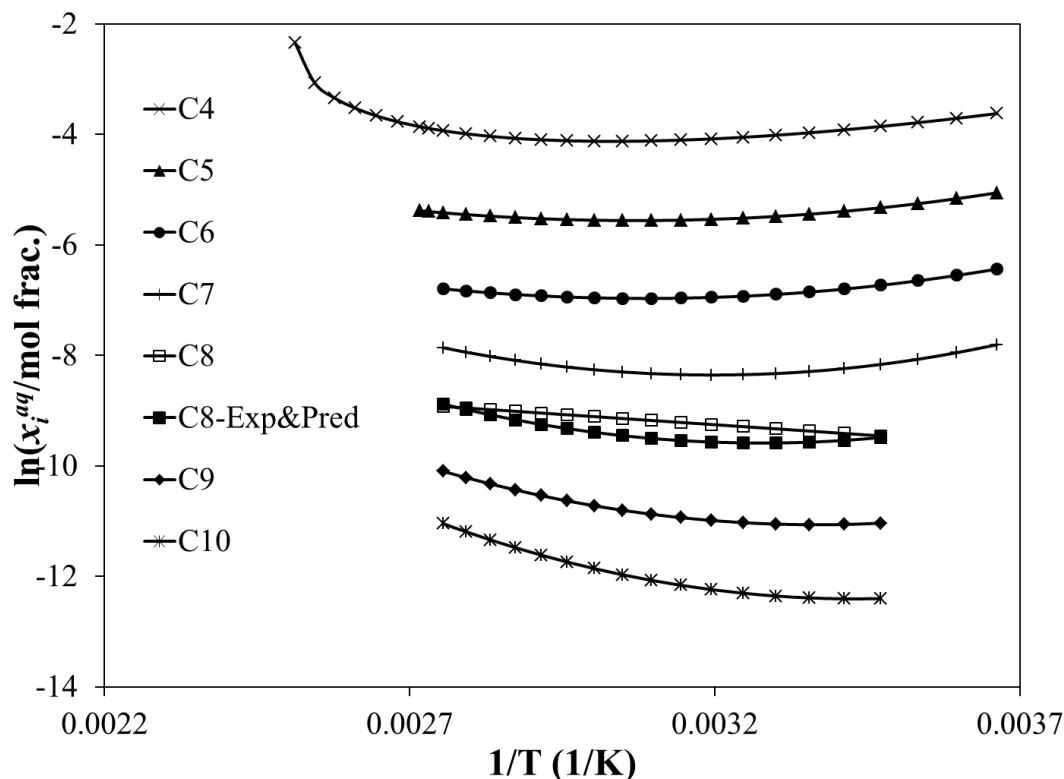


Figure 4.8: n -Alcohol $\ln(x_i^{aq})$ regressions based on experimental data versus inverse temperature. Initial C8 regression based on experimental data (C8) and updated C8 regression based on experimental data and predicted values from the family plot (C8-Exp&Pred) are both shown.

A chemical family plot was used to determine the best k_H and γ_i^∞ data for 1-heptanol (C7). Due to C7 SOLC values ranging between 0.285-0.334 mole fraction over the temperature range of 278.15-333.15 K [58, 59], x_i^{aq} data were not included in the k_H and γ_i^∞ regressions. However, x_i^{aq} data were still helpful in identifying the best datasets. C7 literature values plotted as $\ln(k_H)$ versus inverse temperature are shown in Figure 4.9. The 1-hexanol and 1-octanol (C6 and C8) k_H regressions were also used to identify the best datasets. The data from reference a [47] and the values reported from one of the experimental methods used in reference c [68] agree with the average of the C6 and C8 regressions (Avg C6 & C8). These two references also agree with the literature x_i^{aq} data. Predicted values from the average of the C6 and C8 regressions at

temperatures higher and lower than the available C7 data were included in the regression so the extrapolated behavior better matched the family behavior.

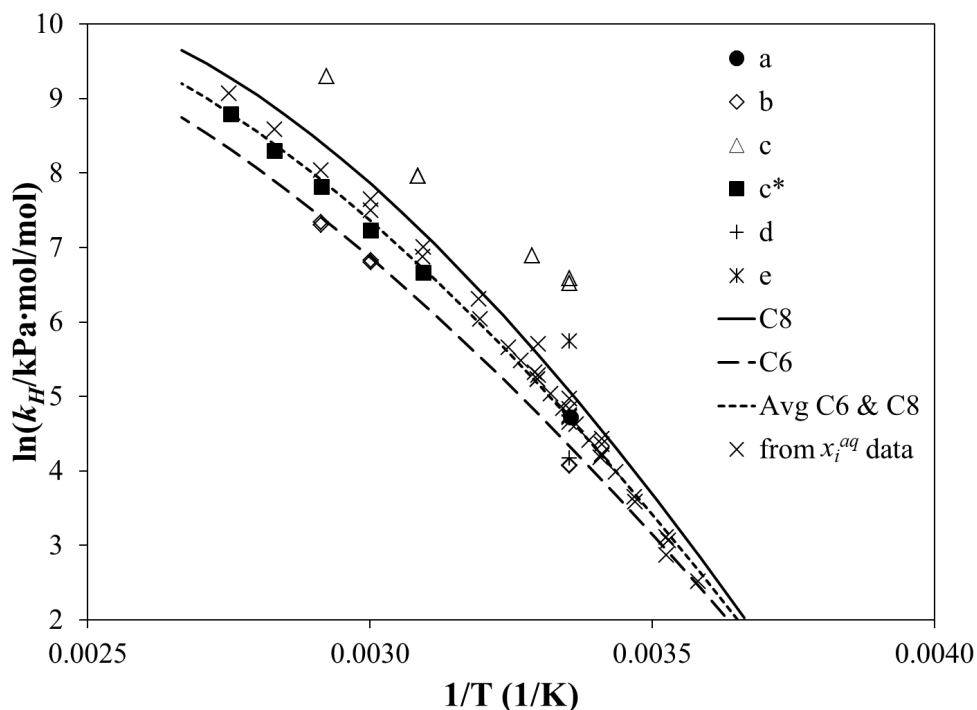


Figure 4.9: 1-Heptanol values plotted as $\ln(k_H)$ versus inverse temperature. 1-Octanol (C8) and 1-hexanol (C6) regressions obtained from experimental data shown along with the average of these two regressions (Avg C6 & C8). References: a [47], b [69], c [68] (*reference reports values using two different experimental methods), d [70], e [71], x_i^{aq} data [44, 72-80]

Figure 4.10 is a plot of $\ln(\gamma_i^\infty)$ regressions versus inverse temperature for the n -alcohols. Values in the figure are plotted within the recommended temperature ranges of each regression; however, extrapolated trends were considered. Initially all of the regressions were developed based solely on experimental data. The larger n -alcohols have fewer data with more scatter than the smaller n -alcohols. As a result, the regressions have larger uncertainties. For 1-octanol (C8), curvature was not significant over the temperature range of the data, so extrapolated values did

not match the shape of the other compounds. Predicted values were combined with the C8 experimental data to obtain curves for γ_i^∞ and k_H (C8-Exp&Pred). As was mentioned, predicted values were also combined with the C7 experimental data to help with extrapolation. The change in the regression due to the predicted points was not as drastic for C7 as with C8, so it is not shown in the plot. As #C increases, γ_i^∞ values increase meaning that deviations from ideality are increasing. As the carbon chain grows, the molecules become more hydrophobic.

Figure 4.11 is a plot of the $\ln(k_H)$ regressions for the *n*-alcohols shown over extrapolated temperature ranges. The same data used to obtain the regressions shown in Figure 4.10 were used to obtain the k_H regressions. As was seen with γ_i^∞ for 1-octanol, the initial regression based only on experimental data (C8) extrapolated differently from the other compounds. The “8-Exp&Pred regression follows the chemical family trend at higher temperatures. The difference between the k_H temperature-dependent behavior of the C8 and C8-Exp&Pred regressions is very noticeable in Figure 4.11. As with the γ_i^∞ values, as #C increases, k_H values increase. However, the influence of temperature is greater on k_H than γ_i^∞ . It has been noted that changes in k_H are small in a chemical family when compared to x_i^{aq} and P_i^{sat} [19]. This is especially obvious at lower temperatures with the *n*-alcohols. Since x_i^{aq} and γ_i^∞ are inversely related for liquids (see Equation 4.4), then changes in γ_i^∞ will also be larger in a chemical family than changes in k_H . These trends are also observed with the *n*-alcohol regressions.

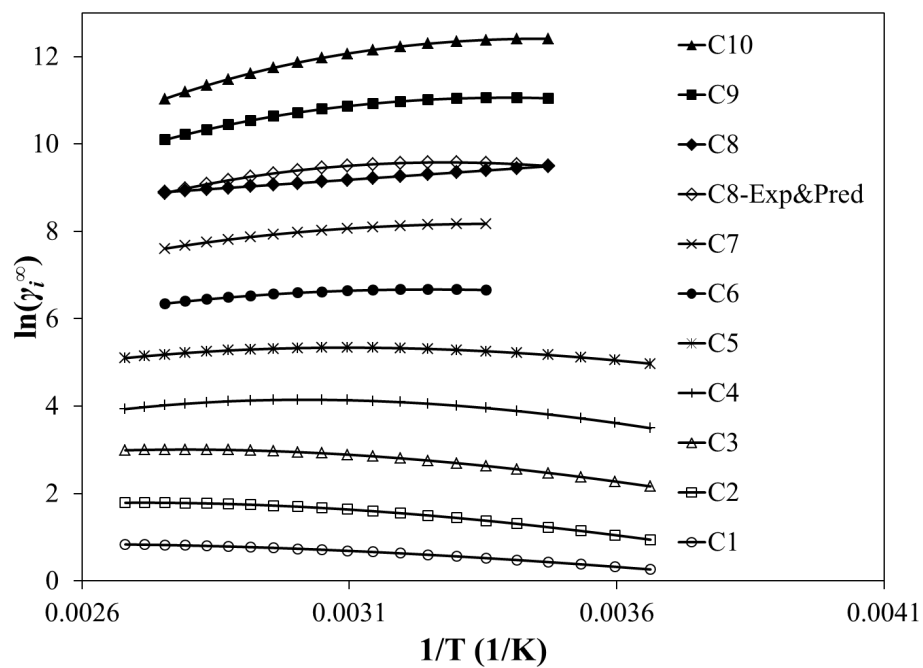


Figure 4.10: n -Alcohol $\ln(\gamma_i^\infty)$ regressions based on experimental data versus inverse temperature

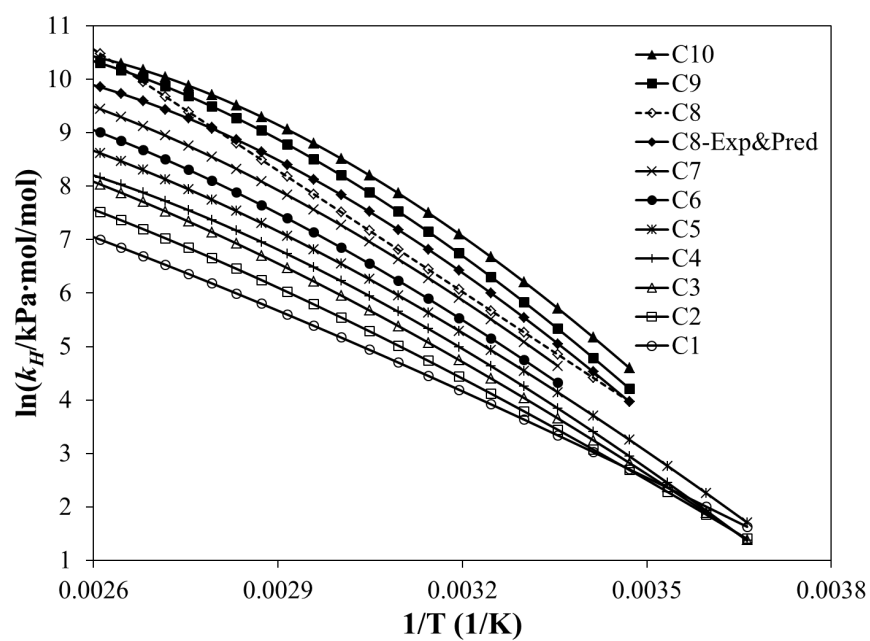


Figure 4.11: n -Alcohol $\ln(k_H)$ regressions based on experimental data versus inverse temperature

4.3.4 Other Data Analysis Notes

For some compounds, γ_i^∞ or k_H values are reported from x_i^{aq} measurements, but the P_i^{sat} and/or Ψ values used in the literature were significantly different from those calculated in this study. In these instances, γ_i^∞ and k_H values were recalculated from the original x_i^{aq} values using DIPPR[®]801 properties and the values reported in the literature were not used.

4.4 Constant Values

Temperature-dependent data are not available for every compound, so chemical family plots were used to evaluate compounds with data only available at 298.15 K. Plots of the natural log of x_i^{aq} or γ_i^∞ often are a linear function of carbon number (#C) [81, 82]. These plots use a group contribution approach (discussed in more detail in Section 7.1). If a plot is created in which a unit on the x-axis represents the addition of a molecular group (which in this study was typically chosen to be $-\text{CH}_2-$), the property will change in incremental values with the number of groups. If there is a functional group that is similar to all compounds, it does not have to be explicitly included on the x-axis. For example, a linear trend can be observed for *n*-alcohols where the x-axis only takes into account the #C. Since all compounds have a hydroxyl group on C1, this functional group is inherently accounted for in the plot. These types of family plots were used to either smooth values as a function of #C or designate the best experimental values based on the family trend. The property that was chosen for the family plots was the property that had the best linear correlation (highest R^2 value) for the natural log of the property plotted as a function of #C.

4.4.1 Chemical Family Plot Regressions

Within a chemical family, some of the compounds had enough data to perform temperature-dependent evaluations (see Section 4.3) while other compounds only had data at a single temperature. In these instances, “Smoothed” values (determined by evaluating the temperature-dependent regressions at 298.15 K) were combined with the available literature data at 298.15 K of the unevaluated compounds to obtain a regression of the natural log of a property as a function of #C. Recommended values were then determined for each unevaluated #C using the regression. This method was used when most of the compounds in a chemical family already had recommended values from evaluating temperature-dependent data. In other instances, all of the original data at 298.15 K were used within a chemical family to obtain a regression of the natural log of a property as a function of #C. Since every chemical family was handled uniquely due to the availability of data, notes about the data used to determine recommended values from family plots were included in the database with the “Smoothed” values. Examples of this data evaluation process and chemical families with interesting behavior are included below.

The constant value family plot for the 1-alkenes is an example of using all of the literature data in the evaluation process. Temperature-dependent evaluations were not performed for any of the liquid 1-alkenes due to the limited amount of temperature-dependent data. Literature k_H and x_i^{aq} data at 298.15 K for the 1-alkenes are plotted as $\ln(x_i^{aq})$ versus #C in Figure 4.12. The data show a linear relationship between $\ln(x_i^{aq})$ and #C ($R^2 = 0.997$). A linear regression was used to obtain recommended x_i^{aq} values at 298.15 K and labeled as “Smoothed” for each #C of the chemical family. The 95% confidence intervals of the mean (as determined by the regression) were used to determine the uncertainty of the “Smoothed” values. Values were predicted for C10-C14 by interpolation and labeled as “Predicted”. The recommended 1-

alkene x_i^{aq} values were used to determine values for k_H and γ_i^∞ using the relationships in Section 4.1. Figure 4.13 is a plot of the $\ln(k_H)$ values for this same family obtained from the “Smoothed” and “Predicted” values of this study compared to the predicted values using the HENRYWIN bond-contribution method [83]. There is good agreement between the HENRYWIN values and the values obtained in this study.

“Smoothed” values from the temperature-dependent evaluations of this study were used in combination with literature data to evaluate constant value data for some chemical families. For the 1-iodoalkanes, temperature-dependent data were evaluated for methyl iodide, ethyl iodide, and *n*-propyl iodide. However, *n*-butyl iodide and *n*-hexyl iodide each only have one reported experimental value. The “Smoothed” values obtained from the temperature-dependent regressions were used with the two literature experimental values to determine “Smoothed” values for *n*-butyl and *n*-hexyl iodide. A plot of $\ln(k_H)$ versus #C for the five points shows a linear trend ($R^2 = 0.9866$). However, when the data are plotted as $\ln(\gamma_i^\infty)$ versus #C, there is a superior linear trend ($R^2 = 0.9995$). Because of this, a regression was performed using $\ln(\gamma_i^\infty)$ versus #C to obtain “Smoothed” γ_i^∞ values for *n*-butyl iodide and *n*-hexyl iodide. These “Smoothed” values were converted to k_H and x_i^{aq} . Figure 4.14 shows the “Smoothed” k_H values obtained from the temperature-dependent regression, the two experimental points of the unevaluated compounds in terms of k_H , the “Smoothed” k_H values obtained from the constant value family plot, and the predicted k_H values from the HENRYWIN bond-contribution method [83].

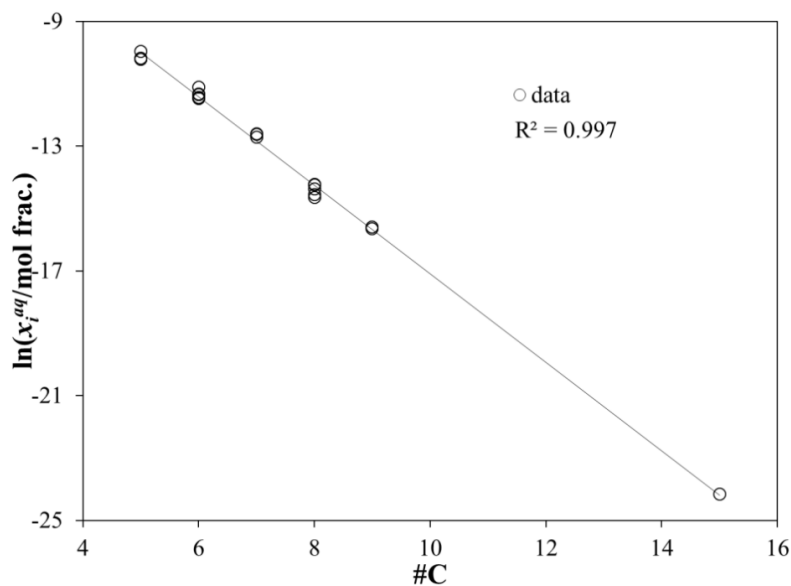


Figure 4.12: 1-Alkene k_H and x_i^{aq} data at 298.15 K plotted as $\ln(x_i^{aq})$ versus carbon number (#C).
References: [47, 72, 84-87]

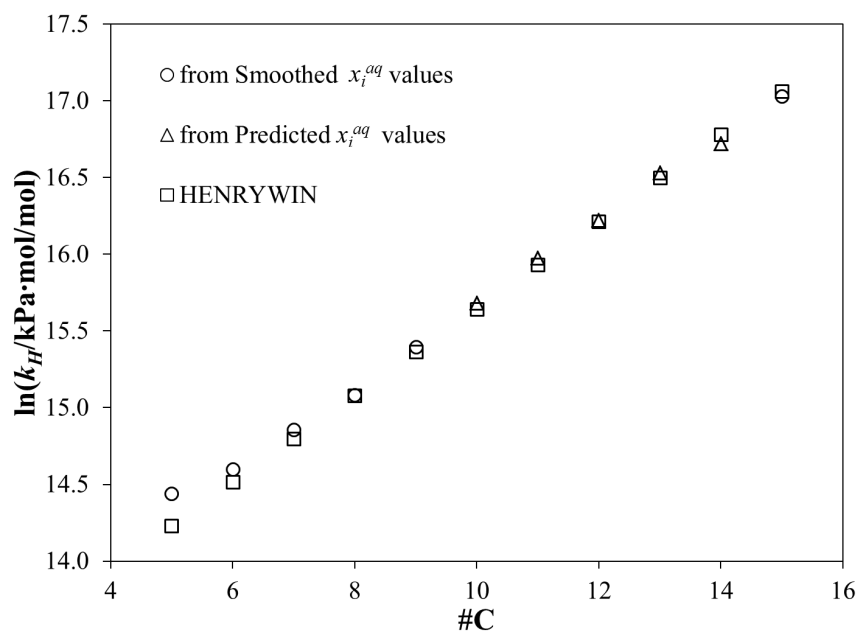


Figure 4.13: 1-Alkene $\ln(k_H)$ values at 298.15 K from this study (“Smoothed” and “Predicted”) and HENRYWIN bond-contribution prediction method [83] plotted as a function of carbon number (#C)

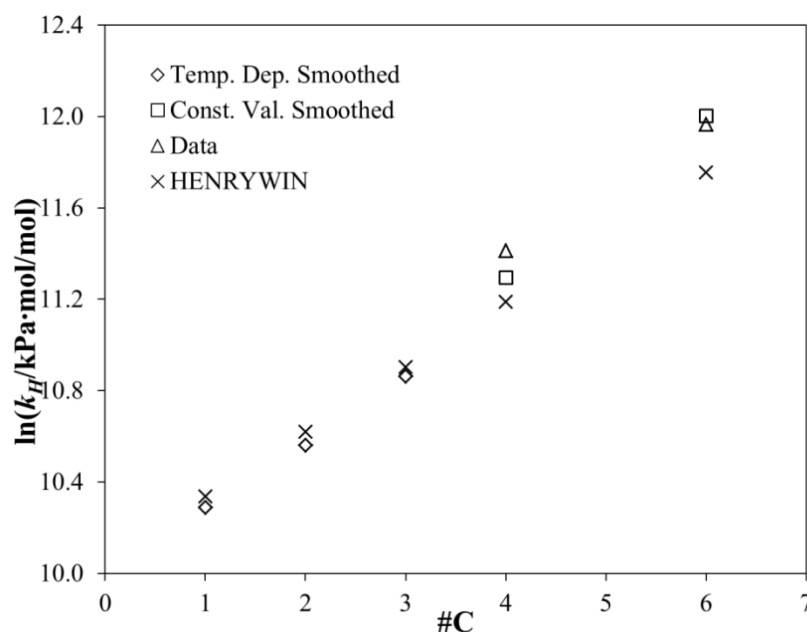


Figure 4.14: 1-Iodoalkane $\ln(k_H)$ values at 298.15 K from temperature-dependent regressions (Temp. Dep. Smoothed), constant value family plot (Const. Val. Smoothed), the literature (data) [47], and HENRYWIN bond-contribution method [83] plotted as a function of carbon number (#C)

Figure 4.15 shows “Smoothed” values obtained from temperature-dependent evaluations for the C4-C10 *n*-alcohols and literature values for the C11 and larger *n*-alcohols. Data at temperatures ranging between 277.15 K and 307.15 K were included due to the limited data available for the larger compounds. The values that are at 298.15 K have solid backgrounds. Possible reasons for the scatter in the literature data include difficulties with accurately measuring small concentrations, the addition of trace amounts of salts by some authors in order to prevent microbial degradation of the alcohol (references c [88] and f [89]), and varying temperatures (x_i^{aq} values are expected to be smaller at lower temperatures). The C16 and C17 data of reference d [90] suggests that the x_i^{aq} values eventually plateau as a function of #C. However, the authors of reference d [90] report that the C17 x_i^{aq} value is less than or equal to the

value shown on the plot. The actual x_i^{aq} was below the detection limits of the method used by the authors, so this point cannot be given serious weight. The values from the other authors do not provide any evidence of the x_i^{aq} values plateauing before C18. Because of all of these factors, x_i^{aq} values for C11-C18 were predicted by extrapolating the regression obtained from the “Smoothed” values (shown in the figure).

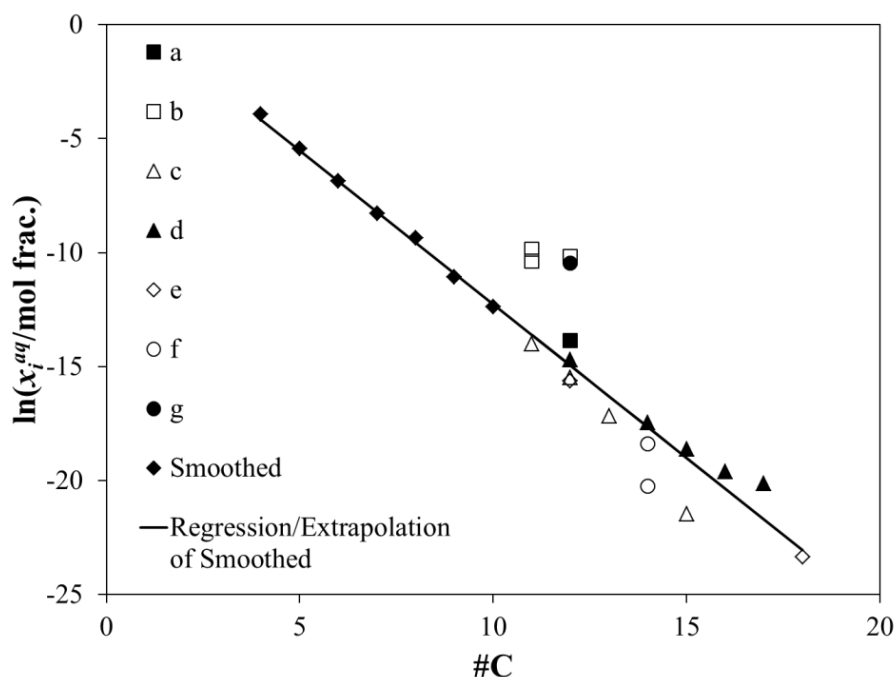


Figure 4.15: *n*-Alcohol “Smoothed” values from temperature-dependent evaluations and literature values plotted as $\ln(x_i^{aq})$ versus carbon number (#C). Solid symbols represent values at 298.15 K. Other values are at temperatures ranging from 277.15 K to 307.15 K. References: a [70], b [91], c [88], d [90], e [92], f [89], g [93]

Figure 4.16 shows the “Smoothed” values from temperature-dependent evaluations and the literature values above C10 for the *n*-alkanes. There is a large degree of scatter in reported values due to the difficulties in accurately measuring the x_i^{aq} values of the larger *n*-alkanes due to problems caused by adsorption, contamination, and detection limitations. Tsonopoulos [81] has

noted that $\ln(x_i^{aq})$ for the *n*-alkanes is a linear function of #C up until *n*-nonane. As #C increases beyond C9, $\ln(x_i^{aq})$ continues to decrease but at a smaller rate. Most of the experimental data indicate that there is an eventual plateau in x_i^{aq} values. One theory is that the folding of the larger chain *n*-alkanes causes this plateau. Ferguson et al.[94] disagree with this theory and claim that the experimental data are erroneous. They used molecular dynamic simulations to study the x_i^{aq} of *n*-alkanes and found that there is an exponential decrease in x_i^{aq} up to C20. This theory better matches the trend observed with the *n*-alcohols (Figure 4.15). Due to the agreement of the molecular simulations and the *n*-alcohol trend, the data of Tolls et al. [95] (points are highlighted in Figure 4.16) were recommended in this study. Future experimental work would be helpful in better understanding this chemical family trend.

Figure 4.17 shows “Smoothed” and literature x_i^{aq} values for the *n*-alkylbenzenes plotted as a function of #C. Tsonopolous has noted slight curvature as $\ln(x_i^{aq})$ data are plotted as a function of #C for the *n*-alkylbenzenes [82]. Because of this, a quadratic equation was used to smooth the data instead of linear equation. The value at C19 suggests that the quadratic trend does not continue after C16. The x_i^{aq} values of large hydrocarbons are difficult to measure due to the small concentration, so more x_i^{aq} values of the large *n*-alkylbenzenes need to be measured in order to confidently understand the curvature after C16. “Smoothed” values obtained from temperature-dependent regressions were included in the constant value regression for C6-C12. The available data point at C16 was also included, but the data points at C13 and C14 were rejected because they do not agree with the family plot. Values for C13-C16 were determined from smoothing these combined values.

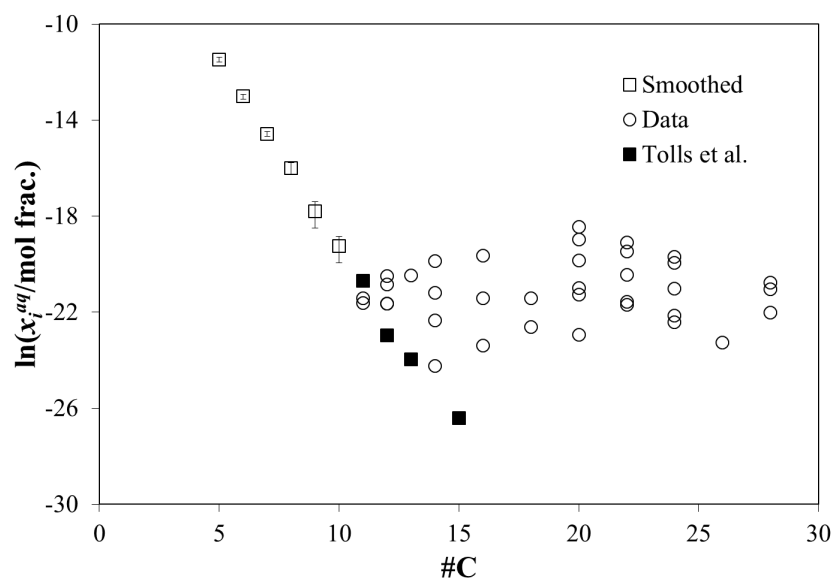


Figure 4.16: *n*-Alkane “Smoothed” values from temperature-dependent evaluations and literature (Data) x_i^{aq} values at 298.15 K. References: [85, 95-104]

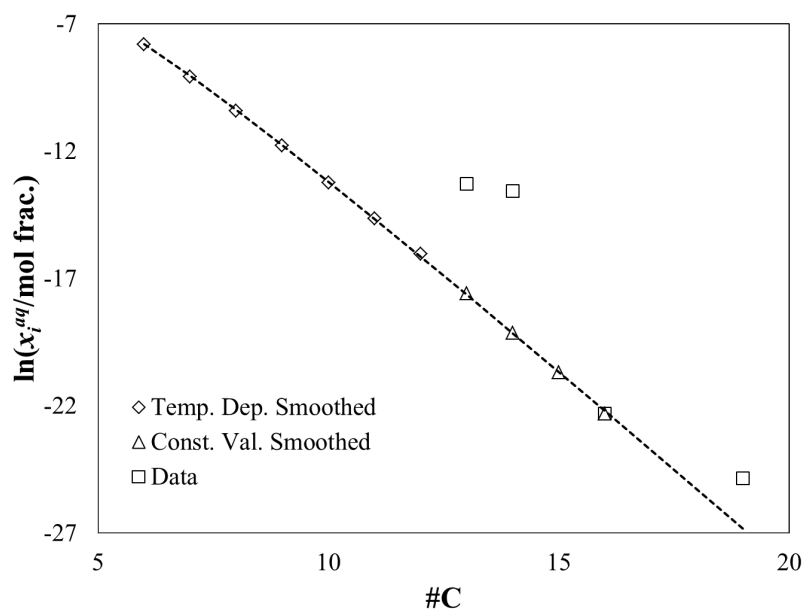


Figure 4.17: *n*-Alkylbenzene $\ln(x_i^{aq})$ values at 298.15 K from temperature-dependent regressions (Temp. Dep. Smoothed), constant value family plot (Const. Val. Smoothed), and the literature (Data) plotted as a function of carbon number (#C). References: [105, 106]

When analyzing the di-*n*-alkyl phthalates, only three of the compounds (dimethyl, diethyl, and di-*n*-butyl phthalates) had temperature-dependent data. These three phthalates were evaluated using the procedure discussed in Section 4.3. The remaining phthalates only have literature x_i^{aq} data at ambient conditions. Figure 4.18 is a plot of $\ln(x_i^{aq})$ at 298.15 K as a function of the number of carbons on the alkyl chains (#C). The linear relationship between $\ln(x_i^{aq})$ and #C was used to provide recommended values for di-*n*-propyl and di-*n*-hexyl phthalates, #C = 6 and #C = 12, respectively. A linear regression was performed with the temperature-dependent “Smoothed” values and the data shown in Figure 4.18. Two literature values were rejected because they did not follow the linear relationship like the remaining data. This regression was used to determine “Smoothed” values for di-*n*-propyl and di-*n*-hexyl phthalates.

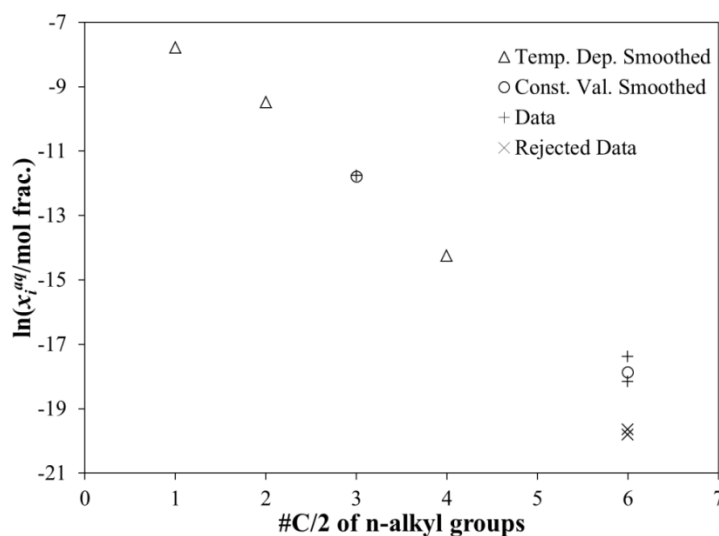


Figure 4.18: di-*n*-Alkyl phthalate $\ln(x_i^{aq})$ values at 298.15 K from temperature-dependent regressions (Temp. Dep. Smoothed), constant value family plot (Const. Val. Smoothed), and the literature (Data and Rejected Data) plotted as a function of the number of carbons on the alkyl chains (#C). References: [107-115]

As in other instances, Equations 4.1 and 4.4 were used with the “Smoothed” x_i^{aq} values to calculate k_H and γ_i^∞ , respectively. A plot of $\ln(k_H)$ versus #C for the di-*n*-alkyl phthalates yields a surprising trend, shown in Figure 4.19. The values from this study do not show a linear trend like the bond-contribution values predicted by HENRYWIN [83]. Data for diethyl phthalate (#C = 4) were reviewed in an effort to identify potential explanations.

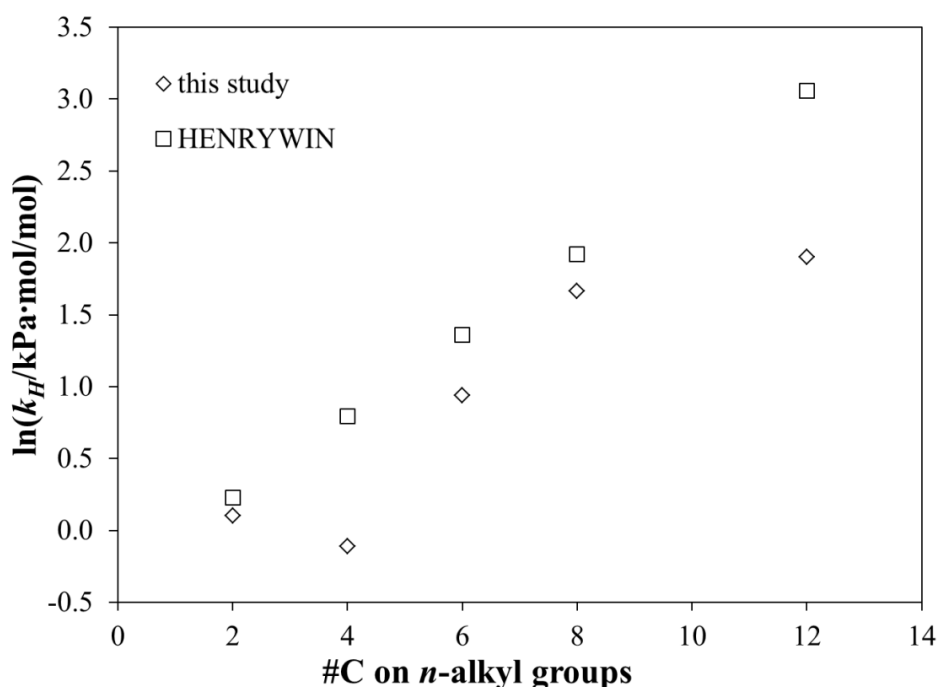


Figure 4.19: di-*n*-Alkyl phthalate “Smoothed” $\ln(k_H)$ values at 298.15 K from this study and from HENRYWIN[83] bond-contribution method

The literature data for diethyl phthalate plotted as $\ln(k_H)$ are shown in Figure 4.20. The experimental value reported in HENRYWIN was calculated from the x_i^{aq} value reported by Howard et al. [110] (reference (e) in the figure). However, the P_i^{sat} value used by HENRYWIN ($2.8 \cdot 10^{-4}$ kPa) to calculate k_H is different from the DIPPR[®] 801 P_i^{sat} value ($6.64 \cdot 10^{-5}$ kPa). The

experimental value reported in HENRYWIN is shown on Figure 4.20 with the bond-contribution value. Both HENRYWIN values are significantly different from the values used in this study. Values in this study are considered superior due to the reviewed P_i^{sat} values available in DIPPR[®] 801. As can be seen by this example, the quality of k_H values determined using x_i^{aq} is largely dependent on the quality of P_i^{sat} values used.

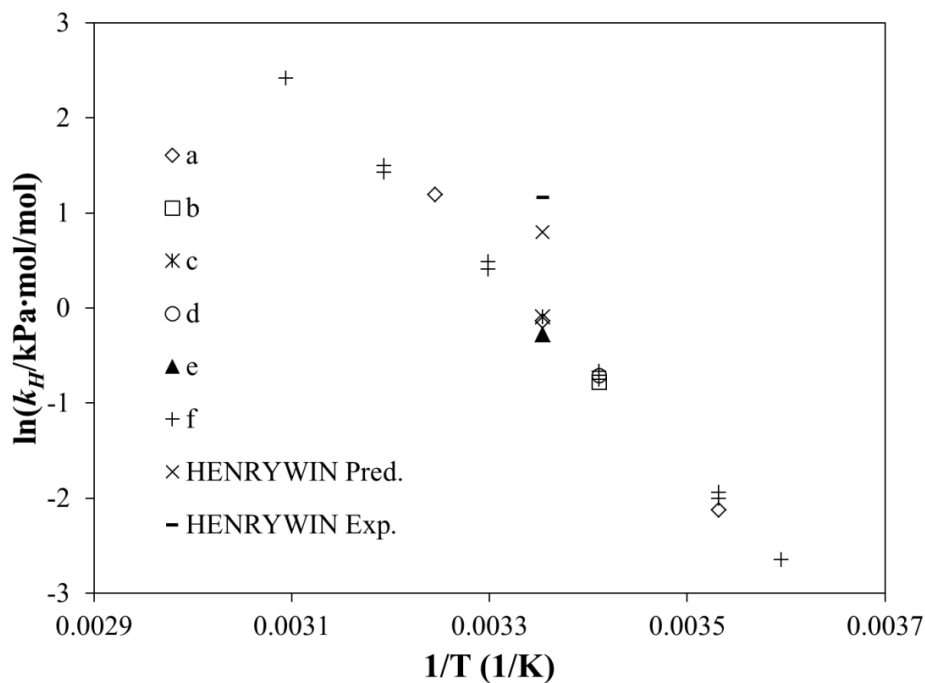


Figure 4.20: Literature data of diethyl phthalate at 298.15 K plotted as $\ln(k_H)$ compared with HENRYWIN bond-contribution value and reported experimental value [83]. References: a [111], b [116], c [109], d [112], e [110], f [107]

The discrepancy between P_i^{sat} values for diethyl phthalate does not fully explain the interesting k_H trend observed in this study. Taking the natural log of Equation 4.1 results in

$$\ln(k_H) = \ln(P_i^{sat}) - \ln(x_i^{aq}) . \quad 4.20$$

Equation 4.20 explains how a nonlinear trend is obtained from two relatively linear trends.

Figure 4.21 shows $\ln(P_i^{sat})$ and $\ln(x_i^{aq})$ on the same plot. Both properties have linear trends with $R^2 > 0.99$. For the two smallest compounds, $\ln(P_i^{sat})$ and $\ln(x_i^{aq})$ are very similar to each other meaning that $\ln(k_H)$ is close to 0. For all of the compounds except diethyl phthalate, $\ln(P_i^{sat})$ is larger than $\ln(x_i^{aq})$ resulting in a positive $\ln(k_H)$. For diethyl phthalate, $\ln(P_i^{sat})$ is slightly smaller than $\ln(x_i^{aq})$ which leads to a negative $\ln(k_H)$. If $\ln(P_i^{sat})$ and $\ln(x_i^{aq})$ were completely linear functions of #C, then $\ln(k_H)$ would exhibit a completely linear trend. The values obtained in this study were accepted despite the unusual trend. Either more experimental data are necessary to find a better trend, or k_H values for this chemical family are not modeled very well using a group contribution approach.

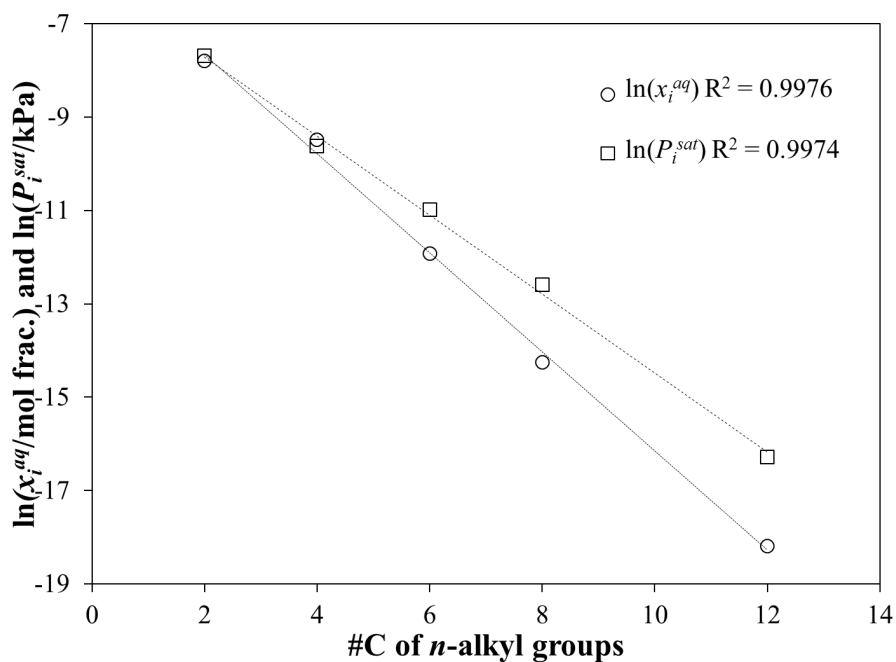


Figure 4.21: Smoothed $\ln(x_i^{aq})$ and $\ln(P_i^{sat})$ [117] of di-*n*-alkyl phthalates at 298.15 K

Temperature-dependent k_H and x_i^{aq} data were available for the C5, C6, and C8 cycloalkanes, allowing for the determination of temperature-dependent regressions. Only limited x_i^{aq} data are available for C7. Values obtained from the regressions at 298.15 K were analyzed with the data for C7. A $\ln(x_i^{aq})$ family plot was used to predict x_i^{aq} for C7 which was then used to calculate k_H and γ_i^∞ values for this compound. (The reported x_i^{aq} value at 298.15 K did not agree with the family plot, so it was rejected.) Similarly, for the C5-C8 cycloalkenes, all but C7 had temperature-dependent k_H and x_i^{aq} values. A family plot showed that the reported x_i^{aq} value for C7 at 298.15 K agrees with the values for the other compounds. This x_i^{aq} value was accepted and used to determine k_H and γ_i^∞ at 298.15 K. The $\ln(x_i^{aq})$ values for the cycloalkanes and cycloalkenes are shown in Figure 4.22 with $\ln(P_i^{sat})$ values shown on the same figure. The cycloalkanes and cycloalkenes both become less soluble in water as #C increases. The cycloalkanes are less soluble than the cycloalkenes which is because of the presence of the double bond in the cycloalkenes.

Figure 4.23 shows cycloalkane and cycloalkene $\ln(k_H)$ values accepted from this study compared to HENRYWIN bond-contribution values [83]. The values in this study do not increase with #C as predicted by HENRYWIN; the values are fairly constant for each of the families. This is explained by the $\ln(x_i^{aq})$ and $\ln(P_i^{sat})$ trends shown in Figure 4.22. The two properties have fairly parallel trends to each other for each chemical family (and even between families). The difference between two parallel lines is a constant. Since, $\ln(k_H)$ is the difference between these parallel trends (see Equation 4.20), the $\ln(k_H)$ values are fairly constant. As shown with the *n*-alcohol example (Section 4.3.3), the difference between k_H values within a chemical family is much smaller than the difference in x_i^{aq} or γ_i^∞ .

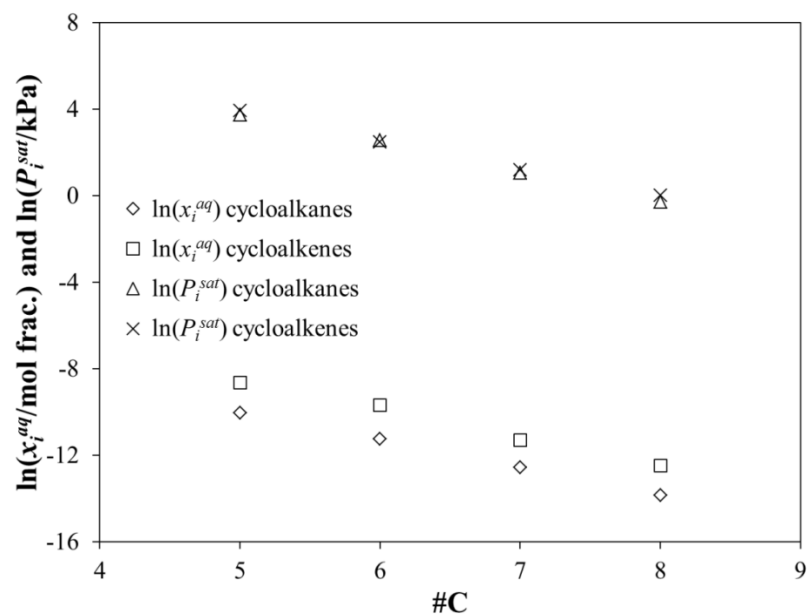


Figure 4.22: Cycloalkane and cycloalkene $\ln(x_i^{aq})$ and $\ln(P_i^{sat})$ values at 298.15 K plotted as a function of carbon number (#C). References: [86, 117]

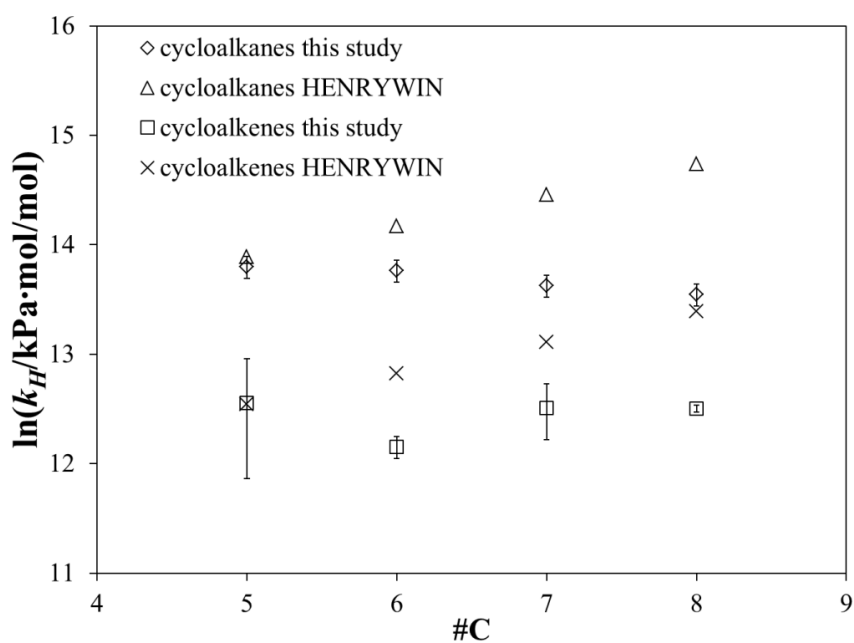


Figure 4.23: Cycloalkane and cycloalkene $\ln(k_H)$ values at 298.15 K compared to HENRYWIN bond-contribution values [83]

4.4.2 Averaging

For some compounds it was difficult or impossible to do a family plot due to the unavailability of data for compounds with similar structures. For some compounds, the reported values at 298.15 K were averaged when multiple reported values are in good agreement with each other. In this case, the error was related to the standard deviation of the average. For example, k_H and γ_i^∞ values are only available for triethylamine at 298.15 K. These values look reasonable compared to x_i^{aq} data at that temperature (Figure 4.24). However, x_i^{aq} data were not used to obtain temperature-dependent equations for k_H and γ_i^∞ due to the relatively high SOLC and the lower critical solution behavior. Instead, averages were taken of the available k_H and γ_i^∞ values after conversion to the same property. The standard deviation of the literature data at this temperature is less than 5%.

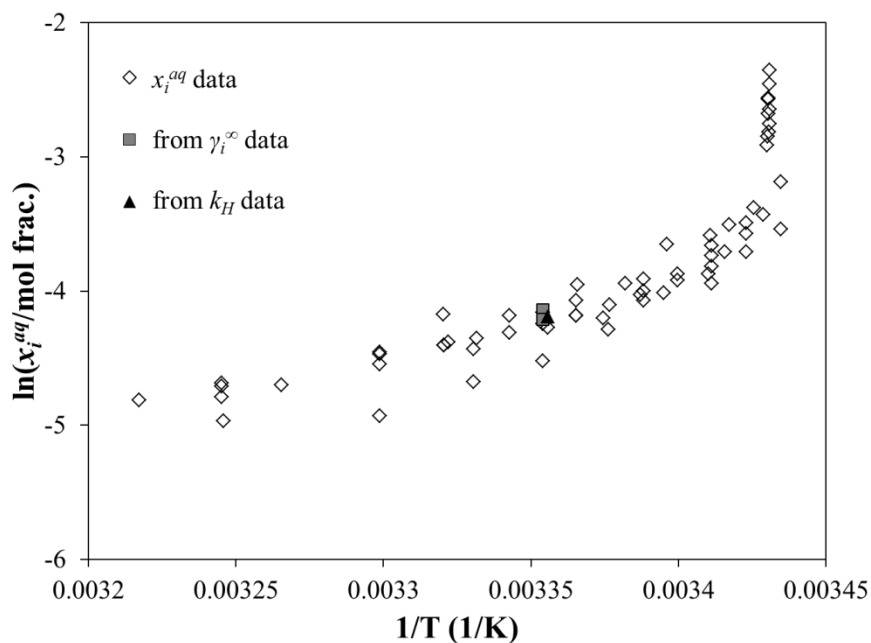


Figure 4.24: Triethylamine x_i^{aq} , γ_i^∞ , and k_H literature data. References: [118-127]

Another example of this data analysis technique is with 2,3,4-trimethylpentane. Only x_i^{aq} data are reported for this compound. Of the available x_i^{aq} data, only one reference provides temperature-dependent data. However, the value at 298.15 K from the reference with temperature-dependent data does not line up with three additional references. In this instance, the three x_i^{aq} values at 298.15 K that are in agreement with each other were averaged and used to derive k_H and γ_i^∞ .

4.5 American Petroleum Institute Recommendations

The American Petroleum Institute (API) Technical Data Book [128] provides recommended temperature-dependent k_H values for various compounds. For gases, API recommendations are based on values measured at multiple pressures. Due to the structure of the 801E database, data were entered only as a function of temperature and not pressure. Only data for gases at a pressure of 1 atm were included. The wider range of data for gases used in the API regressions renders their recommendations more reliable than what could be done in this study. In addition, the API evaluations for select liquids were determined to be reliable. As such, API recommendations for 31 compounds (26 gases and 5 liquids) were accepted in this study and are represented in the compound totals given in Table 3.2 and Table 3.3.

API temperature dependent coefficients were reported for k_H in the form of DIPPR[®] 801 Equation 101 (see Table 4.3). Using the relationship between k_H and x_i^{aq} for gases leads to

$$\ln(k_H) = \ln(101.325) - \ln(x_i^{aq}). \quad 4.21$$

Since $\ln(101.325)$ is a constant, coefficients can be derived from the API coefficients to develop a temperature dependent equation for x_i^{aq} . A similar relationship was given for liquids in Equation 4.20. Since P_i^{sat} is in the form of DIPPR[®] Equation 101 as well, the API k_H and

DIPPR[®] 801 P_i^{sat} coefficients were combined to determine x_i^{aq} coefficients for liquids.

Equations obtained from API equations were classified as “Derived” instead of “Smoothed”.

API regression errors are reported in various forms. The errors had to be converted to the error notation used in DIPPR[®] 801. If the API error was reported as an average absolute deviation (AAD) then the regression was assigned the corresponding error in the database. For example, if AAD = 5.88%, then the error reported in the database is < 10%. If the API error was reported on a log scale (i.e. standard deviation in $\ln(x_i^{aq})$ is 1.8%) then the property was calculated at N points evenly distributed along the temperature range where N is the number of values that were used to obtain the regression. The upper and lower uncertainty limits were calculated at these points on the normal scale, and the AAD was determined at each point. These AADs were averaged and an overall error code was assigned based on this average. The same error was assigned to x_i^{aq} and k_H regressions of the same compound.

API recommendations were accepted without ensuring that all of the data used in the original regression were present in the 801E database. Many of the references included in the API database were included in this study. However, the API database used high pressure data which were purposefully excluded from this study. Uncertainties of the datasets in this database were determined from the reported regression uncertainty in the same manner that uncertainties were assigned in Section 4.3.1.

4.6 Comparison with Literature Recommendations and Predicted Values

The temperature regressions obtained in this study were compared with available recommendations such as the IUPAC-NIST Solubility Data Series [129-165], API Technical Data Book [128], and other literature reviews. This was done to validate the evaluation

procedure used in this study and to ensure that the recommended regressions of this study reflect the data included in the 801E database. Except for the API regressions mentioned in Section 4.5, outside regressions were not accepted so that a future database user can view the datasets that were used to develop the regression.

The IUPAC-NIST equation uses empirical constants that were obtained from simultaneous regression of all of the x_i^{aq} data for compounds within a certain chemical family [166]. Data were then classified as “Recommended”, “Tentative”, “Doubtful”, or “Rejected” based on the error between the regression equation and the actual literature data [150]. These classifications are given per point and not per reference; whereas, in this study, all of the points from a given reference were accepted or rejected unless there was a valid reason to split a data set (for example, the experimental method is not as accurate in a certain temperature or concentration region). In this study, temperature regressions were based solely on literature data.

If the agreement between the regressions of this study and those published was within the regression uncertainty, the parameters in this study were immediately accepted. In instances where the disagreement was larger than the regression uncertainty, the regressions were compared more judiciously. Insights gained through thermodynamic relationships, chemical family trends, and predicted values (as previously discussed) were useful in determining the best values when there were discrepancies.

In order to ensure that the analyses performed in this study were reasonable when derivative properties were excluded, recommended values determined in this study were compared to recommendations given in the literature that included derivative properties. As an example, Figure 4.25 shows methanol literature γ_i^∞ and k_H values plotted as $\ln(\gamma_i^\infty)$, the regression obtained in this study, and the regression recommended by Dohnal et al. [21]. The 95%

confidence intervals for the regression from this study are also shown. In the analysis performed by Dohnal et al. [21], the temperature-dependent parameters were obtained by simultaneously fitting γ_i^∞ , limiting partial molar excess enthalpy, and limiting partial molar heat capacity data. Their analysis included more properties than this study. However, their regression is within the uncertainty of the regression from this study. As an additional example, the evaluations by Plyasunov et al. [21-26] included derivative properties. Values at 298.15 K from this study were compared to their recommendations for approximately 150 compounds. The recommendations of this study are all within the reported uncertainties. These comparisons provided confidence that the properties and evaluation procedure used in this study produce reliable regressions.

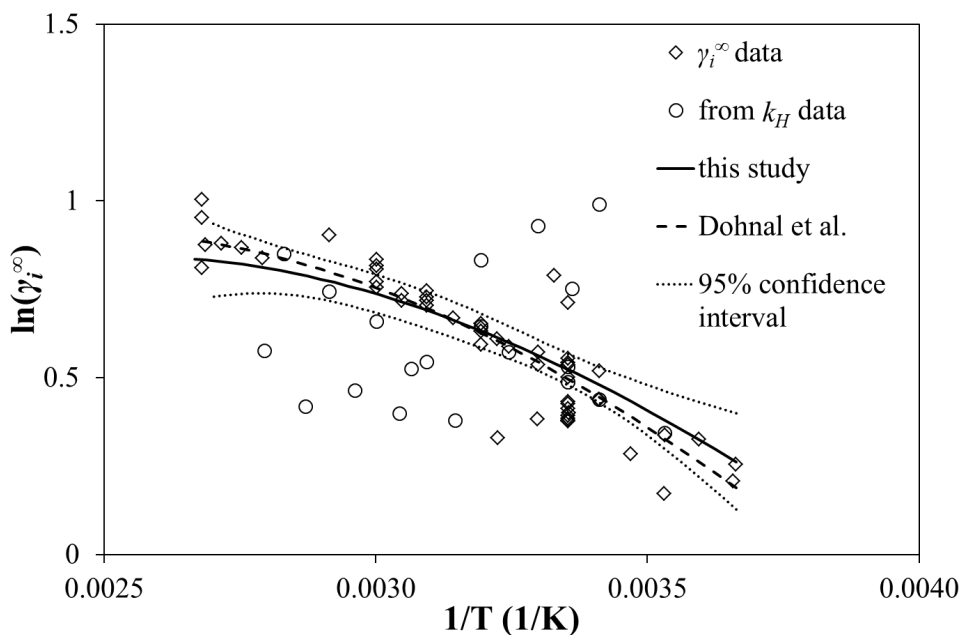


Figure 4.25: Methanol γ_i^∞ regressions from this study and Dohnal et al. [21]. References: [45, 48, 69, 70, 77, 167-182]

The regressions of this study do not always match the regressions reported in the IUPAC-NIST solubility data series. Figure 4.26 shows literature x_i^{aq} data for *n*-propyl formate, the regression obtained in this study, and the regression reported in the IUPAC-NIST Solubility Data Series [159]. The confidence intervals correspond to the average absolute error of the regression obtained in this study (see Section 4.3.1). In addition, the figure shows x_i^{aq} data for ethyl formate for comparison purposes. The IUPAC-NIST regression extrapolates very differently from the regression in this study. However, over the temperature range of experimental data (which is the recommended range of the regression) the regressions are not statistically different. The regression of this study was accepted in the database. Similar trends were observed with additional compounds, and these compounds were handled in the same manner.

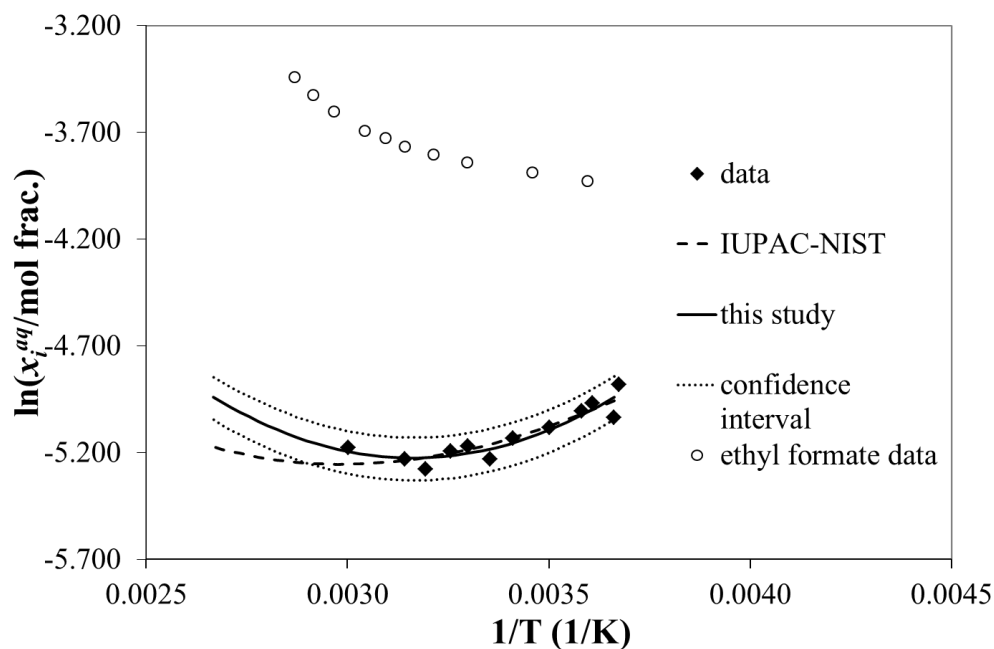


Figure 4.26: *n*-Propyl formate x_i^{aq} data, regression from this study, and IUPAC-NIST [159] regression compared to ethyl formate x_i^{aq} data. References: [183, 184]

Figure 4.27 is an example that shows how data from the combined properties help in the data analysis process. The x_i^{aq} data for di-*n*-propyl ether are shown along with x_i^{aq} values calculated from literature k_H and γ_i^∞ data. All of the x_i^{aq} data were reported in the same paper, but two experimental methods were used. The x_i^{aq} dataset that does not agree with the k_H and γ_i^∞ data were measured using a less reliable method, so those points were rejected. The k_H , γ_i^∞ , and remaining x_i^{aq} data were used to obtain the regression shown in the plot. The IUPAC-NIST [158] trend is also shown, and it is significantly different over the entire temperature range. All of the IUPAC-NIST ether trends use theoretical coefficients, and all ether systems (except one) are forced to exhibit a minimum at 360 K [185]. The regression of this study is recommended because it matches experimental data from multiple references and properties.

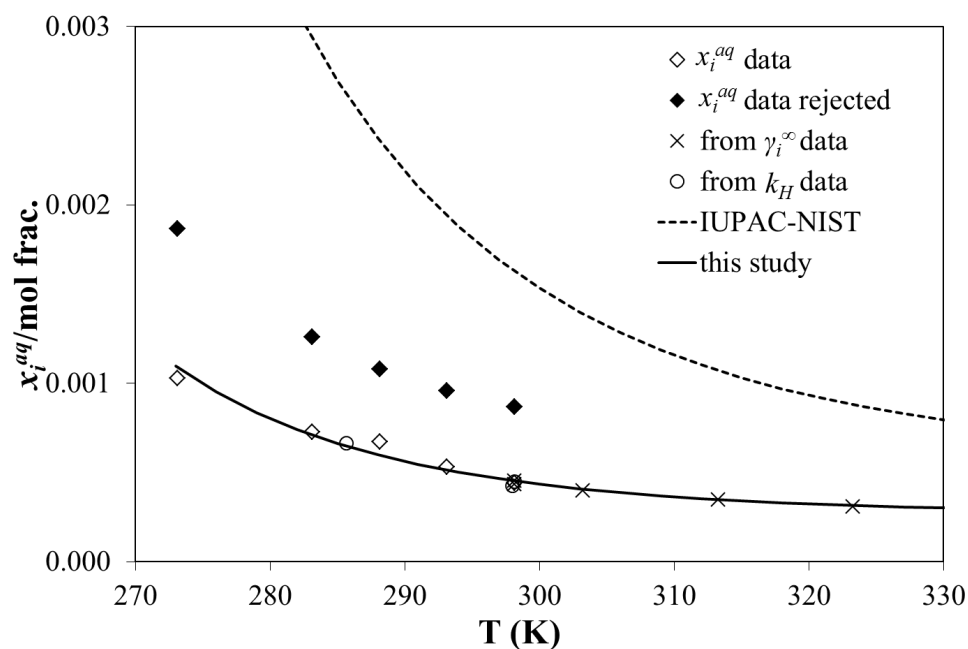


Figure 4.27: di-*n*-Propyl ether x_i^{aq} regressions from this study and from IUPAC-NIST [158].
References: [47, 172, 186-189]

Another example is shown below (Figure 4.28) where the regressions are significantly different over part of the data range. The figure shows literature x_i^{aq} data for 3-methyl-3-pentanol, the regression obtained in this study with the average confidence intervals, and the regression reported in the IUPAC-NIST Solubility Data Series [150]. Some of the data were rejected because the data are old (measured in 1938), and the experimental and analytical methods used are not as reliable as the methods used with the more recent data. The IUPAC-NIST regression shows significantly different behavior as temperature increases. The regression from this study was accepted because it follows the available experimental data.

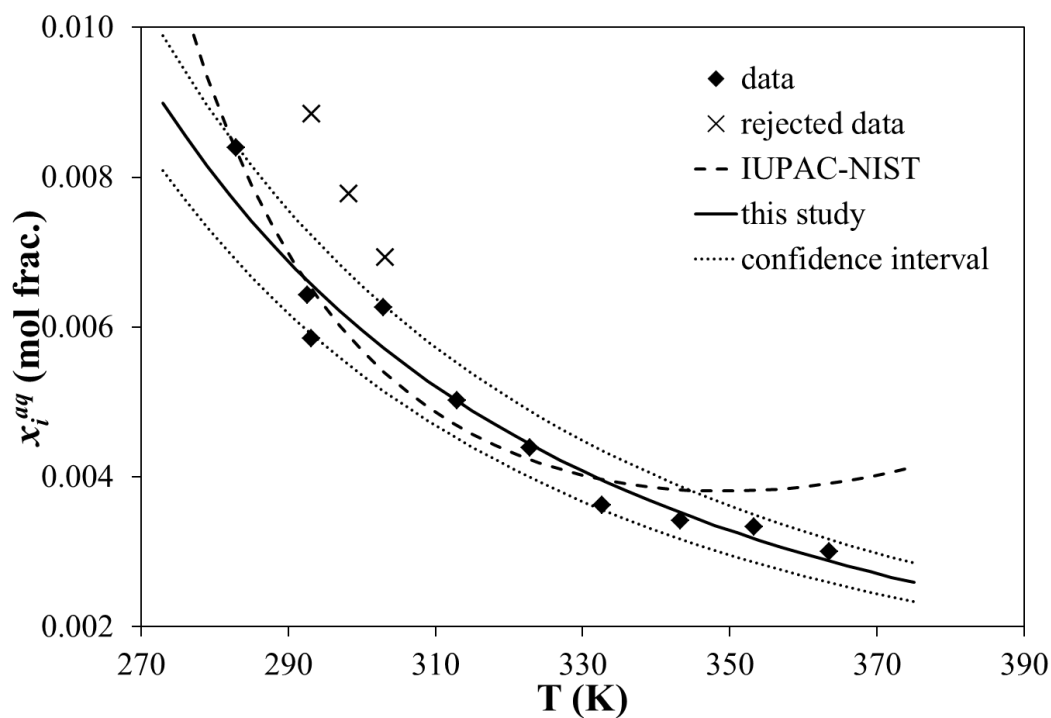


Figure 4.28: 3-Methyl-3-pentanol x_i^{aq} regressions from this study and from IUPAC-NIST [150].
References: [44, 190]

In the following examples, k_H values predicted using the group contribution method of Sedlbauer et al. [191] were used to help identify the best datasets. The Sedlbauer et al. [191] method was developed using data for multiple hydrocarbons, so it can be used reliably to identify the best datasets for similar types of compounds. Values were converted between x_i^{aq} and k_H using Equation 4.1 and DIPPR[®] 801 [117] P_i^{sat} data.

When the data evaluation methods already discussed were not helpful in distinguishing between conflicting data sets for a compound, prediction methods (discussed in Chapter 7) were helpful in identifying the best datasets. As an example, with cumene there are multiple x_i^{aq} temperature trends shown by literature data (Figure 4.29). More data are available in the literature for this compound than are shown in the figure; however, there is enough scatter in these additional data that they do not aid in identifying the best trend. These additional datasets were included in data analysis, but they are not shown in the figure for viewing purposes. The predicted Sedlbauer et. al [191] values (k_H converted to x_i^{aq}) and IUPAC-NIST [136] curve are the closest to the data from reference b [192] while the API [128] regression is based on the data of reference c [193]. The IUPAC-NIST [136] trend is based on x_i^{aq} data for multiple hydrocarbons. The prediction method of Sedlbauer et al. [191] was developed using data for multiple aromatic hydrocarbons, so the trend inherently takes into account chemical family trends. The relative agreement between the Sedlbauer et al. [191] and IUPAC-NIST [136] recommendations is encouraging because both used information gained from examining multiple compounds with similar structures. Because of this, the data from reference b [192] are recommended.

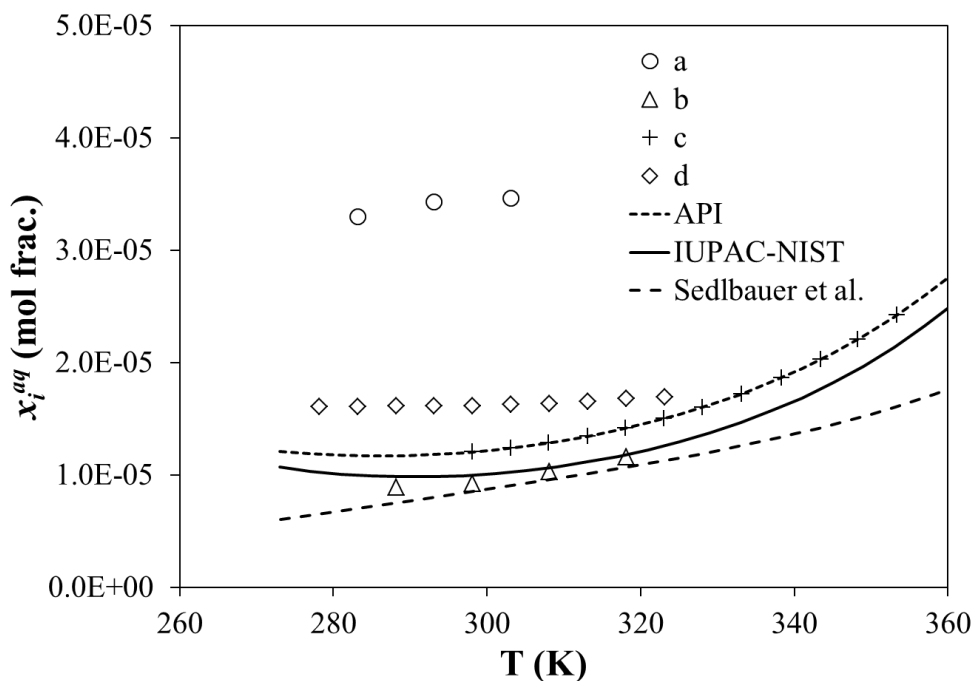


Figure 4.29: Cumene x_i^{aq} data compared with Sedlbauer et. al [191] predicted values and API [128] and IUPAC-NIST [136] recommendations. References: a [194], b [192], c [193], d [105]

As another example, limited data are available for 2,2,5-trimethylhexane. The available literature x_i^{aq} values are plotted as $\ln(k_H)$ versus inverse temperature in Figure 4.30 with k_H values predicted using the group contribution method developed by Sedlbauer et al. [191]. Data for this compound were not included in the Sedlbauer et al. [191] method development; however, data for other aliphatic hydrocarbons were included making the method applicable for this compound. The difference between the two reported values at 298.15 K (0.00335 K^{-1} on the inverse temperature scale) make it difficult to identify the best values by looking at the data alone. The predicted values are in agreement with the data points reported by reference a [195], so these values were accepted.

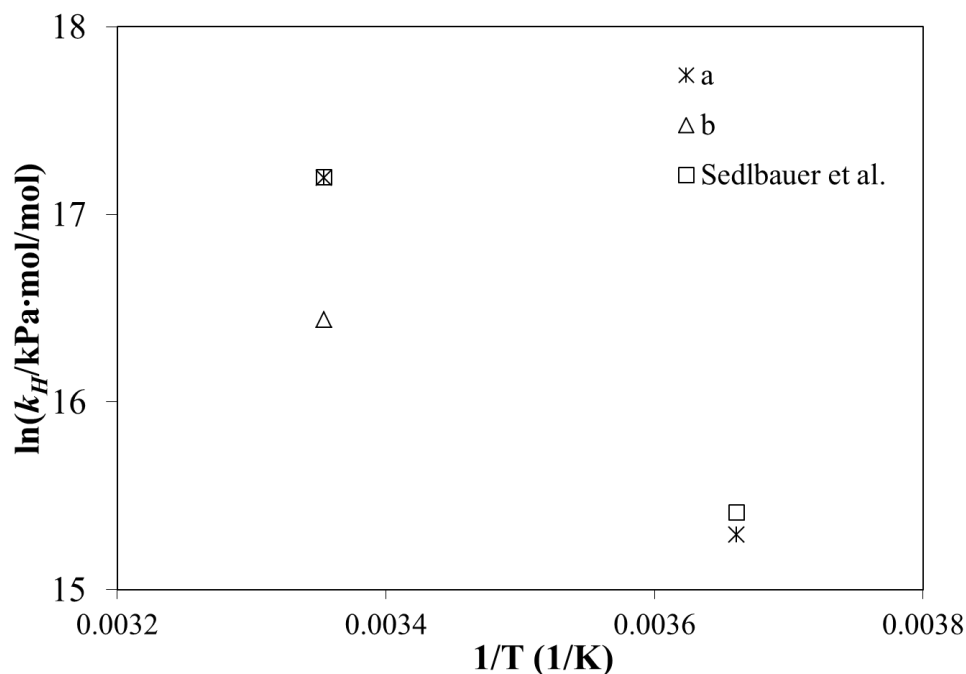


Figure 4.30: 2,2,5-Trimethylhexane x_i^{aq} data plotted as $\ln(k_H)$ compared with Sedlbauer et. al [191] predicted values. References: a [195], b [86]

4.7 Summary

A database containing recommended values for k_H , γ_i^∞ and x_i^{aq} of industrially important chemicals has been compiled for aqueous systems. Recommendations have been provided for at least one of these properties for 509 compounds. The structure of this database is compatible with the existing DIPPR[®] 801 database and DIADEM interface. Thermodynamic relationships, chemical family trends, and predicted values were important data analysis tools. Data from the various properties were related to each other using thermodynamic relationships and reliable DIPPR[®] 801 pure component property values. For hydrophobic compounds, the data from the three properties were combined to obtain recommendations. In instances where the data could not be combined, the thermodynamic relationships were helpful in identifying the best datasets.

Chemical family plots were used with temperature-dependent and single-temperature data to check for consistency within chemical families and to select the best data. Predicted values were used to select the best datasets when there were discrepancies in the literature data. Notes in the database provide information about the procedure and data used to provide recommendations for each compound and property.

5 Experimental Methods

The inert gas stripping (IGS) method has been used to measure both k_{HS} and γ_i^∞ s [37, 196]. In this study, k_{HS} and γ_i^∞ s were measured for various compounds at temperatures between 8°C and 50°C using the IGS method. Compounds in this study were selected based on the absence of experimental data or inconsistencies between reported experimental values. Some of the inconsistencies occur between experimentally determined k_{HS} and calculated k_{HS} (Equation 4.1).

5.1 Materials

The compounds used as solutes are listed in Table 5.1 with the suppliers and stated purities. Chemicals were used as received without further purification. Deionized distilled water was used as the solvent.

Table 5.1: Compounds and purities

Chemical	CAS	Supplier	Purity
Toluene	108-88-3	Sigma-Aldrich	99.8%
1-Butanol	71-36-3	Macron Chemicals	≥99.4%
Anisole	100-66-3	Sigma-Aldrich	≥99.0%
1,2,3-Trichlorobenzene	87-61-6	Sigma-Aldrich	99%
1,2-Difluorobenzene	367-11-3	Sigma-Aldrich	98%
4-Bromotoluene	106-38-7	Sigma-Aldrich	98%
2,4-Dichlorotoluene	95-73-8	Sigma-Aldrich	99%

5.2 Inert Gas Stripping Theory

The IGS method uses gas chromatography (GC) to analyze the change in composition with time of the vapor phase produced from stripping a dilute solution. The IGS methods of Leroi et al. (1977) and Mackay et al. (1979) were developed separately for γ_i^∞ and k_H , respectively [37, 196]; however, the experimental methods are the same and were derived using the same equations and assumptions. Mackay's method uses the relationship between k_H and γ_i^∞ (Equation 4.3). The equation developed by Leroi has been expanded by multiple authors [197-199], but an analogous expansion has not been reported for k_H . The following is the theory behind the IGS method as described by Krummen et al. [198]. The equations have also been written in terms of k_H in this study to easily apply them to solid solutes.

In vapor-liquid equilibrium, the following is used to describe the solute

$$y_i \hat{\phi}_i^V P = \gamma_i x_i f_i^L, \quad 5.1$$

where y_i is the mole fraction in the vapor phase, $\hat{\phi}_i^V$ is the partial molar fugacity coefficient of species i in the vapor phase, P is the total pressure, γ_i is the activity coefficient relative to the pure-component standard state, x_i is the mole fraction in the liquid phase, and f_i^L is the fugacity of pure species i in the liquid phase. In this study, it was assumed that $\hat{\phi}_i^V = 1$. Since the solute concentration is very small, γ_i can be replaced by γ_i^∞ .

For liquids, the liquid fugacity of pure species i is calculated at the system temperature using

$$f_i^L = \phi_i^{sat} P_i^{sat}, \quad 5.2$$

where P_i^{sat} is the vapor pressure and ϕ_i^{sat} is the vapor-phase fugacity coefficient of pure species i at its saturation pressure. This equation assumes the Poynting factor is unity which is valid at atmospheric pressure. Substituting Equation 5.2 into 5.1 gives

$$y_i P = x_i \gamma_i^\infty \phi_i^{sat} P_i^{sat} . \quad 5.3$$

The vapor-liquid equilibrium can be written in terms of k_H by substituting Equation 4.2 into 5.1 which results in Equation 2.3 given in Section 2.1.3 :

$$y_i P = x_i k_H . \quad 2.3$$

This relationship can be used for liquid and solid solutes. The solvent can be described by

$$y_{solv} P = P_{solv}^{sat} , \quad 5.4$$

where y_{solv} is the mole fraction of water in the vapor phase and P_{solv}^{sat} is the solvent vapor pressure. This equation assumes that the Poynting factor and vapor-phase corrections are negligible and that the activity coefficient of solvent is unity which is valid because $x_{solv} \approx 1$.

The equations used to determine k_H and γ_i^∞ using the IGS method are based on mass balances around the system. The mass balance assumes that only solute is stripped from the stripping cell which is accomplished by saturating the stripping gas with solvent before it enters the stripping cell. Additionally, these balances assume ideal gas behavior (valid since experiments were conducted at atmospheric pressure), negligible dissolution of the carrier gas in the system, additive gas volumes (valid at constant temperatures and pressures in unreactive systems), and small concentrations of solute in the headspace ($y_i \approx 0$). Figure 5.1 depicts the inlets and outlets to the cells used in the mass balances.

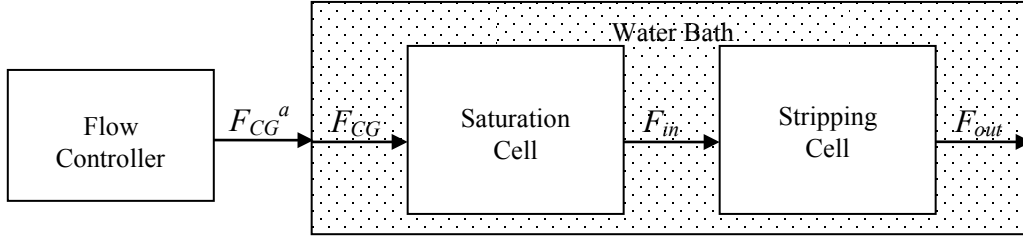


Figure 5.1: Inlets and outlets of saturation and stripping cells used in mass balances

The flow rate of carrier gas entering the saturation cell (F_{CG}) is determined from the measured volumetric flow rate at ambient conditions (F_{CG}^a), cell temperature (T), and ambient temperature (T^a) using

$$F_{CG} = F_{CG}^a \cdot \frac{T}{T^a} . \quad 5.5$$

The volumetric flow rate entering the stripping cell (F_{in}) is equal to the sum of the carrier gas flow rate and the flow rate due to solvent saturation (F_{solv}):

$$F_{in} = F_{CG} + F_{solv} . \quad 5.6$$

The flow rate of solvent leaving the saturation cell and entering the stripping cell is described by

$$F_{solv} = F_{in} y_{solv} = F_{in} \frac{P_{solv}^{sat}}{P} . \quad 5.7$$

Krummen et al. [198] use F_{CG} instead of F_{in} in Equation 5.7 which is incorrect. Combining Equations 5.6 and 5.7 and rearranging to solve for F_{in} yields

$$F_{in} = \frac{F_{CG}}{\left(1 - \frac{P_{solv}^{sat}}{P}\right)} . \quad 5.8$$

The flow rate leaving the stripping cell is the sum of the flow entering the cell and the flow rate of stripped solute (F_i):

$$F_{out} = F_{in} + F_i . \quad 5.9$$

Using an Equation analogous to 5.7 for the stripping cell yields

$$F_i = F_{out} y_i , \quad 5.10$$

where y_i is the mole fraction of solute exiting the stripping cell. This assumes that only solute is stripped from the stripping cell (accomplished by first saturating the carrier gas with solvent) and ideal gas behavior. Combining 5.9 and 5.10 and solving for F_{out} yields

$$F_{out} = \frac{F_{in}}{1 - y_i} . \quad 5.11$$

F_i can also be described by

$$F_i = - \frac{RT}{P} \frac{dn_i}{dt} , \quad 5.12$$

where n_i is the moles of solute in the stripping cell which changes with time (t) and R is the universal gas constant. The derivative term is determined assuming ideal gas behavior:

$$\frac{dn_i}{dt} = -y_i \frac{PF_{out}}{RT} . \quad 5.13$$

Substituting Equation 5.11 into 5.13 and using the phase equilibrium equation for y_i obtained from Equation 5.3 yields

$$\frac{dn_i}{dt} = -x_i \gamma_i^\infty \phi_i^{sat} P_i^{sat} \frac{1}{RT} \frac{F_{in}}{1 - \frac{x_i \gamma_i^\infty \phi_i^{sat} P_i^{sat}}{P}} . \quad 5.14$$

The total moles of solute in the stripping cell is the sum of the moles of solute in the liquid (L) and vapor (V) phases:

$$n_i = n_i^L + n_i^V . \quad 5.15$$

Assuming that the total number of moles in the cell is equal to the number of moles of solvent in the cell (which is valid since $n_{solv} \gg n_i$) and assuming that the vapor phase can be described by an ideal gas yields:

$$n_i = x_i n_{solv} + y_i \frac{PV_g}{RT} , \quad 5.16$$

where V_g is the volume of the headspace in the stripping cell. The total cell volume was determined by filling the cell completely with water and measuring the mass, and then calculating volume using water density. The volume of water added to the cell for each experiment (calculated from the known mass and density) was subtracted from the total cell volume to obtain V_g . Substituting Equation 5.3 into 5.16 and rearranging results in an equation for x_i :

$$x_i = \frac{n_i}{n_{solv} + \gamma_i^\infty \phi_i^{sat} P_i^{sat} \frac{V_g}{RT}} . \quad 5.17$$

Combining 5.14 and 5.17 yields:

$$\frac{dn_i}{dt} = -x_i \gamma_i^\infty \phi_i^{sat} P_i^{sat} \frac{1}{RT} \frac{F_{in}}{1 - \frac{\gamma_i^\infty \phi_i^{sat} P_i^{sat}}{P}} \cdot \frac{n_i}{n_{solv} + \gamma_i^\infty \phi_i^{sat} P_i^{sat} \frac{V_g}{RT}} . \quad 5.18$$

In this study it was assumed $y_i \ll 1$ so that

$$\frac{1}{1-y_i} \approx 1. \quad 5.19$$

With this assumption, 5.18 simplifies to

$$\frac{dn_i}{dt} = - \frac{n_i}{n_{solv} + \gamma_i^\infty \phi_i^{sat} P_i^{sat} \frac{V_g}{RT}} \gamma_i^\infty \phi_i^{sat} P_i^{sat} \frac{F_{in}}{RT}. \quad 5.20$$

The only variable that is a function of time is n_i . Integration yields

$$\ln\left(\frac{n_i}{n_i^0}\right) = - \frac{\gamma_i^\infty \phi_i^{sat} P_i^{sat}}{n_{solv} + \gamma_i^\infty \phi_i^{sat} P_i^{sat} \frac{V_g}{RT}} \frac{F_{in}}{RT} t, \quad 5.21$$

where the superscript 0 indicates the initial condition. Assuming that concentration is proportional to GC peak area (A) gives:

$$\ln\left(\frac{A}{A^0}\right) = - \frac{\gamma_i^\infty \phi_i^{sat} P_i^{sat}}{n_{solv} + \gamma_i^\infty \phi_i^{sat} P_i^{sat} \frac{V_g}{RT}} \frac{F_{in}}{RT} t. \quad 5.22$$

A plot of $\ln(A/A^0)$ versus t yields a straight line with slope, a :

$$a = \frac{\ln\left(\frac{A}{A^0}\right)}{t}. \quad 5.23$$

Substituting 5.23 into 5.22 and rearranging gives:

$$\gamma_i^\infty = \frac{-RTn_{solv}}{\phi_i^{sat} P_i^{sat} \left(V_g + \frac{F_{CG}}{\left(1 - \frac{P_{solv}^{sat}}{P}\right)a} \right)}. \quad 5.24$$

This equation is slightly different from that reported by Krummen et al. [198] due to the change in Equation 5.7. Equation 5.24 refers specifically to liquid solutes where the liquid fugacity of the pure species can be calculated using Equation 5.2. Using Equation 2.3 to describe the phase equilibrium and the methodology outlined above results in an equation for k_H :

$$k_H = \frac{-RTn_{solv}}{V_g + \frac{F_{CG}}{\left(1 - \frac{P_{solv}^{sat}}{P}\right)a}}. \quad 5.25$$

In order to determine ϕ_i^{sat} , the virial equation truncated after the second virial coefficient, B , was used,

$$Z = \frac{PV}{RT} = 1 + \frac{BP}{RT}, \quad 5.26$$

where Z is the compressibility factor and V is volume.

At a constant temperature, ϕ_i^{sat} is determined by:

$$\ln \phi_i^{sat} = \int_0^{P_i^{sat}} (Z_i^V - 1) \frac{dP}{P}. \quad 5.27$$

Using 5.26 results in:

$$\phi_i^{sat} = \exp\left(\frac{B_{ii}P_i^{sat}}{RT}\right). \quad 5.28$$

Temperature dependent equations for B_{ii} were obtained from the DIPPR[®] 801 database[117]. In instances where data were not available within the experimental temperature range, ϕ_i^{sat} was assumed to be unity. For compounds that were solids, Ψ was used to calculate γ_i^∞ as outlined in Section 4.1.

5.3 Apparatus and Procedure

Dilute solutions (on the order of 10^{-3} mole fraction or less) of each of the individual compounds listed in Table 5.1 were made in a glass cell with a diameter of 5 cm and total volume of 414 mL. Literature x_i^{aq} values were referenced to ensure the concentration in the cell was below the solubility limits for the compound in question. The solution was allowed to mix for several hours, typically overnight, to ensure equilibrium was reached. A similar glass cell was filled with deionized, distilled water and was used as the saturation cell. Both cells were threaded and sealed with threaded Teflon caps and Viton or Kalrez o-rings. All the fittings (316 stainless steel) were threaded into the Teflon lid to ensure a leak tight fit. Swagelok[®] quick-connects allowed easy removal of both cells from the system for cleaning purposes. The temperature of the stripping cell was measured using an RTD digital thermometer (Omega, Model HH804) while the pressure was measured using a pressure transducer (Paroscientific, Model 740). Cell pressure varied slightly from ambient pressure with higher flow rates.

Nitrogen ($\geq 99.998\%$) was used as the carrier gas. The flow rate was controlled by a mass flow controller (Aalborg, GFC17). Two controllers were used with ranges of 0-12 and 0-200 mL/min and accuracies of $\pm 1.5\%$ FS. Nitrogen was first bubbled through the saturation cell that contained only water. The saturated nitrogen was then bubbled through the dilute solution in the stripping cell. Small bubbles were formed in both cells using a 2 μm porous 316L stainless steel filter element (Mott). The stripping cell was continuously mixed with a magnetic Teflon stir bar. Both cells were in a water bath controlled by a temperature controller (Omega, CSi32). At temperatures of 298.15 K or below, a cooling loop (Neslab, RTE-8DD) was used in addition to the heating element for temperature control. The temperature of the bath was periodically

checked using a mercury-in-glass thermometers (scale of 0.1 K), and it stayed within ± 0.1 K of the set point.

Gas exiting the stripping cell entered a transfer line that was heated to 403.15 K using a fiberglass heating cord (Omega, #HTC-120) to prevent condensation in the transfer line. The transfer line was made of 1/8" (3.175 mm) 304 stainless steel tubing and was wrapped in aluminum, the heating cord, and then fiber glass insulation. Transfer line temperature was measured and controlled using a type-K thermocouple and temperature controller (Omega, Model CN7500). The transfer line was connected to the gas sampling valve of a GC (Agilent, 7890A) equipped with a flame ionization detector (FID). Helium ($\geq 99.995\%$) was used as the GC carrier gas. Hydrogen ($\geq 99.95\%$) and ultra zero air were used with the FID. An Equity-5 column (Supelco, #28089-U) was installed in the GC. A schematic of the apparatus is given in Figure 5.2.

In order to develop the GC methods, dilute solutions of each compound were made in test tubes. An additional solution was made that also contained acetone. Acetone was used to clean the cells between experiments. In many experiments, trace amounts of acetone were present despite efforts to remove it during the cleaning process (described later in this section). Manual liquid injections were performed using both solutions to develop the methods. Conditions were changed in order to elute sharp peaks and optimize the separation between acetone and the compound of interest. The method had to separate the compound and acetone peaks to eliminate any potential interference. The GC conditions used throughout experimentation that were the same for every method are listed in Table 5.2. The conditions that were changed for each compound are listed in Table 5.3.

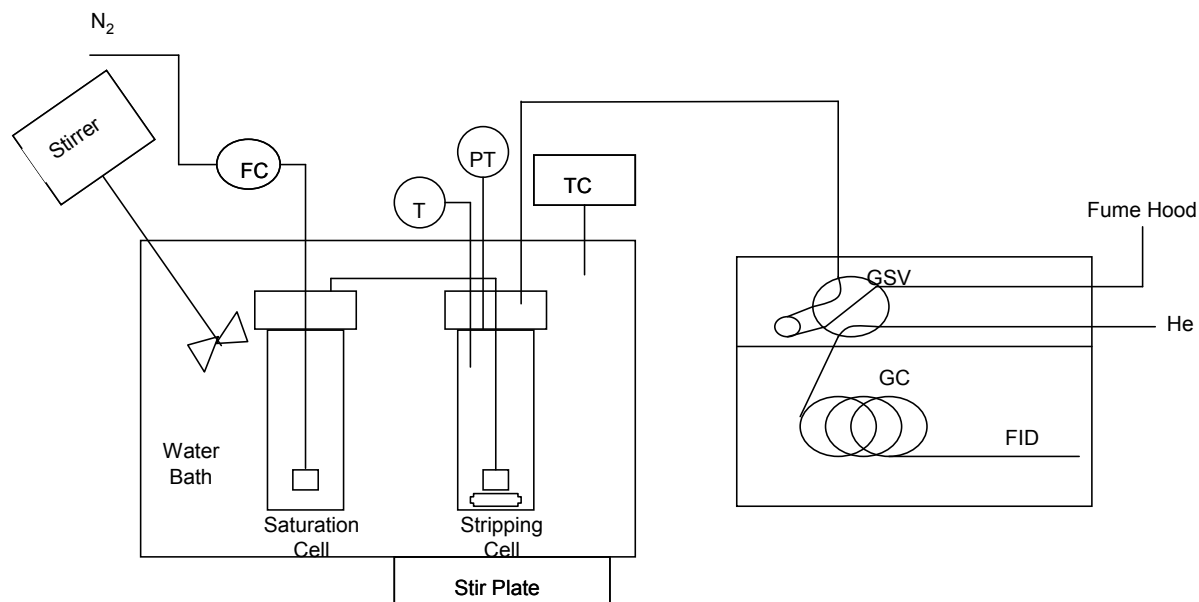


Figure 5.2: IGS apparatus

Table 5.2: Constant GC conditions

Component	Conditions
GSV	130°C
FID	H ₂ : 40 mL/min Air: 450 mL/min Makeup He: 25.388 mL/min

Table 5.3: GC conditions for each compound

Compound	Inlet	FID	Oven Temperature Program
Toluene	130°C Split ratio: 100	250°C	130°C for 0.5 min; ramp at rate of 20°C/min up to 175°C; hold 1 min
1-Butanol	130°C Split ratio: 100	250°C	100°C for 0.5 min; ramp at rate of 15°C/min up to 140°C
4-Bromotoluene	100°C Split ratio: 100	250°C	100°C for 0.5 min; ramp at rate of 10°C/min up to 120°C; hold 0.1 min
Anisole	155°C Split ratio: 100	275°C	150°C for 0.5 min; ramp at rate of 10°C/min up to 165°C; hold 1.5 min
1,2,3-Trichlorobenzene	130°C Split ratio: 40	275°C	200°C for 0.5 min; ramp at rate of 20°C/min up to 250°C; hold 0.1 min
2,4-Dichlorotoluene	130°C Split ratio: 100	275°C	185°C for 0.5 min; ramp at rate of 10°C/min up to 208°C

Since relative concentrations are required instead of actual concentrations, each stripping cell was used for multiple experiments. Initially, one temperature was chosen and the flow rate was changed in a randomized order to determine an optimal flow rate for the experiments over the temperature range of interest. It was assumed that equilibrium would be attained at the other temperatures because a conservative flow rate was chosen. Once the flow rate was selected, the experimental temperature was randomized in measuring the determined number of replicates at each temperature. The amount of time required between runs with the same cell varied depending on the variables that were changed. If temperature was changed, the cell was allowed to mix for at least 30 minutes once it reached the new temperature set point. In addition, the transfer line was flushed between runs to remove any chemical from the previous experiment.

When experiments were completed with a given stripping cell, the cell was thoroughly cleaned before the next experiment. After proper disposal of the aqueous waste, the stripping cell and frit were rinsed with acetone to remove any residual chemical before the cell was washed in the sink since some of the chemicals used are potentially harmful to the environment. Everything was then washed off with soapy water and rinsed with water. A minimum of 30 mL of acetone was then forced through the frit using a syringe. All of the cell components (including all stainless steel tubing) were then rinsed again with acetone. About 100 mL of deionized water was then forced through the frit using a syringe in an effort to remove the acetone. All cell components were then given a final rinse with deionized water and allowed to dry completely before a new cell was prepared. N₂ was used to speed up the drying process of the frit and portions of the tubing. The saturation cell was cleaned in a similar manner. It was not cleaned as often since the only compound entering the cell was N₂. The saturation cell did

not have to dry completely before it was used since it was unnecessary to know the amount of water in the cell.

6 Experimental Results and Discussion

The apparatus was first validated by measuring the k_{HS} and γ_i^∞ s of toluene and 1-butanol at various temperatures. Table 6.1 compares the values obtained in this study with literature values. The percent standard deviations of the average k_H values (s_{kH}) and corresponding γ_i^∞ values (s_{γ^∞}) are included in addition to the number of measurements, N . Also given are the 95% confidence intervals (CIs) of the average k_H calculated using a Student's T distribution. The CIs are reported as a percentage of the average. CIs cannot be calculated using a Student's T distribution when $N = 1$. The reported γ_i^∞ standard deviations, s_{γ^∞} , were determined using the s_{kH} values and propagation of error. In cases where only one measurement was performed, the reported s is an estimated uncertainty. Literature values calculated from γ_i^∞ and P_i^{sat} are noted in the table. There have been reported problems with using IGS to measure k_{HS} of n -alkanols (including 1-butanol) [68]. Reported n -alcohol values using IGS are higher than values measured using a static equilibration method, and this difference has been attributed to evaporation in the stripping cell and adsorption to the bubble surface. However, the values obtained in this study are comparable to other literature values. The largest discrepancy is at the lowest temperature measured. This corresponds to a low k_H meaning it was more difficult to measure a change in concentration with time. The use of toluene and 1-butanol to validate the method established a broad range of k_{HS} for which this method is applicable.

For each compound, multiple flow rates were tested to ensure equilibrium was reached because the measured k_H should not be a function of flow rate if the system is at equilibrium. If the flow rate was too high, k_H would appear lower because there was not as much time for the bubbles to equilibrate with the solution resulting in a lower concentration of solute in the bubble. Problems were also observed if the flow rate was too small. For compounds with lower k_H values, if the flow rate was too small, it was difficult to detect a change in concentration with time which resulted in noisy GC data. Table 6.2 gives the flow rates that were tested for each compound and the temperature of the flow rate experiments. Equilibrium was reached at flow rates that resulted in k_H values that were not statistically different (analysis of variance p-values > 0.05). The bolded values are the flow rates that were selected for subsequent experiments at other temperatures. The flow rates that resulted in lower k_H values are also listed. For toluene, 4-bromotoluene, and 2,4-dichlorotoluene, all of the flow rates tested resulted in equilibrium. For anisole and 1,2,3-trichlorobenzene, flow rates were tested at multiple temperatures due to the noise in the GC data at some of the flow rates. Higher flow rates typically resulted in higher R^2 values, but equilibrium was not always reached.

Table 6.1: Toluene and 1-butanol results at ambient pressure (approximately 12.5 psia) from this study and literature

T/K	k_H/kPa	$s_{kH}/\%$	N	CI/% $\alpha=0.05$	Lit. k_H/kPa	ϕ_i^{sat}	γ_i^∞	$S_{\text{res}}/\%$	Lit. γ_i^∞
Toluene									
298.15	$3.35 \cdot 10^4$	12	13	7	$3.69 \cdot 10^4$ [192]	0.996	$8.84 \cdot 10^3$	13	$9.20 \cdot 10^3$ [200]
308.15	$5.46 \cdot 10^4$	5.2	2	47	$5.44 \cdot 10^4$ [201] ^a	0.994	$8.81 \cdot 10^3$	7.3	$8.72 \cdot 10^3$ [200]
1-Butanol									
298.15	69.4	8.0	1	NA	49.4 [202]	0.996	77.4	26	48.4 [168]
308.15	105	4.1	1	NA	115 [203]	0.995	58.9	26	57.2 [168]
313.15	140	4.3	1	NA	123 [180] ^a	0.993	56.5	26	49.5 [180]
					146 [168] ^a				58.9 [168]
323.15	261	10	2	92	274 [204]	0.991	57.5	27	59.4 [168]

^afrom $\gamma_i^\infty \cdot P_i^{\text{sat}}$

Table 6.2: Flow rates tested and used in this study

Compound	Flow Rates Resulting in Equilibrium (mL/min)	Flow Rates Not Resulting in Equilibrium (mL/min)	T/K
Toluene	12, 40 , 75	NA	298.15
1-Butanol	200	Data too noisy at lower flow rates	298.15
Anisole	125, 160	200	281.02 & 293.15
1,2-Difluorobenzene	20, 40	12, 60, 80	298.22
4-Bromotoluene	20, 40 , 60, 80	NA	308.10
1,2,3-Trichlorobenzene	20 20, 40 40, 60	40, 60 60, 80 NA	281.01 293.07 322.93
2,4-Dichlorotoluene	20, 40 , 60 , 80	NA	293.09

During the method validation process, it was discovered that allowing the frit to soak overnight in the solution resulted in small bubbles (diameter < 1 mm). If the frit is not completely wetted, then there are a small number of paths with less resistance where all of the nitrogen exits resulting in larger bubbles (diameter > 1 mm). In the instances of larger bubbles, the measured k_H values were low because of mass transfer limitations to the equilibration. The solutions were made the night before to allow the frit to soak and to ensure complete mixing. To ensure that the mixing time did not result in erroneous values due to adsorption effects, the equilibration time was varied from 3 hours to 2 days for 1,2-difluorobenzene. There was no difference within the results, so adsorption effects in the cell due to equilibration time were determined to be negligible. Periodically the cells were bypassed allowing only nitrogen to go through the transfer line to ensure there was no chemical holdup in the line. The GC analysis from these trials confirmed there was no holdup.

Using the k_H values obtained in this study and Equation 2.3, values of y_i were calculated based on experimental conditions to determine the validity of the assumption that $y_i \approx 0$ used in the IGS theory. Table 6.3 lists the maximum y_i values that were encountered at any time during

any of the runs for each compound. The most significant y_i is from the 1,2-difluorobenzene experiments. However, this contribution was still considered negligible. Since the cells were used for multiple, randomized experiments, the majority of the runs associated with a given temperature had a much lower initial concentration. There was not a statistically significant difference between the k_H values from the experiment with the starting concentration listed in Table 6.3 and the experiments that had smaller starting concentrations. The k_H values from runs that started at a more dilute condition were not statistically different from those that started at higher concentrations.

Table 6.3: Maximum headspace concentrations possible in this study

Compound	Max y_i (mole fraction)
Toluene	$3.09 \cdot 10^{-2}$
1-Butanol	$2.86 \cdot 10^{-3}$
Anisole	$1.06 \cdot 10^{-2}$
1,2-Difluorobenzene	$1.08 \cdot 10^{-1}$
4-Bromotoluene	$2.19 \cdot 10^{-3}$
1,2,3-Trichlorobenzene	$1.46 \cdot 10^{-4}$
2,4-Dichlorotoluene	$3.05 \cdot 10^{-3}$

Solutions of varying concentrations of both 1,2-difluorobenzene and toluene were made in order to verify the linearity of the detector (FID). A linear response was observed for both compounds. Peak areas for the vapor samples observed during experimentation fell within the peak areas of the liquid samples used to check linearity. For the other compounds, it was assumed that the detector response was linear because small concentrations were used.

During the experiments, it was observed that adsorption effects in the apparatus became significant after the concentration in the cell dropped below a lower limit. A wide range of k_H

values were obtained for the same compound at the same temperature and flow rate even though the plots of $\ln(A/A^0)$ versus time had R^2 values greater than 0.99. Plots were made of the slope (Equation 5.23) as a function of time to determine how the cumulative slope changed with the addition of a new point. In some cases, the slope equilibrated to a constant value whereas in other instances it never settled as illustrated with 1,2-difluorobenzene data in Figure 6.1 and Figure 6.2. Figure 6.1 shows GC data from two separate experiments at the same conditions. Both experiments resulted in high R^2 values (>0.99); however, they gave different slopes resulting in significantly different k_H values. Figure 6.2 shows the change of slope with time for the GC data shown in Figure 6.1. The cumulative slope of Run A never settled to a constant value. In all instances where the slope continued to change with time for each compound, the k_H values were lower than when the slope reached a constant value. The changing slope only occurred when the peak areas (solute concentration) dropped below a value that was compound dependent. It is suspected that adsorption effects in the cell became significant once the solution concentration dropped below a certain value. Since calibration is not required for this method, the exact lower limit concentrations were not determined. The lower limit was estimated to be below 1% of the solubility limit with 1,2-difluorobenzene. The plots of the cumulative slope versus time allowed determination of when this lower limit was passed for each compound. Data were not used to determine k_H values from any experiments where adsorption was significant.

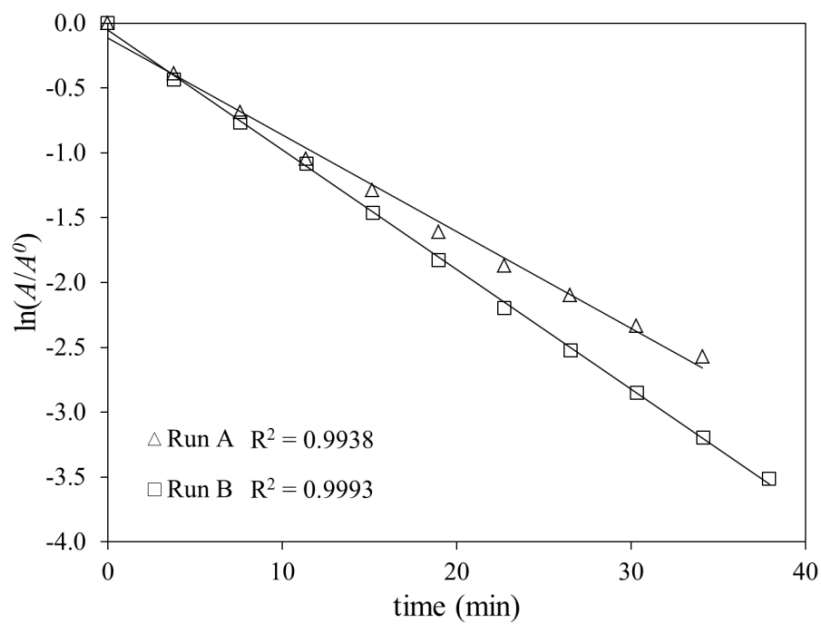


Figure 6.1: GC data for two separate 1,2-difluorobenzene experiments at 323.15 K and a N_2 flow rate of 40 mL/min that resulted in different slopes (a)

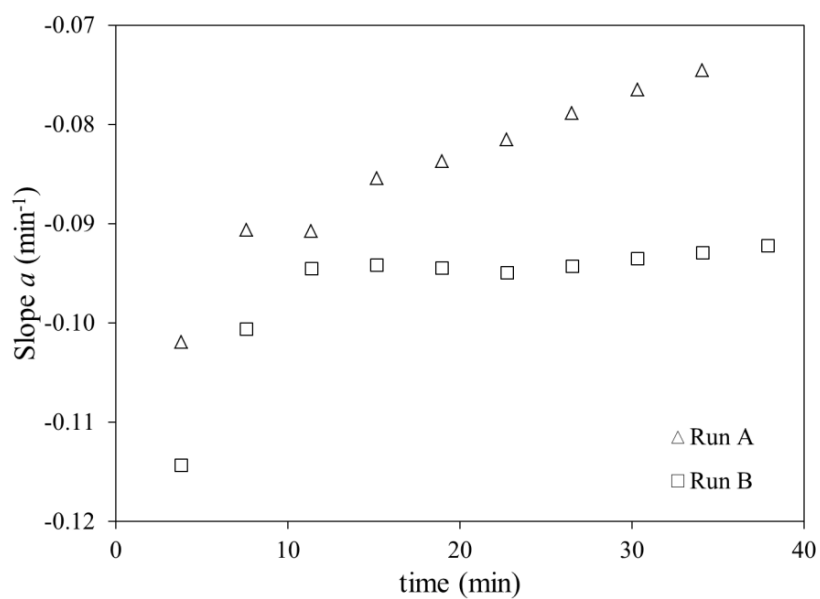


Figure 6.2: Cumulative slopes (a) from GC data shown in Figure 6.1 as a function of time

The average experimental k_H values from this study and corresponding γ_i^∞ s are summarized in Table 6.4 along with ϕ_i^{sat} and Ψ values used in the calculations when the solute is a liquid or solid, respectively. The number of experiments at each temperature, N , and the standard deviation of the k_H average, s_{kH} , are included. Also given are the average k_H 95% confidence intervals (CIs) calculated using a Student's T distribution. The CIs are reported as a percentage of the average. For some compounds, B_{ii} values were unavailable from the DIPPR[®]801 [117] database at the experimental temperatures, so in these cases ϕ_i^{sat} was assumed to be unity. The ϕ_i^{sat} values for 1,2-difluorobenzene indicate that this assumption was valid since the values are close to unity. The reported γ_i^∞ standard deviations, s_{γ^∞} , were determined using the s_{kH} values and propagation of error. For compounds where ϕ_i^{sat} was assumed to be unity, the ϕ_i^{sat} error was assumed to be 10% in the propagation of error calculations. This is a conservative error estimation since the 1,2-difluorobenzene values indicate a difference of 1-2% from unity. In some instances the errors associated with γ_i^∞ are significantly larger than those associated with k_H due to the uncertainty of ϕ_i^{sat} or Ψ . Details about each experimental run are included in Appendix D.

Figure 6.3 shows the 1,2,3-trichlorobenzene k_H values obtained in this study along with those found in the literature. There are many literature values for 1,2,3-trichlorobenzene and considerable scatter in the literature values. There are also inconsistent temperature trends among the extant data. The measurements made in this study provide a strong linear trend with small uncertainty in the temperature dependence. The temperature trend of this study is in good agreement with the data of reference h [205]. While the temperature-dependent slope of the values reported by reference c [206] is consistent with the results obtained here, their absolute

values appear too high at any given temperature. Much of the other literature data are scattered about the values reported here but do not provide a clear temperature-dependent trend owing to the magnitude of that scatter.

Table 6.4: k_H and γ_i^∞ values at ambient pressure (approximately 12.5 psia) from this study and literature

T/K	k_H /kPa	$s_{KH}/\%$	N	CI/% $\alpha = 0.05$	Lit. k_H /kPa	ϕ_i^{sat}	Ψ	γ_i^∞	$s_{\gamma^\infty}/\%$
Anisole									
280.95	$8.35 \cdot 10^2$	10	5	12		1 ^a	-	$5.60 \cdot 10^3$	14
293.16	$1.48 \cdot 10^3$	4.4	6	4.6	$1.48 \cdot 10^3$ [206] ^b	1 ^a	-	$4.23 \cdot 10^3$	11
308.15	$2.95 \cdot 10^3$	5.6	4	8.8		1 ^a	-	$3.32 \cdot 10^3$	12
322.90	$5.72 \cdot 10^3$	3.6	5	4.5		1 ^a	-	$2.83 \cdot 10^3$	11
1,2-Difluorobenzene									
298.19	$4.35 \cdot 10^4$	2.3	4	3.6	$3.95 \cdot 10^4$ [207] ^c	0.991	-	$6.13 \cdot 10^3$	10
308.12	$6.65 \cdot 10^4$	3.6	3	9.0		0.988	-	$5.89 \cdot 10^3$	11
317.90	$9.33 \cdot 10^4$	2.7	3	6.7		0.984	-	$5.41 \cdot 10^3$	10
322.89	$1.05 \cdot 10^5$	6.4	5	7.9		0.982	-	$4.94 \cdot 10^3$	12
4-Bromotoluene									
293.10	$1.18 \cdot 10^4$	1.6	3	4.0		-	0.870	$9.04 \cdot 10^4$	26
298.16	$1.71 \cdot 10^4$	6.0	3	15	$1.77 \cdot 10^4$ [208] ^c	-	0.965	$9.41 \cdot 10^4$	26
308.10	$2.76 \cdot 10^4$	5.1	12	3.2		1 ^a	-	$8.30 \cdot 10^4$	12
318.03	$4.03 \cdot 10^4$	11	2	95		1 ^a	-	$6.92 \cdot 10^4$	15
322.91	$5.28 \cdot 10^4$	2.3	3	5.7		1 ^a	-	$6.99 \cdot 10^4$	11
1,2,3-Trichlorobenzene									
280.95	$3.36 \cdot 10^3$	8.9	3	22	$3.01 \cdot 10^3$ [205] ^{b,d}	-	0.440	$4.34 \cdot 10^5$	17
293.06	$6.84 \cdot 10^3$	5.0	4	7.9		-	0.535	$3.27 \cdot 10^5$	11
308.14	$1.44 \cdot 10^4$	6.4	5	7.9		-	0.703	$2.34 \cdot 10^5$	8.8
322.99	$2.81 \cdot 10^4$	12	8	10		-	0.946	$1.83 \cdot 10^5$	13
2,4-Dichlorotoluene									
280.94	$7.14 \cdot 10^3$	21	4	33		1 ^a	-	$4.39 \cdot 10^5$	23
293.09	$1.68 \cdot 10^4$	15	14	8.6		1 ^a	-	$3.99 \cdot 10^5$	18
308.15	$3.25 \cdot 10^4$	24	5	30	$3.38 \cdot 10^4$ [8] ^c	1 ^a	-	$2.66 \cdot 10^5$	27
323.10	$7.54 \cdot 10^4$	20	4	32		1 ^a	-	$2.41 \cdot 10^5$	23

^a ϕ_i^{sat} error of 10% assumed for propagation of error

^b 291.15 K

^c from P_i^{sat}/x_i^{aq} at 298.15 K

^d 277.15 K

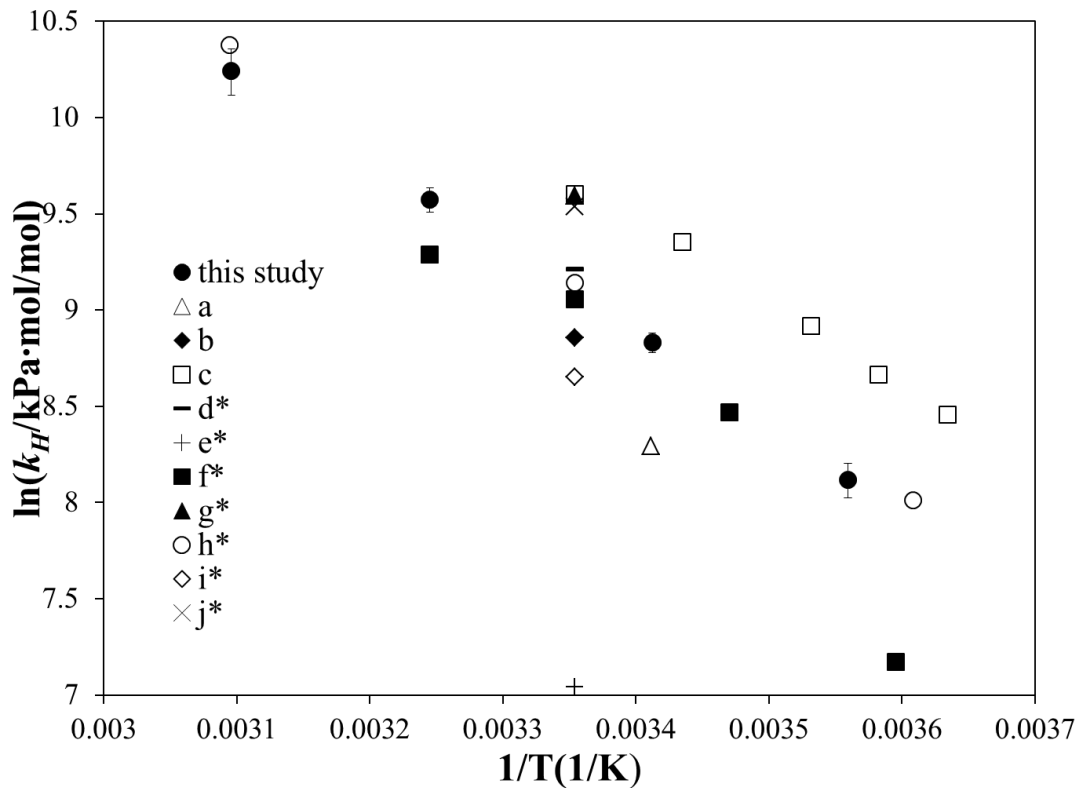


Figure 6.3: 1,2,3-Trichlorobenzene k_H values from this study and from literature. Values from references marked with an asterisk (*) are x_i^{aq} data converted to k_H values using Equation 4.1. References: a [209], b [71], c [206], d* [210], e* [211], f* [212], g* [213], h* [205], i* [207], j* [214]

Figure 6.4 shows the anisole k_H values obtained in this study along with those found in the literature. The measurements from this study show good agreement with the anisole x_i^{aq} data at higher temperatures and match the directly reported k_H values at the lower temperatures. The k_H values obtained from the literature x_i^{aq} data are considered to be low in the lower temperature range. This means that the literature anisole x_i^{aq} values at the lower temperatures have higher errors than the rest of the x_i^{aq} data set.

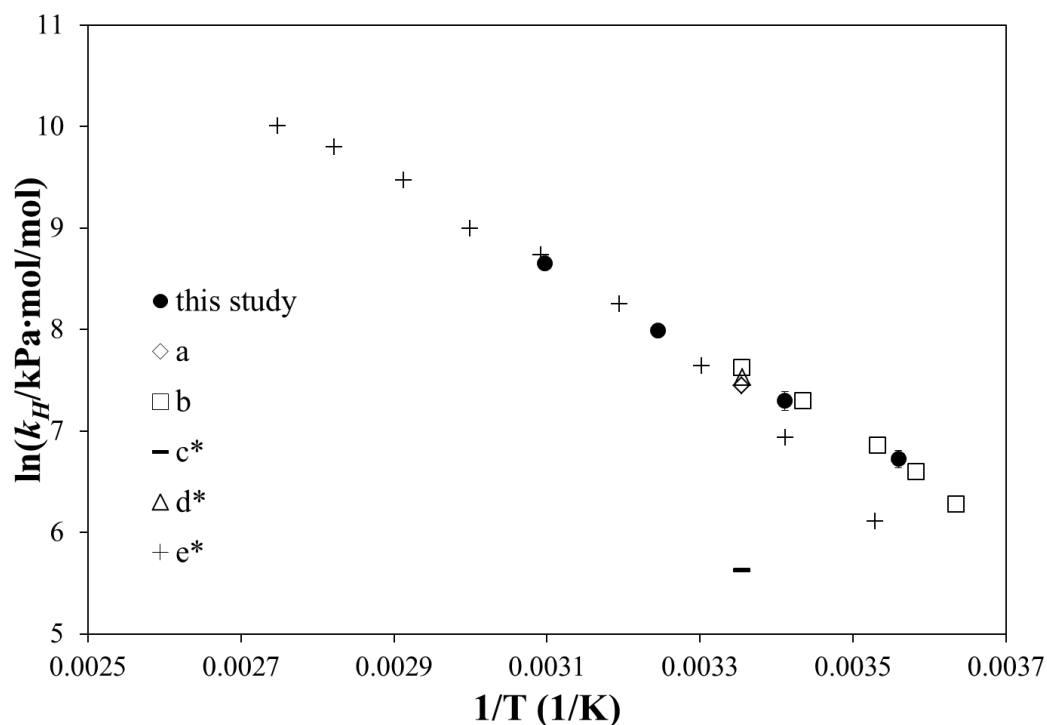


Figure 6.4: Anisole k_H values from this study and from the literature. Values from references marked with an asterisk (*) are x_i^{aq} data converted to k_H values using Equation 4.1. References: a [172], b [206], c* [215], d* [118], e* [216]

The 2,4-dichlorotoluene results are shown in Figure 6.5. The results from this study fall within the trend of the literature x_i^{aq} data. The remaining compounds have very little literature data available for comparison. Figure 6.6 shows the 1,2-difluorobenzene results from this study with the single literature x_i^{aq} value. The difference between the literature value and the value from this study at the same temperature could reasonably be due to experimental uncertainty. The 4-bromotoluene results are shown in Figure 6.7. The data of this study agree the best with the literature x_i^{aq} value.

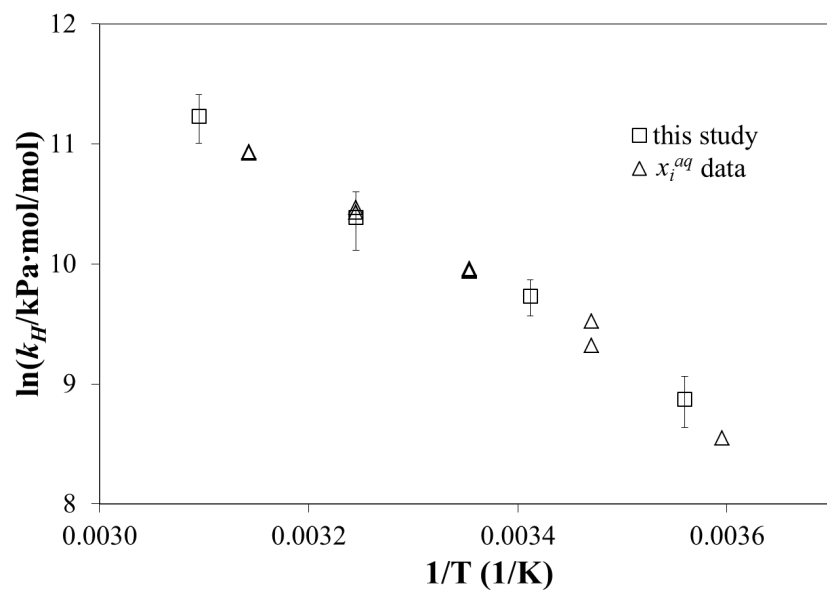


Figure 6.5: 2,4-Dichlorotoluene k_H values from this study and from the literature. Values from x_i^{aq} data were converted to k_H values using Equation 4.1. Reference: [8]

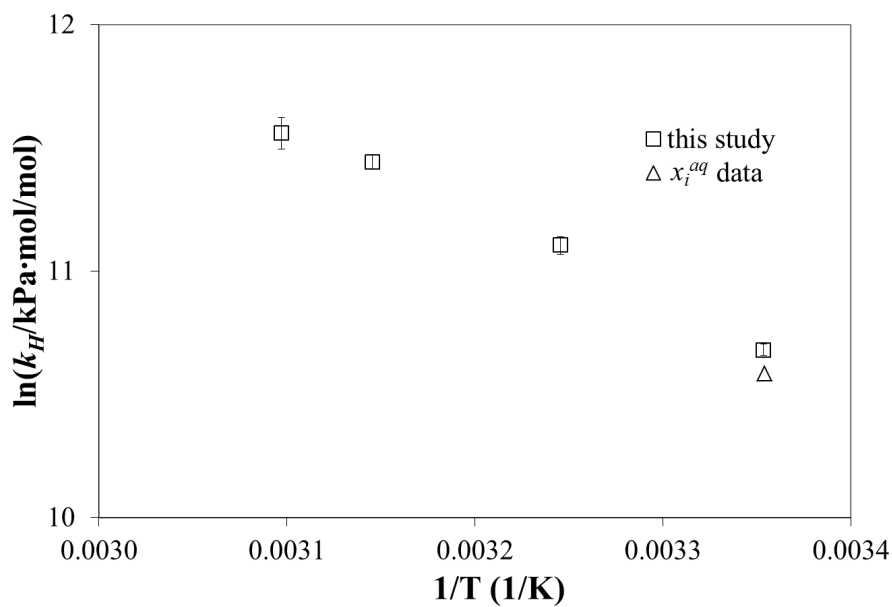


Figure 6.6: 1,2-Difluorobenzene k_H values from this study and from the literature. Values from x_i^{aq} data were converted to k_H values using Equation 4.1. Reference: [207]

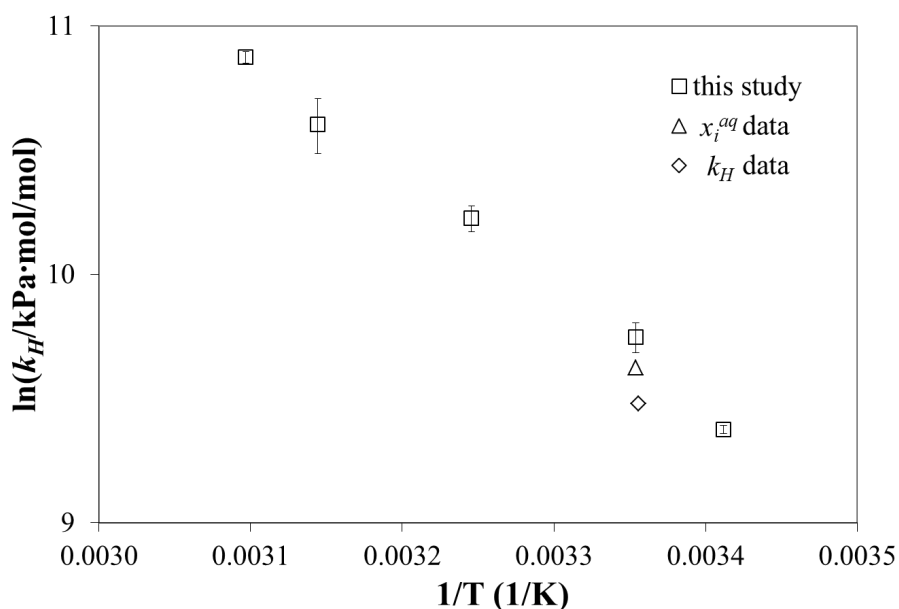


Figure 6.7: 4-Bromotoluene k_H values from this study and from the literature. Values from x_i^{aq} data were converted to k_H values using Equation 4.1. References: [47, 208]

The experimental results from this study clarify discrepancies between literature values for 1,2,3-trichlorobenzene and anisole. Temperature-dependent values were determined for 2,4-dichlorotoluene, 1,2-difluorobenzene, and 4-bromotoluene which only had limited literature data. Properties in the DIPPR[®] 801 database allowed for the determination of γ_i^∞ values for each compound. A fugacity ratio was required to determine γ_i^∞ values for the solid solutes. In addition, a procedure was identified for determining when adsorption effects become significant during IGS experiments.

7 Prediction Methods

7.1 Overview of Prediction Method Types

This section provides general information about the types of prediction methods available in the literature. Property-property relationships (PPRs) relate the property of interest to other properties of the chemical [24]. These relationships can be thermodynamically based or strictly empirical. PPRs are often restricted to the chemical classes from which they were derived [217].

Group-contribution methods (GCMs) parse the chemical's structural formula into functional groups and fragments, and the property of a compound is estimated by summing together the contributions of each segment [218]. First-order methods use groups that do not take into account interactions with neighboring compounds. Second-order methods are more complex because neighboring atoms are included in the group definitions. Limitations of these methods are their inability to differentiate between isomers and large errors for compounds with multiple non-alkyl functional groups. Empirical corrections are often used to offset these deviations. Group-contribution methods fail when a compound contains a functional group that was not contained in the original training set of the method [14, 219]. Bond contribution methods (BCMs) work in a similar fashion; they add up the contributions from each bond in the molecule.

Quantitative structure-property relationships (QSPRs) are based on fragment contributions and structural descriptors [24]. Descriptors (such as topological indices, molecular

volume, molecular surface area, etc.) are calculated from molecular structure [220]. There are various software tools and methods for obtaining and selecting descriptors with some being more complex than others. Another name associated with methods of this type is quantitative structure-activity relationships (QSARs).

Linear Solvation Energy Relationships (LSERs) relate the Gibbs energy of solution of a solute to a series of empirical parameters that describe the cavity formation and the solute-solvent interaction processes that occur when a gaseous solute dissolves in a liquid [221]. LSERs are a type of QSPR [222]. Parameters used (often called “solvatochromic parameters”) include solute dipolarity/polarizability, hydrogen bond donor acidity, and hydrogen bond acceptor basicity. These parameters originate from experimental data which are not always available for every compound.

Molecular simulations (such as Monte Carlo) and quantum mechanical methods (such as continuum-solvation models) have high computational costs making them difficult to use for predicting properties of large groups of compounds [219, 223]. A continuum-solvation model (CSM) that has become more prominent is the Conductor-like Screening Model for Real Solvents (COSMO-RS). COSMO-RS is a surface interaction model that uses molecular charge densities obtained by molecular quantum chemical calculations [224, 225]. An advantage of COSMO-RS is that only a few adjustable parameters are required [226].

Neural networks (NN) have become more popular in prediction method development. Neural networks consist of many processing units arranged in a network [227]. The networks have an input layer, an output layer, and intermediate hidden layers. Each network unit is influenced by connected units, and the extent of influence can be changed by adjusting the weights of the connections. Learning algorithms are used to modify the system behavior by

changing the values of these weights. While these methods have been shown to perform well, they are impossible to duplicate from literature articles. The model would have to be obtained from the author in order to use the method.

7.2 Prediction Method Selection

Most k_H prediction methods only predict values at 298.15 K. Table 7.1 lists some of the available prediction methods for k_H at 298.15 K. UNIFAC expansions were excluded from the table due to the large number of authors that have added to this method. Dearden and Schüürmann [14] compared 12 of these methods and found that the EPA's HENRYWIN bond contribution method [83] outperformed the other methods. Values predicted from this method have been included as constant value entries in the database. The actual method was not programmed into DIADEM because it uses hundreds of bond contributions. The software is available without cost from the EPA website. The predicted values were imported into the database from the software.

There are significantly fewer k_H methods that predict temperature dependence, and it was determined that the greatest contribution can be made to these methods. An initial review was conducted on temperature-dependent prediction methods to determine the methods to include in this study. Temperature-dependent prediction methods that were excluded are listed in Table 7.2 with a brief reason for exclusion. The method of Yezdimer, Sedlbauer, and Wood (2000) [276] applies to some chemical families that were not included in the database analysis. Of the groups that applied to compounds included in this study, a similar method was published a few years later with superior contributions (see Section 7.2.1). The methods of Goss [277] and Abraham et al. [278] both require molecular descriptors that are based on experimental data. These methods

were excluded due to the time that would be required to find reported descriptor values in the literature for the compounds in the database. The method of Hilal et al. [279, 280] uses two independent models to predict x_i^{aq} and P_i^{sat} and then calculates k_H using Equation 4.1. There are publications concerning the method performance, but no publications have been found on the actual method parameters preventing the inclusion of this method in this study.

Table 7.1: k_H prediction methods at 298.15 K

Authors	Method Type	Authors	Method Type	Authors	Method Type
Hine and Mookerjee (1975) [23, 228]	(1) BCM (2) GCM	Cramer and Truhlar (1992) [262]	CSM	Dearden et al. (2000) [249]	QSPR
Cramer (1980) [230]	QSPR	Jorgensen and Nguyen [263] (1993)	Monte Carlo	Meylan and Howard (2000) [219]	(1) BC & (2) GC
Cabani (1981) [232]	GCM	Myrdal (1994) [265]	PPR	Öberg (2001) [252]	QSPR
Shiu and Mackay (1986) [234]	PPR	Abraham et al. (1994) [267]	LSER	Ferreira (2001) [254]	QSPR
Jin et al. (1986) [236]	GCM	Schüürmann (1995) [269]	CSM	Yao et al. (2002) [256]	QSPR
Nirmalakhandan and Speece (1988) [238]	QSPR	Chambers et al. (1996) [271]	CSM	Lin and Sandler (2002) [218]	GCM
Hawker (1989) [23, 240]	QSPR	Dearden et al. (1997) [272]	QSPR	Yaffe et al. (2003) [259]	QSPR with NN
Sabljić and Gusten (1989) [23, 242]	QSPR	Schüürmann (1997) [274]	software	Modarresi et al. (2005) [261]	QSPR
Brunner et al. (1990) [23, 248]	QSPR	Barone et al. (1997) [229]	CSM	Modarresi et al. (2007) [219]	QSPR
Crovetto et al. (1990) [250]	PPR	Hawkins et al. (1997) [231]	CSM	Schröder et al. (2010) [264]	COSMO-RS
Meylan and Howard (1991) [251]	BCM	Hawkins et al. (1998) [233]	CSM	Gharagheizi et al. (2010) [266]	GCM with NN
Yaws et al. (1991) [13], (2010) [253]	Must be bought	Katritzky et al. (1998) [235]	QSPR	Goodarzi et al. [268]	QSPR with NN
Suzuki et al. (1992) [255]	GCM & QSPR	Famini et al. (1999) 32 [237]	LSER	Peterson (2011) [270]	QSPR
Russell et al. (1992) [257]	QSPR	Cash (1999) [239]	QSPR	Gharagheizi et al. (2010) [266]	GCM with NN
Dunnivant et al. (1992) [258]	QSPR	Gramatica et al. (1999) [241]	QSPR with NN	Raventos-Duran (2010) [273]	GCM
Cramer and Truhlar 36 [260](1992)	CSM	Plyasunov and Shock (2000) [243] (2004) [244] (2005) [245] (2006) [246, 247]	GCM	Sahoo (2011) [275]	QSPR

Table 7.2: k_H temperature-dependent prediction methods excluded from this study

Author	Method Description	Reason for Exclusion
Yezdimer, Sedlbauer, and Wood (2000) [276]	GCM; need water equation of state	Similar, superior method published later
Goss (2006) [277]	LSER	Difficulty in obtaining descriptors
Abraham et al. (2007) [278]	LSER	Difficulty in obtaining descriptors
Nirmalakhandan et al. (1997) [281]	QSPR	Does not apply to very many compounds
Huang et al. (2009) [282]	Molecular Simulation	Computational cost and difficulty obtaining models
Hilal et al. (2008) [279, 280]	SPARC-solvation model	Publications on method performance but not method parameters

Table 7.3 lists the temperature-dependent prediction methods that have been programmed into DIADEM along with the type of method and the applicable compounds. The GCM and BCM methods developed by Lau et al. [39] are applicable for different chemical families. The chemical families that do not overlap between the methods are italicized. Initially the plan was to add to one of these existing methods. However, it was decided to develop a completely different method. The methods listed in Table 7.3 are described in more detail along with reasons they were not expanded in this study. A quantitative comparison of the methods is provided in Section 7.2.5.

Table 7.3: k_H temperature-dependent prediction methods chosen for evaluation

Author	Method Description	Applicable compounds	
Sedlbauer et al. (2002) [191], (2007) [283] and (2008) [22]	GCM; need water equation of state	alkanes, alkenes, alkyl phenols, alkyl anilines, aromatic hydrocarbons, cycloalkanes, H ₂ S, and CO ₂	
Plyasunov and Shock (2003) [284]	GCM	substructural parameters with atom types C, H, N, O, F, Cl, Br, I, and S	
Kühne, Ebert, and Schüürmann (2005) [285]	GCM & BCM; need a reference value	substructural parameters with atom types C, H, N, O, F, Cl, Br, I, and S	
Lau, Rogers, and Zei (2010) [39]	Separate GCM & BCM; need a reference value	GCM: alkanes, cycloalkanes, <i>cycloalkenes</i> , alkylbenzenes, <i>primary alcohols</i> , <i>secondary alcohols</i> , aliphatic chlorides, ketones	BCM: alkanes, <i>alkenes</i> , cycloalkanes, alkylbenzenes, aliphatic chlorides, ketones

7.2.1 Method of Sedlbauer et al. [22, 191, 283]

This method was initially published for aqueous hydrocarbons by Sedlbauer, Bergin, and Majer in 2002 [191]. In 2008, the same authors published additional parameters for CH₄, CO₂ and H₂S [22]. Parameters for alkylphenols and alkylanilines were developed in 2007 by Čenský, Sedlbauer, Majer, and Růžička [283] using the hydrocarbon parameters developed in 2002. Because these parameters can all be used together, they have been lumped together as one method even though the parameters were published in three separate papers.

In addition to predicting k_H , this method also predicts models for derivative properties (Equations 2.4-2.7). Experimental data related to the derivative properties were used in addition to k_H , x_i^{aq} , and γ_i^∞ data to develop this prediction method. The experimental data were simultaneously fit to a semitheoretical hydration model that uses the derivative property relationships. Group contribution values are used to calculate a reference value at T_{ref} and P_{ref} (298.15 K and 0.1 MPa), and a separate set of group contribution values are used to model the temperature and pressure dependency. The Plyasunov and Shock [243] group contribution values at reference conditions were used for hydrocarbons. The hydration model requires properties of water that can be calculated using an equation of state. Sedlbauer et al. [22, 191, 283] used the equation of state by Hill [286]. The IAPWS-95 [287] equation of state was used when programming this method into DIADEM. This difference should not affect method performance.

This method applies to hydrocarbons up to temperatures and pressures of approximately 570 K and 100 MPa, respectively. This broad range allows the prediction method to be applicable for systems encountered in the oil and natural gas industries. Since this method is both a function of temperature and pressure and DIADEM allows for one dependent variable, the

variable of pressure was “removed” by using the DIPPR[®] 801 liquid density equation for water (which only has the dependent variable of temperature) in all calculations that required density. This results in thermodynamic properties of water along the saturation line.

This method was not chosen for expansion in this study because derivative properties were not included in the database. In addition, this model extends beyond the temperature and pressure scope of the data evaluated in this study.

7.2.2 Method of Plyasunov and Shock [284]

The method of Plyasunov and Shock [284] is a group contribution method that predicts values of the vapor-liquid distribution coefficient (K_D) from 273 K to the critical temperature of water (647.1 K). K_D is related to k_H by:

$$k_H = K_D \phi_i^\infty P_{sol}^{sat},$$

where ϕ_i^∞ is the fugacity coefficient of the solute at infinite dilution and P_{sol}^{sat} is the vapor pressure of water. The method includes group contribution values of ϕ_i^∞ allowing for k_H to be calculated. High-temperature calculations are based on the Krichevskii parameter. This method uses water vapor pressure, and liquid and vapor water densities obtained from the IAPWS-95 equation of state [287]. This model was developed for hydrothermal geochemical applications which are interested in high-temperature and high-pressure values.

The Plyasunov and Shock method [284] was not the most accurate of the prediction methods evaluated in this study which is possibly due to the target temperature range of applicability. This method may be more accurate at near-critical conditions compared to methods that were developed using data closer to ambient conditions. However, the prediction

capabilities of this method at high temperatures and pressures were not determined because these conditions are beyond the scope of the data evaluated in this study. This method was not chosen for additional development or refinement due to the temperature scope of data included in this project. In addition, any development of this method would require Krichevskii parameters which were not included in the database.

7.2.3 Methods of Lau et al. [39]

Lau et al. [39] published separate bond and group contribution methods in the same paper. From a statistical standpoint, these methods are equivalent in their prediction capabilities. There are differences in the chemical families covered by both methods as indicated in Table 7.3. The chemical families that do not overlap between the methods are italicized. Temperature dependence is modeled using a variation of DIPPR[®] 801 Equation 101 (see Table 4.3),

$$\ln\left(\frac{k_H}{k_{H,ref}}\right) = \left(\sum_i B_i n_i\right) \left(\frac{1}{T_{ref}} - \frac{1}{T}\right) + \left(\sum_i C_i n_i\right) \left(\ln\left(\frac{T}{T_{ref}}\right)\right), \quad 7.1$$

where B_i and C_i are group contribution values for fragment i , n_i is the number of times a group occurs, T is temperature, and superscript *ref* refers to a reference value obtained from experimental data or a prediction method. Since $\ln(k_H)$ is related to ΔG_{hyd}^∞ (Equation 2.4), coefficients “ B ” and “ C ” are related to ΔH_{hyd}^∞ (through integration of Equation 2.5) and $\Delta C p_{hyd}^\infty$ (through integration of Equation 2.6), respectively. Both the bond and group contribution methods predict coefficients “ B ” and “ C ” given in Equation 7.1. This method was not chosen for expansion in this study due to the requirement of a reference value. In addition, some of the

group (and/or bond) contributions were developed on limited data. More data for similar compounds are available in the 801E database.

7.2.4 Method of Kühne, Ebert, and Schüürmann [285]

The method of Kühne et al. [285] predicts temperature-dependence using an abbreviated form of Equation 7.1. This method uses group contribution values and correction factors to predict coefficient “ B ”, so it only predicts a linear relationship between $\ln(k_H)$ and inverse temperature. Therefore, it can only be applied over a more limited range of temperature than the method of Lau et al. [39]. Out of the temperature-dependent methods programmed into DIADEM, this method has the widest chemical family applicability. This method was not chosen for expansion due to its limited temperature applicability and the requirement of a reference value.

7.2.5 Existing Method Comparison

The prediction methods were compared using data for 157 compounds. Some of the methods apply to compounds outside of the chemical families included in the comparison. Only the compounds included in later method development were included to make the comparisons more meaningful. The number of applicable compounds (n_C) and the average absolute deviations (AAD) for each chemical family and method are summarized in Table 7.4. Also included is a breakdown of the AAD based on the error of the experimental data.

The methods of Plyasunov and Shock [284] and Kühne et al. [285] apply to all of the compounds included in the comparison, but the method of Kühne et al. [285] has a smaller overall AAD. The Lau et al. [39] methods have comparable overall performances, but the group-

contribution method applies to more of the compounds included in the comparison. The Lau et al. [39] methods are better than the Kühne et al. [285] method, and this can be attributed to the additional temperature-dependent coefficient. The Sedlbauer et al. [22, 191, 283] method applies to the smallest group of compounds. It does not perform as well as the Lau et al. [39] methods. However, the Lau et al. [39] methods use an experimental reference value whereas the Sedlbauer et al. [22, 191, 283] method is completely predictive.

7.3 Prediction Method Development

While the Kühne et al. [285] and Lau et al. [39] methods have reasonable prediction capabilities, they both require a reference value. The reference value can be an experimental value or it can be predicted using a separate prediction method. Prediction methods that can be used to calculate a reference value are not available that use the same group definitions as Kühne et al. [285] or Lau et al. [39]. It was decided that a prediction method would be developed that uses the same group definitions for all parameters including the reference value. This simplifies the calculations because each compound only has to be parsed once.

Initially efforts were made to develop a group contribution method that predicted coefficients “*A*”, “*B*”, and “*C*” of DIPPR[®] Equation 101 simultaneously. This did not work well due to the high number of parameters. Instead, it was decided to use the method of Plyasunov et al. [243] for the reference values, and then develop a model using the same group definitions to predict coefficients “*B*” and “*C*” in Equation 7.1. Sedlbauer et al. [191] uses the method of Plyasunov et al. [243] as a reference point in their prediction method. Plyasunov et al. [243] did a thorough literature review of k_H and derivative property data for hydrocarbons. They provided recommended values for each compound and developed their prediction methods based on these

values. While this method is more limited in chemical applicability than the HENRYWIN bond-contribution method [83], the method is more reliable due to the data used for method development.

Table 7.4 Temperature-dependent k_H prediction method comparison for select compounds

Method	Plyasunov and Shock AAD%	nC	Kühne	nC	Lau GCM AAD%	nC	Lau BCM AAD%	Comps	Sedlbauer AAD%	nC
Overall	139.8	157	40.4	157	16.9	98	13.8	69	30.1	46
Acetates	71.8	14	15.2	14	-	-	-	-	-	-
Aliphatic ethers	176.8	11	18.6	11	-	-	-	-	-	-
Dimethylalkanes	87.4	6	113.8	6	12.7	6	12.6	6	65.7	6
Formates	84.3	7	14.9	7	-	-	-	-	-	-
Ketones	91.9	24	34.6	24	39.5	21	23.6	22	-	-
Methylalkanes	43.9	8	18.3	8	3.9	8	4.0	8	20.8	8
<i>n</i> -alcohols	62.9	10	29.7	10	14.5	10	-	-	-	-
<i>n</i> -alkanes	64.3	14	102.6	14	11.9	14	12.2	14	32.1	14
<i>n</i> -alkylbenzenes	139.2	8	18.5	8	7.8	8	7.7	8	19.0	8
Other aliphatic alcohols	87.9	21	20.5	21	11.5	21	-	-	-	-
Other alkanes	82.0	2	14.8	2	11.2	2	2.7	2	24.2	2
Other alkylbenzenes	949.3	9	145.0	9	10.5	8	10.1	9	22.0	8
Other ethers/diethers	92.3	5	10.9	5	-	-	-	-	-	-
Other polyfunctional C, H, O	97.9	2	10.6	2	-	-	-	-	-	-
Other saturated aliphatic esters	84.0	3	31.0	3	-	-	-	-	-	-
Propionates and butyrates	93.6	13	26.4	13	-	-	-	-	-	-
< 1%	81.1	1	11.6	1	2.1	1	4.7	1	24.2	1
< 3%	74.2	11	11.9	11	4.5	8	4.3	7	11.3	7
< 5%	112.3	27	23.8	27	6.3	19	6.0	12	14.9	9
< 10%	119.0	77	36.2	77	10.4	52	8.4	34	21.1	19
< 25%	124.0	71	51.8	71	33.7	43	28.5	28	41.0	16
< 50%	161.8	34	49.7	34	36.5	27	35.0	19	48.1	15
< 100%	513.8	15	65.3	15	76.2	9	75.4	8	77.5	6

Initially Plyasunov et al. [243] developed a first-order group contribution prediction method for hydrocarbons and alcohols. Additional functional group contribution terms were added to their method in later publications [245-247, 288]. In these later publications, second-order group contribution parameters were also developed. However, second-order hydrocarbon groups were not defined for all of the hydrocarbons included in the original first-order method. In this study, both the first- and second-order methods were expanded.

The method of Plyasunov et al. [243] uses the relationship between k_H and ΔG_{hyd}^∞ (Equation 2.4). Plyasunov et al. [243] define the standard state in terms of molality (standard molality of 1 mol/1000 g at a P_{ref} of 100 kPa), so a unit conversion must be used with the definition of k_H in this study,

$$RT \ln \left(\frac{k_H}{P_{ref} \cdot N_W} \right) = \Delta G_{hyd}^\infty, \quad 7.2$$

where $N_W = (1000 \text{ g/1 mol}) / (18.0153 \text{ g/mol}) \approx 55.5084 \text{ (mol/mol)}$. The group contribution values determined by Plyasunov et al. [243] are given in terms of ΔG_{hyd}^∞ . For the first- and second-order methods of this study, values for $k_{H,ref}$ are calculated using Equation 7.2. Using a group contribution scheme, any property (Y) can be determined by:

$$Y = Y^0 + \sum_i n_i Y_i \quad 7.3$$

where n_i is the number of a times group i appears in the compound, Y_i is the contribution of the group to the overall property, and Y^0 is a constant. Plyasunov and Shock [243] defined a Y^0 value of 7.95 kJ/mol for ΔG_{hyd}^∞ based on theoretical models (the contribution of a material point). The Y^0 values for B and C are 0 for the first- and second-order methods of this study.

7.3.1 Data Selection

The first step was selecting the data to use in method development. The recommended values at 298.15 K of this study were used in constant value (or reference value) method development. With temperature-dependent models, there are multiple ways that data are used.

1. Regression coefficients are determined from the data, and the prediction method is then developed from the regression coefficients (A , B , C , etc.) and not directly from data. Examples are the methods of Kühne et al. [285], Nagvekar [289], Hsu [290].
2. Regression coefficients are determined from the data, and values are calculated at equal temperature intervals (such as every 10 K) using the regression. The prediction method is developed from these values. Examples include prediction methods developed by Elbro et al. [291] and Harrison [292].
3. Prediction method is developed directly from data.

The first two options were not used due to the different temperature ranges of available data. The coefficients obtained in the data analysis of this study do not refer to the same temperature ranges. In addition, a second temperature-dependent coefficient is not statistically significant for some compounds due to the limited data. A combination of Options 2 and 3 was determined to be the best method. The data were not used directly due to the extreme distribution of the number of points per compound in addition to the experimental error. For example, benzene has 341 experimental points whereas 1,2,3,4-tetramethylbenzene has 3 experimental points. Smoothed values were calculated from the regressions obtained in the data analysis process at every unique temperature found within the data of a given compound. For example, if data are available for a compound at temperatures of 275 K, 298 K, 298 K, and 330 K, values calculated from the regression at 275 K, 298 K, and 330 K were included in the

method development. The uncertainties assigned to the regression (determined from 95% confidence intervals during data analysis) were then assigned to these smoothed points. Compounds with more temperature-dependent data will also inherently have more weight due to the existence of more unique temperatures.

A similar process was used for the data of compounds in which the API regression was accepted (see Section 4.5). The unique temperatures of the data in the 801E database were determined for a given compound. The Smoothed values were determined using the API regressions.

A weighted least-squares procedure was used to determine the group contribution values. The objective function, O , to be minimized is defined by

$$O = \sum_i \left[\frac{\ln k_H^{\text{mod},i} - \ln k_H^{\text{exp},i}}{\sigma_i} \right]^2, \quad 7.4$$

where $\ln k_H^{\text{mod},i}$ are the natural log values calculated from the model, $\ln k_H^{\text{exp},i}$ are the natural log experimental values, and σ_i is the uncertainty (on a natural log scale) of the Smoothed value. By using weighted values, Smoothed values with smaller uncertainty will be given greater weight in the regression. In this study, the weighted least-squares procedure was completed using SAS[®] software.

7.3.2 Validation Techniques

A general procedure was used multiple times throughout method development. Approximately 10% of the compounds were assigned to a validation set. The remaining compounds (training set) were then randomly divided into 10 subsets to perform a 10-fold cross-validation. In the 10-fold cross-validations, a single subset of the training set was reserved as a

prediction set and the remaining 9 were used as a learning set. A preliminary model was developed using the learning set and applied to the prediction set. This process was repeated so that each subset was used as the prediction set once. The statistics from the 10 iterations were averaged to provide a better estimation of the overall method accuracy. The entire training set was then used to develop model parameters. The model was finally applied to the validation set as an additional check on the method performance.

The average absolute deviation (AAD) is given by:

$$AAD = \sum_{i=1}^n \frac{|k_H^{\text{mod},i} - k_H^{\text{exp},i}|}{k_H^{\text{exp},i}}. \quad 7.5$$

The AAD is calculated on a normal scale and not a log scale to provide better insight of the method performance. Also reported is the Pearson's product moment correlation coefficient, R^2 , (the R^2 value returned by the Microsoft Excel[®] RSQ function) given by

$$R^2 = \frac{\sum_i^n (\ln k_H^{\text{mod},i} - \ln k_H^{\text{mod},\text{avg}})(\ln k_H^{\text{exp},i} - \ln k_H^{\text{exp},\text{avg}})}{\sqrt{\sum_i^n (\ln k_H^{\text{mod},i} - \ln k_H^{\text{mod},\text{avg}})^2 \sum_i^n (\ln k_H^{\text{exp},i} - \ln k_H^{\text{exp},\text{avg}})^2}}, \quad 7.6$$

where $\ln k_H^{\text{exp},\text{avg}}$ is the average natural log experimental value used in the training set (or learning set during cross-validation) and $\ln k_H^{\text{mod},\text{avg}}$ is the average natural log model value used in the validation set (or prediction set during cross-validation). A reliable method will have R^2 values that approach unity and low AADs for both the training and validation sets.

7.3.3 First-Order Group Contribution Method

Evaluated temperature-dependent data were available for 48 hydrocarbons. More data are available for compounds at 298.15 K; however, these compounds were not used because they would only test the validity of the Plyasunov et al. [243] method and not the temperature-dependent method. Compounds were classified as alkanes, alkenes, cycloalkane, cycloalkene, or benzene derivative, and one or two compounds were randomly chosen from each classification for the validation set. This ensured a diverse set of groups were represented in the validation set. The remaining 42 compounds (training set) were randomly divided into 10 subsets to perform a 10-fold cross-validation. The group contribution parameters were determined using the entire training set, and the prediction method was then applied to the six compounds that were initially set aside in the validation set. Results of the 10-fold cross-validation and overall method performance are summarized in Table 7.5. The statistics indicate that the method has reasonable predictive capabilities. The results are shown in Figure 7.1.

Table 7.5: Summary of performance for first-order group contribution temperature-dependent method for hydrocarbons

Set	#Compounds	AAD	R ²	10-Fold Cross-Validation	Avg. AAD	Avg. R ²
Training	42	14%	0.994	Learning	14%	0.994
				Prediction	27%	0.951
Validation	6	13%	0.993	-	-	-

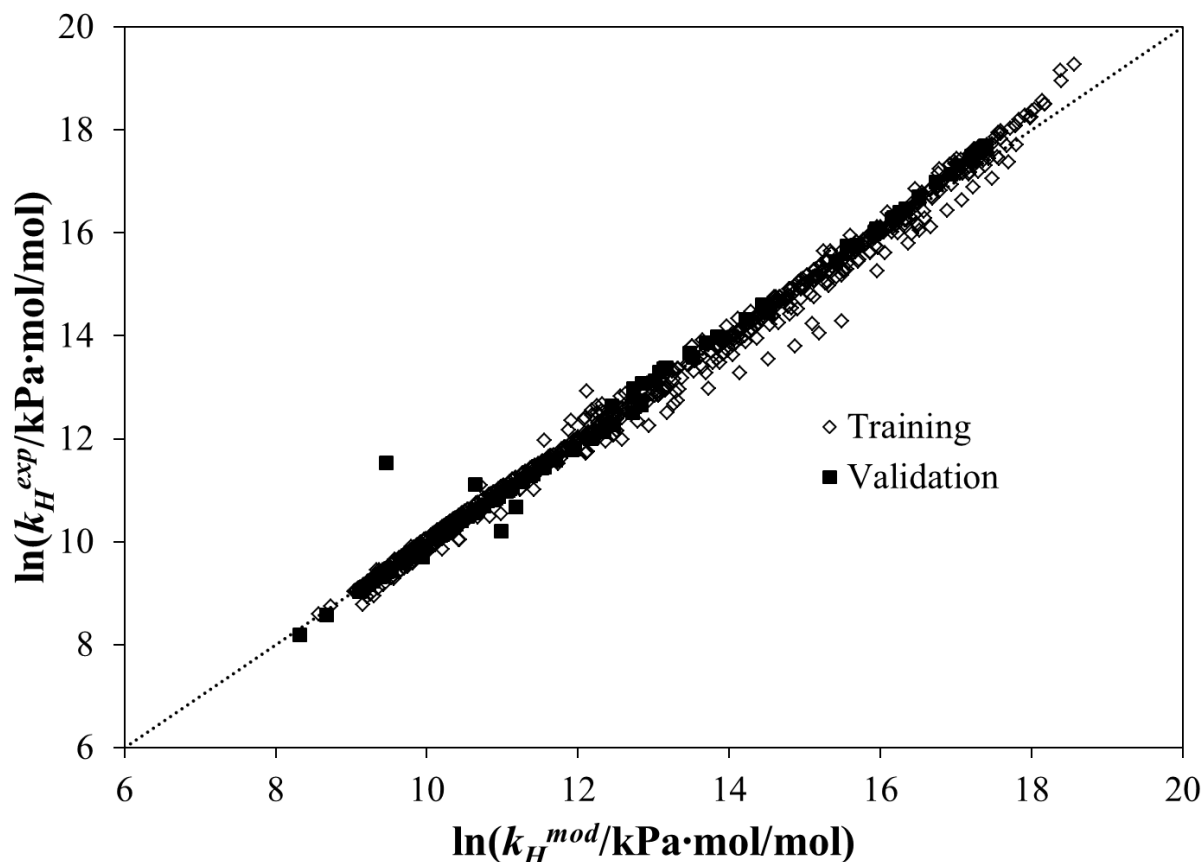


Figure 7.1: Plot of experimental versus model values for the first-order temperature-dependent group contribution method for hydrocarbons

The compound with the biggest discrepancy in the validation set is 1,4-cyclohexadiene. It is the only cycloalkene with more than one double bond. (It randomly ended up in the validation group instead of the training group.) The cyclic compounds also have the largest error in the training set. There are less data for these compounds and more uncertainty in the available data compared to the alkanes and benzene derivatives.

The method was expanded for aliphatic alcohols, ketones, and formates. Data were found for 61 compounds. Seven compounds were randomly chosen for the validation set. The remaining 54 training set compounds were divided into 10 subgroups. A 10-fold cross validation

was performed as previously described. The group contribution parameters were then determined using the entire training set, and the prediction method was then applied to the seven compounds in the validation set. Results of the 10-fold cross-validation and overall method performance are summarized in Table 7.6. The results are shown in Figure 7.2.

Table 7.6: Summary of performance for first-order group contribution temperature-dependent method for alcohols, ketones, and formates

Set	#Compounds	AAD	R ²	10-Fold Cross-Validation	Avg. AAD	Avg. R ²
Training	54	24%	0.920	Learning	24%	0.921
				Prediction	29%	0.934
Validation	7	33%	0.969	-	-	-

The two compounds with the largest discrepancies are acetophenone and acetylacetone. Acetophenone is the only aromatic ketone included in the entire dataset, and acetylacetone is the only di-ketone included in the entire dataset. The large errors could be attributed to experimental error and/or the structures. More data from similar compounds would be necessary to determine the exact cause. The statistics given in Table 7.6 include the results of these two compounds. It is recommended that this method should only be used for alcohols, ketones, and formates that are aliphatic. In addition, correction factors might be necessary for multifunctional compounds.

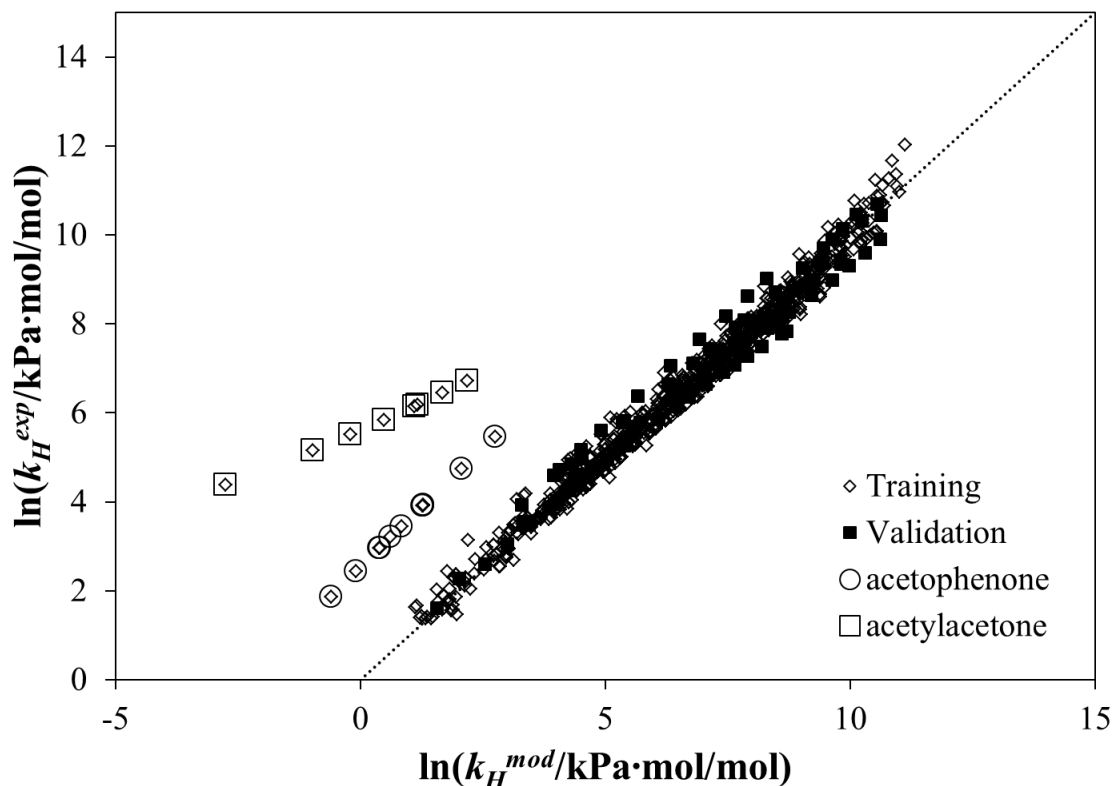


Figure 7.2: Plot of experimental versus model values for the first-order temperature-dependent group contribution method for alcohols, ketones, and formates

The temperature-dependent coefficients determined in this study are summarized in Table 7.7 along with the standard errors and two-tailed p-values. The table also includes the reference values developed by Plyasunov et al. [243-247], the number of compounds that contained a group (nC), the total number of temperature-dependent points that contained a group (nG), and the temperature range of the data used for the individual groups. Multiple coefficients have large standard errors and are not statistically significant (p-values > 0.05). In general, the groups corresponding to these coefficients do not have as much data compared to the groups with statistically significant coefficients resulting in higher uncertainty. When Plyasunov et al. initially developed the group contribution value for ketones, they did not include a correction

factor [288]. A correction factor was added later for tertiary carbons attached to a ketone group [244]. The only compound influenced by this correction in the training and validation sets was diisopropyl ketone. The reference value used in method development included the correction factor. However, the correction factor was not included in the temperature-dependent development because it was not needed. Sample calculations are included in Appendix D.

Table 7.7: Summary of parameter estimates for the first-order method

Group	$\Delta G_{hyd,i}^{\infty}$ (kJ/mol) ^a	B_i			C_i			Training		Val- idation		Overall Temp. Range (K)
		Value (K)	SE ^b	p-value ^c	Value	SE ^b	p-value ^c	nC	nG	nC	nG	
c-C=C ^d	-9.47	94550	16375	< 0.001	-326	55	< 0.001	3	24	1	12	273-318
c-CH ^d	-1.03	335	990	0.74	-3.6	3.0	0.22	4	78	0	0	273-444
c-CH ₂ ^d	0.83	3275	276	< 0.001	-8.26	0.82	< 0.001	10	681	2	39	273-444
C	-4.51	1203	2676	0.65	-8.1	8.6	0.34	3	19	1	4	273-424
CH	-1.72	1987	1305	0.13	-8.0	4.2	0.06	6	56	0	0	273-423
CH ₂	0.70	2962	291	< 0.001	-8.43	0.90	< 0.001	17	935	1	92	273-456
CH ₃	3.67	3683	557	< 0.001	-8.73	1.76	< 0.001	33	1034	4	170	262-568
C _{ar} ^e	-3.85	-102	586	0.86	0.05	1.8	0.98	13	388	1	90	262-568
CH _{ar} ^e	-0.65	1849	26	< 0.001	-4.16	0.07	< 0.001	13	1940	1	450	262-568
C=C	-10.23	9750	23970	0.68	-34	80	0.67	3	59	1	6	273-361
H ^f	3.91	-153	6566	0.98	2.3	22	0.92	6	246	2	30	273-361
I(C-C) ^g	-1.01	6047	4279	0.16	-19.2	13.9	0.17	5	46	0	0	273-353
C=O ^h	-22.74	4520	1010	< 0.001	-4.0	3.1	0.20	19	332	3	34	265-453
OH	-25.4	5094	1039	< 0.001	0.2	3	0.96	29	480	3	48	273-375
HCOO ⁱ	-15.33	-1650	5960	0.78	14	20	0.48	6	62	1	12	272-364
Y ^j	7.95	0	-	-	0	-	-	-	-	-	-	-

^aFrom Plyasunov et al. [243-247]

^bStandard error

^cTwo-tailed p-values

^dCyclic (nonaromatic) group

^eAromatic carbon

^fHydrogen bound to alkene group

^gCorrection for nearest-neighbor interactions of two -CH₃ or -CH₂- groups attached to the benzene ring or to the cyclic ring for cis-isomers

^hKetone

ⁱFormate

Efforts to expand the model to include esters and ethers were unsuccessful because the first-order contributions were not always adequate in modeling the behavior of compounds with functional groups [244-247]. Due to these limitations, a different model was pursued.

7.3.4 Second-Order Group Contribution Method

Plyasunov et al. [244-247] developed a second-order group contribution model due to the first-order limitations. A second-order method takes into account the nearest neighbor interactions for particular groups. This requires defining more groups than a first-order group contribution method. The group definitions as outlined by Domalski et al. [293] were used for parsing the compounds. Plyasunov et al. [244-247] only define four hydrocarbon second-order contributions. Additional hydrocarbon contributions were first established at 298.15 K before developing temperature-dependent contributions.

7.3.4.1 Reference Value Additions

The values of the four hydrocarbon groups defined by Plyasunov et al. [244-247] were used without modification. Values for four alkylbenzene groups were developed in this study. Accepted k_H values at 298.15 K were found for 17 alkylbenzenes. Two of these compounds were randomly chosen for the validation set. The 15 training set compounds were divided into 10 subgroups. A 10-fold cross validation was performed as previously described. Only three of the four parameters could be determined in one of the iterations due to the learning model not being full rank. Of the remaining 9 iterations, the relative absolute standard deviation of the group contribution values did not exceed 5.6%. Overall parameters were determined using the full training set, and the model was applied to the validation set. The AAD of the overall training set was 18% and the AAD of the validation set was 14%.

An additional ether term was also developed, C-(H)₃(O), for a reference value at 298.15 K. A total of 7 compounds were found with this group. One of the compounds was randomly set aside for validation. A leave-one-out cross-validation was performed on the remaining 6 compounds (similar idea as the 10-fold cross-validation except each compound is a subset). The value of the group contribution did not change significantly during cross-validation (standard deviation of 2%). The AAD of the overall training set (6 compounds) was 21% and the AAD of the validation set (1 compound) was 1%. Table 7.8 is a summary of the second-order group contribution values for the prediction method at 298.15 K that were determined in this study and by Plyasunov et al. [244-247]. The table includes the number of compounds with each group (nC) and the number of times each group appeared in a set (nG).

7.3.4.2 Temperature-Dependent Contributions

A total of 31 (aliphatic and aromatic) hydrocarbons defined by the groups in Table 7.8 have temperature-dependent data. Four were randomly selected by subfamily (2 alkanes, 2 benzene derivatives) for validation. The training set of 27 compounds was divided into 10 groups for 10-fold cross-validation. Overall parameters were determined using the full training set. Results of the 10-fold cross-validation and overall method performance are summarized in Table 7.9 and shown in Figure 7.3.

Table 7.8: Summary of second-order group contributions for reference values at 298.15 K and 0.1 MPa

Group ^a	Training		Validation		$\Delta G_{hyd,i}^{\infty}$		
	nC ^b	nG ^c	nC ^b	nG ^c	kJ/mol	SE ^d	p-value ^e
C-(H) ₂ (C)(Cb)	6	6	0	0	1.02	0.31	0.007
C-(H) ₃ (Cb)	8	21	1	3	4.00	0.33	< 0.0001
Cb-(C)(Cb) ₂	15	28	1	3	-4.53	0.36	< 0.0001
Cb-(H)(Cb) ₂	15	62	2	9	-0.60	0.04	< 0.0001
C-(H) ₃ (O)	6	7	1	1	3.48	0.11	< 0.0001
C-(H) ₃ (C) ^f	7	8	0	0	3.72	-	-
C-(H) ₂ (C) ₂ ^f	5	18	0	0	0.68	-	-
C-(H)(C) ₃ ^f	1	1	0	0	-1.93	-	-
C-(C) ₄ ^f	0	0	0	0	-4.60	-	-
[CH ₃ -CH ₂ -O-CH ₂] corr ^{f,g}	-	-	-	-	-1.42	-	-
[O-(CH ₂) ₂ -O] corr ^{f,g}	-	-	-	-	2.22	-	-
C-(C)(H) ₂ (O) ^f	-	-	-	-	0.77	-	-
C-(C) ₂ (H)(O) alcohol ^f	-	-	-	-	-1.64	-	-
C-(C) ₂ (H)(O) ester ^f	-	-	-	-	-1.46	-	-
C-(C) ₂ (H)(O) ether ^f	-	-	-	-	-2.82	-	-
C-(C) ₃ (O) alcohol ^f	-	-	-	-	-4.58	-	-
C-(C) ₃ (O) ester ^f	-	-	-	-	-2.58	-	-
C-(C) ₃ (O) ether ^f	-	-	-	-	-7.77	-	-
C-(CO)(C) ₃ ^f	-	-	-	-	-2.50	-	-
C-(CO)(H)(C) ₂ ^f	-	-	-	-	-0.88	-	-
C-(CO)(H) ₂ (C) ^f	-	-	-	-	1.15	-	-
CO-(C) ₂ ^f	-	-	-	-	-23.46	-	-
COO-(C) ₂ ^f	-	-	-	-	-20.53	-	-
HCOO-(C) ^f	-	-	-	-	-15.43	-	-
O-(C)(H) ^f	-	-	-	-	-25.46	-	-
O-(C) ₂ ^f	-	-	-	-	-15.52	-	-
Y ^{0 f}	-	-	-	-	7.95	-	-

^aCb is an aromatic carbon (benzene)

^bnC is the number of compounds with the group

^cnG is the number of times the group appeared in the set

^dStandard error

^eTwo-tailed p-value

^fDetermined by Plyasunov et al. [244-247]

^gether corrections; used in addition to group contributions

Table 7.9: Performance summary of the second-order temperature-dependent group contribution method for alkylbenzenes

Set	#Compounds	AAD	R ²	10-Fold Cross-Validation	Avg. AAD	Avg. R ²
Training	27	10%	0.998	Learning	10%	0.998
				Prediction	19%	0.950
Validation	4	14%	0.997	-	-	-

Temperature-dependent parameters were then determined for functional groups using the hydrocarbon backbone. In order for the model to be full rank, either structural groups had to be redefined or group contribution values had to be defined. There are three carbon groups associated with the O-(C)(H) group (hydroxyl group). The hydroxyl group will always be found in a 1:1 ratio with these three groups, so the hydroxyl group cannot be considered an independent variable. The groups could be redefined to include the hydroxyl group which would cause the model to be full rank. The groups as defined by Plyasunov et al. [244-247] and possible new group definitions are given in Table 7.10. However, as originally defined the C-(C)(H)₂(O) group applies to ethers and esters in addition to alcohols, so an additional group would have to be added for ethers and esters if these new group definitions were used. In order to still use the original group definitions, the hydroxyl contributions were defined as 0. This is essentially treating the carbon groups as if they were the new groups given in Table 7.10 without formally redefining them. The C-(C)(H)₂(O) group is then still applicable for other chemical families.

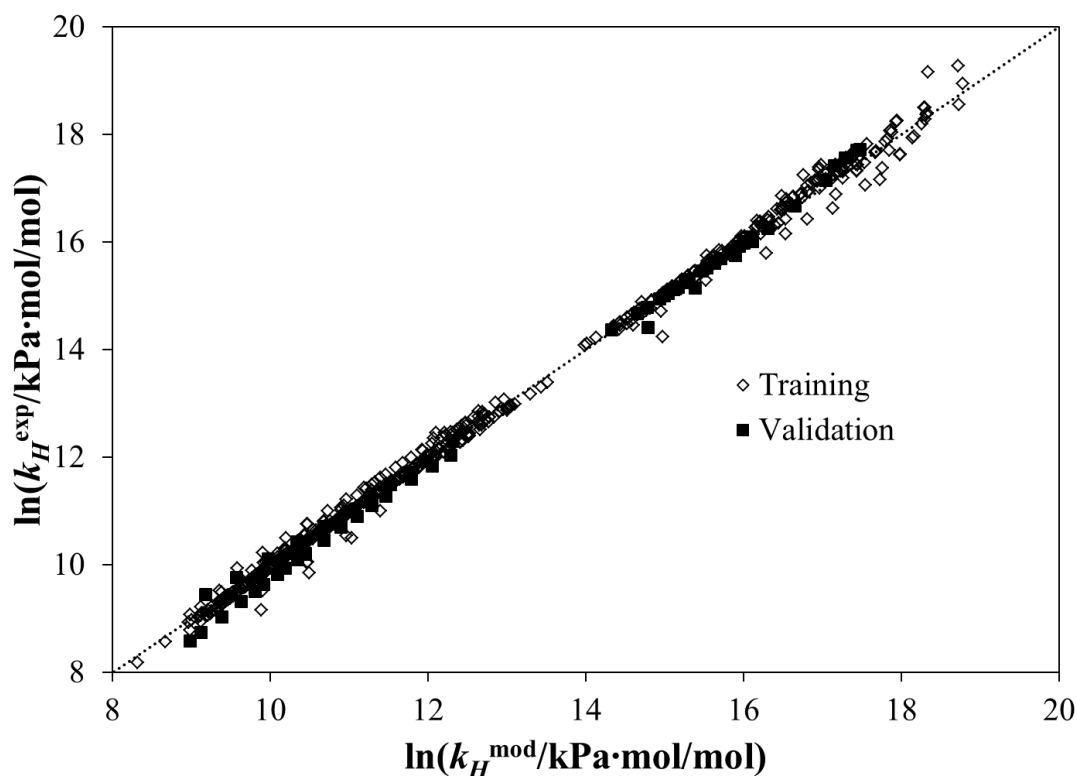


Figure 7.3: Plot of experimental versus model values for the second-order temperature-dependent group contribution method for alkylbenzenes and alkanes

Table 7.10 Second-order alcohol groups

Plyasunov et al. [244-247] group	"New" group
C-(C)(H) ₂ (O)	C[OH]-(C)(H) ₂ ^a
C-(C) ₂ (H)(O) alcohol	C[OH]-(C) ₂ (H)
C-(C) ₃ (O) alcohol	C[OH]-(C) ₃
O-(C)(H)	NA

^aOnly for alcohols. Group treated as original definition for other compounds

If a group was only represented in 1 or 2 compounds, the compounds were added to the training set. The remaining compounds were divided into subsets (COO, CH₃O, COOH, CO, OH, and O), and compounds were randomly chosen from each subset to ensure multiple groups were represented in the validation set. The training and validation sets contained data for 83 and

11 compounds, respectively. During model development, it was discovered that the model showed a predictable bias. Figure 7.4 shows the natural log of the ratio of the experimental versus model values normalized by the reference value. Also shown in the figure is the line that corresponds to a perfect model. A correction factor was added to the model to improve the results.

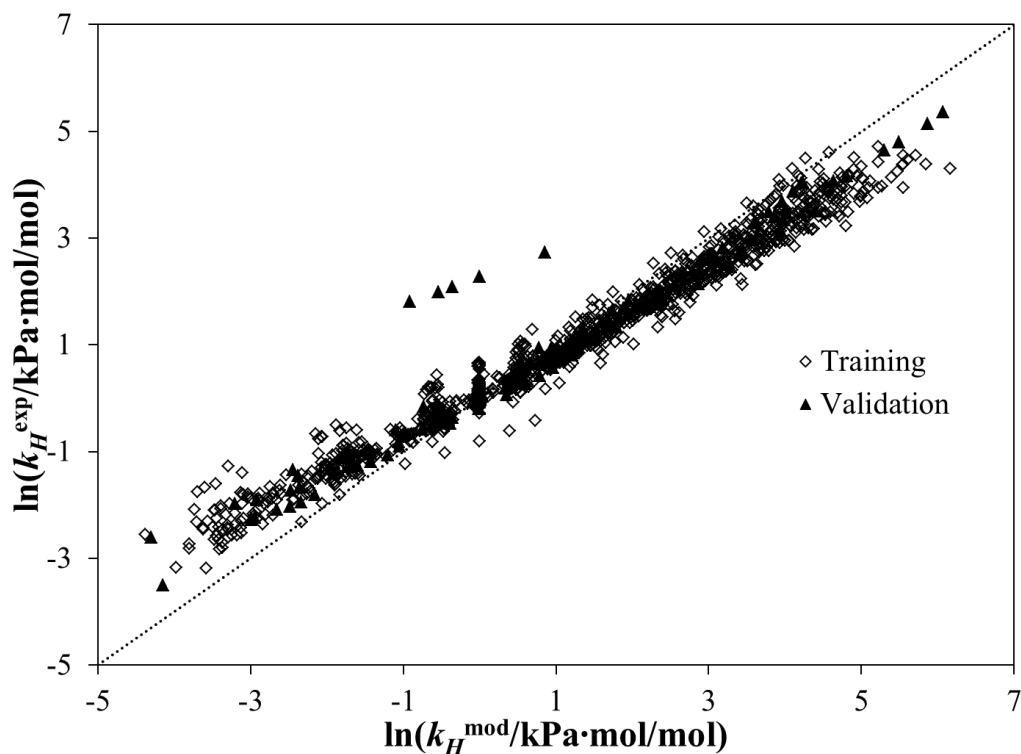


Figure 7.4: Initial second-order group contribution model performance for functional groups

The correction factor was obtained by determining the slope, D , of the training set values as plotted in Figure 7.4. The modified model is then given by:

$$\ln\left(\frac{k_H}{k_{H,ref}}\right) = D\left[\left(\sum_i B_i n_i\right)\left(\frac{1}{T_{ref}} - \frac{1}{T}\right) + \left(\sum_i C_i n_i\right)\left(\ln\left(\frac{T}{T_{ref}}\right)\right)\right]. \quad 7.7$$

The correction factor is only necessary for alcohols, esters, ethers, and ketones. For hydrocarbons, D equals unity. Figure 7.5 shows the corrected natural log of the ratio of the experimental versus model values normalized by the reference value. The outliers in Figure 7.4 and Figure 7.5 are data points for 2-butoxyethanol which is a multifunctional compound. Plyasunov et al. [247] found it necessary to provide a correction for diethers ($[\text{O}-(\text{CH}_2)_2-\text{O}]$). 2-Butoxyethanol has a similar structure except that one ether group is replaced by a hydroxyl group. A correction factor was not added in this study due to the limited amount of data. However, the data from this compound suggest that correction factors may be necessary for multifunctional compounds.

The training set was divided into 10 subgroups for 10-fold cross-validation. With three of the folds, the learning sets were not full rank so parameters could not be determined. The results of the remaining seven iterations were averaged for the cross-validation statistics. Overall parameters were determined using the full training set. The model was then applied to the validation set. Results of the 10-fold cross-validation and overall method performance are summarized in Table 7.11. Results are shown in Figure 7.6.

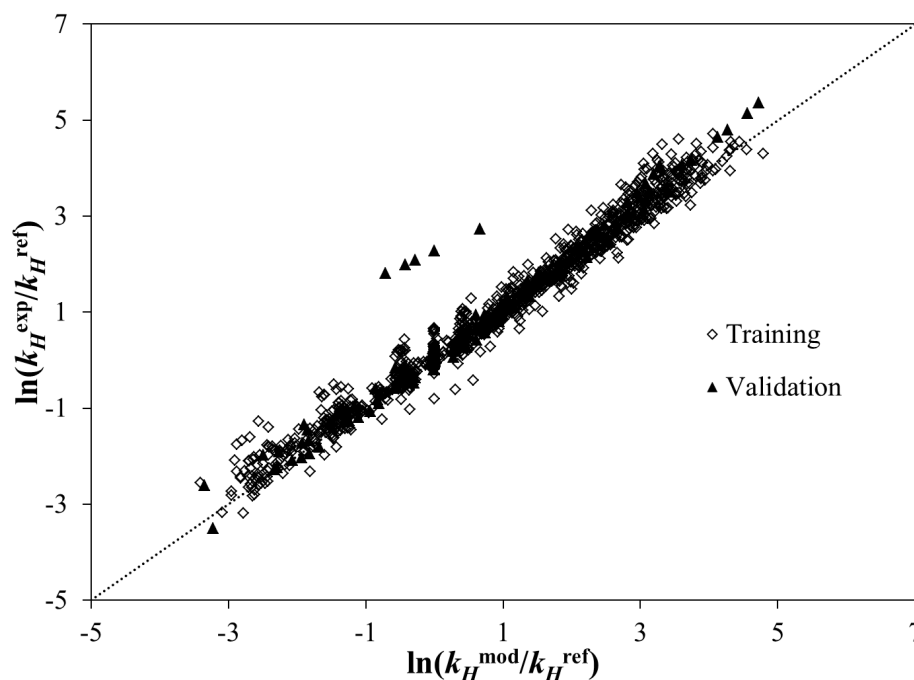


Figure 7.5: Corrected second-order group contribution model performance for functional groups

Table 7.11: Summary of performance for second-order temperature-dependent group contribution method for esters, ethers, ketones, and alcohols

Set	#Compounds	AAD	R ²	10-Fold Cross-Validation ^a	Avg. AAD	Avg. R ²
Training	83	17%	0.987	Learning	18%	0.986
				Prediction	21%	0.982
Validation	11	19%	0.970	-	-	-

^aOnly includes results of the seven iterations that were full rank

The second-order group contribution method works slightly better for hydrocarbons than compounds with functional groups. This is not surprising due to the increased complexity of the compound structures. In addition, there are typically less temperature-dependent data available for the compounds with functional groups, and the data generally had higher experimental uncertainties.

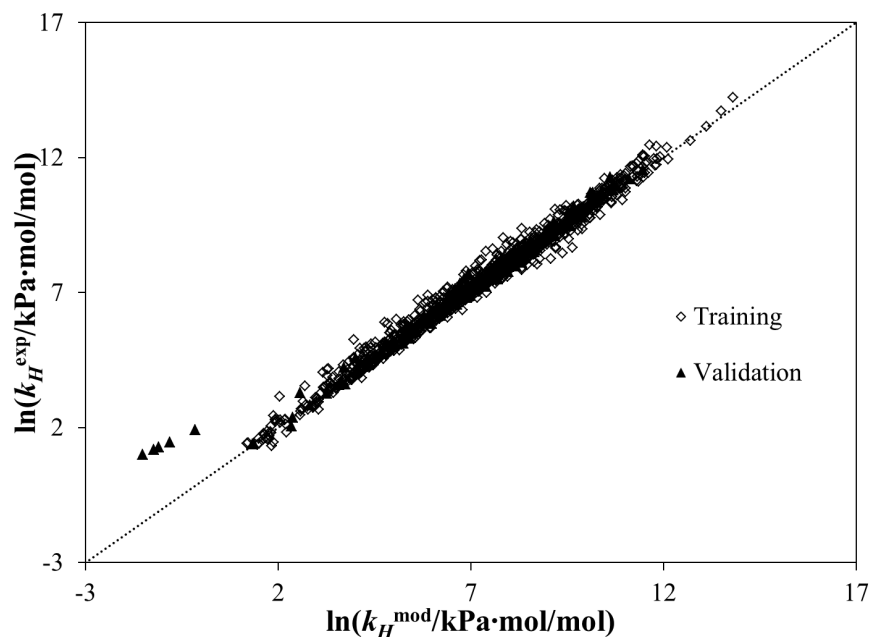


Figure 7.6: Plot of experimental versus model values for the second-order temperature-dependent group contribution method for esters, ethers, ketones, and alcohols

All of the temperature-dependent groups developed in the two regressions (hydrocarbon and functional groups) and the coefficient standard errors (SE) and two-tailed p-values are summarized in Table 7.12. Only the first four hydrocarbon groups were used in the functional group regression. The table also includes the number of times a group appeared in a compound (nC) and the number of times a group appeared in all of the temperature-dependent data (nG) in the training and validation sets for each regression. The minimum and maximum temperatures of the data corresponding to each group found in the training and validation sets is also included. The groups with the smallest amount of data typically have the largest standard errors and p-values. The second-order correction factor is summarized in Table 7.13. Sample calculations are included in Appendix D.

Table 7.12: Summary of second-order temperature-dependent model values

Group	B_i			C_i			Training		Validation		Temp. Range
	Value/K	SE	p-value	Value	SE	p-value	nC	nG	nC	nG	K
Hydrocarbon Regression											
C-(C) ₄	-5451	2205	0.01	13.2	7.1	0.06	4	23	0	0	273-424
C-(H)(C) ₃	-4267	1166	0.0003	11.8	3.8	0.002	5	45	1	11	273-423
C-(H) ₂ (C) ₂	868	292	0.003	-1.89	0.90	0.04	13	830	3	56	273-456
C-(H) ₃ (C)	6820	542	< 0.0001	-18.6	1.7	< 0.0001	19	645	3	90	273-456
C-(H) ₂ (C)(Cb)	19962	33762	0.55	-65	111	0.56	4	115	1	11	273-373
C-(H) ₃ (Cb)	16423	33753	0.63	-52	111	0.64	7	303	1	38	262-568
Cb-(C)(Cb) ₂	-13358	33753	0.69	45	111	0.69	12	423	2	49	262-568
Cb-(H)(Cb) ₂	1870	20	< 0.0001	-4.28	0.06	< 0.0001	13	2229	2	131	262-568
Y^0	0	-	-	0	-	-	-	-	-	-	-
Functional Group Regression											
[CH ₃ -CH ₂ -O-CH ₂] corr ^a	1106	5853	0.85	-6	19	0.76	2	35	0	0	269-364
[O-(CH ₂) ₂ -O] corr ^a	-26034	17351	0.14	72	56	0.20	1	11	0	0	273-373
C-(C)(H) ₂ (O)	22467	1574	< 0.0001	-49.8	4.9	< 0.0001	39	608	7	103	269-375
C-(C) ₂ (H)(O) alcohol	19591	994	< 0.0001	-39.3	3.1	< 0.0001	11	174	2	20	273-373
C-(C) ₂ (H)(O) ester	18533	6657	0.006	-38	21	0.07	2	11	0	0	273-364
C-(C) ₂ (H)(O) ether	12226	5167	0.02	-15	17	0.37	1	40	0	0	273-334
C-(C) ₃ (O) alcohol	18816	2525	< 0.0001	-36.8	7.9	< 0.0001	3	42	0	0	273-371
C-(C) ₃ (O) ester	34714	6219	< 0.0001	-90	20	< 0.0001	2	21	0	0	273-354
C-(C) ₃ (O) ether	12359	9442	0.19	-15	30	0.62	3	72	1	9	273-363
C-(CO)(C) ₃	-7084	15452	0.65	25	49	0.61	1	10	0	0	273-364
C-(CO)(H)(C) ₂	-14006	4324	0.001	46	14	0.0007	3	28	0	0	273-364
C-(CO)(H) ₂ (C)	-9550	2434	< 0.0001	32.2	7.6	< 0.0001	12	128	4	42	273-364
CO-(C) ₂	17819	805	< 0.0001	-37.8	2.5	< 0.0001	19	306	1	13	265-453
COO-(C) ₂	-5096	2902	0.08	13	9	0.17	21	259	4	42	273-365
HCOO-(C)	1855	5873	0.75	-14	19	0.47	6	50	1	14	272-364
O-(C)(H)	0	-	-	0	-	-	27	425	4	54	273-375
O-(C) ₂	-29379	5765	< 0.0001	68.9	18.7	0.0002	9	173	2	14	269-373
C-(H) ₃ (O)	28004	7327	0.0001	-69.9	23.3	0.003	6	112	1	6	273-373

^aether corrections; used in addition to group contributions

Table 7.13: Second-order method correction factor

Chemical Families	<i>D</i>	
	Value	SE
Hydrocarbons	1	-
Alcohols, Ketones, Esters, Ethers	0.7758	0.0034

7.3.5 Method Application Summary

The general procedure required to use the first- and second-order bond contribution methods of this study are described in the following steps.

1. Calculate $k_{H,ref}$ using Equation 7.2 and the ΔG_{hyd}^{∞} contributions listed in
2. Table 7.7 for the first-order method and Table 7.8 for the second-order method.
3. Calculate coefficients *B* and *C* using the group contribution values listed in
4. Table 7.7 for the first-order method and Table 7.12 for the second-order method.
5. Determine coefficient *D* using Table 7.13 if using the second-order method.
6. Determine the final equation by combining the coefficients and reference value using Equation 7.1 for the first-order method and Equation 7.7 for the second-order method.

7.3.6 Method Comparison and Recommendations

The methods of this study were compared with the literature methods using data from 144 compounds used in the first- and second-order method development. The reference point used in the Lau et al. [39] and Kühne et al. [285] methods is the recommended Smoothed or Experimental value at 298.15 K from this study. The results are summarized in Table 7.14.

The first- and second-order methods from this study are comparable to the methods of Lau et al. [39] and better than the method of Kühne et al. [285] for hydrocarbons. However, in general the methods of Kühne et al. [285] and Lau et al. [39] (when applicable) are more accurate for compounds with functional groups. This is partially due to the fact that these methods used an experimental value as a reference point instead of a predicted value. The first-order group contribution method from this study is recommended for hydrocarbons, alcohols, ketones, and formats where the functional group is not directly attached to a benzene ring. It is easier to use than the Lau et al. [39] methods because it does not require an additional prediction method or experimental value. The Kühne et al. [285] method is recommended for compounds with functional groups over a limited temperature range that have reliable reference values. The second-order group contribution method developed in this study is recommended for compounds without a reliable experimental reference point.

Table 7.14: Comparison of methods from this study with existing methods

Method	Kühne AAD%	nC	Lau GCM AAD%	nC	Lau BCM AAD%	nC	This study 1 st order AAD%	nC	This study 2 nd order AAD%	nC
Overall	49.9	144	18.6	101	11.6	50	24.2	108	24.5	125
1-Alkenes	12.2	2	21.0	2	-	-	7.4	2	-	-
Acetates	14.9	10	-	-	-	-	-	-	26.3	10
Aliphatic ethers	16.9	9	-	-	-	-	-	-	25.7	9
Alkylcyclohexanes	258.4	4	11.1	4	-	-	48.0	4	-	-
Cycloaliphatic alcohols	9.5	1	7.0	1	-	-	37.7	1	-	-
Cycloalkanes	21.8	4	14.6	4	-	-	52.7	4	-	-
Cycloalkenes	8.1	3	14.7	3	-	-	40.6	3	-	-
Dialkenes	10.8	1	7.4	1	-	-	65.6	1	-	-
Dimethylalkanes	159.3	4	9.8	4	11.2	4	23.9	4	22.8	4
Formates	14.9	7	-	-	-	-	20.9	7	33.0	7
Ketones	35.8	22	38.1	22	11.7	20	30.2	22	28.3	21
Methylalkanes	67.4	2	13.8	2	13.3	2	14.1	2	14.8	2
Methylalkenes	6.3	2	45.1	2	-	-	7.5	2	-	-
<i>n</i> -Alcohols	29.7	10	14.5	10	-	-	21.5	10	17.6	10
<i>n</i> -Alkanes	172.9	8	14.1	8	14.5	8	16.1	8	15.6	8
<i>n</i> -Alkylbenzenes	21.2	7	8.9	7	8.7	7	10.1	7	8.8	7
Other aliphatic alcohols	20.5	21	11.5	21	-	-	22.1	21	22.7	21
Other alkanes	28.7	1	21.6	1	4.4	1	20.3	1	16.7	1
Other alkylbenzenes	163.2	8	10.9	8	11.4	8	14.6	8	29.8	8
Other ethers/diethers	13.1	1	-	-	-	-	-	-	35.1	1
Other monoaromatics	6.2	1	6.9	1	-	-	6.0	1	-	-
Other polyfunctional C, H, O	10.3	1	-	-	-	-	-	-	89.7	1
Other saturated aliphatic esters	27.6	2	-	-	-	-	-	-	37.9	2
Propionates and butyrates	26.4	13	-	-	-	-	-	-	25.5	13
< 3%	15.4	8	13.4	6	5.5	4	5.6	6	11.6	7
< 5%	41.4	25	7.5	21	7.1	9	16.6	22	17.7	21
< 10%	43.5	71	12.1	53	9.8	29	19.3	56	19.5	66
< 25%	52.1	76	31.3	51	20.1	25	36.4	54	30.7	66
< 50%	58.0	29	43.5	23	45.9	13	46.9	23	46.7	27
< 100%	68.3	15	71.1	11	75.4	8	88.0	11	71.4	13

8 Summary

A database containing recommended values for Henry's law constants, infinite dilution activity coefficients and solubility data of industrially important chemicals has been compiled for aqueous systems. These properties are important in predicting the fate and transport of chemicals in the environment. The structure of this database is compatible with the existing DIPPR[®] 801 database and DIADEM interface. Recommendations have been provided for 498 compounds for at least one of the properties.

Thermodynamic relationships, chemical family trends, and predicted values were important data analysis tools. Data from the various properties were related to each other using thermodynamic relationships and reliable DIPPR[®] 801 pure component property values. For hydrophobic compounds, the data from the three properties were combined to obtain recommendations. In instances where the data could not be combined, the thermodynamic relationships were helpful in identifying the best datasets. Chemical family plots were used with temperature-dependent and single-temperature data to check for consistency within chemical families and to select the best data. Predicted values were used to select the best datasets when there were discrepancies in the literature data. Notes in the database provide information about the procedure and data used to provide recommendations for each compound and property.

Henry's law constants were measured for seven compounds at various temperatures and atmospheric pressure using the inert gas stripping method. Two of these compounds were used to validate the experimental method, and good agreement was obtained between results from this study and literature values. The remaining compounds were chosen to clarify discrepancies between literature values or to provide values for compounds that have not been measured. Infinite dilution activity coefficients were determined for each compound. A fugacity ratio was required to determine infinite dilution activity coefficients for the solid solutes. It was obtained from the heats of fusion and the liquid and solid heat capacities for the pure solutes as obtained from the DIPPR[®] 801 database. The experimental work was published [294].

A first-order temperature-dependent group contribution method was developed using the method of Plyasunov et al. [243] as a reference point. The method works for hydrocarbons, alcohols, ketones, and formats where none of the functional groups are directly attached to a benzene ring. Attempts to add group contributions for ethers and esters were unsuccessful. A second-order temperature-dependent group contribution method was developed using the groups defined by Plyasunov et al. [244-247]. Additional second-order hydrocarbon groups were first defined at the reference condition of 298.15 K and 100 kPa before developing temperature-dependent parameters. The method is applicable for hydrocarbons, esters, ethers, ketones, and alcohols. The first-order group contribution method from this study is recommended when applicable. It is easier to use than the Lau et al. [39] methods because it does not require an additional prediction method or experimental value. The Kühne et al. method [285] is recommended for esters and ethers below 353 K when a reliable experimental reference value is available. The second-order group contribution method developed in this study is recommended for esters and ethers when a reliable experimental value is not available for a reference point.

8.1 Recommendations for Future Work

The 801E database could be expanded to include additional properties. As mentioned in Section 2.1.3, derivative properties (ΔH_{hyd}^{∞} , $\Delta C p_{hyd}^{\infty}$, and V^{∞}) are thermodynamically related to k_H . These properties could be added to the database to further aid in the designation of recommended values and the development of prediction methods.

The current database does not take into account the effects of pressure which are significant for gases. Because of this, API recommendations were accepted (see Section 4.5). The database could be modified to include pressure as a dependent variable. This would allow data at intermediate and high pressures to be included. The unit conversions could be modified in order to take into account non-idealities due to high temperatures and pressures. More data at high temperatures and pressures could also be included for compounds that are liquids and solids.

Although SOLC data were included in the database, no efforts were made to provide recommended values for this property. In the future, SOLC and x_i^{aq} data could be analyzed simultaneously to obtain the entire solubility curves for compounds with critical solution temperatures.

Further temperature-dependent k_H experiments could be performed to continue measuring compounds with discrepancies in the literature and/or missing literature data. If temperature-dependent regressions have been assigned high uncertainties, then the available data have significant noise. Additional measurements could provide a better regression. In addition, measurements could be performed for compounds with interesting chemical family trends (i.e. cycloalkanes, cycloalkenes, *n*-alkanes) to provide further insights into the trends. More data would also be useful for compounds with multiple functional groups.

Additional second-order group contribution reference values can be added to the prediction method developed in this study to broaden the chemical applicability. Prediction methods have the largest errors for compounds with multiple functional groups. Unfortunately, limited data are available for multifunctional compounds. As mentioned, additional experimental work could provide more data for multifunctional compounds allowing for the development of better prediction methods.

9 References

1. U.S. Environmental Protection Agency, *Chemistry Assistance Manual for Premanufacture Notification Submitters*, 1997, Office of Pollution Prevention and Toxics, U.S. Environmental Protection Agency Washington, DC.
2. *REACH*. Feb. 19, 2009. European Commission. Retrieved 2009 April 8. <http://ec.europa.eu/environment/chemicals/reach/reach_intro.htm>.
3. *The cost of doing business just got higher*. AIChE. Retrieved 2009 April 8. <<http://www.aiche.org/DIPPR/products/REACH.aspx>>.
4. *Physico-chemical data requirements*. Department of Chemistry of the French National Centre for Scientific Research. Retrieved 2009 April 7. <http://www.prc.cnrs-gif.fr/reach/en/physicochemical_data.html>.
5. *SPARC Performs Automated Reasoning in Chemistry*. 10 Jan. 2013. United States Environmental Protection Agency. 24 Jan. 2013. <<http://www.epa.gov/athens/research/sparc.html>>.
6. Kline, A.A., et al., An overview of compiling, critically evaluating, and delivering reliable physical property data from AIChE DIPPR® Projects 911 and 912. *Fluid Phase Equilib.*, 1998. **150-151**: p. 421-428.
7. Hemond, H.F. and E.J. Fechner-Levy, *Chemical fate and transport in the environment*. 2000, San Diego: San Diego : Academic Press.
8. Ma, J.H.Y., et al., Temperature Dependence of the Aqueous Solubility of Selected Chlorobenzenes and Chlorotoluenes. *J. Chem. Eng. Data*, 2001. **46**(3): p. 619-622.
9. Keeley, D.F., M.A. Hoffpauir, and J.R. Meriwether, Solubility of aromatic hydrocarbons in water and sodium chloride solutions of different ionic strengths: C2-Substituted benzenes. *J. Chem. Eng. Data*, 1991. **36**(4): p. 456-459.
10. Pinal, R., et al., Cosolvency of partially miscible organic solvents on the solubility of hydrophobic organic chemicals. *Environ. Sci. Technol.*, 1990. **24**(5): p. 639-647.

11. Sawamura, S., K. Kitamura, and Y. Taniguchi, *Effect of pressure on the solubilities of benzene and alkylbenzenes in water*. J. Phys. Chem., 1989. **93**(12): p. 4931-4935.
12. Nielsen, F., E. Olsen, and A. Fredenslund, Henry's Law Constants and Infinite Dilution Activity Coefficients for Volatile Organic Compounds in Water by a Validated Batch Air Stripping Method. Environ. Sci. Technol., 1994. **28**(12): p. 2133-2138.
13. Yaws, C., H.-C. Yang, and X. Pan, *Henry's Law Constants for 362 Organic Compounds in Water*. Chem. Eng., 1991. **98**(11): p. 179-85.
14. Dearden, J.C. and G. Schüürmann, Quantitative structure-property relationships for predicting Henry's law constant from molecular structure. Environ. Toxicol. Chem., 2003. **22**(8): p. 1755-1770.
15. Moore, R.C., *Henry's Law Constant for Environmentally Significant Compounds*. Appl. Biochem. Biotechnol., 1992. **34-5**: p. 671-680.
16. Smith, F.L. and A.H. Harvey, *Avoid Common Pitfalls When Using Henry's law*. Chem. Eng. Prog., 2007. **103**(9): p. 33-39.
17. Prausnitz, J.M., R.N. Lichtenthaler, and E. Gomes de Azevedo, *Molecular Thermodynamics of Fluid-Phase Equilibria*. 3rd ed. 1999, Upper Sadle River: Prentice Hall PTR.
18. Smith, J.M., H.C. Van Ness, and M.M. Abbott, *Introduction to Chemical Engineering Thermodynamics*. 7th ed. Chemical Engineering Series. 2005, New York: McGraw-Hill. 817.
19. Mackay, D. and W.Y. Shiu, *A Critical Review of Henry's Law Constants for Chemicals of Environmental Interest*. J. Phys. Chem. Ref. Data, 1981. **10**(4): p. 1175-1199.
20. Ben-Naim, A., *Solvation Thermodynamics*. 1987, New York: Plenum Press. pp. 246.
21. Dohnal, V., D. Fenclova, and P. Vrbka, Temperature Dependences of Limiting Activity Coefficients, Henry's Law Constants, and Derivative Infinite Dilution Properties of Lower (C1-C5) 1-Alkanols in Water. Critical Compilation, Correlation, and Recommended Data. J. Phys. Chem. Ref. Data, 2006. **35**(4): p. 1621-1651.
22. Majer, V., J. Sedlbauer, and G. Bergin, Henry's law constant and related coefficients for aqueous hydrocarbons, CO₂ and H₂S over a wide range of temperature and pressure. Fluid Phase Equilib., 2008. **272**: p. 65-74.
23. Staudinger, J. and P.V. Roberts, *A critical review of Henry's law constants for environmental applications*. Crit. Rev. Environ. Sci. Technol., 1996. **26**(3): p. 205-297.
24. Baum, E.J., *Chemical Property Estimation: Theory and Application*. 1998, Boca Raton: CRC Press LLC.

25. Schwarz, F.P. and S.P. Wasik, A fluorescence method for the measurement of the partition coefficients of naphthalene, 1-methylnaphthalene, and 1-ethylnaphthalene in water. J. Chem. Eng. Data, 1977. **22**(3): p. 270-273.
26. Sarraute, S., et al., Aqueous solubility, Henry's law constants and air/water partition coefficients of n-octane and two halogenated octanes. Chemosphere, 2004. **57**(10): p. 1543-1551.
27. Betterton, E.A. and M.R. Hoffmann, *Henry's law constants of some environmentally important aldehydes*. Environ. Sci. Technol., 1988. **22**(12): p. 1415-1418.
28. Sandler, S.I., Infinite dilution activity coefficients in chemical, environmental and biochemical engineering. Fluid Phase Equilib., 1996. **116**(1-2): p. 343-353.
29. Myrdal, P., et al., AQUAFAC 1: AQUEOUS FUNCTIONAL GROUP ACTIVITY COEFFICIENTS; APPLICATION TO HYDROCARBONS. Chemosphere, 1992. **24**(8): p. 1047-1061.
30. Tiegs, D., et al., *Activity coefficients at infinite dilution*. Chemistry Data Series, ed. D. Tiegs. 1986: Frankfurt/Main : DECHEMA ; Great Neck, N.Y. : Distributed by Scholium International.
31. Clark, M.M., Transport Modeling for Environmental Engineers and Scientists. 1996, New York: John Wiley & Sons, Inc.
32. Bahadur, N.P., et al., Temperature Dependence of Octanol-Water Partition Coefficient for Selected Chlorobenzenes. J. Chem. Eng. Data, 1997. **42**(4): p. 685-688.
33. Daylight Chemical Information Systems, I. *Medchem Database*. 2008 BioByte Corp. Retrieved 2010 July 12. <http://www.daylight.com/products/medchem.html>.
34. *LOGKOW[®]: A databank of evaluated octanol-water partition coefficients (Log P)*. 6 Jun 2013. Sangster Research Laboratories,. Retrieved 2013 Jun 13. <http://logkow.cisti.nrc.ca/logkow/index.jsp>.
35. Dohnal, V., Measurement of Limiting Activity Coefficients Using Analytical Tools, in Measurement of the Thermodynamic Properties of Multiple Phases, R.D. Weir and T.W.d. Loos, Editors. 2005, Elsevier: Amsterdam. p. 359-381.
36. Tummers, B. *Datahief III*. 2006; 1.5. Available from: <<http://datathief.org/>>.
37. Mackay, D., W.Y. Shiu, and R.P. Sutherland, *Determination of air-water Henry's law constants for hydrophobic pollutants*. Environ. Sci. Technol., 1979. **13**(3): p. 333-337.
38. Fu, J.-K. and R.G. Luthy, *Aromatic compound solubility in solvent/water mixtures*. J. Environ. Eng. (N. Y.), 1986. **112**(2): p. 328-45.

39. Lau, K., T.N. Rogers, and D.A. Zei, Modeling the temperature dependence of the Henry's law constant of organic solutes in water. *Fluid Phase Equilib.*, 2010. **290**(1–2): p. 166-180.
40. Van Noort, P.C.M., Fugacity ratio estimations for high-melting rigid aromatic compounds. *Chemosphere*, 2004. **56**(1): p. 7-12.
41. Mackay, D., W.Y. Shiu, and K.C. Ma, Henry's Law Constant, in *Handbook of Property Estimation Methods for Chemicals: Environmental and Health Sciences*. 2000, Lewis Publishers. p. 69-87.
42. Prausnitz, J.M. and F.H. Shair, *A Thermodynamic Correlation of Gas Solubilities*. *AIChE J.*, 1961. **7**(4): p. 682-7.
43. Ginnings, P.M. and R. Baum, *Aqueous Solubilities of the Isomeric Pentanols*. *J. Am. Chem. Soc.*, 1937. **59**(6): p. 1111-1113.
44. Stephenson, R., J. Stuart, and M. Tabak, *Mutual Solubility of Water and Aliphatic Alcohols*. *J. Chem. Eng. Data*, 1984. **29**(3): p. 287-290.
45. Kühne, R., et al., Hydrophobicity and hydrophobic interactions. II. Differentiation of surface area effects on several thermodynamic transfer data of hydrophobic solutes. *Eur. J. Med. Chem. - Chim. Ther.*, 1981. **16**(1): p. 7-12.
46. Fenclová, D., et al., Temperature Dependence of Limiting Activity Coefficients, Henry's Law Constants, and Related Infinite Dilution Properties of Branched Pentanols in Water. Measurement, Critical Compilation, Correlation, and Recommended Data. *J. Chem. Eng. Data*, 2010. **55**(9): p. 3032-3043.
47. Abraham, M.H., et al., *Thermodynamics of solute transfer from water to hexadecane*. *J. Chem. Soc., Perkin Trans. 2*, 1990(2): p. 291-300.
48. Dohnal, V. and D. Ondo, Refined non-steady-state gas-liquid chromatography for accurate determination of limiting activity coefficients of volatile organic compounds in water - Application to C1-C5 alkanols. *J. Chromatogr., A*, 2005. **1097**(1-2): p. 157-164.
49. Sherman, S.R., et al., Compilation and Correlation of Limiting Activity Coefficients of Nonelectrolytes in Water. *Ind. Eng. Chem. Res.*, 1996. **35**(4): p. 1044-58.
50. Noubigh, A., et al., Solubility of Gallic Acid, Vanillin, Syringic Acid, and Protocatechuic Acid in Aqueous Sulfate Solutions from (293.15 to 318.15) K. *J. Chem. Eng. Data*, 2008. **53**(7): p. 1675-1678.
51. Dehn, W.M., COMPARATIVE SOLUBILITIES IN WATER, IN PYRIDINE AND IN AQUEOUS PYRIDINE. *J. Am. Chem. Soc.*, 1917. **39**(7): p. 1399-1404.
52. Mange, C.E. and O. Ehler, *Solubilities of Vanillin*. *Ind. Eng. Chem.*, 1924. **16**(12): p. 1258-1260.

53. Cartwright, L.C., Vanilla-Like Synthetics: Solubility and Volatility of Propenyl Guaethol, Bourbonal, Vanillin, and Coumarin. *J. Agric. Food. Chem.*, 1953. **1**(4): p. 312-314.
54. Noubigh, A., M. Abderrabba, and E. Provost, Temperature and salt addition effects on the solubility behaviour of some phenolic compounds in water. *J. Chem. Thermodynamics*, 2007. **39**(2): p. 297-303.
55. Bozdag, O. and J.A. Lamb, Dependence of the mutual solubilities of (2-butanol+water) upon pressure and temperature. *J. Chem. Thermodynamics*, 1983. **15**(2): p. 165-171.
56. Hafter, G.T., A.F.M. Barton, and A. Chand, Semi-automated apparatus for the determination of liquid solubilities: mutual solubilities of water and butan-2-ol. *J. Chem. Soc., Faraday Trans.*, 1991. **87**(4): p. 591-596.
57. Ochi, K., T. Saito, and K. Kojima, *Measurement and Correlation of Mutual Solubilities in 2-Butanol + Water*. *J. Chem. Eng. Data*, 1996. **41**(2): p. 361-364.
58. Moriyoshi, T., et al., *Mutual solubility of 2-butanol + water under high pressure*. *J. Chem. Thermodynamics*, 1975. **7**(6): p. 537-545.
59. Evans, T.W., *The Hill Method for Solubility Determinations*. *Ind. Eng. Chem. Anal. Ed.*, 1936. **8**(3): p. 206-208.
60. Letcher, T.M. and P.M. Siswana, Liquid-liquid equilibria for mixtures of an alkanol + water + a methyl substituted benzene at 25°C. *Fluid Phase Equilib.*, 1992. **74**(0): p. 203-217.
61. Letcher, T.M., S. Ravindran, and S. Radloff, *Liquid-liquid equilibria for mixtures of an alkanol + diisopropyl ether + water at 25°C*. *Fluid Phase Equilib.*, 1992. **71**(1-2): p. 177-188.
62. De Santis, R., L. Marrelli, and P.N. Muscetta, Liquid—liquid equilibria in water—aliphatic alcohol systems in the presence of sodium chloride. *Chem. Eng. J.*, 1976. **11**(3): p. 207-214.
63. Jaoui, M., M. Luszczek, and M. Rogalski, Liquid–Liquid and Liquid–Solid Equilibria of Systems Containing Water and Selected Chlorophenols. *J. Chem. Eng. Data*, 1999. **44**(6): p. 1269-1272.
64. Sidgwick, N.V. and S.L. Turner, *The Solubility of the Chlorophenols*. *J. Chem. Soc.*, 1922. **121**: p. 2256-2263.
65. Achard, C., et al., Aqueous Solubilities of Phenol Derivatives by Conductivity Measurements. *J. Chem. Eng. Data*, 1996. **41**(3): p. 504-507.
66. Blackman, G.E., M.H. Parke, and G. Garton, The physiological activity of substituted phenols. II. Relationships between physical properties and physiological activity. *Arch. Biochem. Biophys.*, 1955. **54**(1): p. 55-71.

67. Ma, K.C., W.Y. Shiu, and D. Mackay, *Aqueous Solubilities of Chlorinated Phenols at 25°C*. J. Chem. Eng. Data, 1993. **38**(3): p. 364-366.
68. Lei, Y.D., C. Shunthirasingham, and F. Wania, Comparison of Headspace and Gas-Stripping Techniques for Measuring the Air-Water Partitioning of Normal Alkanols (C4 to C10): Effect of Temperature, Chain Length, and Adsorption to the Water Surface. J. Chem. Eng. Data, 2007. **52**(1): p. 168-179.
69. Tochigi, K., M. Uchiyama, and K. Kojima, Measurement of infinite-dilution activity coefficients of alcohols in water using relative gas-liquid chromatographic method. Korean J. Chem. Eng., 2000. **17**(5): p. 502-505.
70. Altschuh, J., et al., Henry's law constants for a diverse set of organic chemicals: experimental determination and comparison of estimation methods. Chemosphere, 1999. **39**(11): p. 1871-1887.
71. Shiu, W.-Y. and D. Mackay, Henry's Law Constants of Selected Aromatic Hydrocarbons, Alcohols, and Ketones. J. Chem. Eng. Data, 1997. **42**(1): p. 27-30.
72. Tewari, Y.B., et al., Aqueous solubility and octanol/water partition coefficient of organic compounds at 25.0°C. J. Chem. Eng. Data, 1982. **27**(4): p. 451-454.
73. Addison, C.C., 26. The properties of freshly formed surfaces. Part IV. The influence of chain length and structure on the static and the dynamic surface tensions of aqueousalcoholic solutions. J. Chem. Soc., 1945: p. 98-106.
74. Li, A. and A.W. Andren, Solubility of polychlorinated biphenyls in water/alcohol mixtures. 1. Experimental data. Environ. Sci. Technol., 1994. **28**(1): p. 47-52.
75. Kinoshita, K., H. Ishikawa, and K. Shinoda, *Solubility of Alcohols in Water Determined by the Surface Tension Measurements*. Bull. Chem. Soc. Jpn., 1958. **31**(9): p. 1081-1082.
76. Hill, D.J.T. and L.R. White, *The Enthalpies of Solution of Hexan-1-ol and Heptan-1-ol in Water*. Aust. J. Chem., 1974. **27**: p. 1905-1916.
77. Butler, J.A.V., D.W. Thomson, and W.H. MacLennan, 173. The free energy of the normal aliphatic alcohols in aqueous solution. Part I. The partial vapour pressures of aqueous solutions of methyl, n-propyl, and n-butyl alcohols. Part II. The solubilities of some normal aliphatic alcohols in water. Part III. The theory of binary solutions, and its application to aqueous-alcoholic solutions. J. Chem. Soc., 1933: p. 674-686.
78. Vochten, R. and G. Petre, Study of the heat of reversible adsorption at the air-solution interface. II. Experimental determination of the heat of reversible adsorption of some alcohols. J. Colloid Interface Sci., 1973. **42**(2): p. 320-327.
79. Shchelokova, E.A., et al., Solubility of monohydric aliphatic alcohols in water and aqueous HCl solutions. Russ. J. Appl. Chem., 2012. **85**(3): p. 465-469.

80. Rao, K.S., M.V.R. Rao, and C.V. Rao, Ternary Liquid Equilibria: Acetone-water-n-heptanol & Acetone-water-n-octanol Systems. *J. Sci. Industr. Res.*, 1961. **20B**: p. 283-6.
81. Tsonopoulos, C., Thermodynamic analysis of the mutual solubilities of normal alkanes and water. *Fluid Phase Equilib.*, 1999. **156**(1-2): p. 21-33.
82. Tsonopoulos, C., Thermodynamic analysis of the mutual solubilities of hydrocarbons and water. *Fluid Phase Equilib.*, 2001. **186**(1-2): p. 185-206.
83. HENRYWIN--Version3.20. 2011.
84. Ryu, S.-A. and S.-J. Park, Determination of the air/water partition coefficient for environmental pollutants by using the modified EPICS method and the relationship between the partition coefficient and physical properties. *Hwahak Konghak*, 1998. **36**(4): p. 554-561.
85. Coates, M., D.W. Connell, and D.M. Barron, *Aqueous solubility and octan-1-ol-water partition coefficients of aliphatic hydrocarbons*. *Environ. Sci. Technol.*, 1985. **19**(7): p. 628-632.
86. McAuliffe, C., Solubility in Water of Paraffin, Cycloparaffin, Olefin, Acetylene, Cycloolefin, and Aromatic Hydrocarbons. *J. Phys. Chem.*, 1966. **70**(4): p. 1267-1275.
87. Leinonen, P.J. and D. Mackay, *The multicomponent solubility of hydrocarbons in water*. *Can. J. Chem. Eng.*, 1973. **51**(2): p. 230-233.
88. Letinski, D.J., et al., Slow-stir water solubility measurements of selected alcohols and diesters. *Chemosphere*, 2002. **48**(3): p. 257-265.
89. Hoffman, C.S. and E.W. Anacker, *Water solubilities of tetradecanol and hexadecanol*. *J. Chromatogr.*, 1967. **30**(2): p. 390-6.
90. Robb, I.D., Determination of the aqueous solubility of fatty acids and alcohols. *Aust. J. Chem.*, 1966. **19**(12): p. 2281-2284.
91. Stephenson, R. and J. Stuart, *Mutual binary solubilities: water-alcohols and water-esters*. *J. Chem. Eng. Data*, 1986. **31**(1): p. 56-70.
92. Krause, F.P. and W. Lange, Aqueous solubilities of n-dodecanol, n-hexadecanol, and n-octadecanol by a new method. *J. Phys. Chem.*, 1965. **69**(9): p. 3171-3.
93. Kırbaşlar, Ş.İ., S. Şahin, and M. Bilgin, *(Liquid + liquid) equilibria of (water + propionic acid + alcohol) ternary systems*. *J. Chem. Thermodynamics*, 2006. **38**(12): p. 1503-1509.
94. Ferguson, A.L., P.G. Debenedetti, and A.Z. Panagiotopoulos, *Solubility and Molecular Conformations of n-Alkane Chains in Water*. *J. Phys. Chem. B*, 2009. **113**(18): p. 6405-6414.

95. Tolls, J., et al., Aqueous Solubility–Molecular Size Relationships: A Mechanistic Case Study Using C10- to C19-Alkanes. *The Journal of Physical Chemistry A*, 2002. **106**(11): p. 2760-2765.
96. Mackay, D., W.Y. Shiu, and A.W. Wolkoff, *Gas Chromatographic Determination of Low Concentrations of Hydrocarbons in Water by Vapor Phase Extraction*. ASTM Spec. Tech. Publ., 1975. **573**(Water Quality Parameters Symposium): p. 251-8.
97. McAuliffe, C., *Solubility in water of normal C9 and C10 alkane hydrocarbons*. Science (New York, N.Y.), 1969. **163**(3866): p. 478-9.
98. Sutton, C. and J.A. Calder, Solubility of higher-molecular-weight normal-paraffins in distilled water and sea water. *Environ. Sci. Technol.*, 1974. **8**(7): p. 654-657.
99. Franks, F., Solute-water interactions and the solubility behavior of long-chain paraffin hydrocarbons. *Nature (London, U. K.)*, 1966. **210**(5031): p. 87-8.
100. McBain, J.W. and P.H. Richards, *Solubilization of Insoluble Organic Liquids by Detergents*. *Ind. Eng. Chem.*, 1946. **38**(6): p. 642-646.
101. Peake, E. and G.W. Hodgson, Alkanes in aqueous systems. I. Exploratory investigations on the accommodation of C20-C33 n-alkanes in distilled water and occurrence in natural water systems. *J. Am. Oil Chem. Soc.*, 1966. **43**(4): p. 215-22.
102. Baker, E.G., *Origin and migration of oil*. Science (Washington, DC, U. S.), 1959. **129**: p. 871-4.
103. Hellinger, S. and S.I. Sandler, *Liquid-Liquid Equilibria of Fuel Oxygenate + Water + Hydrocarbon Mixtures*. 2. *J. Chem. Eng. Data*, 1995. **40**(1): p. 321-325.
104. Krasnoshchekova, R.Y. and M. Gubergrits, *Solubility of paraffin hydrocarbons in fresh and salt water*. *Neftekhimiya*, 1973. **13**(6): p. 885-8.
105. Ben-Naim, A. and J. Wilf, Solubilities and hydrophobic interactions in aqueous solutions of monoalkylbenzene molecules. *J. Phys. Chem.*, 1980. **84**(6): p. 583-586.
106. Sherblom, P.M., P.M. Gschwend, and R.P. Eganhouse, Aqueous solubilities, vapor pressures, and 1-octanol-water partition coefficients for C9-C14 linear alkylbenzenes. *J. Chem. Eng. Data*, 1992. **37**(4): p. 394-9.
107. Řehák, K., et al., (Liquid + liquid) equilibrium for (dimethyl phthalate + water), (diethyl phthalate + water), and (dibutyl phthalate + water). *J. Chem. Thermodynamics*, 2000. **32**(3): p. 393-400.
108. Kırbaşlar, Ş.İ., S. Şahin, and M. Bilgin, *(Liquid+liquid) equilibria of (water + butyric acid + esters) ternary systems*. *J. Chem. Thermodynamics*, 2007. **39**(9): p. 1279-1285.

109. Wolfe, N.L., W.C. Steen, and L.A. Burns, *Phthalate ester hydrolysis: Linear free energy relationships*. Chemosphere, 1980. **9**(7-8): p. 403-408.
110. Howard, P.H., S. Banerjee, and K.H. Robillard, Measurement of water solubilities, octanol/water partition coefficients and vapor pressures of commercial phthalate esters. Environ. Toxicol. Chem., 1985. **4**(5): p. 653-661.
111. Thomsen, M., L. Carlsen, and S. Hvidt, Solubilities and surface activities of phthalates investigated by surface tension measurements. Environ. Toxicol. Chem., 2001. **20**(1): p. 127-132.
112. Leyder, F. and P. Boulanger, Ultraviolet absorption, aqueous solubility, and octanol-water partition for several phthalates. Bull. Environ. Contam. Toxicol., 1983. **30**(2): p. 152-7.
113. Russell, D.J., Transport and Fate of Organic Pollutants in the Environment with Emphasis on Phthalate Esters, Ph.D., State University of New York at Binghamton, New York. 1983.
114. Defoe, D.L., et al., Solubility and toxicity of eight phthalate esters to four aquatic organisms. Environ. Toxicol. Chem., 1990. **9**(5): p. 623-636.
115. Ellington, J.J., Octanol/Water Partition Coefficients and Water Solubilities of Phthalate Esters†. J. Chem. Eng. Data, 1999. **44**(6): p. 1414-1418.
116. Fishbein, L. and P.W. Albro, *Chromatographic and biological aspects of the phthalate esters*. J. Chromatogr., 1972. **70**(2): p. 365-412.
117. Rowley, R.L., et al., *DIPPR® Data Compilation of Pure Chemical Properties*, Design Institute for Physical Properties, AIChE, New York, NY (2011).
118. Vesala, A., Thermodynamics of transfer of nonelectrolytes from light to heavy water. I. Linear free energy correlations of free energy of transfer with solubility and heat of melting of a nonelectrolyte. Acta Chem. Scand., Ser. A, 1974. **A28**(8): p. 839-45.
119. Stephenson, R.M., *Mutual solubility of water and aliphatic amines*. J. Chem. Eng. Data, 1993. **38**(4): p. 625-629.
120. Hales, B.J., G.L. Bertrand, and L.G. Hepler, *Effects of Third Components on Critical Mixing in the Water-Triethylamine System*. J. Phys. Chem., 1966. **70**(12): p. 3970-3975.
121. De Santis, R., Phase equilibriums of the water-sodium chloride-triethylamine system. J. Chem. Eng. Data, 1976. **21**(3): p. 328-330.
122. Kitamura, N., et al., Laser-induced liquid-to-droplet extraction of chlorophenol: photothermal phase separation of aqueous triethylamine solutions. Anal. Chem., 2005. **77**(18): p. 6055-61.

123. Mendonça, Â.F.S.S., S.N.R. Pereira, and I.M.S. Lampreia, *Solubility of Triethylamine in Calcium Chloride Aqueous Solutions from 20 to 35°C*. J. Solution Chem., 2003. **32**(12): p. 1033-1044.
124. Ishiguro, T., et al., Equilibria of the systems amine-water-caustic alkali. II. Mutual solubilities of Et₃N-H₂O-NaOH and Et₃N-H₂O-KOH systems. Yakugaku Zasshi, 1955. **75**: p. 188-92.
125. Wenzel, J., et al., *Kinetics of phase separation in binary liquid mixtures*. J. Phys. Chem., 1980. **84**(15): p. 1991-1995.
126. Davidson, R.R., W.H. Smith, and D.W. Hood, *Structure and Amine-Water Solubility in Desalination by Solvent Extraction*. J. Chem. Eng. Data, 1960. **5**(4): p. 420-423.
127. Kohler, F. and O.K. Rice, *Coexistence Curve of the Triethylamine-Water System*. J. Chem. Phys., 1957. **26**(6): p. 1614-1618.
128. *API Technical Data Book--Version 9.0*, 2006, The American Petroleum Institute and EPCON International.
129. Horvath, A.L. and F.W. Getzen, IUPAC-NIST Solubility Data Series 68. Halogenated Aliphatic Hydrocarbon Compounds C₃-C₁₄ With Water. J. Phys. Chem. Ref. Data, 1999. **28**(3): p. 649-777.
130. Horvath, A.L., F.W. Getzen, and Z. Maczynska, *IUPAC-NIST Solubility Data Series 67. Halogenated Ethanes and Ethenes with Water*. J. Phys. Chem. Ref. Data, 1999. **28**(2): p. 395-627.
131. Sazonov, V.P., K.N. Marsh, and G.T. Hefter, *IUPAC-NIST Solubility Data Series 71. Nitromethane with Water or Organic Solvents: Binary Systems*. J. Phys. Chem. Ref. Data, 2001. **29**(5): p. 1165-1354.
132. Sazonov, V.P., et al., *IUPAC-NIST Solubility Data Series. 78. Acetonitrile Binary Systems*. J. Phys. Chem. Ref. Data, 2002. **31**(4): p. 989-1133.
133. Sazonov, V.P., K.N. Marsh, and G.T. Hefter, IUPAC-NIST solubility data series 71. Nitromethanes with water or organic solvents: binary systems. [Erratum to document cited in CA134:301421]. J. Phys. Chem. Ref. Data, 2002. **31**(2): p. 699.
134. Sazonov, V.P., D.G. Shaw, and K.N. Marsh, IUPAC-NIST solubility data series. 77. C₂+ nitroalkanes with water or organic solvents: binary and multicomponent systems. J. Phys. Chem. Ref. Data, 2002. **31**(1): p. 1-121.
135. Clever, H.L., et al., IUPAC-NIST Solubility Data Series. 80. Gaseous fluorides of boron, nitrogen, sulfur, carbon, and silicon and solid xenon fluorides in all solvents. [Erratum to document cited in CA143:160456]. J. Phys. Chem. Ref. Data, 2005. **34**(4): p. 2347-2349.

136. Shaw, D.G., et al., IUPAC-NIST Solubility Data Series. 81. Hydrocarbons with Water and Seawater-Revised and Updated. Part 8. C₉ Hydrocarbons with Water. J. Phys. Chem. Ref. Data, 2005. **34**(4): p. 2299-2345.
137. Shaw, D.G., et al., IUPAC-NIST Solubility Data Series. 81. Hydrocarbons with Water and Seawater Revised and Updated. Part 7. C₈H₁₂-C₈H₁₈ Hydrocarbons with Water. J. Phys. Chem. Ref. Data, 2005. **34**(4): p. 2261-2298.
138. Shaw, D.G., et al., IUPAC-NIST Solubility Data Series. 81. Hydrocarbons with water and seawater-revised and updated. Part 6. C₈H₈-C₈H₁₀ Hydrocarbons with water. J. Phys. Chem. Ref. Data, 2005. **34**(3): p. 1489-1553.
139. Maczynski, A., et al., IUPAC-NIST Solubility Data Series. 81. Hydrocarbons with water and seawater-revised and updated. Part 5. C₇ Hydrocarbons with water and heavy water. J. Phys. Chem. Ref. Data, 2005. **34**(3): p. 1399-1487.
140. Maczynski, A., et al., IUPAC-NIST Solubility Data Series. 81. Hydrocarbons with water and seawater-Revised and updated. Part 4. C₆H₁₄ Hydrocarbons with water. J. Phys. Chem. Ref. Data, 2005. **34**(2): p. 709-753.
141. Maczynski, A., et al., IUPAC-NIST Solubility Data Series. 81. Hydrocarbons with water and seawater-revised and updated. Part 3. C₆H₈ - C₆H₁₂ hydrocarbons with water and heavy water. J. Phys. Chem. Ref. Data, 2005. **34**(2): p. 657-708.
142. Maczynski, A., et al., IUPAC-NIST Solubility Data Series. 81. Hydrocarbons with water and seawater-revised and updated. Part 2. Benzene with water and heavy water. J. Phys. Chem. Ref. Data, 2005. **34**(2): p. 477-552.
143. Maczynski, A., et al., IUPAC-NIST Solubility Data Series. 81. Hydrocarbons with Water and Seawater-Revised and updated Part 1. C₅ Hydrocarbons with Water. J. Phys. Chem. Ref. Data, 2005. **34**(2): p. 441-476.
144. Shaw, D.G., et al., IUPAC-NIST Solubility Data Series. 81. Hydrocarbons with Water and Seawater-Revised and Updated Part 12. C₅-C₂₆ Hydrocarbons with Seawater. J. Phys. Chem. Ref. Data, 2006. **35**(2): p. 785-838.
145. Shaw, D.G., et al., IUPAC-NIST Solubility Data Series. 81. Hydrocarbons with Water and Seawater-Revised and Updated. Part 11. C₁₃-C₃₆ Hydrocarbons with Water. J. Phys. Chem. Ref. Data, 2006. **35**(2): p. 687-784.
146. Shaw, D.G., et al., IUPAC-NIST Solubility Data Series. 81. Hydrocarbons with Water and Seawater-Revised and Updated. Part 10. C₁₁ and C₁₂ Hydrocarbons with Water. J. Phys. Chem. Ref. Data, 2006. **35**(1): p. 153-203.
147. Shaw, D.G., et al., IUPAC-NIST Solubility Data Series. 81. Hydrocarbons with Water and Seawater-Revised and Updated. Part 9. C₁₀ Hydrocarbons with Water. J. Phys. Chem. Ref. Data, 2006. **35**(1): p. 93-151.

148. Maczynski, A., et al., IUPAC-NIST Solubility Data Series. 82. Alcohols with Water-Revised and Updated: Part 5. C₈-C₁₇ Alcohols with Water. J. Phys. Chem. Ref. Data, 2007. **36**(3): p. 685-731.
149. Maczynski, A., et al., IUPAC-NIST Solubility Data Series. 82. Alcohols with water-revised and updated: part 4. C₇ Alcohols with water. J. Phys. Chem. Ref. Data, 2007. **36**(2): p. 445-484.
150. Maczynski, A., et al., IUPAC-NIST Solubility Data Series. 82. Alcohols with water-revised and updated: Part 3. C₆ Alcohols with water. J. Phys. Chem. Ref. Data, 2007. **36**(2): p. 399-443.
151. Maczynski, A., et al., IUPAC-NIST Solubility Data Series. 82. Alcohols with Water-Revised and Updated: Part 1. C₄ Alcohols with Water. J. Phys. Chem. Ref. Data, 2007. **36**(1): p. 59-132.
152. Maczynski, A., et al., IUPAC-NIST Solubility Data Series. 82: Alcohols with Water-Revised and Updated: Part 2. C₅ Alcohols with Water. J. Phys. Chem. Ref. Data, 2007. **36**(1): p. 133-190.
153. Maczynski, A., et al., IUPAC-NIST Solubility Data Series. 86. Ethers and ketones with water. Part 6. C₇-C₁₂ Ketones with water. J. Phys. Chem. Ref. Data, 2008. **37**(3): p. 1611-1653.
154. Maczynski, A., et al., IUPAC-NIST Solubility Data Series. 86. Ethers and ketones with water. Part 5. C₆ Ketones with water. J. Phys. Chem. Ref. Data, 2008. **37**(3): p. 1575-1609.
155. Maczynski, A., et al., IUPAC-NIST Solubility Data Series. 86. Ethers and ketones with water. Part 4. C₄ and C₅ Ketones with water. J. Phys. Chem. Ref. Data, 2008. **37**(3): p. 1517-1574.
156. Maczynski, A., et al., IUPAC-NIST Solubility Data Series. 86. Ethers and Ketones with Water. Part 3. C₇-C₁₄ Ethers with Water. J. Phys. Chem. Ref. Data, 2008. **37**(2): p. 1169-1182.
157. Maczynski, A., et al., IUPAC-NIST Solubility Data Series. 86. Ethers and ketones with water. Part 2. C₆ Ethers with water. J. Phys. Chem. Ref. Data, 2008. **37**(2): p. 1147-1168.
158. Maczynski, A., et al., IUPAC-NIST Solubility Data Series. 86. Ethers and ketones with water. Part 1. C₂-C₅ Ethers with water. J. Phys. Chem. Ref. Data, 2008. **37**(2): p. 1119-1146.
159. Goral, M., et al., IUPAC-NIST Solubility Data Series. 88. Esters with water-revised and updated. Part 1. C₂ to C₄ Esters. J. Phys. Chem. Ref. Data, 2009. **38**(4): p. 1093-1127.
160. Goral, M., et al., IUPAC-NIST Solubility Data Series. 88. Esters with water-revised and updated. Part 4. C₁₀ to C₃₂ Esters. J. Phys. Chem. Ref. Data, 2010. **39**(3): p. 033107/1-033107/38.
161. Harvey, A.H. and M. Salomon, *Editorial: IUPAC-NIST Solubility Data Series*. J. Phys. Chem. Ref. Data, 2010. **39**(2): p. 020401/1.

162. Goral, M., et al., IUPAC-NIST Solubility Data Series. 88. Esters with water. Revised and updated. Part 3. C₇ to C₉ Esters. J. Phys. Chem. Ref. Data, 2010. **39**(2): p. 023102/1-023102/70.
163. Goral, M., et al., IUPAC-NIST Solubility Data Series. 88. Esters with Water-Revised and Updated. Part 2: C₅ and C₆ Esters. J. Phys. Chem. Ref. Data, 2010. **39**(1): p. 013102/1-013102/58.
164. Goral, M., et al., IUPAC-NIST Solubility Data Series. 91. Phenols with Water. Part 2. C₈ to C₁₅ Alkane Phenols with Water. J. Phys. Chem. Ref. Data, 2011. **40**(3): p. 033103/1-033103/60.
165. Goral, M., et al., IUPAC-NIST Solubility Data Series. 91. Phenols with Water. Part 1. C₆ and C₇ Phenols with Water and Heavy Water. J. Phys. Chem. Ref. Data, 2011. **40**(3): p. 033102/1-033102/46.
166. Goral, M., B. Wisniewska-Gocłowska, and A. Maczynski, *Recommended Liquid--Liquid Equilibrium Data. Part 4. 1-Alkanol--Water Systems*. J. Phys. Chem. Ref. Data, 2006. **35**(3): p. 1391-1414.
167. O'Farrell, C.E. and W.E. Waghorne, *Henry's Law Constants of Organic Compounds in Water and n-Octane at T = 293.2 K*. J. Chem. Eng. Data, 2010. **55**(4): p. 1655-1658.
168. Vrbka, P., et al., Measurement of infinite dilution activity coefficients of 1-alkanols (C1–C5) in water as a function of temperature (273–373 K). Fluid Phase Equilib., 2005. **237**(1–2): p. 123-129.
169. Burnett, M.G., Determination of partition coefficients at infinite dilution by the gas chromatographic analysis of the vapor above dilute solutions. Anal. Chem., 1963. **35**(11): p. 1567-70.
170. Snider, J.R. and G.A. Dawson, Tropospheric light alcohols, carbonyls, and acetonitrile: Concentrations in the southwestern United States and Henry's Law data. J. Geophys. Res. Atmos., 1985. **90**(D2): p. 3797-3805.
171. Gupta, A.K., et al., Henry's constants of n-alkanols (methanol through n-hexanol) in water at temperatures between 40°C and 90°C. Fluid Phase Equilib., 2000. **170**(2): p. 183-192.
172. Li, J. and P.W. Carr, Measurement of water-hexadecane partition coefficients by headspace gas chromatography and calculation of limiting activity coefficients in water. Anal. Chem., 1993. **65**(10): p. 1443-50.
173. Pividal, K.A., A. Birtigh, and S.I. Sandler, Infinite dilution activity coefficients for oxygenate systems determined using a differential static cell. J. Chem. Eng. Data, 1992. **37**(4): p. 484-487.
174. Dalager, P., Vapor-liquid equilibriums of binary systems of water with methanol and ethanol at extreme dilution of the alcohols. J. Chem. Eng. Data, 1969. **14**(3): p. 298-301.

175. Landau, I., A.J. Belfer, and D.C. Locke, *Measurement of limiting activity coefficients using non-steady-state gas chromatography*. Ind. Eng. Chem. Res., 1991. **30**(8): p. 1900-1906.
176. Lebert, A. and D. Richon, Infinite dilution activity coefficients of n-alcohols as a function of dextrin concentration in water-dextrin systems. J. Agric. Food. Chem., 1984. **32**(5): p. 1156-1161.
177. Butler, J.A.V., C.N. Ramchandani, and D.W. Thomson, The solubility of nonelectrolytes. I. Free energy of hydration of some aliphatic alcohols. J. Chem. Soc., 1935: p. 280-5.
178. Kaneko, T., P.Y. Wang, and A. Sato, Partition coefficients of some acetate esters and alcohols in water, blood, olive oil, and rat tissues. Occup. Environ. Med., 1994. **51**(1): p. 68-72.
179. Kojima, K., et al., Determination of vapor-liquid equilibriums from boiling point curves. Kagaku Kogaku, 1968. **32**(2): p. 149-53.
180. Hofstee, T., A. Kwantes, and G.W.A. Rijnders, *Determination of activity coefficients at infinite dilution by gas-liquid chromatography*. Proc. Intern. Symposium Distillation, Brighton, England, 1960: p. 105-9.
181. Hardy, C.J., Determination of activity coefficients at infinite dilution from gas chromatographic measurements. J. Chromatogr., 1959. **2**: p. 490-8.
182. Rytting, J.H., L.P. Huston, and T. Higuchi, *Thermodynamic group contributions for hydroxyl, amino, and methylene groups*. J. Pharm. Sci., 1978. **67**(5): p. 615-618.
183. Rius, A. and C. Alfonso, *Liquid-liquid equilibriums. Systems water-propyl formate and formic acid-propyl formate*. An. R. Soc. Esp. Fis. Quim., Ser. B, 1955. **51B**: p. 649-58.
184. Kendall, J. and L.E. Harrison, Part II. Intermediate addition-compounds and chain reactions. Compound formation in ester-water systems. Trans. Faraday Soc., 1928. **24**(0): p. 588-596.
185. Goral, M., A. Maczynski, and P. Oracz, *Recommended Liquid-Liquid Equilibrium Data. Part 5. Ether-Water Systems*. J. Phys. Chem. Ref. Data, 2007. **36**(4): p. 1399-1415.
186. Atik, Z., et al., Measurement of Activity Coefficients at Infinite Dilution of Benzene, Toluene, Ethanol, Esters, Ketones, and Ethers at Various Temperatures in Water Using the Dilutor Technique. J. Chem. Eng. Data, 2004. **49**(5): p. 1429-1432.
187. Li, J., et al., Measurement of large infinite dilution activity coefficients of nonelectrolytes in water by inert gas stripping and gas chromatography. Anal. Chem., 1993. **65**(22): p. 3212-3218.
188. Hartkopf, A. and B.L. Karger, *Study of the interfacial properties of water by gas chromatography*. Acc. Chem. Res., 1973. **6**(6): p. 209-216.

189. Bennett, G.M. and W.G. Philip, Influence of Structure on the Solubilities of Ethers. Part I. Aliphatic Ethers. J. Chem. Soc., 1928: p. 1930-7.
190. Ginnings, P.M. and R. Webb, *Aqueous Solubilities of Some Isomeric Hexanols*. J. Am. Chem. Soc., 1938. **60**(6): p. 1388-1389.
191. Sedlbauer, J., G. Bergin, and V. Majer, *Group contribution method for Henry's Law constant of aqueous hydrocarbons*. AIChE J., 2002. **48**(12): p. 2936-2959.
192. Sanemasa, I., et al., Solubility measurements of benzene and the alkylbenzenes in water by making use of solute vapor. Bull. Chem. Soc. Jpn., 1982. **55**(4): p. 1054-62.
193. Glew, D.N. and R.E. Robertson, The Spectrophotometric Determination of the Solubility of Cumene in Water by a Kinetic Method. J. Phys. Chem., 1956. **60**: p. 332-337.
194. Howe, G.B., M.E. Mullins, and T.N. Rogers, Evaluation and prediction of Henry's law constants and aqueous solubilities for solvents and hydrocarbon fuel components. Volume 1. Technical discussion, 1987, Research Triangle Inst., Research Triangle Park, NC, USA. pp. 97.
195. Polak, J. and B.C.Y. Lu, *Mutual Solubilities of Hydrocarbons and Water at 0 and 25 °C*. Can. J. Chem., 1973. **51**(24): p. 4018-23.
196. Leroi, J.C., et al., Accurate measurement of activity coefficients at infinite dilution by inert gas stripping and gas chromatography. Ind. Eng. Chem., Process Des. Dev., 1977. **16**(1): p. 139-44.
197. Duhem, P. and J. Vidal, Extension of the Dilutor Method to Measurement of High Activity Coefficients at Infinite Dilution. Fluid Phase Equilib., 1978. **2**: p. 231-235.
198. Krummen, M., D. Gruber, and J. Gmehling, Measurement of Activity Coefficients at Infinite Dilution in Solvent Mixtures Using the Dilutor Technique. Ind. Eng. Chem. Res., 2000. **39**(6): p. 2114-2123.
199. Hovorka, Š. and V. Dohnal, Determination of Air–Water Partitioning of Volatile Halogenated Hydrocarbons by the Inert Gas Stripping Method. J. Chem. Eng. Data, 1997. **42**(5): p. 924-933.
200. Dohányosová, P., et al., Measurement of Aqueous Solubility of Hydrophobic Volatile Organic Compounds by Solute Vapor Absorption Technique: Toluene, Ethylbenzene, Propylbenzene, and Butylbenzene at Temperatures from 273 K to 328 K. J. Chem. Eng. Data, 2001. **46**(6): p. 1533-1539.
201. Sanemasa, I., et al., Solubilities of benzene and the alkylbenzenes in water. Method for obtaining aqueous solutions saturated with vapors in equilibrium with organic liquids. Chem. Lett., 1981(2): p. 225-8.

202. Buttery, R.G., L. Ling, and D.G. Guadagni, *Food volatiles. Volatilities of aldehydes, ketones, and esters in dilute water solution*. J. Agr. Food Chem., 1969. **17**(2): p. 385-9.
203. Butler, J.A.V. and W.S. Reid, The solubility of non-electrolytes. Part III. The entropy of hydration. J. Chem. Soc., 1936: p. 1171-1173.
204. Hovorka, Š., et al., Determination of temperature dependence of limiting activity coefficients for a group of moderately hydrophobic organic solutes in water. Fluid Phase Equilib., 2002. **201**(1): p. 135-164.
205. Shiu, W.-Y., et al., Temperature Dependence of Aqueous Solubility of Selected Chlorobenzenes, Polychlorinated Biphenyls, and Dibenzofuran. J. Chem. Eng. Data, 1997. **42**(2): p. 293-297.
206. Dewulf, J., H. Van Langenhove, and P. Everaert, Determination of Henry's law coefficients by combination of the equilibrium partitioning in closed systems and solid-phase microextraction techniques. J. Chromatogr. A, 1999. **830**(2): p. 353-363.
207. Yalkowsky, S.H., R.J. Orr, and S.C. Valvani, *Solubility and Partitioning. 3. The Solubility of Halobenzenes in Water*. Ind. Eng. Chem. Fundam., 1979. **18**(4): p. 351-353.
208. Dreisbach, R.R., Physical Properties Of Chemical Compounds, in Physical Properties of Chemical Compounds. 1961, American Chemical Society. p. 3-523.
209. Ten Hulscher, T.E.M., L.E. van der Velde, and W.A. Bruggeman, Temperature dependence of Henry's law constants for selected chlorobenzenes, polychlorinated biphenyls and polycyclic aromatic hydrocarbons. Environ. Toxicol. Chem., 1992. **11**: p. 1595-1603.
210. Banerjee, S., *Solubility of organic mixtures in water*. Environ. Sci. Technol., 1984. **18**(8): p. 587-591.
211. Chin, Y.P., W.J. Weber, Jr., and T.C. Voice, Determination of partition coefficients and aqueous solubilities by reverse phase chromatography - II. Evaluation of partitioning and solubility models. Water Res., 1986. **20**(11): p. 1443-50.
212. Finizio, A. and A. Di Guardo, Estimating temperature dependence of solubility and octanol-water partition coefficient for organic compounds using RP-HPLC. Chemosphere, 2001. **45**(6-7): p. 1063-1070.
213. Miller, M.M., et al., Aqueous Solubilities, Octanol/Water Partition Coefficients, and Entropies of Melting of Chlorinated Benzenes and Biphenyls. J. Chem. Eng. Data, 1984. **29**: p. 184-190.
214. Boyd, E.M., et al., Toxicity of chlorobenzenes to a lux-marked terrestrial bacterium, *Pseudomonas fluorescens*. Environ. Toxicol. Chem., 1998. **17**(11): p. 2134-2140.

215. Booth, H.S. and H.E. Everson, *Hydrotropic Solubilities: Solubilities in 40 Per Cent Sodium Xylenesulfonate*. J. Ind. Eng. Chem., 1948. **40**(8): p. 1491-1493.
216. Stephenson, R.M., Mutual solubilities: water-ketones, water-ethers, and water-gasoline-alcohols. J. Chem. Eng. Data, 1992. **37**(1): p. 80-95.
217. Brüggemann, R. and J. Altschuh, A validation study for the estimation of aqueous solubility from n-octanol/water partition coefficients. Sci. Total Environ., 1991. **109-110**: p. 41-57.
218. Lin, S.-T. and S.I. Sandler, Henry's law constant of organic compounds in water from a group contribution model with multipole corrections. Chem. Eng. Sci., 2002. **57**(14): p. 2727-2733.
219. Modarresi, H., H. Modarress, and J.C. Dearden, QSPR model of Henry's law constant for a diverse set of organic chemicals based on genetic algorithm-radial basis function network approach. Chemosphere, 2007. **66**(11): p. 2067-2076.
220. Huuskonen, J., J. Rantanen, and D. Livingstone, Prediction of aqueous solubility for a diverse set of organic compounds based on atom-type electrotopological state indices. Eur. J. Med. Chem., 2000. **35**(12): p. 1081-8.
221. Hait, M.J., et al., *Space predictor for infinite dilution activity coefficients*. Ind. Eng. Chem. Res., 1993. **32**(11): p. 2905-14.
222. Katritzky, A.R., et al., A General Treatment of Solubility. 1. The QSPR Correlation of Solvation Free Energies of Single Solutes in Series of Solvents. J. Chem. Inf. Comput. Sci., 2003. **43**(6): p. 1794-1805.
223. Delaney, J.S., *Predicting aqueous solubility from structure*. Drug Discovery Today, 2005. **10**(4): p. 289-295.
224. Schroeder, B., et al., *Prediction of aqueous solubilities of solid carboxylic acids with COSMO-RS*. Fluid Phase Equilib., 2010. **289**(2): p. 140-147.
225. Putnam, R., et al., *Prediction of Infinite Dilution Activity Coefficients Using COSMO-RS*. Ind. Eng. Chem. Res., 2003. **42**(15): p. 3635-3641.
226. Grensemann, H. and J. Gmehling, Performance of a Conductor-Like Screening Model for Real Solvents. Model in Comparison to Classical Group Contribution Methods. Ind. Eng. Chem. Res., 2005. **44**(5): p. 1610-1624.
227. Bodor, N., A. Harget, and M.J. Huang, *Neural network studies. 1. Estimation of the aqueous solubility of organic compounds*. J. Am. Chem. Soc., 1991. **113**(25): p. 9480-3.

228. Hine, J. and P.K. Mookerjee, Structural effects on rates and equilibria. XIX. Intrinsic hydrophilic character of organic compounds. Correlations in terms of structural contributions. *J. Org. Chem.*, 1975. **40**(3): p. 292-298.
229. Barone, V., M. Cossi, and J. Tomasi, A new definition of cavities for the computation of solvation free energies by the polarizable continuum model. *J. Chem. Phys.*, 1997. **107**(8): p. 3210-3221.
230. Cramer, R.D., BC(DEF) parameters. 2. An empirical structure-based scheme for the prediction of some physical properties. *J. Am. Chem. Soc.*, 1980. **102**(6): p. 1849-1859.
231. Hawkins, G.D., C.J. Cramer, and D.G. Truhlar, Parametrized Model for Aqueous Free Energies of Solvation using Geometry-Dependent Atomic Surface Tensions with Implicit Electrostatics. *J. Phys. Chem. B*, 1997. **101**(36): p. 7147-7157.
232. Cabani, S., et al., Group contributions to the thermodynamic properties of nonionic organic solutes in dilute aqueous solution. *J. Solution Chem.*, 1981. **10**(8): p. 563-95.
233. Hawkins, G.D., C.J. Cramer, and D.G. Truhlar, *Universal Quantum Mechanical Model for Solvation Free Energies Based on Gas-Phase Geometries*. *J. Phys. Chem. B*, 1998. **102**(17): p. 3257-3271.
234. Shiu, W.Y. and D. Mackay, A critical review of aqueous solubilities, vapor pressures, Henry's law constants, and octanol-water partition coefficients of the polychlorinated biphenyls. *J. Phys. Chem. Ref. Data*, 1986. **15**(2): p. 911-29.
235. Katritzky, A.R., et al., QSPR Studies on Vapor Pressure, Aqueous Solubility, and the Prediction of Water-Air Partition Coefficients. *J. Chem. Inf. and Modeling*, 1998. **38**(4): p. 720-725.
236. Jin, G., J.M. Walsh, and M.D. Donohue, A group-contribution correlation for predicting thermodynamic properties with the perturbed-soft-chain theory. *Fluid Phase Equilib.*, 1986. **31**(2): p. 123-46.
237. Famini, G.R., et al., *Computational parameters in correlation analysis: gas-water distribution coefficient*. *Collect. Czech. Chem. Commun.*, 1999. **64**(11): p. 1727-1747.
238. Nirmalakhandan, N.N. and R.E. Speece, *QSAR model for predicting Henry's constant*. *Environ. Sci. Technol.*, 1988. **22**(11): p. 1349-1357.
239. Cash, G.G., Prediction of physicochemical properties from Euclidean distance methods based on electrotopological state indices. *Chemosphere*, 1999. **39**(14): p. 2583-2591.
240. Hawker, D.W., Vapor pressures and Henry's law constants of polychlorinated biphenyls. *Environ. Sci. Technol.*, 1989. **23**(10): p. 1250-3.

241. Gramatica, P., N. Navas, and R. Todeschini, Classification of organic solvents and modelling of their physico-chemical properties by chemometric methods using different sets of molecular descriptors. *TrAC, Trends Anal. Chem.*, 1999. **18**(7): p. 461-471.
242. Sabljic, A. and H. Guستن, *Predicting Henry's law constants for polychlorinated biphenyls*. *Chemosphere*, 1989. **19**(10-11): p. 1503-11.
243. Plyasunov, A.V. and E.L. Shock, *Thermodynamic functions of hydration of hydrocarbons at 298.15 K and 0.1 MPa*. *Geochim. Cosmochim. Acta*, 2000. **64**(3): p. 439-468.
244. Plyasunov, A.V., N.V. Plyasunova, and E.L. Shock, Group Contribution Values for the Thermodynamic Functions of Hydration of Aliphatic Esters at 298.15 K, 0.1 MPa. *J. Chem. Eng. Data*, 2004. **49**(5): p. 1152-1167.
245. Plyasunova, N.V., A.V. Plyasunov, and E.L. Shock, Group Contribution Values for the Thermodynamic Functions of Hydration at 298.15 K, 0.1 MPa. 2. Aliphatic Thiols, Alkyl Sulfides, and Polysulfides. *J. Chem. Eng. Data*, 2005. **50**(1): p. 246-253.
246. Plyasunov, A.V., N.V. Plyasunova, and E.L. Shock, Group Contribution Values for the Thermodynamic Functions of Hydration at 298.15 K, 0.1 MPa. 4. Aliphatic Nitriles and Dinitriles. *J. Chem. Eng. Data*, 2006. **51**(5): p. 1481-1490.
247. Plyasunov, A.V., N.V. Plyasunova, and E.L. Shock, Group Contribution Values for the Thermodynamic Functions of Hydration at 298.15 K, 0.1 MPa. 3. Aliphatic Monoethers, Diethers, and Polyethers. *J. Chem. Eng. Data*, 2006. **51**(1): p. 276-290.
248. Brunner, S., et al., Henry's law constants for polychlorinated biphenyls: experimental determination and structure-property relationships. *Environ. Sci. Technol.*, 1990. **24**(11): p. 1751-4.
249. Dearden, J.C., et al., *QSPR prediction of Henry's law constant: improved correlation with new parameters*. *Mol. Model. Predict. Bioact.*, [Proc. Eur. Symp. Quant. Struct.-Act. Relat.], 12th, 2000: p. 273-274.
250. Harvey, A.H., R. Crovetto, and J.M.H.L. Sengers, *Limiting vs. Apparent Critical Behavior of Henry's Constants and K Factors*. *AIChE J.*, 1990. **36**(12): p. 1901-1904.
251. Meylan, W.M. and P.H. Howard, *Bond contribution method for estimating Henry's law constants*. *Environ. Toxicol. Chem.*, 1991. **10**(10): p. 1283-93.
252. Öberg, T., Prediction of physical properties for PCB congeners from molecular descriptors. *Internet J. Chem.*, 2001. **4**: p. No pp given, Article No 11.
253. Yaws, C., Calculating H₂O solubility, Henry's Law constant for cycloalkanes in crude. *Oil Gas J.*, 2010. **108**(4): p. 54-57.

254. Ferreira, M.M.C., *Polycyclic aromatic hydrocarbons: a QSPR study*. Chemosphere, 2001. **44**(2): p. 125-146.
255. Suzuki, T., K. Ohtaguchi, and K. Koide, Application of Principal Components Analysis to Calculate Henry's Constant from Molecular Structure. Comput. Chem. (Oxford), 1992. **16**(1): p. 41-52.
256. Yao, X., et al., Radial basis function network-based quantitative structure-property relationship for the prediction of Henry's law constant. Anal. Chim. Acta, 2002. **462**(1): p. 101-117.
257. Russell, C.J., S.L. Dixon, and P.C. Jurs, Computer-assisted study of the relationship between molecular structure and Henry's law constant. Anal. Chem., 1992. **64**(13): p. 1350-1355.
258. Dunnivant, F.M., et al., Quantitative structure-property relationships for aqueous solubilities and Henry's law constants of polychlorinated biphenyls. Environ. Sci. Technol., 1992. **26**(8): p. 1567-1573.
259. Yaffe, D., et al., A Fuzzy ARTMAP-Based Quantitative Structure-Property Relationship (QSPR) for the Henry's Law Constant of Organic Compounds. J. Chem. Inf. Comput. Sci., 2003. **43**(1): p. 85-112.
260. Cramer, C.J. and D.G. Truhlar, An SCF Solvation Model for the Hydrophobic Effect and Absolute Free Energies of Aqueous Solvation. Science (New York, N.Y.), 1992. **256**(5054): p. 213-7.
261. Modarresi, H., H. Modarress, and J.C. Dearden, Henry's law constant of hydrocarbons in air-water system: The cavity ovality effect on the non-electrostatic contribution term of solvation free energy. SAR QSAR Environ. Res., 2005. **16**(5): p. 461-482.
262. Cramer, C.J. and D.G. Truhlar, AM1-SM2 and PM3-SM3 parameterized SCF solvation models for free energies in aqueous solution. J. Comput.-Aided Mol. Des., 1992. **6**(6): p. 629-66.
263. Jorgensen, W.L. and T.B. Nguyen, Monte Carlo simulations of the hydration of substituted benzenes with OPLS potential functions. J. Comput. Chem., 1993. **14**: p. 149-205.
264. Schröder, B., et al., Prediction of environmental parameters of polycyclic aromatic hydrocarbons with COSMO-RS. Chemosphere, 2010. **79**(8): p. 821-829.
265. Myrdal, P. and S.H. Yalkowsky, A Simple Scheme for Calculating Aqueous Solubility, Vapor Pressure and Henry's Law Constant: Application to the Chlorobenzenes. SAR QSAR Environ. Res., 1994. **2**: p. 17-28.
266. Gharagheizi, F., R. Abbasi, and B. Tirandazi, Prediction of Henry's Law Constant of Organic Compounds in Water from a New Group-Contribution-Based Model. Ind. Eng. Chem. Res., 2010. **49**(20): p. 10149-10152.

267. Abraham, M.H., et al., Hydrogen bonding. Part 34. The factors that influence the solubility of gases and vapors in water at 298 K, and a new method for its determination. J. Chem. Soc., Perkin Trans. 2, 1994(8): p. 1777-91.
268. Goodarzi, M., et al., Linear and non-linear relationships mapping the Henry's law parameters of organic pesticides. Atmos. Environ., 2010. **44**(26): p. 3179-3186.
269. Schüürmann, G., Quantum chemical approach to estimate physicochemical compound properties: application to substituted benzenes. Environ. Toxicol. Chem., 1995. **14**(12): p. 2067-76.
270. Peterson, B.K., *Property Estimation Using Analogous Series*. Ind. Eng. Chem. Res., 2011. **50**(12): p. 7696-7704.
271. Chambers, C.C., C.J. Cramer, and D.G. Truhlar, *Model for Aqueous Solvation Based on Class IV Atomic Charges and First Solvation Shell Effects*. J. Phys. Chem., 1996. **100**(40): p. 16385-16398.
272. Dearden, J.C., et al., *The prediction of Henry's law constant: a QSPR from fundamental considerations*. Quant. Struct.-Act. Relat. Environ. Sci.-VII, Proc. QSAR 96, 1997: p. 135-142.
273. Raventos-Duran, T., et al., Structure-activity relationships to estimate the effective Henry's law constants of organics of atmospheric interest. Atmos. Chem. Phys., 2010. **10**(16): p. 7643-7654.
274. Schüürmann, G., et al. A software system for automatic chemical property estimation from molecular structure. in Quantitative Structure-Activity Relationships in Environmental Sciences-VII, Proceedings of QSAR 96. 1997. Elsinore, Den.: SETAC Press.
275. Sahoo, S., S. Patel, and B.K. Mishra, Quantitative structure property relationship for Henry's law constant of some alkane isomers. Thermochim. Acta, 2011. **512**(1-2): p. 273-277.
276. Yezdimer, E.M., J. Sedlbauer, and R.H. Wood, Predictions of thermodynamic properties at infinite dilution of aqueous organic species at high temperatures via functional group additivity. Chem. Geol., 2000. **164**(3-4): p. 259-280.
277. Goss, K.-U., Prediction of the temperature dependency of Henry's law constant using poly-parameter linear free energy relationships. Chemosphere, 2006. **64**(8): p. 1369-1374.
278. Abraham, M.H. and W.E. Acree, Jr., Prediction of gas to water partition coefficients from 273 to 373 K using predicted enthalpies and heat capacities of hydration. Fluid Phase Equilib., 2007. **262**(1-2): p. 97-110.
279. Hilal, S.H., S.N. Ayyampalayam, and L.A. Carreira, Air-Liquid Partition Coefficient for a Diverse Set of Organic Compounds: Henry's Law Constant in Water and Hexadecane. Environ. Sci. Technol., 2008. **42**(24): p. 9231-9236.

280. Hilal, S.H., A.N. Saravanaraj, and L.A. Carreira, *Estimation of Henry's law constant for a diverse set of organic compounds from molecular structure*. Prepr. Ext. Abstr. - ACS Natl. Meet., Am. Chem. Soc., Div. Environ. Chem., 2008. **48**(2): p. 717-721.
281. Nirmalakhandan, N., R.A. Brennan, and R.E. Speece, *Predicting Henry's Law constant and the effect of temperature on Henry's Law constant*. Water Res., 1997. **31**(6): p. 1471-1481.
282. Huang, Y.-L., et al., Henry's Law Constant from Molecular Simulation: A Systematic Study of 95 Systems. Int. J. Thermophys., 2009. **30**(6): p. 1791-1810.
283. Čenský, M., et al., Standard partial molal properties of aqueous alkylphenols and alkylanilines over a wide range of temperatures and pressures. Geochim. Cosmochim. Acta, 2007. **71**(3): p. 580-603.
284. Plyasunov, A.V. and E.L. Shock, Prediction of the vapor-liquid distribution constants for volatile nonelectrolytes in water up to its critical temperature. Geochim. Cosmochim. Acta, 2003. **67**(24): p. 4981-5009.
285. Kühne, R., R.-U. Ebert, and G. Schüürmann, *Prediction of the Temperature Dependency of Henry's Law Constant from Chemical Structure*. Environ. Sci. Technol., 2005. **39**(17): p. 6705-6711.
286. Hill, P.G., A unified fundamental equation for the thermodynamic properties of water. J. Phys. Chem. Ref. Data, 1990. **19**(5): p. 1233-74.
287. Wagner, W. and A. Pruss, The IAPWS Formulation 1995 for the Thermodynamic Properties of Ordinary Water Substance for General and Scientific Use. J. Phys. Chem. Ref. Data, 2002. **31**(2): p. 387-535.
288. Plyasunov, A.V. and E.L. Shock, Group Contribution Values of the Infinite Dilution Thermodynamic Functions of Hydration for Aliphatic Noncyclic Hydrocarbons, Alcohols, and Ketones at 298.15 K and 0.1 MPa. J. Chem. Eng. Data, 2001. **46**(5): p. 1016-1019.
289. Nagvekar, M. and T.E. Daubert, *A group contribution method for liquid thermal conductivity*. Ind. Eng. Chem. Res., 1987. **26**: p. 1362-5.
290. Hsu, H.-C., Y.-W. Sheu, and C.-H. Tu, Viscosity estimation at low temperatures ($T_r < 0.75$) for organic liquids from group contributions. Chem. Eng. J. (Amsterdam, Neth.), 2002. **88**: p. 27-35.
291. Elbro, H.S., A. Fredenslund, and P. Rasmussen, Group contribution method for the prediction of liquid densities as a function of temperature for solvents, oligomers, and polymers. Ind. Eng. Chem. Res., 1991. **30**: p. 2576-82.
292. Harrison, B.K. and W.H. Seaton, *Solution to missing group problem for estimation of ideal gas heat capacities*. Ind. Eng. Chem. Res., 1988. **27**: p. 1536-40.

293. Domalski, E.S. and E.D. Hearing, *Estimation of the Thermodynamic Properties of C-H-N-O-S-Halogen Compounds at 298.15 K*. J. Phys. Chem. Ref. Data, 1993. **22**(4): p. 805-1159.
294. Brockbank, S.A., et al., Infinite dilution activity coefficients and Henry's law constants of compounds in water using the inert gas stripping method. Fluid Phase Equilib., 2013. **348**: p. 45-51.

Appendix A. Experimental Method Details

Table A.1: x_i^{aq} experimental methods

Experiment	Description	Assumptions/Limits
Column elution method [1, 2]	Water is used to elute a substance that is coated on an inert carrier material (glass beads, silica gel, sand, etc.) contained in a small column. Solubility is determined when the mass concentration of the exiting stream becomes constant with time.	Good method for $x_i^{aq} < 10^{-2}$ g/L. Not to be used for volatile chemicals because they can evaporate with the solvent during loading of the carrier material
HPLC correlation [3, 4]	Chemicals injected into an HPLC column have various retention times. Reference values are used to develop a relationship between the property (x_i^{aq}) and retention time. This relationship is used to find the property of other compounds based on retention time.	Reference substance is needed, so this is sometimes considered an estimation method. This method should not be used for strong acids and bases, metal complexes, or compounds that react with either the eluent or column packing.
Shake flask method [1] or batch contacting [5] or equilibrium cell	An excess amount of solute is added to a flask and shaken or vigorously stirred. It is then centrifuged or allowed to settle, and the concentration is determined in the aqueous phase (or in both phases) using an analytical method (ex. GC, HPLC, UV, etc.).	Good method for $x_i^{aq} > 10^{-2}$ g/L. Problems that can arise with less soluble substances include: incomplete equilibration, dispersion rather than dissolution if the solute is a solid, suspension of microcrystals in solution, adsorptive losses on equipment surfaces, and interference from impurity and/or water signals. This method should not be used for volatile chemicals which tend to be less soluble.
Radiometric method [6] (similar to shake flask method)	Radioagent is dissolved in organic solvent before pure water is added. Mixture is shaken vigorously. Some organic solvent dissolves in water, which causes a volume reduction of the organic phase. The radioagent does not dissolve in water. The specific activities of the radioagent are measured initially and finally in order to estimate the decreased volume of the organic phase.	

Table A.1: x_i^{aq} experimental methods (cont.)

Experiment	Description	Assumptions/Limits
Generator column or dynamic coupled column liquid chromatographic technique [7-12]	Water is pumped through the generator column that is packed with a solid support coated with the test compound to produce a saturated aqueous solution. The compound is then extracted from the saturated solution onto an extractor column. It is then eluted with a solvent/water mixture and analyzed with HPLC or GC. The water solubility is the concentration of the compound in the generator column effluent. (similar to column elution method)	This method avoids errors that could result from sorption on glassware and the presence of colloids when measuring small concentrations. Recommended range is 1 ppb-500 ppm.
Nephelometric or turbidity method [13]	Several dilutions are made of a stable suspension of known concentration. A plot of concentration verses turbidity produces a line. x_i^{aq} is the concentration at 0 turbidity, and it can be estimated through a linear regression of the turbidity data.	This method cannot distinguish between solute and impurities. Recommended for $x_i^{aq} > 10^{-3}$ g/L if the solute forms stable suspensions in water.
Slow stir method [13]	The sample is dissolved in a volatile solvent. This solution is deposited on the walls (not the bottom) of a glass container, and the solvent is evaporated using nitrogen. Water and a magnetic stir bar are added, and the system is mixed until equilibrium is reached. The concentration in the aqueous phase is then determined using an appropriate analytical technique.	Recommended for $x_i^{aq} < 10^{-3}$ g/L. This method avoids emulsions.
Gas solubility methods	The apparatus is a complex system that uses manometers and multiple vessels. A known mass of solvent is used. Equilibration can be accomplished through stirring, bubbling, shaking, or static contact. The final concentration can be determined by mass, pressure, volume, or an analytical technique.	
Thermostatic (for SOLC) [14]	In various solvents, the solubility of silver perchlorate increases in the presence of trace amounts of water. The solubility behavior can be used to determine how much water is present.	
Vapor Saturation [15]	Solvent is equilibrated with solute vapor.	
Residue volume or volumetric method or absorbed volume [16, 17]	An excess of solute is added and the mixture is centrifuged. The residue of solute is then measured volumetrically. The data is then plotted as grams of solute added vs. volume of undissolved residue. The value at zero volume of undissolved residue is then extrapolated.	
Synthetic method [17]	Weighed quantities of solute and solvent are sealed up together. The temperature at which turbidity appears/disappears is determined in a temperature bath.	This cannot distinguish between solute and impurities.
Multiple equilibration [18]	See Table A.2 for description.	
Dialysis tubing method [19]	Dialysis tubing containing pure water is knotted on one end, connected to glass tubing on the other end, and sealed with a plastic cap. This tubing is then put in a test tube containing solute. Injections are taken and analyzed with a GC.	

Table A.1: x_i^{aq} experimental methods (cont.)

Experiment	Description	Assumptions/Limits
Surface tension method [20]	Surface tension is measured by the drop weight method. From the surface tension and the logarithm of the concentration, the saturation points are determined.	
Solute vapor absorption [21]	This method is similar to the EXPSAT method (see Table A.2). Inert gas is saturated with solute and solvent and then bubbled through solvent until the solvent is saturated with solute.	
Cloud point or isothermal method	Solute is added to solvent until a precipitate is observed.	
Equilibrium cell with gas agitation [22]	Excess solute is added to the equilibrium cell. Mixing is accomplished by bubbling N_2 through the cell. Flow is stopped and layers are allowed to separate and equilibrate. Concentration is determined using an analytical method.	
Isopiestic method [23]	See Table A.3.	
Laser Multiphoton Ionization [24]	Combines microdroplet sampling and multiphoton ionization-based fast-conductivity techniques. At the onset of equilibrium solubility, a sharp increase in the detected photocharges is recorded.	
Supersaturation [25]	Solute is dissolved in water at an elevated temperature and brought down to temperature of interest	
Static Cell [26]	Solute and solvent are allowed to equilibrate without shaking or stirring.	

Table A.2: k_H experimental methods

Experiment	Description	Assumptions/Limits
Single equilibration [27] or equilibrium cell [28]	Solute and solvent are equilibrated in the liquid and vapor phases in a single cell. Concentration is measured in one or both phases. If only one phase is measured, the concentration in the other phase is calculated using a mass balance. This could be used as a general description for some of the methods described below.	This method is good for compounds with high vapor pressures and x_i^{aq} s [29, 30]. For compounds with low x_i^{aq} s and vapor pressures, it is difficult to determine the concentrations in either phase using analytical techniques. Other methods are preferred that utilize relative changes in detector response.
Internal standard	A dilute solution is made with two compounds, one of which has a known k_H . The solution is equilibrated with air, and concentrations are determined. Ratio of k_H values is related to the ratio of peak areas. (similar to RGLC method for γ_i^∞)	

Table A.2: k_H experimental methods (cont.)

Experiment	Description	Assumptions/Limits
Headspace solid-phase microextraction (SPME) [31, 32]	This is a single equilibration method that uses SPME as the analytical technique.	The SPME fiber must be calibrated in the standard gas atmosphere at a temperature and air humidity identical to those in the experiment.
Multiple Equilibration Technique [27, 33] or Indirect HGLC	A liquid sample with known volume and concentration is introduced into a variable volume vessel such as a syringe. The system is allowed to equilibrate with air. One of the phases is analyzed, and then the air is removed. The same amount of air is added to the vessel again and allowed to equilibrate. The selected phase is analyzed. This process can be repeated until the solute is undetectable. A plot of the log of the concentration in the gas phase versus the number of equilibrations produces a straight line. k_H is related to the slope.	The k_H error is reduced as the number of equilibrations increases and by using a larger gas/liquid volume ratio. This requires relatively large initial aqueous concentrations so that the concentration is still detectable after multiple equilibrations.
Headspace multiple-injection interrupted-flow (MIIF) technique [34, 35]	The first part of this method is like the x_i^{aq} shake flask method. MIIF is an analytical headspace technique. Vapor removal in a closed system disturbs the equilibrium. Vapor samples are taken at known times, and a linear extrapolation of the GC response is used to determine the GC response at equilibrium. This is a modified method of Massaldi and King [36].	Solutes must have appreciable vapor pressures.
Equilibrium Partitioning in Closed Systems (EPICS) [37] and modified EPICS [38]	Known masses are added to two systems of different volumes at the same temperature, and k_H is determined from the ratio of the concentrations in one phase. The original EPICS calculations assumed that equal masses of solute were added to each container, but the modified EPICS only requires the relative masses to be known. If a stock solution is made, the concentration does not need to be known, only the relative masses added to each container.	The systems must be at the same temperature, and they must be allowed to equilibrate before measurement. The technique has the maximum sensitivity when one system has a low liquid volume and the other a high. Adsorption can be a problem [39]. A recommended range is $0.1 \leq k_H$ (unitless molar concentration basis) ≤ 1 with a possible precision of 5% [40].
Variable headspace or static headspace or phase ratio variation method [40, 41]	The relative air concentration is measured from aliquots of the same solution in multiple containers having different headspace-to-liquid ratios. Plots of the reciprocal of the GC peak area versus headspace-to-liquid volume ratios results in a linear plot. The slope and y-intercept are related to k_H . (similar to EPICS method)	Neither the exact concentrations of the compound or its matrix need to be known. This method is applicable for volatile and semivolatile organic compounds. This method can handle smaller liquid-phase loadings than EPICS which makes it work in lower regions (around 0.001 unitless molar concentration basis).

Table A.2: k_H experimental methods (cont.)

Experiment	Description	Assumptions/Limits
Batch air stripping [30] or inert gas stripping or gas stripping or air stripping or exponential dilutor method	A dilute solution is stripped using an inert gas (usually N_2). The change in concentration with time is related to k_H . Some experiments first saturate the stripping gas with water. There are many variations of this method. The original equation has been expanded upon by multiple authors to include various corrections. Some authors take into account different corrections while others do not.	When compared to the single equilibration method, adsorption is less likely to be an issue due to the dynamic principle and turbulent conditions [39]. Absolute detector response is unimportant; it just needs to be linear [42]. It can be difficult to determine if equilibrium is reached [37]. The depth/height of the column can be increased. If the value does not change, then equilibrium was reached. However, a taller tower makes it harder to ensure complete mixing. Equilibrium can be verified by changing the flow rate of the stripping gas and seeing if k_H remains constant [43]. A range of system volumes and purge gas flow rates can also be used to verify the assumptions of complete mixing and equilibrium [44]. Non-equilibrium leads to biased results [27, 39]. This method works for chemicals with low solubilities and vapor pressures [29, 45], but problems can arise if the volatility is too low because it is difficult to detect a change in concentration after air stripping [44]. It is a good method for systems with a high limiting relative volatility and involatile solvents [46]. Bias has been reported for this method for semivolatile organic chemicals[47]. The authors claim there can be significant absorption on the surface of the bubbles. When they pop on the surface, they release excess chemical into the headspace. Recommended ranges are $0.0001 < k_H$ (unitless molar concentration basis) < 1 [31]; relative volatility > 1000 [48]; $10 \text{ kPa} < k_H < 10^5 \text{ kPa}$ and $P_2^{\text{sat}} < 50 \text{ kPa}$ [40] with 1% uncertainty possible in this range. The diameter of the bubble should be below 1 mm, and they should not coalesce [40, 49] A good relative elution rate is 0.01 min^{-1} or lower. Minimize the vapor space, and use as dilute a solution as possible.
EXPSAT (exponential saturator)[50, 51]	Contaminated air/ N_2 (gas first saturated with solute) is bubbled through water to resemble what happens in atmosphere.	An advantage over gas stripping is that dissolution of hydrophobic compound in water to make dilute solution is not required [40]. Recommended ranges: $500 < \gamma_i^\infty < 10^5$ or $1 \text{ MPa} < k_H < 100 \text{ MPa}$ with precision possible of 1-2%.
Modified gas stripping[52]	Differences between this and inert gas stripping are that chemical is introduced into stripping cell using a separate loop. Solute is in a separate flask. The line between the flask and stripping cell is "vacuumed". Line is opened and water saturated carrier is used to carry solute into cell (similar idea to EXPSAT). The concentration is monitored real time.	This is faster than the inert gas stripping method. This can be applied to a broader range of compounds than normal gas stripping.

Table A.2: k_H experimental methods (cont.)

Experiment	Description	Assumptions/Limits
Purge and trap[53] or gas sparging [54] or dynamic equilibrium system [55]	An inert gas is bubbled through a purge chamber. The exiting solute is deposited on a cooled trap or cartridge. The trap is then heated in order to inject the solute into the GC or solute can be extracted from the cartridge with solvent in order to inject into the GC.	
Packed column [56]	Similar to the inert gas stripping method. Column just has packing in it.	Packing eliminates problem of aerosols/bursting bubbles at surface
Counterflow packed column[56, 57]	Water-saturated air flows up through a packed column. The test solution flows down through the column. The exiting solution is collected and analyzed for concentration.	This can be used for compounds with low to moderate volatilities. Recommended range is $1 \cdot 10^{-4} \leq k_H$ (unitless molar concentration basis) $\leq 5 \cdot 10^{-2}$. The stripping factor must be less than 1. If the stripping factor is too small, it can be difficult to measure differences between inlet and outlet concentration.
Concurrent flow or wetted wall column [58, 59]	Solute-saturated water enters the wetted-wall column where it is in contact with a concurrent flow of gas (air). The two streams are separated at the bottom of the column and analyzed using GC.	The system must be validated to ensure full equilibrium is reached within the column [27].
Horizontal Flow Reactor [60]	This method is a cross between the wetted wall column method and evaporation from a water surface. N_2 is blown through a horizontal tube. A container of water is sitting stagnant on the bottom of the tube.	
Evaporation from a water surface [61]	Air is blown across the top of a solution at a known flow rate to evaporate the solute. The concentrations of solute in the water and gas phases are measured. Equilibrium is not achieved, so estimated mass transfer coefficients are used to determine the departure from equilibrium.	This method is for substances with low air-water partition coefficients that aggregate in solution, ionize, or display surface activity. Recommended range is k_H (unitless molar concentration basis) $< 10^{-3}$.
Diffusion in air and water boundary layers [62]	k_H is obtained from two partition coefficients between polydimethylsiloxane (PDMS) and air and between PDMS and water. Partition coefficients with PDMS are determined by measuring the mass transfer rate constants in the boundary layer.	

Table A.3: γ_i^∞ experimental methods

Experiment	Description	Assumptions/Limits
VLE extrapolation [63]	VLE data can be fit using a Gibbs excess model (i.e. Margules, Wilson, NRTL, etc.) which will give activity coefficients as a function of composition, and γ_i^∞ s are obtained from these models by extrapolating to infinite dilution [43, 63]	This assumes that extrapolating is valid.
Relative GLC (RGLC) [63, 64]	This uses an internal standard during measurement, so one γ_i^∞ of a solute must be known for each solvent. Two solutes are run through the GC simultaneously. The ratio of γ_i^∞ s is determined by measurement of the retention times of the components.	This method is applicable to a wide range of solvent volatilities[40]. The solute must be more volatile than solvent. One reference states this method can have a precision of $\geq 5\%$ [64]. Errors that go into the calculation include vapor pressure errors and the error of the reference value.
Isopiestic Method [23]	The partial pressure of water is controlled using calcium chloride solutions (or another drying agent). Equilibrium with the organic-rich phase is obtained through the vapor phase. γ_i^∞ is determined from measurement of water in the organic phase.	
Non-Steady-State Gas-Liquid Chromatography (NSGC) [63] or NSGLC	A slug of liquid solvent with an appreciable vapor pressure is injected into the GC column. The solvent condenses on the solid support and then equilibrates with the carrier gas. The solvent then bleeds out of the column at a steady rate. Solute is repeatedly injected over a period of time. The retention times of the solute will decrease at the same rate as the rate of the solvent evaporation. The retention times are related to a retention differential parameter which is used to determine γ_i^∞ .	The exact amount of the stationary phase or solvent does not need to be known, so it avoids the greatest source of error for measurement of γ_i^∞ with volatile solvents. Effects of adsorption probably cancel out due to the use of retention differences [43]. It is only applicable for systems in which the solute is more volatile than the solvent, and it cannot be used to study mixed solvents. This method is considered less accurate than others [40]. If a compressibility correction is used, the accuracy can get to 3 %. A practical upper bound may be a relative volatility of 100. The applicability for water as the solvent is the range 23–350 K.
Gas-Liquid Chromatography (GLC) [63, 65]	A column is prepared with a solvent supported on an inert solid. An inert carrier gas is passed through the column, and the retention behavior of the solute is measured. Some authors include a gas phase correction.	This method does not work as well with volatile solvents [63, 66]. The major source of error in this method is the weight of solvent in the column because it is difficult to determine accurately. Some researchers claim precisions of 1-3% are possible [40, 43]. Adsorption problems have been noted for aqueous systems [48]. Solvent purity is critical in this method, but solute purity is not due to separation in the chromatograph [65]. This is one of the fastest methods. A recommended range is $0.0005 < k_H$ (unitless molar concentration basis) < 0.05 .

Table A.3: γ_i^∞ experimental methods (cont.)

Experiment	Description	Assumptions/Limits
Headspace GLC (HGLC) [40, 48, 67]	The headspace above a solution is analyzed using GC and the response is related to γ_i^∞ .	The major limitation is the lack of high quality experimental gas phase virial coefficients needed to correct for pressure. A recommended range is $20 < k_H$ (unitless molar concentration basis) $< 20,000$. For insoluble solutes or low k_H s, the RGLC or gas stripping methods are preferred. Under good conditions, HGLC can give precision within 1-5%.
Indirect HGLC [68]	Indirect HGLC measures the hexadecane-water partition coefficient and relates it to γ_i^∞ . The solute GC peak area of the vapor phase in equilibrium with a dilute aqueous solution is measured, and then it is measured again after the addition of a known volume of hexadecane. The partition coefficient is determined from the ratio of GC peak areas and the volume of hexadecane which is then related to γ_i^∞ .	This method does not require a standard solution of the solute or calibration of the detector. However, it does require that the air-hexadecane partition coefficient be known. The solute must be volatile enough to be detected in the headspace of the water-hexadecane system.
Liquid-Liquid Chromatography (LLC) [66]	LLC is used to determine the γ_i^∞ ratio of a solute between a mobile and stationary phase which is a liquid on a solid support. It is necessary to determine γ_i^∞ of the solute in one of the phases using another technique in order to determine γ_i^∞ in the other phase.	This method is only accurate if the mutual solubilities of the stationary phase and mobile phase are negligible. The accuracy of this method is not as high as the GC techniques [43]. There have been reported adsorption problems for aqueous systems [48].
Gas stripping	See description in Table A.2.	
EXPSAT method [40]	Exponential saturator method. See description in Table A.2.	
Differential distillation [40] or transpiration method	A small amount of solute is distilled off (using an inert gas) of a dilution solution so that its composition is essentially unchanged. Concentrations of samples in the distillate and batch solution are determined.	This is recommended for relative volatilities $\ll 1$. Relatively large batches of solution are required, so this is a good method for water and inexpensive solvents.
Differential Ebulliometry [69]	Differential ebulliometry consists of measuring the difference in boiling points of a pure solvent and the solvent with dissolved solute [70]. The data can be related back to the γ_i^∞ through thermodynamics.	Steady state can be achieved fairly quickly [71]; however, steady state may not represent true equilibrium. This method requires characterization of the ebulliometer and accurate determination of the vaporization ratio. Ebulliometry yields accurate results for systems where the limiting relative volatility is near unity [43, 46]. Another recommended range is for relative volatilities between 0.3-20 [48]. The accuracy is not as good for high-boiling-point systems and highly viscous systems [43].

Table A.3: γ_i^∞ experimental methods (cont.)

Experiment	Description	Assumptions/Limits
Rayleigh distillation method [40, 46] (similar to differential distillation)	It is based on a single stage distillation. A dilute solution is prepared in a glass equilibrium cell. An inert gas is introduced into the solution at a flow rate that will ensure complete saturation. When enough solution has been distilled off, the cell is weighed, and the solution is analyzed using chromatography. The GC peak response and mass ratios are related to γ_i^∞ .	Dilute solutions must be used. This method can reach precision of 1%. A flow rate of about $0.2 \text{ cm}^3/\text{s}$ is required for aqueous solutions. A recommended range for relative volatilities is 1-70.
Differential static cell method [43, 72]	The differences in equilibrium vapor pressure are measured at constant temperature between a dilute solution and the pure solvent as a function of composition in the dilute solution. For solutes with similar volatility as water, liquid composition is determined gravimetrically. Since composition is not measured directly, a vaporization correction must be made for volatile solutes since they vaporize into the headspace.	This method has been applied to measure solubility without a compositional analysis. A correction must be made to measured data, but this method is better than others (such as ebulliometry) for solvents with poor boiling properties (such as water or polymer solutions) or for mixtures with high relative volatility.
Circulation Still [40, 73]	VLE circulation still operated at constant pressure used to provide samples of corresponding vapor and liquid compositions. No measurement of temperature needed as the boiling temp of the solution is indistinguishable from that of the neat solvent.	Recommended ranges are 30 K below the normal boiling temperature of the solvent and a relative volatility between 0.1-30. An uncertainty of 3-5% is possible.
Dew-point method [43, 72, 74]	The dew point method is similar to the ebulliometric method. The change in dew temperature of the vapor of a solvent is measured as a small amount of solute is added.	This method works well for systems with relative volatilities below about 0.3. GLC, GS, HGLC, ebulliometry, and the differential static methods all have difficulties measuring γ_i^∞ at volatilities this low. This method is typically limited to binary systems of low volatility.
Molecular Beams [75]	The vapors from each of two vapor-liquid equilibrium cells are collimated into molecular beams. The beams are alternately chopped before being detected by a mass spectrometer. The output signal from the mass spectrometer will be representative of the difference in partial pressure for any component in the two cells. Differences in pressure between a pure component and that containing a species at infinite dilution can be easily detected.	

Table A.4: KOW experimental methods

Experiment	Description	Assumptions/Limits
Shake flask method [45, 76, 77]	Solute is added to a flask with octanol/water and shaken. The concentrations are determined in the phases using an analytical method.	The recommend range is approximately - $2 \geq \log KOW \geq 4$. This method should typically not be used for polycyclic aromatic substances without polar substituents; halogenated hydrocarbons; or large, non-polar chemicals.
Generator column [78]	A generator column is packed with a support that is coated in the test chemical at a fixed n-octanol concentration. The solute is eluted from the column with water. The extractor column extracts solute from the aqueous phase, and it is then eluted and analyzed using HPLC or GC.	This can be used for $1 < \log KOW < 6$.
Slow stir method [45, 79]	This method is similar to the shake flask method. The main difference is that the octanol and water phases are slowly stirred instead of shaken while the system reaches equilibrium.	This method prevents the formation of emulsions. It can be used for $5 < \log KOW < 8.2$.
HPLC correlation (indirect) [3, 80, 81]	Chemicals injected into an HPLC column have various retention times. Reference values are used to develop a relationship between the property (x_i^{aq} or KOW) and retention time. This equation is used to find the property of other compounds based on retention time.	Reference substances are needed, so this is sometimes considered an estimation method. Recommended ranges are $0 < \log KOW < 6$ according to OPPTS and OECD and $0 < \log KOW < 8$ according to ASTM. This method should not be used for strong acids and bases, metal complexes, or compounds that react with either the eluent or column packing.

Appendix A References

1. Organisation for Economic Co-operation and Development, *OECD Guideline for the Testing of Chemicals: 105 Water Solubility*, 1995.
2. U.S. Environmental Protection Agency Office of Prevention Pesticides and Toxic Substances, *Product Properties Test Guidelines: OPPTS 830.7840 Water Solubility: Column Elution Method; Shake Flask Method*, 1996.
3. Organisation for Economic Co-operation and Development, *OECD Guidelines for the Testing of Chemicals: 117 Partition Coefficient (n-octanol/water), High Performance Liquid Chromatography (HPLC) Method*, 2004.
4. Chin, Y.P., W.J. Weber, Jr., and T.C. Voice, *Determination of partition coefficients and aqueous solubilities by reverse phase chromatography - II. Evaluation of partitioning and solubility models*. *Water Res.*, 1986. **20**(11): p. 1443-50.
5. Beneš, M. and V. Dohnal, *Limiting Activity Coefficients of Some Aromatic and Aliphatic Nitro Compounds in Water*. *J. Chem. Eng. Data*, 1999. **44**(5): p. 1097-1102.
6. Lo, J.M., C.L. Tseng, and J.Y. Yang, *Radiometric method for determining solubility of organic solvents in water*. *Anal. Chem.*, 1986. **58**(7): p. 1596-1597.
7. U.S. Environmental Protection Agency Office of Prevention Pesticides and Toxic Substances, *Product Properties Test Guidelines: OPPTS 830.7860 Water Solubility (Generator Column Method)*, 1996.
8. May, W.E., S.P. Wasik, and D.H. Freeman, *Determination of the solubility behavior of some polycyclic aromatic hydrocarbons in water*. *Anal. Chem.*, 1978. **50**(7): p. 997-1000.
9. May, W.E., S.P. Wasik, and D.H. Freeman, *Determination of the aqueous solubility of polynuclear aromatic hydrocarbons by a coupled column liquid chromatographic technique*. *Anal. Chem.*, 1978. **50**(1): p. 175-9.
10. DeVoe, H., M.M. Miller, and S.P. Wasik, *Generator columns and high pressure liquid chromatography for determining aqueous solubilities and octanol-water partition coefficients of hydrophobic substances*. *J. Res. Natl. Bur. Stand. (U. S.)*, 1981. **86**(4): p. 361-6.
11. Wasik, S.P., et al., *Octanol/water partition coefficients and aqueous solubilities of organic compounds*, 1981, *Natl. Bur. Stand.: Washington, D.C.* pp. 68.
12. Mackay, D. and W.Y. Shiu, *A Critical Review of Henry's Law Constants for Chemicals of Environmental Interest*. *J. Phys. Chem. Ref. Data*, 1981. **10**(4): p. 1175-1199.
13. ASTM, *E 1148 Standard Test Method for Measurements of Aqueous Solubility*, 1987 (1993), ASTM International: West Conshohocken, PA.
14. Hill, A.E., *Mutual solubility of liquids. I. Mutual solubility of ethyl ether and water. II. Solubility of water in benzene*. *J. Am. Chem. Soc.*, 1923. **45**: p. 1143-55.
15. Nelson, H.D. and C.L. de Ligny, *The determination of the solubilities of some n-alkanes in water at different temperatures, by means of gas chromatography*. *Recl. Trav. Chim. Pays-Bas*, 1968. **87**(5): p. 528-544.
16. Booth, H.S. and H.E. Everson, *Hydrotropic Solubilities: Solubilities in 40 Per Cent Sodium Xylenesulfonate*. *J. Ind. Eng. Chem.*, 1948. **40**(8): p. 1491-1493.
17. Bennett, G.M. and W.G. Philip, *Influence of Structure on the Solubilities of Ethers. Part I. Aliphatic Ethers*. *J. Chem. Soc.*, 1928: p. 1930-7.

18. Mackay, D., W.Y. Shiu, and A.W. Wolkoff, *Gas Chromatographic Determination of Low Concentrations of Hydrocarbons in Water by Vapor Phase Extraction*. ASTM Special Technical Publication, 1975. **573**(Water Quality Parameters Symposium): p. 251-8.
19. Etzweiler, F., E. Senn, and H.W.H. Schmidt, *Method for Measuring Aqueous Solubilities of Organic Compounds*. Anal. Chem., 1995. **67**: p. 655-658.
20. Kinoshita, K., H. Ishikawa, and K. Shinoda, *Solubility of Alcohols in Water Determined by the Surface Tension Measurements*. Bull. Chem. Soc. Jpn., 1958. **31**(9): p. 1081-1082.
21. Dohányosová, P., et al., *Measurement of Aqueous Solubility of Hydrophobic Volatile Organic Compounds by Solute Vapor Absorption Technique: Toluene, Ethylbenzene, Propylbenzene, and Butylbenzene at Temperatures from 273 K to 328 K*. J. Chem. Eng. Data, 2001. **46**(6): p. 1533-1539.
22. Franks, F., M. Gent, and H.H. Johnson, *The Solubility of Benzene in Water*. J. Chem. Soc., 1963: p. 2716-2723.
23. Christian, S.D., et al., *Control and measurement of water activity*. J. Chem. Educ., 1963. **40**(8): p. 419-21.
24. Gridin, V.V., et al., *Determination of Aqueous Solubility and Surface Adsorption of Polycyclic Aromatic Hydrocarbons by Laser Multiphoton Ionization*. Anal. Chem., 1998. **70**(13): p. 2685-2692.
25. Chapin, E.M. and J.M. Bell, *The Solubility of Oxalic Acid in Aqueous Solutions of Hydrochloric Acid*. J. Am. Chem. Soc., 1931. **53**: p. 3284-3287.
26. Smith, R.R., N.W. Charon, and W.J. Canady, *Thermodynamics of Solution of Aromatic Hydrocarbons in Water and in Water-Ethanol Solutions: Comparison of Some Methodologies*. J. Phys. Chem., 1989. **93**: p. 5938-5943.
27. Staudinger, J. and P.V. Roberts, *A critical review of Henry's law constants for environmental applications*. Crit. Rev. Environ. Sci. Technol., 1996. **26**(3): p. 205-297.
28. Leighton, D.T. and J.M. Calo, *Distribution coefficients of chlorinated hydrocarbons in dilute air-water systems for groundwater contamination applications*. J. Chem. Eng. Data, 1981. **26**(4): p. 382-385.
29. Alae, M., R.M. Whittall, and W.M.J. Strachan, *The effect of water temperature and composition on Henry's law constant for various PAH's*. Chemosphere, 1996. **32**(6): p. 1153-1164.
30. Mackay, D., W.Y. Shiu, and R.P. Sutherland, *Determination of air-water Henry's law constants for hydrophobic pollutants*. Environ. Sci. Technol., 1979. **13**(3): p. 333-337.
31. Böhme, A., et al., *Determination of Temperature-Dependent Henry's Law Constant of Four Oxygenated Solutes in Water Using Headspace Solid-Phase Microextraction Technique*. J. Chem. Eng. Data, 2008. **53**(12): p. 2873-2877.
32. Omur-Ozbek, P., *Determination of temperature-dependent constants of odorous contaminants and their application to human perception*. Environ. Sci. Technol., 2005. **39**: p. 3957-3963.
33. McAullife, C., *GC [gas-chromatographic] determination of solutes by multiple phase equilibration*. Chem. Technol., 1971(Jan.): p. 46-51.
34. Keeley, D.F., M.A. Hoffpauir, and J.R. Meriwether, *Solubility of aromatic hydrocarbons in water and sodium chloride solutions of different ionic strengths: benzene and toluene*. J. Chem. Eng. Data, 1988. **33**(2): p. 87-89.

35. Keeley, D.F., M.A. Hoffpauir, and J.R. Meriwether, *Solubility of aromatic hydrocarbons in water and sodium chloride solutions of different ionic strengths: C2-Substituted benzenes*. J. Chem. Eng. Data, 1991. **36**(4): p. 456-459.
36. Massaldi, H.A. and C.J. King, *Simple technique to determine solubilities of sparingly soluble organics. Solubility and activity coefficients of d-limonene, butylbenzene, and n-hexyl acetate in water and sucrose solutions*. J. Chem. Eng. Data, 1973. **18**(4): p. 393-7.
37. Gosset, J.M., et al., *Mass transfer coefficients and Henry's constants for packed-tower air stripping of volatile organics: measurements and correlation*, 1985, Dep. Civ. Eng., Cornell Univ., Ithaca, NY, USA. pp. 294.
38. Gossett, J.M., *Measurement of Henry's law constants for C1 and C2 chlorinated hydrocarbons*. Environ. Sci. Technol., 1987. **21**(2): p. 202-8.
39. Nielsen, F., E. Olsen, and A. Fredenslund, *Henry's Law Constants and Infinite Dilution Activity Coefficients for Volatile Organic Compounds in Water by a Validated Batch Air Stripping Method*. Environ. Sci. Technol., 1994. **28**(12): p. 2133-2138.
40. Dohnal, V., *Measurement of Limiting Activity Coefficients Using Analytical Tools*, in *Measurement of the Thermodynamic Properties of Multiple Phases*, R.D. Weir and T.W.d. Loos, Editors. 2005, Elsevier: Amsterdam. p. 359-381.
41. Robbins, G.A., S. Wang, and J.D. Stuart, *Using the static headspace method to determine Henry's law constants*. Anal. Chem., 1993. **65**(21): p. 3113-18.
42. Li, J., et al., *Measurement of large infinite dilution activity coefficients of nonelectrolytes in water by inert gas stripping and gas chromatography*. Anal. Chem., 1993. **65**(22): p. 3212-3218.
43. Kojima, K., S. Zhang, and T. Hiaki, *Measuring methods of infinite dilution activity coefficients and a database for systems including water*. Fluid Phase Equilib., 1997. **131**(1-2): p. 145-179.
44. Gossett, J.M. and A.H. Lincoff, *Solute-gas equilibriums in multi-organic aqueous systems*, 1981, Dep. Environ. Eng., Cornell Univ., Ithaca, NY, USA. pp. 37.
45. U.S. Environmental Protection Agency, *Chemistry Assistance Manual for Premanufacture Notification Submitters*, 1997.
46. Dohnal, V. and I. Horakova, *A new variant of the Rayleigh distillation method for the determination of limiting activity coefficients*. Fluid Phase Equilib., 1991. **68**: p. 173-85.
47. Shunthirasingham, C., Y.D. Lei, and F. Wania, *Evidence of Bias in Air–Water Henry's Law Constants for Semivolatile Organic Compounds Measured by Inert Gas Stripping*. Environ. Sci. Technol., 2007. **41**(11): p. 3807-3814.
48. Sherman, S.R., et al., *Compilation and Correlation of Limiting Activity Coefficients of Nonelectrolytes in Water*. Ind. Eng. Chem. Res., 1996. **35**(4): p. 1044-58.
49. Richon, D., P. Antoine, and H. Renon, *Infinite dilution activity coefficients of linear and branched alkanes from C1 to C9 in n-hexadecane by inert gas stripping*. Ind. Eng. Chem. Process Des. Dev., 1980. **19**(1): p. 144-7.
50. Kames, J., S. Schweighoefer, and U. Schurath, *Henry's Law constant and hydrolysis of peroxyacetyl nitrate (PAN)*. J. Atmos. Chem., 1991. **12**(2): p. 169-80.
51. Dohnal, V. and S. Hovorka, *Exponential Saturator: A Novel Gas-Liquid Partitioning Technique for Measurement of Large Limiting Activity Coefficients*. Ind. Eng. Chem. Res., 1999. **38**(5): p. 2036-2043.

52. Richon, D., *New equipment and new technique for measuring activity coefficients and Henry's constants at infinite dilution*. Rev. Sci. Instrum., 2011. **82**(2): p. 025108/1-025108/8.
53. Ruiz-Bevia, F. and M.J. Fernandez-Torres, *Determining the Henry's Law constants of THMs in seawater by means of purge-and-trap gas chromatography (PT-GC): the influence of seawater as sample matrix*. Anal. Sci., 2010. **26**(6): p. 723-6.
54. Dacey, J.W.H., S.G. Wakeham, and B.L. Howes, *Henry's law constants for dimethylsulfide in freshwater and seawater*. Geophys. Res. Lett., 1984. **11**(10): p. 991-994.
55. Poulain, L., et al., *In-cloud multiphase behaviour of acetone in the troposphere: gas uptake, Henry's law equilibrium and aqueous phase photooxidation*. Chemosphere, 2010. **81**(3): p. 312-20.
56. Johnson, B.J., E.A. Betterton, and D. Craig, *Henry's law coefficient of formic and acetic acids*. J. Atmos. Chem., 1996. **24**(2): p. 113-119.
57. Caires, J.P., M. Suzuki, and T.C.G. Kibbey, *Counterflow Method for Measurement of Henry's Law Constants*. J. Environ. Eng., 2003. **129**: p. 1169-1175.
58. Fendinger, N.J. and D.E. Glotfelty, *A laboratory method for the experimental determination of air-water Henry's law constants for several pesticides*. Environ. Sci. Technol., 1988. **22**(11): p. 1289-93.
59. Emmert, R.E. and R.L. Pigford, *Interfacial resistance. A study of gas absorption in falling liquid films*. Chem. Eng. Prog., 1954. **50**: p. 87-93.
60. Strekowski, R.S. and C. George, *Measurement of Henry's Law Constants for Acetone, 2-Butanone, 2,3-Butanedione, and Isobutyraldehyde Using a Horizontal Flow Reactor*. J. Chem. Eng. Data, 2005. **50**(3): p. 804-810.
61. Li, H., D. Ellis, and D. Mackay, *Measurement of low air-water partition coefficients of organic acids by evaporation from a water surface*. J. Chem. Eng. Data, 2007. **52**(5): p. 1580-1584.
62. Lee, H., H.-J. Kim, and J.-H. Kwon, *Determination of Henry's Law Constant Using Diffusion in Air and Water Boundary Layers*. J. Chem. Eng. Data, 2012. **57**(11): p. 3296-3302.
63. *Activity coefficients at infinite dilution*, ed. D. Tiegs. 1986, Frankfurt/Main : Great Neck, N.Y.: Frankfurt/Main : DECHEMA ; Great Neck, N.Y. : Distributed by Scholium International.
64. Orbey, H. and S.I. Sandler, *Relative measurements of activity coefficients at infinite dilution by gas chromatography*. Ind. Eng. Chem. Res., 1991. **30**(8): p. 2006-11.
65. Thomas, E.R., et al., *Limiting activity coefficients of nonpolar and polar solutes in both volatile and nonvolatile solvents by gas chromatography*. J. Chem. Eng. Data, 1982. **27**(4): p. 399-405.
66. Alessi, P. and I. Kikic, *Activity coefficients determination by liquid-liquid chromatography*. Gazz. Chim. Ital., 1974. **104**(7-8): p. 739-46.
67. Hussam, A. and P.W. Carr, *Rapid and precise method for the measurement of vapor/liquid equilibria by headspace gas chromatography*. Anal. Chem., 1985. **57**(4): p. 793-801.

68. Li, J. and P.W. Carr, *Measurement of water-hexadecane partition coefficients by headspace gas chromatography and calculation of limiting activity coefficients in water*. Anal. Chem., 1993. **65**(10): p. 1443-50.
69. Bernauer, M. and V. Dohnal, *Temperature Dependence of Air-Water Partitioning of N-Methylated (C1 and C2) Fatty Acid Amides*. J. Chem. Eng. Data, 2008. **53**(11): p. 2622-2631.
70. Moore, R.C., *Henry's Law Constant for Environmentally Significant Compounds*. Appl. Biochem. Biotechnol., 1992. **34-5**: p. 671-680.
71. Raal, J.D. and D. Ramjugernath, *Measurement of Limiting Activity Coefficients: Non-Analytical Tools*, in *Measurement of the Thermodynamic Properties of Multiple Phases*, R.D. Weir and T.W.d. Loos, Editors. 2005, Elsevier: Amsterdam. p. 339-357.
72. Wright, D.A., S.I. Sandler, and D. DeVoll, *Infinite dilution activity coefficients and solubilities of halogenated hydrocarbons in water at ambient temperatures*. Environ. Sci. Technol., 1992. **26**(9): p. 1828-1831.
73. Ondo, D. and V. Dohnal, *Temperature dependence of limiting activity coefficients and Henry's law constants of cyclic and open-chain ethers in water*. Fluid Phase Equilib., 2007. **262**(1-2): p. 121-136.
74. Trampe, D.B. and C.A. Eckert, *A dew point technique for limiting activity coefficients in nonionic solutions*. AIChE J., 1993. **39**(6): p. 1045-50.
75. Schmidt, T.W., *Determination of infinite dilution activity coefficients (γ_∞) using molecular beams*, 1980, Phillips Petroleum Co.: United States.
76. U.S. Environmental Protection Agency Office of Prevention Pesticides and Toxic Substances, *Product Properties Test Guidelines: OPPTS 830.7550 Partition Coefficient (n-Octanol/Water), Shake Flask Method*, 1996.
77. Organisation for Economic Co-operation and Development, *OECD Guidelines for the Testing of Chemicals: 107 Partition Coefficient (n-octanol/water): Shake Flask Method*, 1995.
78. U.S. Environmental Protection Agency Office of Prevention Pesticides and Toxic Substances, *Product Properties Test Guidelines: OPPTS 830.7560 Partition Coefficient (n-Octanol/Water), Generator Column Method*, 1996.
79. Organisation for Economic Co-operation and Development, *OECD Guidelines for the Testing of Chemicals: 123 Partition Coefficient (1-Octanol/Water): Slow-Stirring Method*, 2006.
80. U.S. Environmental Protection Agency Office of Prevention Pesticides and Toxic Substances, *Product Properties Test Guidelines: OPPTS 830.7570 Partition Coefficient (n-Octanol/Water), Estimation By Liquid Chromatography*, 1996.
81. Veith, G.D. and R.T. Morris, *A Rapid Method for Estimating Log P*, 1978: Duluth, MN.

Appendix B. Fugacity Error Propagation

The total error of a function can be determined by taking the partial derivatives of the function with respect to all of the variables. In general terms, y is a function of a, b, c , etc.,

$$y = f(a, b, c, \dots).$$

The variance in y , s_y , can be determined using the partial derivatives of the function with respect to all of the variables (a, b, c , etc.) and the variances of the variables (s_a, s_b, s_c , etc.)

$$s_y^2 = \left(\frac{\partial x}{\partial a} \right)^2 s_a^2 + \left(\frac{\partial x}{\partial b} \right)^2 s_b^2 + \left(\frac{\partial x}{\partial c} \right)^2 s_c^2 + \dots$$

The error associated with the fugacity ratio, Ψ , must be estimated in order to determine the uncertainty of γ_i^∞ . The natural log of equation 4.15 yields

$$\ln(\Psi_i) = \int_{T_{m_i}}^T \frac{\Delta H_{f,T_m} + \int_{T_{m_i}}^T C p_i^L dT - \int_{T_{m_i}}^T C p_i^S dT}{RT^2} dT.$$

Separating the terms yields:

$$\ln(\Psi_i) = \int_{T_{m_i}}^T \frac{\Delta H_{f,T_m}}{RT^2} dT + \int_{T_{m_i}}^T \frac{\int_{T_{m_i}}^T C p_i^L dT}{RT^2} dT - \int_{T_{m_i}}^T \frac{\int_{T_{m_i}}^T C p_i^S dT}{RT^2} dT.$$

In order to simplify the error estimation, it has been assumed that Cps are constant and that the

error in Tm_i is negligible (this was not assumed in the actual calculation). Integrating and combining terms yields:

$$\ln(\Psi_i) = \frac{-\Delta H_{f,Tm}}{R} \left(\frac{1}{T} - \frac{1}{Tm_i} \right) + \frac{Cp_i^L}{R} \left[\ln \left(\frac{T}{Tm_i} \right) + \frac{Tm_i}{T} - 1 \right] + \frac{Cp_i^S}{R} \left[-\ln \left(\frac{T}{Tm_i} \right) - \frac{Tm_i}{T} + 1 \right].$$

Due to the assumptions made, the error in $\ln(\Psi_i)$ is determined from:

$$s_{\ln \Psi} = \left(\frac{\partial \ln \Psi_i}{\partial \Delta H_{f,Tm}} \right)^2 s_{\Delta H_{f,Tm}}^2 + \left(\frac{\partial \ln \Psi_i}{\partial Cp_i^L} \right)^2 s_{Cp_i^L}^2 + \left(\frac{\partial \ln \Psi_i}{\partial Cp_i^S} \right)^2 s_{Cp_i^S}^2,$$

where

$$\frac{\partial \ln \Psi_i}{\partial \Delta H_{f,Tm}} = -\frac{1}{R} \left(\frac{1}{T} - \frac{1}{Tm_i} \right),$$

$$\frac{\partial \ln \Psi_i}{\partial Cp_i^L} = \frac{1}{R} \left[\ln \left(\frac{T}{Tm_i} \right) + \frac{Tm_i}{T} - 1 \right],$$

and

$$\frac{\partial \ln \Psi_i}{\partial Cp_i^S} = \frac{1}{R} \left[-\ln \left(\frac{T}{Tm_i} \right) - \frac{Tm_i}{T} + 1 \right].$$

The uncertainties for the individual properties were the uncertainties assigned in DIPPR[®] 801.

Appendix C. Summary of Recommended Values

The regression coefficients determined in this study are included in this appendix. Coefficients are not included for the API recommendations that were accepted in this study. Abbreviations used throughout the tables are explained in Table C.1.

Table C.1: Abbreviations used in regression tables

Abbreviation	Explanation
D	Data Type
E	Experimental
EP	Experimental and Predicted
P	Predicted
S	Smoothed
s	Uncertainty of regression
Tmin	Minimum temperature of regression (K)
Tmax	Maximum temperature of regression (K)
U	Unknown

The compounds within each table are sorted by chemical family. The DIPPR[®] equation number (Eq) is included for each set of coefficients (see Table 4.3). DIPPR[®] equation 100 was also used for “regressions” that are only valid at one temperature:

$$Y = A + BT + CT^2 + DT^3 + ET^4.$$

In these instances, only coefficient “A” was defined.

Table C.2: k_H regression summary

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
1-ALKENES											
1-PENTENE	109-67-1	100	1.867E+06	0	0	0	0	298.15	298.15	< 25%	S
1-HEXENE	592-41-6	100	2.190E+06	0	0	0	0	298.15	298.15	< 25%	S
1-HEPTENE	592-76-7	100	2.828E+06	0	0	0	0	298.15	298.15	< 10%	S
1-OCTENE	111-66-0	100	3.555E+06	0	0	0	0	298.15	298.15	< 10%	S
1-NONENE	124-11-8	100	4.854E+06	0	0	0	0	298.15	298.15	< 25%	S
1-DECENE	872-05-9	100	6.459E+06	0	0	0	0	298.15	298.15	< 25%	P
1-UNDECENE	872-05-9	100	8.669E+06	-1.003E+04	-3.126E+01	0	0	298.15	298.15	< 25%	P
1-DODECENE	872-05-9	100	1.108E+07	-5.042E+04	-2.173E+02	2.501E-01	1	298.15	298.15	< 25%	P
1-TRIDECENE	592-41-6	100	1.513E+07	-1.203E+04	-3.844E+01	0	0	298.15	298.15	< 50%	P
1-TETRADECENE	872-05-9	100	1.827E+07	0	0	0	0	298.15	298.15	< 50%	P
1-PENTADECENE	13360-61-7	100	2.493E+07	0	0	0	0	298.15	298.15	< 50%	S
2,3,4-ALKENES											
cis-2-PENTENE	627-20-3	100	1.262E+06	0	0	0	0	298.15	298.15	< 5%	E
trans-2-PENTENE	646-04-8	100	1.293E+06	0	0	0	0	298.15	298.15	< 5%	E
cis-2-HEPTENE	6443-92-1	100	2.341E+06	0	0	0	0	298.15	298.15	< 10%	E
trans-2-HEPTENE	14686-13-6	100	2.292E+06	0	0	0	0	298.15	298.15	< 10%	E
ACETATES											
METHYL ACETATE	79-20-9	101	4.579E+02	-2.509E+04	-6.446E+01	0	0	293.15	343.25	< 10%	E
ETHYL ACETATE	141-78-6	101	2.419E+02	-1.524E+04	-3.230E+01	0	0	288.15	353.15	< 10%	E
n-PROPYL ACETATE	109-60-4	101	2.912E+02	-1.778E+04	-3.939E+01	0	0	298	343.25	< 10%	E
n-BUTYL ACETATE	123-86-4	101	2.499E+02	-1.619E+04	-3.305E+01	0	0	273.15	363.65	< 10%	E
ISOBUTYL ACETATE	110-19-0	101	8.284E+01	-8.464E+03	-8.209E+00	0	0	273.15	323.15	< 25%	E
ISOPENTYL ACETATE	123-92-2	101	3.240E+02	-1.974E+04	-4.386E+01	0	0	273.15	363.85	< 10%	E

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
ISOPROPYL ACETATE	108-21-4	101	-6.279E+01	-2.050E+03	1.353E+01	0	0	273.15	310.2	< 25%	E
sec-BUTYL ACETATE	105-46-4	101	2.287E+02	-1.494E+04	-2.998E+01	0	0	273.15	363.65	< 10%	E
VINYL ACETATE	108-05-4	101	1.816E+02	-1.163E+04	-2.364E+01	0	0	273.35	333.15	< 3%	E
n-PENTYL ACETATE	628-63-7	101	4.256E+02	-2.473E+04	-5.879E+01	0	0	273.15	364.5	< 25%	E
n-HEXYL ACETATE	142-92-7	101	4.425E+02	-2.542E+04	-6.131E+01	0	0	273.15	363.35	< 25%	E
n-HEPTYL ACETATE	112-06-1	101	3.811E+03	1.191E+02	0	0	0	298.15	298.15	< 25%	P
n-OCTYL ACETATE	112-14-1	101	5.471E+03	-1.036E+04	-3.420E+01	0	0	298.15	298.15	< 25%	P
tert-BUTYL ACETATE	540-88-5	101	2.451E+02	-1.574E+04	-3.227E+01	0	0	273.15	353.65	< 10%	E
CYCLOHEXYL ACETATE	622-45-7	101	2.617E+02	-1.729E+04	-3.465E+01	0	0	273.15	363.45	< 10%	E
PHENYL ACETATE	122-79-2	101	2.503E+02	-1.713E+04	-3.319E+01	0	0	273.15	363.95	< 10%	E
ALDEHYDES											
ACETALDEHYDE	75-07-0	101	2.462E+01	-5.570E+03	0	0	0	273	318.15	< 10%	E
PROPANAL	123-38-6	101	2.821E+02	-1.701E+04	-3.844E+01	0	0	278.15	318.15	< 5%	E
BUTANAL	123-72-8	101	2.664E+01	-6.032E+03	0	0	0	278.15	318.15	< 10%	E
2-METHYLPROPANAL	78-84-2	100	1.093E+03	0	0	0	0	298.15	298.15	< 5%	S
PENTANAL	110-62-3	101	2.851E+01	-6.494E+03	0	0	0	278.15	318.15	< 10%	E
HEPTANAL	111-71-7	101	3.256E+01	-7.502E+03	0	0	0	283.15	318.15	< 25%	E
HEXANAL	66-25-1	101	2.848E+01	-6.372E+03	0	0	0	283.15	318.15	< 25%	E
OCTANAL	124-13-0	101	3.253E+01	-7.342E+03	0	0	0	283.15	318.15	< 10%	E
NONANAL	124-19-6	101	3.145E+01	-6.837E+03	0	0	0	283.15	318.15	< 25%	E
2-ETHYLHEXANAL	123-05-7	101	2.662E+01	-5.414E+03	0	0	0	273.15	363.15	< 25%	E
2-ETHYL-2-HEXENAL	645-62-5	101	2.647E+01	-5.815E+03	0	0	0	273.15	363.15	< 50%	E
DECANAL	112-31-2	101	3.897E+01	-8.912E+03	0	0	0	298.15	318.15	< 50%	E
2-METHYL-BUTYRALDEHYDE	96-17-3	101	2.641E+01	-5.574E+03	0	0	0	273.15	363.15	< 25%	P
ACROLEIN	107-02-8	101	2.385E+01	-5.131E+03	0	0	0	273	298	< 100%	E

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
trans-CROTONALDEHYDE	123-73-9	101	1.928E+02	-1.287E+04	-2.545E+01	0	0	273.15	363.15	< 25%	EP
METHACROLEIN	78-85-3	101	2.256E+01	-4.605E+03	0	0	0	273.15	333.15	< 10%	E
p-TOLUALDEHYDE	104-87-0	101	2.876E+01	-7.189E+03	0	0	0	278	298	< 25%	E
BENZALDEHYDE	100-52-7	101	5.844E+02	-3.018E+04	-8.394E+01	0	0	273.08	323.15	< 10%	E
m-TOLUALDEHYDE	620-23-5	101	2.911E+01	-7.166E+03	0	0	0	278	298	< 25%	E
ALIPHATIC ETHERS											
DIETHYL ETHER	60-29-7	101	2.854E+02	-1.727E+04	-3.841E+01	0	0	273.35	333.25	< 5%	E
DIISOPROPYL ETHER	108-20-3	101	-1.401E+01	-4.538E+03	6.784E+00	0	0	273.15	334.15	< 10%	E
DI-n-BUTYL ETHER	142-96-1	101	3.269E+01	-6.711E+03	0	0	0	273.15	363.65	< 25%	E
METHYL tert-BUTYL ETHER	1634-04-4	101	1.038E+02	-9.286E+03	-1.129E+01	0	0	273.15	343.15	< 10%	E
METHYL ISOBUTYL ETHER	625-44-5	101	3.182E+01	-6.669E+03	0	0	0	273.15	298.15	< 5%	E
DI-n-HEXYL ETHER	112-58-3	101	8.915E+04	-7.488E+03	-2.548E+01	0	0	298.15	298.15	< 25%	P
METHYL n-BUTYL ETHER	628-28-4	101	7.536E+02	-3.836E+04	-1.081E+02	0	0	273.15	298.15	< 25%	E
DI-n-PENTYL ETHER	693-65-2	101	5.151E+04	-5.574E+03	0	0	0	298.15	298.15	< 25%	P
METHYL tert-PENTYL ETHER	994-05-8	101	2.294E+02	-1.545E+04	-2.965E+01	0	0	273.15	352.35	< 10%	E
tert-BUTYL ETHYL ETHER	637-92-3	101	3.193E+01	-6.851E+03	0	0	0	278.15	318.15	< 25%	E
ETHYL tert-PENTYL ETHER	919-94-8	101	3.404E+01	-7.373E+03	0	0	0	288.15	318.15	< 25%	E
DI-n-PROPYL ETHER	111-43-3	101	4.575E+02	-2.541E+04	-6.361E+01	0	0	273.15	333.25	< 5%	E
n-BUTYL ETHYL ETHER	628-81-9	101	2.585E+02	-1.600E+04	-3.439E+01	0	0	273.15	363.85	< 25%	E
ALKYLCYCLOHEXANES											
METHYLCYCLOHEXANE	108-87-2	101	4.101E+02	-2.147E+04	-5.678E+01	0	0	293.15	443.95	< 25%	E
ETHYLCYCLOHEXANE	1678-91-7	101	1.847E+03	-6.190E+04	-3.046E+02	3.719E-01	1	273.15	443.95	< 10%	E
cis-1,2-DIMETHYLCYCLO- HEXANE	2207-01-4	101	3.907E+02	-2.062E+04	-5.388E+01	0	0	273.15	313.15	< 25%	E
trans-1,2-DIMETHYL-	6876-23-9	101	1.558E+03	-5.359E+04	-2.548E+02	2.970E-01	1	273.15	443.25	< 5%	E

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
CYCLOHEXANE											
ALKYLCYCLOPENTANES											
METHYL-CYCLOPENTANE	96-37-7	100	2.005E+06	7.831E+03	2.727E+01	6.321E-06	2	298.15	298.15	< 3%	S
ALKYNES											
ETHYLACETYLENE	107-00-6	100	9.853E+04	0	0	0	0	298.15	298.15	< 25%	S
1-PENTYNE	627-19-0	100	1.436E+05	0	0	0	0	298.15	298.15	< 25%	S
1-HEXYNE	693-02-7	100	2.093E+05	0	0	0	0	298.15	298.15	< 25%	S
1-OCTYNE	629-05-0	100	4.445E+05	0	0	0	0	298.15	298.15	< 25%	S
1-NONYNE	3452-09-3	100	6.477E+05	0	0	0	0	298.15	298.15	< 25%	S
1-HEPTYNE	628-71-7	100	3.050E+05	0	0	0	0	298.15	298.15	< 25%	S
AROMATIC ALCOHOLS											
2,3-XYLENOL	526-75-0	101	2.249E+01	-6.077E+03	0	0	0	349.05	371.65	< 25%	S
2,4-XYLENOL	105-67-9	101	2.115E+01	-5.479E+03	0	0	0	349.05	371.65	< 25%	E
2,5-XYLENOL	95-87-4	101	4.433E+02	-2.706E+04	-6.154E+01	0	0	278	371.65	< 10%	E
2,6-XYLENOL	576-26-1	101	1.892E+02	-1.364E+04	-2.455E+01	0	0	278	371.65	< 10%	E
3,4-XYLENOL	95-65-8	101	2.495E+01	-7.210E+03	0	0	0	293.15	371.65	< 25%	E
3,5-XYLENOL	108-68-9	101	1.997E+02	-1.533E+04	-2.580E+01	0	0	293.15	371.65	< 25%	E
PHENOL	108-95-2	101	5.283E+02	-3.105E+04	-7.428E+01	0	0	278	373.15	< 25%	E
o-CRESOL	95-48-7	101	3.028E+02	-1.968E+04	-4.119E+01	0	0	293.15	371.65	< 5%	E
m-CRESOL	108-39-4	101	7.124E+01	-8.354E+03	-7.310E+00	0	0	298.15	371.65	< 25%	E
p-CRESOL	106-44-5	101	1.167E+02	-1.074E+04	-1.390E+01	0	0	298	371.65	< 3%	E
2-PHENYLETHANOL	60-12-8	101	2.325E+02	-1.641E+04	-3.094E+01	0	0	283.15	373.15	< 10%	P
AROMATIC AMINES											
o-TOLUIDINE	95-53-4	101	2.612E+01	-7.057E+03	0	0	0	298.15	323.15	< 25%	E
m-TOLUIDINE	108-44-1	101	2.644E+01	-7.202E+03	0	0	0	298.15	323.15	< 25%	E
p-TOLUIDINE	106-49-0	101	1.765E+01	-4.381E+03	0	0	0	293.15	313.15	< 5%	P

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
N,N-DIETHYLANILINE	91-66-7	101	2.913E+01	-6.682E+03	0	0	0	273.15	333.15	< 25%	E
PYRIDINE	110-86-1	101	1.987E+01	-4.699E+03	0	0	0	298	374.15	< 25%	E
ANILINE	62-53-3	101	2.395E+02	-1.634E+04	-3.199E+01	0	0	298	374.15	< 10%	E
2-METHYLPYRIDINE	109-06-8	101	2.533E+01	-6.348E+03	0	0	0	298	313.15	< 5%	E
o-ETHYLANILINE	578-54-1	101	2.240E+02	-1.605E+04	-2.933E+01	0	0	283.15	363.15	< 10%	P
DIBENZOPYRROLE	86-74-8	101	1.387E+01	-4.292E+03	0	0	0	278.15	308.15	< 50%	E
2,6-DIMETHYL-PYRIDINE	108-48-5	101	2.870E+01	-7.334E+03	0	0	0	298	313.15	< 25%	E
3-METHYLPYRIDINE	108-99-6	101	2.516E+01	-6.371E+03	0	0	0	298	313.15	< 25%	E
4-METHYLPYRIDINE	108-89-4	101	2.527E+01	-6.484E+03	0	0	0	298	313.15	< 5%	E
2,4-DIMETHYLANILINE	95-68-1	101	3.271E+02	-2.102E+04	-4.451E+01	0	0	273.15	363.15	< 25%	P
N-ETHYLANILINE	103-69-5	101	2.866E+02	-1.928E+04	-3.821E+01	0	0	273.15	363.15	< 10%	E
AROMATIC CHLORIDES											
BENZYL CHLORIDE	100-44-7	101	2.274E+01	-4.447E+03	0	0	0	277.65	303.15	< 25%	E
MONOCHLOROBENZENE	108-90-7	101	3.469E+02	-1.808E+04	-4.849E+01	0	0	274.15	353.16	< 10%	E
o-DICHLOROBENZENE	95-50-1	101	8.041E+02	-3.979E+04	-1.161E+02	0	0	278.15	318.15	< 10%	E
m-DICHLOROBENZENE	541-73-1	101	2.467E+01	-4.435E+03	0	0	0	278.15	318.15	< 10%	E
p-DICHLOROBENZENE	106-46-7	101	5.888E+02	-3.085E+04	-8.350E+01	0	0	278.15	346.55	< 10%	E
HEXACHLOROBENZENE	118-74-1	101	3.108E+01	-6.570E+03	0	0	0	278.15	318.15	< 25%	E
o-CHLOROTOLUENE	95-49-8	101	2.530E+01	-4.528E+03	0	0	0	274.15	318.15	< 25%	E
p-CHLOROTOLUENE	106-43-4	101	2.621E+01	-4.854E+03	0	0	0	278.15	318.15	< 25%	E
2,4-DICHLOROTOLUENE	95-73-8	101	2.660E+01	-4.973E+03	0	0	0	278.15	323.1	< 10%	E
1,2,4-TRICHLOROBENZENE	120-82-1	101	3.085E+01	-6.372E+03	0	0	0	278.15	318.15	< 25%	E
1,2,3-TRICHLOROBENZENE	87-61-6	101	2.446E+01	-4.581E+03	0	0	0	277.15	323.15	< 25%	E
AROMATIC ESTERS											
BENZYL ACETATE	140-11-4	101	2.710E+01	-6.836E+03	0	0	0	273.15	322.95	< 25%	E

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
DI-n-PROPYL PHTHALATE	131-16-8	100	2.556E+00	0	0	0	0	298.15	298.15	< 25%	S
DI-n-HEXYL PHTHALATE	84-75-3	100	6.687E+00	0	0	0	0	298.15	298.15	< 50%	S
METHYL BENZOATE	93-58-3	101	2.728E+02	-1.738E+04	-3.673E+01	0	0	273.15	363.65	< 25%	E
ETHYL BENZOATE	93-89-0	101	1.896E+02	-1.354E+04	-2.429E+01	0	0	273.15	363.45	< 25%	E
BENZYL FORMATE	104-57-4	101	2.700E+01	-7.056E+03	0	0	0	273.15	322.85	< 25%	E
DIETHYL PHTHALATE	84-66-2	101	3.994E+02	-2.656E+04	-5.448E+01	0	0	278.15	374.15	< 10%	E
DI-n-BUTYL PHTHALATE	84-74-2	101	9.165E+02	-5.076E+04	-1.307E+02	0	0	283.15	323.15	< 25%	E
DIMETHYL PHTHALATE	131-11-3	101	2.739E+02	-1.899E+04	-3.688E+01	0	0	283.15	374.15	< 10%	EP
C, H, BR COMPOUNDS											
BROMOMETHANE	74-83-9	101	2.316E+01	-3.762E+03	0	0	0	273.15	305.15	< 25%	E
BROMOETHANE	74-96-4	101	2.373E+01	-3.906E+03	0	0	0	273.15	303.15	< 3%	E
1,1,2,2-TETRABROMOETHANE	79-27-6	101	1.252E+01	-1.268E+03	0	0	0	298.15	323.15	< 50%	E
1-BROMOPROPANE	106-94-5	101	2.624E+01	-4.582E+03	0	0	0	273.15	303.15	< 5%	E
2-BROMOPROPANE	75-26-3	101	3.871E+02	-2.007E+04	-5.419E+01	0	0	273.15	303.15	< 5%	E
1-BROMOBUTANE	109-65-9	100	7.447E+04	0	0	0	0	298.15	298.15	< 25%	S
1-BROMOHEPTANE	629-04-9	100	2.483E+05	0	0	0	0	298.15	298.15	< 25%	S
1,2-DIBROMOETHANE	106-93-4	101	2.246E+01	-4.218E+03	0	0	0	273.15	363.75	< 10%	E
m-DIBROMOBENZENE	108-36-1	100	4.750E+03	0	0	0	0	298.15	298.15	< 25%	E
BROMOBENZENE	108-86-1	101	4.526E+02	-2.357E+04	-6.390E+01	0	0	283.15	343.15	< 10%	E
TRIBROMOMETHANE	75-25-2	101	2.420E+01	-4.824E+03	0	0	0	283	313.15	< 10%	E
DIBROMOMETHANE	74-95-3	101	1.239E+02	-8.812E+03	-1.508E+01	0	0	273.15	363.25	< 5%	E
2-BROMOBUTANE	78-76-2	100	6.319E+04	0	0	0	0	298.15	298.15	< 10%	E
p-BROMOTOLUENE	106-38-7	101	2.572E+01	-4.794E+03	0	0	0	293.1	322.91	< 25%	E
C, H, F COMPOUNDS											
METHYL FLUORIDE	593-53-3	101	1.396E+02	-7.560E+03	-1.805E+01	0	0	273.21	353.4	< 3%	E

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
PERFLUORO-n-HEXANE	355-42-0	100	5.966E+09	1.272E+03	0	0	0	298.15	298.15	< 100%	E
PERFLUORO-n-OCTANE	307-34-6	100	6.880E+10	-2.945E+04	-9.548E+01	0	0	298.15	298.15	< 100%	E
1,2-DIFLUOROBENZENE	367-11-3	101	2.309E+01	-3.710E+03	0	0	0	298.15	322.89	< 25%	E
FLUOROBENZENE	462-06-6	101	1.836E+02	-1.116E+04	-2.381E+01	0	0	273.15	353.15	< 10%	E
HEXAFLUOROBENZENE	392-56-3	101	3.285E+02	-1.839E+04	-4.470E+01	0	0	281.46	339.76	< 3%	E
PERFLUORO-n-PENTANE	678-26-2	100	1.211E+09	5.338E+02	0	0	0	298.15	298.15	< 100%	E
PERFLUORO-n-HEPTANE	335-57-9	100	1.828E+10	-4.822E+03	0	0	0	298.15	298.15	< 100%	E
C, H, I COMPOUNDS											
METHYL IODIDE	74-88-4	101	3.635E+02	-1.838E+04	-5.118E+01	0	0	273.15	322.91	< 5%	E
ETHYL IODIDE	75-03-6	101	3.349E+02	-1.753E+04	-4.661E+01	0	0	273.15	303.15	< 3%	E
n-PROPYL IODIDE	107-08-4	101	2.604E+01	-4.523E+03	0	0	0	273.15	303.15	< 5%	E
ISOPROPYL IODIDE	75-30-9	101	2.673E+01	-4.664E+03	0	0	0	273.15	303.15	< 5%	E
IODOBENZENE	591-50-4	101	1.569E+00	2.185E+03	0	0	0	298	303.15	< 25%	E
n-HEXYL IODIDE	638-45-9	100	1.635E+05	0	0	0	0	298.15	298.15	< 25%	S
n-BUTYL IODIDE	542-69-8	100	8.057E+04	0	0	0	0	298.15	298.15	< 25%	S
C, H, MULTIHALOGEN COMPOUNDS											
DICHLORODIFLUORO-METHANE	75-71-8	101	2.848E+02	-1.478E+04	-3.876E+01	0	0	274	313.9	< 25%	E
TRICHLOROFLUORO-METHANE	75-69-4	101	2.471E+01	-3.438E+03	0	0	0	274	313.9	< 10%	E
1,1,2-TRICHLOROTRIFLUORO-ETHANE	76-13-1	101	2.823E+01	-4.110E+03	0	0	0	275.15	303.15	< 25%	E
C, H, NO2 COMPOUNDS											
NITROMETHANE	75-52-5	101	2.448E+02	-1.397E+04	-3.386E+01	0	0	273.15	363.15	< 10%	E
NITROETHANE	79-24-3	101	1.804E+02	-1.174E+04	-2.377E+01	0	0	273.15	363.25	< 5%	E
1-NITROPROPANE	108-03-2	101	2.032E+02	-1.313E+04	-2.688E+01	0	0	273.15	363.65	< 10%	E
2-NITROPROPANE	79-46-9	101	1.697E+02	-1.138E+04	-2.195E+01	0	0	292.85	363.35	< 3%	E

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
1-NITROBUTANE	627-05-4	100	7.245E+02	0	0	0	0	298.15	298.15	< 25%	S
o-NITROTOLUENE	88-72-2	101	2.493E+01	-5.806E+03	0	0	0	283.15	313.15	< 10%	E
p-NITROTOLUENE	99-99-0	101	2.804E+01	-7.129E+03	0	0	0	281	338	< 10%	E
m-NITROTOLUENE	99-08-1	101	2.866E+01	-6.962E+03	0	0	0	283.15	313.15	< 25%	E
NITROBENZENE	98-95-3	101	2.229E+02	-1.487E+04	-2.953E+01	0	0	283.15	372.85	< 5%	E
2,4-DINITROTOLUENE	121-14-2	101	2.605E+02	-1.830E+04	-3.489E+01	0	0	277.15	334.95	< 5%	E
2,6-DINITROTOLUENE	606-20-2	101	2.793E+01	-7.598E+03	0	0	0	278	314	< 5%	E
3,4-DINITROTOLUENE	610-39-9	101	3.267E+01	-9.586E+03	0	0	0	278.2	313.2	< 5%	E
2,4,6-TRINITROTOLUENE	118-96-7	101	1.351E+02	-1.270E+04	-1.661E+01	0	0	275.55	334.15	< 10%	E
C1/C2 ALIPHATIC CHLORIDES											
CARBON TETRACHLORIDE	56-23-5	101	3.746E+02	-1.968E+04	-5.206E+01	0	0	273.15	366.15	< 5%	E
METHYL CHLORIDE	74-87-3	101	1.775E+02	-9.830E+03	-2.345E+01	0	0	273.15	366.15	< 10%	E
ETHYL CHLORIDE	75-00-3	101	1.936E+02	-1.056E+04	-2.581E+01	0	0	281.15	366.15	< 25%	E
VINYL CHLORIDE	75-01-4	101	3.305E+01	-3.487E+03	-1.681E+00	0	0	273.35	366.15	< 25%	E
DICHLOROMETHANE	75-09-2	101	1.263E+02	-7.983E+03	-1.578E+01	0	0	273.15	366.15	< 10%	E
CHLOROFORM	67-66-3	101	2.287E+02	-1.327E+04	-3.056E+01	0	0	273.15	366.15	< 3%	E
1,1-DICHLOROETHANE	75-34-3	101	2.587E+02	-1.445E+04	-3.508E+01	0	0	273.15	366.15	< 5%	E
1,2-DICHLOROETHANE	107-06-2	101	2.782E+02	-1.560E+04	-3.810E+01	0	0	273.15	366.15	< 10%	E
1,1,2-TRICHLOROETHANE	79-00-5	101	1.159E+02	-8.493E+03	-1.384E+01	0	0	273.15	363.95	< 5%	E
HEXACHLOROETHANE	67-72-1	101	3.026E+01	-6.068E+03	0	0	0	283.15	303.15	< 25%	E
1,1,1-TRICHLOROETHANE	71-55-6	101	7.812E+01	-6.437E+03	-7.910E+00	0	0	273.15	348.15	< 10%	E
1,1,1,2- TETRACHLOROETHANE	630-20-6	101	2.594E+01	-4.909E+03	0	0	0	273.15	323.15	< 10%	E
1,1,2,2- TETRACHLOROETHANE	79-34-5	101	2.448E+01	-4.984E+03	0	0	0	283.15	328.65	< 10%	E
TRICHLOROETHYLENE	79-01-6	101	2.642E+02	-1.488E+04	-3.570E+01	0	0	274.22	366.15	< 5%	E

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
TETRACHLORO-ETHYLENE	127-18-4	101	3.077E+02	-1.717E+04	-4.188E+01	0	0	273.15	364.95	< 10%	E
cis-1,2-DICHLOROETHYLENE	156-59-2	101	2.228E+02	-1.290E+04	-2.975E+01	0	0	281.15	366.15	< 10%	E
trans-1,2-DICHLOROETHYLENE	156-60-5	101	2.410E+01	-3.940E+03	0	0	0	283.15	323.15	< 10%	E
PENTACHLOROETHANE	76-01-7	101	3.521E+01	-7.667E+03	0	0	0	293.15	298	< 25%	E
1,1-DICHLOROETHYLENE	75-35-4	101	3.060E+02	-1.643E+04	-4.191E+01	0	0	275.15	363.65	< 10%	E
C3 & HIGHER ALIPHATIC CHLORIDES											
1,4-DICHLOROBUTANE	110-56-5	101	2.001E+01	-3.599E+03	0	0	0	286.65	298.05	< 10%	E
1,5-DICHLOROPENTANE	628-76-2	101	2.216E+01	-4.045E+03	0	0	0	274.15	295.85	< 50%	E
1,2-DICHLOROPROPANE	78-87-5	101	2.546E+02	-1.434E+04	-3.454E+01	0	0	273.15	353.35	< 10%	E
ISOPROPYL CHLORIDE	75-29-6	101	2.621E+01	-4.393E+03	0	0	0	273.15	303.15	< 10%	E
1,2,3-TRICHLOROPROPANE	96-18-4	101	1.366E+01	-1.857E+03	0	0	0	275.05	318.15	< 25%	E
HEXACHLORO-1,3-BUTADIENE	87-68-3	101	2.978E+01	-5.451E+03	0	0	0	275.15	298.15	< 50%	E
PROPYL CHLORIDE	540-54-5	101	2.625E+01	-4.489E+03	0	0	0	273.15	310.15	< 10%	E
n-BUTYL CHLORIDE	109-69-3	101	2.644E+01	-4.455E+03	0	0	0	273.35	310.15	< 10%	E
sec-BUTYL CHLORIDE	78-86-4	101	2.691E+01	-4.502E+03	0	0	0	275.05	298.05	< 10%	E
1-CHLOROPENTANE	543-59-9	101	4.078E+02	-2.100E+04	-5.714E+01	0	0	275.65	310.15	< 5%	E
1,3-DICHLOROPROPANE	142-28-9	101	1.898E+01	-3.134E+03	0	0	0	275.05	303.15	< 25%	E
CYCLOALIPHATIC ALCOHOLS											
CYCLOHEXANOL	108-93-0	101	2.835E+01	-7.692E+03	0	0	0	298	333.15	< 25%	E
CYCLOALKANES											
CYCLOPENTANE	287-92-3	101	2.262E+02	-1.246E+04	-2.995E+01	0	0	298	426.25	< 10%	E
CYCLOHEXANE	110-82-7	101	-1.486E+03	3.489E+04	2.667E+02	-4.584E-01	1	274.25	424.15	< 25%	E
CYCLOHEPTANE	291-64-5	100	8.267E+05	-3.094E+04	-7.154E+01	0	0	298.15	298.15	< 10%	P

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
CYCLOOCTANE	292-64-8	101	1.860E+02	-1.172E+04	-2.337E+01	0	0	273.15	313.15	< 10%	E
CYCLOALKENES											
CYCLOPENTENE	142-29-0	101	2.369E+01	-3.321E+03	0	0	0	283.15	298.15	< 50%	E
CYCLOHEXENE	110-83-8	101	2.423E+01	-3.602E+03	0	0	0	278.26	318.36	< 25%	E
CYCLOHEPTENE	628-92-2	100	2.705E+05	-8.903E+03	0	1.478E+06	-2	298.15	298.15	< 25%	E
CYCLOOCTENE	931-88-4	101	1.856E+02	-1.135E+04	-2.370E+01	0	0	273.15	313.15	< 5%	E
DIALKENES											
1,4-CYCLOHEXADIENE	628-41-1	101	2.424E+01	-4047.17	0	0	0	278.26	318.36	< 25%	E
DIMETHYLALKANES											
2,2-DIMETHYLBUTANE	75-83-2	100	9.781E+06	0	0	0	0	298.15	298.15	< 25%	S
2,3-DIMETHYLBUTANE	79-29-8	101	3.898E+02	-2.037E+04	-5.366E+01	0	0	273.15	422.65	< 25%	E
2,4-DIMETHYLPENTANE	108-08-7	100	1.937E+07	0	0	0	0	298.15	298.15	< 50%	S
3,3-DIMETHYLPENTANE	562-49-2	101	-1.836E+03	4.668E+04	3.255E+02	-5.334E-01	1	298.15	423.64	< 25%	E
2,2-DIMETHYLHEXANE	590-73-8	101	6.831E+02	-3.353E+04	-9.718E+01	0	0	273.15	313.15	< 5%	E
2,5-DIMETHYLHEXANE	592-13-2	101	8.022E+02	-3.833E+04	-1.153E+02	0	0	273.15	313.15	< 3%	E
DIPHENYL/POLYAROMATICS											
BIPHENYL	92-52-4	101	6.933E+02	-3.653E+04	-9.889E+01	0	0	293.15	337.65	< 10%	E
EPOXIDES											
1,4-DIOXANE	123-91-1	101	1.235E+02	-1.032E+04	-1.500E+01	0	0	273.35	374.15	< 25%	E
TETRAHYDROPYRAN	142-68-7	101	2.170E+02	-1.432E+04	-2.851E+01	0	0	273.35	328.15	< 5%	E
TETRAHYDROFURAN	109-99-9	101	2.492E+01	-5.663E+03	0	0	0	273.35	328.15	< 10%	E
DIBENZOFURAN	132-64-9	101	2.617E+01	-5.831E+03	0	0	0	278.15	318.15	< 25%	E
FORMATES											
METHYL FORMATE	107-31-3	101	1.995E+01	-3.806E+03	0	0	0	285.65	333.25	< 10%	E
ETHYL FORMATE	109-94-4	101	2.076E+01	-3.974E+03	0	0	0	293.15	333.25	< 25%	E
n-PROPYL FORMATE	110-74-7	101	2.356E+01	-4.788E+03	0	0	0	272.15	333.15	< 10%	E

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
n-BUTYL FORMATE	592-84-7	101	2.007E+01	-3.697E+03	0	0	0	273.15	363.75	< 25%	E
ISOBUTYL FORMATE	542-55-2	101	2.564E+01	-5.223E+03	0	0	0	273.15	333.55	< 10%	E
n-PENTYL FORMATE	638-49-3	101	2.720E+02	-1.682E+04	-3.642E+01	0	0	273.15	363.65	< 25%	E
tert-BUTYL FORMATE	762-75-4	101	2.041E+01	-3.621E+03	0	0	0	278.15	308.15	< 5%	E
KETONES											
ACETONE	67-64-1	101	1.327E+02	-1.016E+04	-1.639E+01	0	0	273	373.15	< 10%	E
METHYL ETHYL KETONE	78-93-3	101	2.253E+02	-1.451E+04	-3.000E+01	0	0	273	373.15	< 10%	E
3-PENTANONE	96-22-0	101	2.558E+02	-1.603E+04	-3.433E+01	0	0	273.15	310.2	< 10%	E
METHYL ISOBUTYL KETONE	108-10-1	101	2.306E+02	-1.522E+04	-3.034E+01	0	0	273.15	363.55	< 10%	E
3-HEPTANONE	106-35-4	101	2.236E+02	-1.559E+04	-2.890E+01	0	0	273.15	363.25	< 10%	E
4-HEPTANONE	123-19-3	101	2.766E+02	-1.775E+04	-3.687E+01	0	0	273.15	363.35	< 10%	E
3-HEXANONE	589-38-8	101	3.027E+01	-7.070E+03	0	0	0	293.15	310.2	< 10%	E
2-PENTANONE	107-87-9	101	2.179E+02	-1.450E+04	-2.865E+01	0	0	273.15	363.65	< 10%	E
METHYL ISOPROPYL KETONE	563-80-4	101	2.737E+02	-1.686E+04	-3.702E+01	0	0	265.15	453.15	< 10%	P
2-HEXANONE	591-78-6	101	1.536E+02	-1.206E+04	-1.876E+01	0	0	273.15	312.75	< 10%	E
2-HEPTANONE	110-43-0	101	2.878E+02	-1.827E+04	-3.858E+01	0	0	273.15	363.65	< 5%	E
5-METHYL-2-HEXANONE	110-12-3	101	3.052E+02	-1.904E+04	-4.117E+01	0	0	273.15	362.85	< 5%	E
3,3-DIMETHYL-2-BUTANONE	75-97-8	101	1.694E+02	-1.217E+04	-2.131E+01	0	0	273.15	363.35	< 5%	E
DIISOBUTYL KETONE	108-83-8	101	3.516E+02	-2.165E+04	-4.750E+01	0	0	273.15	363.85	< 10%	E
DIISOPROPYL KETONE	565-80-0	101	1.603E+02	-1.197E+04	-1.972E+01	0	0	273.15	363.45	< 10%	E
5-NONANONE	502-56-7	101	2.901E+02	-1.938E+04	-3.824E+01	0	0	273.15	363.75	< 25%	E
2-NONANONE	821-55-6	101	3.154E+02	-2.001E+04	-4.227E+01	0	0	273.15	364.35	< 25%	E
ACETYLACETONE	123-54-6	101	1.685E+01	-3.780E+03	0	0	0	303.15	373.15	< 25%	E
CYCLOPENTANONE	120-92-3	101	2.341E+01	-5.848E+03	0	0	0	293.2	353.15	< 100%	E

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
CYCLOHEXANONE	108-94-1	101	2.494E+01	-6.365E+03	0	0	0	298	353.15	< 10%	E
2-OCTANONE	111-13-7	101	3.274E+02	-2.068E+04	-4.406E+01	0	0	273.15	364.15	< 25%	E
ACETOPHENONE	98-86-2	101	4.071E+02	-2.464E+04	-5.626E+01	0	0	278	318.15	< 25%	E
3-OCTANONE	106-68-3	101	1.965E+02	-1.427E+04	-2.483E+01	0	0	273.15	364.15	< 10%	E
3-NONANONE	925-78-0	101	2.908E+01	-6.587E+03	0	0	0	273.15	353.35	< 25%	E
2,6,8-TRIMETHYL-4-NONANONE	123-18-2	101	5.209E+03	-4.277E+03	-9.471E+00	0	0	298.15	298.15	< 25%	P
MERCAPTANS											
ISOPROPYL MERCAPTAN	75-33-2	101	2.295E+01	-3.670E+03	0	0	0	288.15	318.15	< 10%	E
n-BUTYL MERCAPTAN	109-79-5	101	4.008E+02	-2.101E+04	-5.613E+01	0	0	288.15	333.15	< 10%	E
METHYLALKANES											
2-METHYLPENTANE	107-83-5	101	4.013E+02	-2.096E+04	-5.529E+01	0	0	273.15	422.65	< 25%	E
3-METHYLPENTANE	96-14-0	100	9.326E+06	1.259E+04	5.099E+01	-4.402E-02	1	298.15	298.15	< 3%	S
2-METHYLHEXANE	591-76-4	100	1.920E+07	0	0	0	0	298.15	298.15	< 10%	E
3-METHYLHEXANE	589-34-4	100	1.716E+07	0	0	0	0	298.15	298.15	< 10%	E
3-METHYLHEPTANE	589-81-1	100	2.094E+07	0	0	0	0	298.15	298.15	< 10%	E
4-METHYLHEPTANE	589-53-7	100	2.066E+07	0	0	0	0	298.15	298.15	< 5%	E
METHYLALKENES											
2-METHYL-2-BUTENE	513-35-9	101	1.486E+02	-8.954E+03	-1.839E+01	0	0	273.16	322.88	< 3%	E
N-ALCOHOLS											
METHANOL	67-56-1	101	1.040E+02	-9.075E+03	-1.233E+01	0	0	273	373.15	< 10%	E
ETHANOL	64-17-5	101	1.745E+02	-1.301E+04	-2.237E+01	0	0	273	373.15	< 10%	E
1-PROPANOL	71-23-8	101	2.216E+02	-1.567E+04	-2.904E+01	0	0	273	373.15	< 10%	E
1-BUTANOL	71-36-3	101	3.189E+02	-2.039E+04	-4.329E+01	0	0	273	373.15	< 10%	E
1-PENTANOL	71-41-0	101	2.743E+02	-1.835E+04	-3.661E+01	0	0	273.35	371.15	< 10%	E
1-HEXANOL	111-27-3	101	2.987E+02	-1.978E+04	-4.003E+01	0	0	298	363.15	< 25%	E

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
1-HEPTANOL	111-70-6	101	3.399E+02	-2.193E+04	-4.593E+01	0	0	273	375	< 10%	EP
1-OCTANOL	111-87-5	101	4.751E+02	-2.858E+04	-6.567E+01	0	0	288.15	363.45	< 10%	EP
1-NONANOL	143-08-8	101	4.823E+02	-2.911E+04	-6.658E+01	0	0	288.15	363.15	< 25%	E
1-DECANOL	112-30-1	101	5.385E+02	-3.155E+04	-7.495E+01	0	0	288.15	363.15	< 10%	E
1-UNDECANOL	112-42-5	100	4.030E+02	0	0	0	0	298.15	298.15	< 50%	P
1-DODECANOL	112-53-8	100	4.668E+02	0	0	0	0	298.15	298.15	< 50%	P
N-ALIPHATIC PRIMARY AMINES											
ETHYLAMINE	75-04-7	100	6.618E+01	1.162E+04	2.974E+01	0	0	298.15	298.15	< 10%	S
n-PROPYLAMINE	107-10-8	100	8.255E+01	1.063E+04	2.880E+01	0	0	298.15	298.15	< 10%	S
n-BUTYLAMINE	109-73-9	100	1.030E+02	1.277E+04	3.289E+01	0	0	298.15	298.15	< 10%	S
n-PENTYLAMINE	110-58-7	100	1.284E+02	8.883E+03	2.125E+01	0	0	298.15	298.15	< 10%	S
n-HEXYLAMINE	111-26-2	100	1.602E+02	0	0	0	0	298.15	298.15	< 10%	S
n-HEPTYLAMINE	111-68-2	100	1.999E+02	0	0	0	0	298.15	298.15	< 25%	S
n-OCTYLAMINE	111-86-4	100	2.493E+02	0	0	0	0	298.15	298.15	< 25%	S
N-ALKANES											
n-PENTANE	109-66-0	101	-1.239E+03	2.706E+04	2.257E+02	-4.087E-01	1	273.15	422.65	< 25%	E
n-HEXANE	110-54-3	101	4.056E+02	-2.109E+04	-5.595E+01	0	0	273.15	443.15	< 10%	E
n-HEPTANE	142-82-5	101	4.213E+02	-2.189E+04	-5.819E+01	0	0	273.15	443.75	< 25%	E
n-OCTANE	111-65-9	101	5.052E+02	-2.622E+04	-7.032E+01	0	0	273.15	456.15	< 25%	E
n-NONANE	111-84-2	101	7.275E+02	-3.721E+04	-1.028E+02	0	0	287.95	409.75	< 50%	E
n-DECANE	124-18-5	100	4.165E+07	0	0	0	0	298.15	298.15	< 50%	S
N-ALKYLBENZENES											
BENZENE	71-43-2	101	6.091E+02	-2.377E+04	-9.619E+01	9.737E-02	1	273.15	527.15	< 3%	E
TOLUENE	108-88-3	101	6.605E+02	-2.545E+04	-1.049E+02	1.101E-01	1	261.7	473.55	< 5%	E
ETHYLBENZENE	100-41-4	101	3.619E+02	-1.988E+04	-4.994E+01	0	0	273.15	373.15	< 5%	E
n-PROPYLBENZENE	103-65-1	101	2.463E+02	-1.450E+04	-3.276E+01	0	0	273.65	328.15	< 10%	E

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
n-BUTYLBENZENE	104-51-8	101	4.132E+02	-2.226E+04	-5.744E+01	0	0	273.65	373.15	< 5%	E
n-HEPTYLBENZENE	1078-71-3	100	2.079E+05	-1.779E+04	-5.443E+01	0	0	298.15	298.15	< 10%	P
n-DECYLBENZENE	104-72-3	100	7.285E+05	0	0	0	0	298.15	298.15	< 10%	S
n-PENTYLBENZENE	538-68-1	101	4.005E+02	-2.240E+04	-5.510E+01	0	0	280.15	318.15	< 10%	E
n-HEXYLBENZENE	1077-16-3	101	3.675E+02	-2.136E+04	-4.985E+01	0	0	278.15	318.15	< 10%	E
n-OCTYLBENZENE	2189-60-8	100	3.005E+05	5.570E+03	0	0	0	298.15	298.15	< 10%	P
n-NONYLBENZENE	1081-77-2	100	3.866E+05	-4.112E+04	-2.080E+02	2.466E-01	1	298.15	298.15	< 25%	P
NAPHTHALENES											
NAPHTHALENE	91-20-3	101	3.592E+02	-2.027E+04	-4.973E+01	0	0	275.05	346.55	< 3%	E
1-METHYL-NAPHTHALENE	90-12-0	101	2.722E+01	-5.758E+03	0	0	0	277.25	304.85	< 10%	E
2-METHYL-NAPHTHALENE	91-57-6	101	2.506E+01	-5.113E+03	0	0	0	277.25	304.15	< 25%	E
1-ETHYL-NAPHTHALENE	1127-76-0	101	2.778E+01	-5.919E+03	0	0	0	281.55	304.85	< 10%	E
NITRILES											
ACETONITRILE	75-05-8	101	-2.253E+00	-3.365E+03	3.205E+00	0	0	273	353.15	< 5%	E
PROPIONITRILE	107-12-0	101	9.456E+01	-7.820E+03	-1.110E+01	0	0	278.15	298.15	< 25%	E
ACRYLONITRILE	107-13-1	101	1.926E+01	-3.863E+03	0	0	0	273.15	343.15	< 3%	E
METHACRYLONITRILE	126-98-7	101	2.729E+02	-1.537E+04	-3.756E+01	0	0	273.15	353.15	< 10%	P
n-BUTYRONITRILE	109-74-0	101	1.333E+02	-1.001E+04	-1.650E+01	0	0	273.15	313.15	< 3%	E
VALERONITRILE	110-59-8	101	2.630E+01	-6.092E+03	0	0	0	273.15	303.15	< 25%	E
ISOBUTYRONITRILE	78-82-0	101	2.382E+01	-5.246E+03	0	0	0	273.15	303.15	< 25%	E
BENZONITRILE	100-47-0	101	3.097E+02	-1.834E+04	-4.267E+01	0	0	273.15	363.15	< 25%	E
VINYLACETONITRILE	109-75-1	101	2.209E+01	-4.916E+03	0	0	0	273.15	303.15	< 5%	E
PHENYLACETONITRILE	140-29-4	101	2.665E+01	-6.778E+03	0	0	0	273.15	303.15	< 100%	E
NITROAMINES											
o-NITROANILINE	88-74-4	101	2.256E+01	-6.802E+03	0	0	0	298.15	313.25	< 50%	E

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
p-NITROANILINE	100-01-6	101	2.228E+01	-8.150E+03	0	0	0	293.15	313.25	< 10%	E
ORGANIC SALTS											
DIETHYL CARBONATE	105-58-8	101	3.022E+02	-1.859E+04	-4.099E+01	0	0	273.15	363.45	< 5%	E
DIMETHYL CARBONATE	616-38-6	101	2.193E+02	-1.373E+04	-2.937E+01	0	0	273.35	333.15	< 3%	E
OTHER ALIPHATIC ALCOHOLS											
ISOPROPANOL	67-63-0	101	2.321E+02	-1.607E+04	-3.060E+01	0	0	273	373.15	< 10%	E
2-METHYL-1-PROPANOL	78-83-1	101	3.840E+02	-2.321E+04	-5.299E+01	0	0	273.35	374.15	< 10%	E
2-BUTANOL	78-92-2	101	2.945E+02	-1.939E+04	-3.956E+01	0	0	273	371.15	< 5%	E
2-METHYL-2-PROPANOL	75-65-0	101	2.378E+02	-1.682E+04	-3.109E+01	0	0	273	371.15	< 10%	E
2-PENTANOL	6032-29-7	101	3.240E+02	-2.093E+04	-4.377E+01	0	0	273.35	371.15	< 5%	E
2-METHYL-2-BUTANOL	75-85-4	101	2.800E+02	-1.898E+04	-3.719E+01	0	0	273.35	371.45	< 5%	E
2-METHYL-1-BUTANOL	137-32-6	101	3.671E+02	-2.288E+04	-5.021E+01	0	0	273.35	363.15	< 5%	E
2,2-DIMETHYL-1-PROPANOL	75-84-3	101	3.360E+02	-2.136E+04	-4.548E+01	0	0	285.15	363.15	< 25%	P
2-HEXANOL	626-93-7	101	3.326E+02	-2.148E+04	-4.486E+01	0	0	273.15	363.35	< 25%	EP
3-HEXANOL	623-37-0	101	3.385E+02	-2.197E+04	-4.560E+01	0	0	273.15	363.45	< 5%	EP
3-METHYL-1-PENTANOL	589-35-5	101	4.097E+02	-2.540E+04	-5.620E+01	0	0	273.15	363.65	< 10%	P
3-METHYL-3-PENTANOL	77-74-7	101	1.770E+02	-1.383E+04	-2.197E+01	0	0	282.95	363.55	< 10%	P
3-PENTANOL	584-02-1	101	3.054E+02	-1.989E+04	-4.110E+01	0	0	273.35	371.15	< 5%	E
3-METHYL-1-BUTANOL	123-51-3	101	4.590E+02	-2.701E+04	-6.392E+01	0	0	273.35	374.15	< 10%	E
3-METHYL-2-BUTANOL	598-75-4	101	3.121E+02	-2.017E+04	-4.208E+01	0	0	273.35	371.15	< 3%	E
2-HEPTANOL	543-49-7	101	3.782E+02	-2.400E+04	-5.135E+01	0	0	273.15	363.35	< 5%	P
3-HEPTANOL	589-82-2	101	3.659E+02	-2.376E+04	-4.933E+01	0	0	273.15	363.35	< 10%	E
4-METHYL-2-PENTANOL	108-11-2	101	2.657E+02	-1.777E+04	-3.520E+01	0	0	273.15	363.35	< 10%	EP
2-OCTANOL	123-96-6	101	5.167E+02	-3.094E+04	-7.154E+01	0	0	273.15	363.65	< 10%	P
2-NONANOL	628-99-9	101	5.400E+02	-3.288E+04	-7.444E+01	0	0	273.15	363.65	< 25%	P

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
2,6-DIMETHYL-4-HEPTANOL	108-82-7	101	2.593E+02	-1.915E+04	-3.313E+01	0	0	273.15	363.65	< 25%	P
OTHER ALIPHATIC AMINES											
DIMETHYLAMINE	124-40-3	100	8.902E+01	0	0	0	0	298.15	298.15	< 50%	S
TRIETHYLAMINE	121-44-8	100	5.961E+02	0	0	0	0	298.15	298.15	< 5%	S
DI-n-PROPYLAMINE	142-84-7	100	2.576E+02	0	0	0	0	298.15	298.15	< 25%	S
DIETHYLAMINE	109-89-7	100	1.514E+02	-1.652E+04	-4.148E+01	0	0	298.15	298.15	< 25%	S
DI-n-BUTYLAMINE	111-92-2	100	4.383E+02	0	0	0	0	298.15	298.15	< 50%	S
TRI-n-BUTYLAMINE	102-82-9	101	3.425E+01	-8.719E+03	0	0	0	273.15	353.15	< 25%	E
OTHER ALKANES											
2,3,4-TRIMETHYLPENTANE	565-75-3	100	1.701E+07	-1.047E+04	-3.146E+01	0	0	298.15	298.15	< 5%	P
2,2,5-TRIMETHYLHEXANE	3522-94-9	101	3.797E+01	-6.194E+03	0	0	0	273.15	298.15	< 10%	E
OTHER ALKYL BENZENES											
o-XYLENE	95-47-6	101	3.284E+02	-1.815E+04	-4.516E+01	0	0	273.15	353.15	< 10%	E
m-XYLENE	108-38-3	101	4.853E+02	-2.121E+04	-7.387E+01	5.787E-02	1	273.15	543.75	< 10%	E
p-XYLENE	106-42-3	101	8.256E+02	-3.283E+04	-1.302E+02	1.238E-01	1	273.15	568.05	< 10%	E
CUMENE	98-82-8	101	2.709E+01	-4.790E+03	0	0	0	288.15	318.15	< 10%	E
1,2,3-TRIMETHYLBENZENE	526-73-8	101	2.534E+01	-4.560E+03	0	0	0	288.15	318.15	< 25%	E
1,2,4-TRIMETHYLBENZENE	95-63-6	101	2.614E+01	-4.676E+03	0	0	0	288.15	318.15	< 5%	E
MESITYLENE	108-67-8	101	2.301E+02	-1.351E+04	-3.055E+01	0	0	283.15	373.15	< 25%	E
p-DIETHYLBENZENE	105-05-5	101	2.841E+04	2.844E+03	0	0	0	293.15	293.15	< 100%	E
1,2,3,4-TETRAMETHYLBENZENE	488-23-3	101	2.968E+01	-5.913E+03	0	0	0	288.15	308.15	< 100%	E
1,2,4,5-TETRAMETHYLBENZENE	95-93-2	100	4.079E+04	0	0	0	0	298.15	298.15	< 10%	E

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
OTHER AMINES, IMINES											
CYCLOHEXYLAMINE	108-91-8	101	2.024E+02	-1.528E+04	-2.596E+01	0	0	298.15	372.82	< 25%	E
TRIALLYLAMINE	102-70-5	101	4.010E+02	-2.393E+04	-5.502E+01	0	0	273.15	363.15	< 25%	E
OTHER CONDENSED RINGS											
FLUORANTHENE	206-44-0	101	2.755E+01	-6.910E+03	0	0	0	281.25	333.15	< 10%	E
FLUORENE	86-73-7	101	2.665E+01	-6.145E+03	0	0	0	279.75	333.15	< 10%	E
ANTHRACENE	120-12-7	101	3.683E+02	-2.185E+04	-5.088E+01	0	0	273.45	347.85	< 5%	E
PHENANTHRENE	85-01-8	101	4.223E+02	-2.462E+04	-5.873E+01	0	0	277.15	346.55	< 5%	E
CHRYSENE	218-01-9	101	4.104E+01	-1.184E+04	0	0	0	279.65	302.15	< 25%	E
PYRENE	129-00-0	101	3.078E+02	-1.937E+04	-4.193E+01	0	0	277.85	347.85	< 5%	E
ACENAPHTHENE	83-32-9	101	2.469E+01	-5.381E+03	0	0	0	277.25	347.85	< 25%	E
ACENAPHTHALENE	208-96-8	101	2.748E+01	-6.263E+03	0	0	0	277.25	304.15	< 25%	E
BENZO[a]PYRENE	50-32-8	101	3.108E+01	-8.515E+03	0	0	0	281.15	308.15	< 10%	E
BENZO[b]FLUORANTHENE	205-99-2	101	1.949E+01	-5.422E+03	0	0	0	283.15	328.15	< 10%	E
OTHER ETHERS/DIETHERS											
METHYLAL	109-87-5	101	2.270E+01	-4.738E+03	0	0	0	273.35	333.15	< 25%	E
1,2-DIMETHOXYETHANE	110-71-4	101	2.511E+02	-1.722E+04	-3.328E+01	0	0	273.35	372.5	< 3%	E
DIETHYLENE GLYCOL DI-n-BUTYL ETHER	112-73-2	101	4.602E+01	-1.280E+04	0	0	0	273.15	293.15	< 100%	P
ANISOLE	100-66-3	101	2.158E+01	-4.179E+03	0	0	0	275.15	363.85	< 5%	E
DIPHENYL ETHER	101-84-8	100	2.903E+03	0	0	0	0	298.15	298.15	< 50%	S
OTHER INORGANICS											
HYDROGEN PEROXIDE	7722-84-1	101	2.142E+01	-7.235E+03	0	0	0	278.15	297.15	< 25%	E
OTHER MONOAROMATICS											
STYRENE	100-42-5	101	2.190E+01	-3.653E+03	0	0	0	293.32	313.02	< 5%	E

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
OTHER POLYFUNCTIONAL C, H, O											
2-BUTOXYETHANOL	111-76-2	101	2.922E+01	-8.275E+03	0	0	0	293.15	303.15	< 25%	E
VANILLIN	121-33-5	101	-3.909E+02	1.144E+04	6.127E+01	0	0	293.15	318.15	< 5%	E
OTHER POLYFUNCTIONAL ORGANICS											
MALATHION	121-75-5	101	2.879E+01	-8.274E+03	0	0	0	283.15	303.15	< 25%	E
OTHER SATURATED ALIPHATIC ESTERS											
ETHYL ISOVALERATE	108-64-5	101	2.443E+02	-1.600E+04	-3.201E+01	0	0	273.15	363.65	< 10%	E
DIETHYL MALONATE	105-53-3	101	2.731E+02	-1.846E+04	-3.660E+01	0	0	273.15	363.75	< 10%	P
DIETHYL SUCCINATE	123-25-1										
ETHYL TRIMETHYL ACETATE	3938-95-2	101	2.619E+02	-1.693E+04	-3.441E+01	0	0	273.15	363.55	< 25%	E
PEROXIDES											
METHYL HYDROPEROXIDE	3031-73-0	101	2.049E+01	-5.244E+03	0	0	0	277.15	297.15	< 3%	E
ETHYL HYDROPEROXIDE	3031-74-1	101	2.288E+01	-5.985E+03	0	0	0	278.15	297.15	< 10%	E
POLYFUNCTIONAL AMIDES/AMINES											
N,N- DIMETHYLFORMAMIDE	68-12-2	101	2.051E+02	-1.570E+04	-2.692E+01	0	0	289.4	372	< 25%	E
N-METHYLFORMAMIDE	123-39-7	101	2.081E+01	-7.161E+03	0	0	0	333.15	372.65	< 5%	E
N,N- DIMETHYLACETAMIDE	127-19-5	101	2.480E+01	-8.014E+03	0	0	0	343.15	372.28	< 10%	E
POLYFUNCTIONAL C, H, N, HALIDE, (O)											
o-CHLOROANILINE	95-51-2	101	1.336E+03	-5.042E+04	-2.173E+02	2.501E-01	1	298.15	431.15	< 25%	P
p- CHLORONITROBENZENE	100-00-5	101	2.414E+01	-5.868E+03	0	0	0	283.15	313.15	< 10%	E
POLYFUNCTIONAL C, H, O, HALIDE											
DI(2- CHLOROETHYL)ETHER	111-44-4	101	2.107E+02	-1.440E+04	-2.767E+01	0	0	273.15	364.85	< 5%	E
ETHYLCHLOROACETATE	105-39-5	101	2.546E+02	-1.630E+04	-3.415E+01	0	0	273.15	333.25	< 10%	E

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
METHYL CHLOROACETATE	96-34-4	101	1.755E+02	-1.239E+04	-2.268E+01	0	0	273.15	323.35	< 5%	E
POLYFUNCTIONAL C, H, O, N											
N-METHYL-2-PYRROLIDONE	872-50-4	101	4.600E+02	-3.011E+04	-6.372E+01	0	0	298.8	372.85	< 5%	E
POLYFUNCTIONAL C, H, O, S											
DIMETHYL SULFOXIDE	67-68-5	101	2.115E+01	-7.789E+03	0	0	0	298.8	337.9	< 25%	E
POLYFUNCTIONAL ESTERS											
METHYL SALICYLATE	119-36-8	101	4.635E+02	-2.852E+04	-6.394E+01	0	0	273.15	363.65	< 25%	E
2-ETHOXYETHYL ACETATE	111-15-9	100	1.850E+01	0	0	0	0	303.15	303.15	< 10%	E
ETHYL ACETOACETATE	141-97-9	101	2.610E+01	-7.239E+03	0	0	0	273.15	303.15	< 25%	EP
ETHYLENE GLYCOL MONOBUTYL ETHER ACETATE	112-07-2	100	3.062E+01	0	0	0	0	298.15	298.15	< 25%	E
PROPIONATES AND BUTYRATES											
METHYL PROPIONATE	554-12-1	101	2.605E+01	-5.719E+03	0	0	0	273.15	310.2	< 5%	E
ETHYL PROPIONATE	105-37-3	101	2.963E+02	-1.786E+04	-4.021E+01	0	0	273.15	353.45	< 25%	E
n-PROPYL PROPIONATE	106-36-5	101	2.299E+02	-1.536E+04	-2.996E+01	0	0	273.15	363.65	< 10%	E
n-BUTYL PROPIONATE	590-01-2	101	2.785E+02	-1.795E+04	-3.694E+01	0	0	273.15	363.55	< 10%	E
n-PROPYL n-BUTYRATE	105-66-8	101	2.587E+02	-1.660E+04	-3.422E+01	0	0	273.15	363.65	< 25%	E
METHYL n-BUTYRATE	623-42-7	101	1.665E+02	-1.196E+04	-2.091E+01	0	0	273.15	363.65	< 25%	E
ETHYL n-BUTYRATE	105-54-4	101	1.949E+02	-1.355E+04	-2.488E+01	0	0	273.15	363.65	< 25%	E
ISOBUTYL ISOBUTYRATE	97-85-8	101	2.887E+02	-1.832E+04	-3.832E+01	0	0	273.15	363.45	< 25%	E
n-BUTYL n-BUTYRATE	109-21-7	101	3.706E+02	-2.260E+04	-5.032E+01	0	0	273.15	363.45	< 10%	E
METHYL ISOBUTYRATE	547-63-7	101	2.024E+02	-1.347E+04	-2.625E+01	0	0	273.15	353.45	< 10%	E
ETHYL ISOBUTYRATE	97-62-1	101	1.871E+02	-1.308E+04	-2.370E+01	0	0	273.15	363.75	< 10%	P
n-PENTYL n-BUTYRATE	540-18-1	101	2.955E+01	-6.393E+03	0	0	0	273.15	363.65	< 25%	E

Table C.2: k_H regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
ISOPENTYL BUTYRATE	106-27-4	101	3.076E+02	-1.973E+04	-4.087E+01	0	0	282.55	363.85	< 100%	E
SULFIDES/THIOPHENES											
DIMETHYL SULFIDE	75-18-3	101	2.083E+02	-1.184E+04	-2.797E+01	0	0	272.35	353.45	< 10%	E
DIMETHYL DISULFIDE	624-92-0	101	2.120E+01	-3.695E+03	0	0	0	273.2	343.15	< 10%	E
2-METHYLTHIOPHENE	554-14-3	101	2.390E+01	-4.287E+03	0	0	0	298.15	343.15	< 10%	E
DIISOPROPYL SULFIDE	625-80-9	101	2.628E+01	-4.889E+03	0	0	0	298.15	343.15	< 10%	E
UNSATURATED ALIPHATIC ESTERS											
ETHYL ACRYLATE	140-88-5	101	2.199E+01	-4.393E+03	0	0	0	298.6	360.2	< 10%	P
METHYL METHACRYLATE	80-62-6	101	2.853E+02	-1.690E+04	-3.880E+01	0	0	273.35	333.15	< 3%	E

Table C.3: γ_i^∞ regression summary

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
1-ALKENES											
1-PENTENE	109-67-1	100	2.2033E+04	0	0	0	0	298.15	298.15	< 25%	S
1-HEXENE	592-41-6	100	9.1053E+04	4.5755E+03	5.6855E+00	0	0	298.15	298.15	< 25%	S
1-HEPTENE	592-76-7	100	3.7625E+05	0	0	0	0	298.15	298.15	< 10%	S
1-OCTENE	111-66-0	100	1.5549E+06	0	0	0	0	298.15	298.15	< 10%	S
1-NONENE	124-11-8	100	6.4250E+06	0	0	0	0	298.15	298.15	< 25%	S
1-DECENE	872-05-9	100	2.6552E+07	0	0	0	0	298.15	298.15	< 25%	P
1-UNDECENE	872-05-9	100	1.0973E+08	0	0	0	0	298.15	298.15	< 25%	P
1-DODECENE	872-05-9	100	4.5341E+08	0	0	0	0	298.15	298.15	< 25%	P
1-TRIDECENE	592-41-6	100	1.8737E+09	0	0	0	0	298.15	298.15	< 50%	P
1-TETRADECENE	872-05-9	100	7.7426E+09	0	0	0	0	298.15	298.15	< 50%	P
1-PENTADECENE	13360-61-7	100	3.1997E+10	0	0	0	0	298.15	298.15	< 50%	S

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
2,3,4-ALKENES											
cis-2-PENTENE	627-20-3	100	1.9178E+04	0	0	0	0	298.15	298.15	< 5%	E
trans-2-PENTENE	646-04-8	100	1.9178E+04	0	0	0	0	298.15	298.15	< 5%	E
cis-2-HEPTENE	6443-92-1	100	3.6334E+05	0	0	0	0	298.15	298.15	< 10%	E
trans-2-HEPTENE	14686-13-6	100	3.6334E+05	0	0	0	0	298.15	298.15	< 10%	E
ACETATES											
METHYL ACETATE	79-20-9	101	4.0958E+02	-1.9719E+04	-5.9724E+01	0	0	293.15	343.25	< 10%	E
ETHYL ACETATE	141-78-6	101	1.8730E+02	-9.2371E+03	-2.6696E+01	0	0	288.15	353.15	< 10%	E
n-PROPYL ACETATE	109-60-4	101	2.2269E+02	-1.0697E+04	-3.1801E+01	0	0	298	343.25	< 10%	E
n-BUTYL ACETATE	123-86-4	101	1.7311E+02	-8.2435E+03	-2.4321E+01	0	0	273.15	363.65	< 10%	E
ISOBUTYL ACETATE	110-19-0	101	2.9865E+01	-1.9167E+03	-2.9174E+00	0	0	273.15	323.15	< 25%	E
ISOPENTYL ACETATE	123-92-2	101	2.3843E+02	-1.1153E+04	-3.3853E+01	0	0	273.15	363.85	< 10%	E
ISOPROPYL ACETATE	108-21-4	101	-1.0523E+02	3.4976E+03	1.7345E+01	0	0	273.15	310.2	< 25%	E
sec-BUTYL ACETATE	105-46-4	101	1.8364E+02	-8.8655E+03	-2.5835E+01	0	0	273.15	363.65	< 10%	E
VINYL ACETATE	108-05-4	101	1.3344E+02	-6.0176E+03	-1.8964E+01	0	0	273.35	333.15	< 3%	E
n-PENTYL ACETATE	628-63-7	101	3.3660E+02	-1.5786E+04	-4.8331E+01	0	0	273.15	364.5	< 25%	E
n-HEXYL ACETATE	142-92-7	101	3.7674E+02	-1.7146E+04	-5.4324E+01	0	0	273.15	363.35	< 25%	E
n-HEPTYL ACETATE	112-06-1	100	6.5861E+04	-1.8462E+04	-3.6599E+01	0	0	298.15	298.15	< 25%	P
n-OCTYL ACETATE	112-14-1	100	2.6455E+05	-2.1364E+04	-4.5483E+01	0	0	298.15	298.15	< 25%	P
tert-BUTYL ACETATE	540-88-5	101	1.6617E+02	-8.2029E+03	-2.3147E+01	0	0	273.15	353.65	< 10%	E
CYCLOHEXYL ACETATE	622-45-7	101	2.0341E+02	-9.4699E+03	-2.8728E+01	0	0	273.15	363.45	< 10%	E
PHENYL ACETATE	122-79-2	101	1.6704E+02	-7.1054E+03	-2.3876E+01	0	0	273.15	363.95	< 10%	E
ALDEHYDES											
PROPANAL	123-38-6	101	2.3549E+02	-1.1978E+04	-3.3870E+01	0	0	278.15	318.15	< 5%	E
BUTANAL	123-72-8	101	1.0100E+01	-1.9064E+03	0	0	0	278.15	318.15	< 10%	E
2-METHYLPROPANAL	78-84-2	100	5.1912E+01	-6.5156E+03	-2.1156E+01	-8.2308E-06	2	298.15	298.15	< 5%	S

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
PENTANAL	110-62-3	101	1.1248E+01	-1.7998E+03	0	0	0	278.15	318.15	< 10%	E
HEPTANAL	111-71-7	101	1.4484E+01	-1.9125E+03	0	0	0	283.15	318.15	< 25%	E
HEXANAL	66-25-1	101	1.0459E+01	-1.1086E+03	0	0	0	283.15	318.15	< 25%	E
OCTANAL	124-13-0	101	1.3249E+01	-1.0346E+03	0	0	0	283.15	318.15	< 10%	E
NONANAL	124-19-6	101	8.9172E+00	8.7039E+02	0	0	0	283.15	318.15	< 25%	E
2-ETHYLHEXANAL	123-05-7	101	8.2330E+00	4.5752E+02	0	0	0	273.15	363.15	< 25%	E
2-ETHYL-2-HEXENAL	645-62-5	101	8.3085E+00	1.0122E+02	0	0	0	273.15	363.15	< 50%	E
DECANAL	112-31-2	101	1.6820E+01	-1.0090E+03	0	0	0	298.15	318.15	< 50%	E
2-METHYL-BUTYRALDEHYDE	96-17-3	101	9.8196E+00	-1.2187E+03	0	0	0	273.15	363.15	< 25%	P
ACROLEIN	107-02-8	101	7.9257E+00	-1.4558E+03	0	0	0	273	298	< 100%	E
trans-CROTONALDEHYDE	123-73-9	101	1.2597E+02	-5.9638E+03	-1.8057E+01	0	0	273.15	363.15	< 25%	EP
METHACROLEIN	78-85-3	101	6.5218E+00	-7.1561E+02	0	0	0	273.15	333.15	< 10%	E
p-TOLUALDEHYDE	104-87-0	101	9.6622E+00	-5.4954E+02	0	0	0	278	298	< 25%	E
BENZALDEHYDE	100-52-7	101	5.1540E+02	-2.1901E+04	-7.6382E+01	0	0	273.08	323.15	< 10%	E
m-TOLUALDEHYDE	620-23-5	101	9.4640E+00	-3.8418E+02	0	0	0	278	298	< 25%	E
ALIPHATIC ETHERS											
DIETHYL ETHER	60-29-7	101	2.3981E+02	-1.2537E+04	-3.3940E+01	0	0	273.35	333.25	< 5%	E
DIISOPROPYL ETHER	108-20-3	101	-4.6536E+01	7.2880E+01	9.2540E+00	0	0	273.15	334.15	< 10%	E
DI-n-BUTYL ETHER	142-96-1	101	1.4973E+01	-1.3729E+03	0	0	0	273.15	363.65	< 25%	E
METHYL tert-BUTYL ETHER	1634-04-4	101	5.7408E+01	-4.2388E+03	-6.7324E+00	0	0	273.15	343.15	< 10%	E
METHYL ISOBUTYL ETHER	625-44-5	101	1.5503E+01	-2.8030E+03	0	0	0	273.15	298.15	< 5%	E
DI-n-HEXYL ETHER	112-58-3	100	1.4510E+07	-1.8280E+04	-5.4709E+01	0	0	298.15	298.15	< 25%	P
METHYL n-BUTYL ETHER	628-28-4	101	7.0311E+02	-3.2841E+04	-1.0296E+02	0	0	273.15	298.15	< 25%	E
DI-n-PENTYL ETHER	693-65-2	100	4.5100E+05	-7.0534E+03	-2.2745E+01	0	0	298.15	298.15	< 25%	P

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
METHYL tert-PENTYL ETHER	994-05-8	101	1.7808E+02	-9.7068E+03	-2.4430E+01	0	0	273.15	352.35	< 10%	E
tert-BUTYL ETHYL ETHER	637-92-3	101	1.5616E+01	-2.8259E+03	0	0	0	278.15	318.15	< 25%	E
ETHYL tert-PENTYL ETHER	919-94-8	101	1.7640E+01	-2.9893E+03	0	0	0	288.15	318.15	< 25%	E
DI-n-PROPYL ETHER	111-43-3	101	4.0286E+02	-1.9387E+04	-5.7942E+01	0	0	273.15	333.25	< 5%	E
n-BUTYL ETHYL ETHER	628-81-9	101	2.0852E+02	-1.0117E+04	-2.9431E+01	0	0	273.15	363.85	< 25%	E
ALKYLCYCLOHEXANES											
METHYLCYCLOHEXANE	108-87-2	101	3.6535E+02	-1.5923E+04	-5.2511E+01	0	0	293.15	443.95	< 25%	E
ETHYLCYCLOHEXANE	1678-91-7	101	1.7556E+03	-5.4303E+04	-2.9260E+02	3.6218E-01	1	273.15	443.95	< 10%	E
cis-1,2-DIMETHYL-CYCLOHEXANE	2207-01-4	101	3.3288E+02	-1.3995E+04	-4.7754E+01	0	0	273.15	313.15	< 25%	E
trans-1,2-DIMETHYL-CYCLOHEXANE	6876-23-9	101	1.4668E+03	-4.6285E+04	-2.4271E+02	2.8645E-01	1	273.15	443.25	< 5%	E
ALKYLCYCLOPENTANES											
METHYL-CYCLOPENTANE	96-37-7	100	1.0935E+05	-7.8306E+03	-2.7270E+01	-6.3208E-06	2	298.15	298.15	< 3%	S
ALKYNES											
1-PENTYNE	627-19-0	100	2.4720E+03	0	0	0	0	298.15	298.15	< 25%	S
1-HEXYNE	693-02-7	100	1.1810E+04	0	0	0	0	298.15	298.15	< 25%	S
1-OCTYNE	629-05-0	100	2.5340E+05	0	0	0	0	298.15	298.15	< 25%	S
1-NONYNE	3452-09-3	100	1.3100E+06	0	0	0	0	298.15	298.15	< 25%	S
1-HEPTYNE	628-71-7	100	5.4810E+04	0	0	0	0	298.15	298.15	< 25%	S
AROMATIC ALCOHOLS											
2,3-XYLENOL	526-75-0	101	3.2073E+00	9.3895E+02	0	0	0	349.05	371.65	< 25%	S
2,4-XYLENOL	105-67-9	101	1.6630E+00	1.5325E+03	0	0	0	349.05	371.65	< 25%	E
2,5-XYLENOL	95-87-4	101	3.4136E+02	-1.5740E+04	-4.9567E+01	0	0	278	371.65	< 10%	E
2,6-XYLENOL	576-26-1	101	1.2864E+02	-4.9837E+03	-1.8451E+01	0	0	278	371.65	< 10%	E
3,4-XYLENOL	95-65-8	101	3.2111E+00	8.9290E+02	0	0	0	293.15	371.65	< 25%	E

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
3,5-XYLENOL	108-68-9	101	1.1755E+02	-4.5606E+03	-1.6823E+01	0	0	293.15	371.65	< 25%	E
PHENOL	108-95-2	101	4.5472E+02	-2.1677E+04	-6.6388E+01	0	0	278	373.15	< 25%	E
o-CRESOL	95-48-7	101	2.1997E+02	-9.7201E+03	-3.1956E+01	0	0	293.15	371.65	< 5%	E
m-CRESOL	108-39-4	101	-1.5582E+01	2.1554E+03	2.4416E+00	0	0	298.15	371.65	< 25%	E
p-CRESOL	106-44-5	101	1.5458E+02	-6.2545E+03	-2.2485E+01	0	0	298	371.65	< 3%	E
2-PHENYLETHANOL	60-12-8	101	1.7589E+02	-7.4876E+03	-2.5479E+01	0	0	283.15	373.15	< 10%	P
AROMATIC AMINES											
o-TOLUIDINE	95-53-4	101	5.6946E+00	3.2271E+01	0	0	0	298.15	323.15	< 25%	E
m-TOLUIDINE	108-44-1	101	5.6591E+00	8.1355E+01	0	0	0	298.15	323.15	< 25%	E
p-TOLUIDINE	106-49-0	101	6.3431E-01	1.6356E+03	0	0	0	293.15	313.15	< 5%	P
N,N-DIETHYLANILINE	91-66-7	101	9.2931E+00	4.2183E+02	0	0	0	273.15	333.15	< 25%	E
PYRIDINE	110-86-1	101	3.2354E+00	-4.9087E+01	0	0	0	298	374.15	< 25%	E
ANILINE	62-53-3	101	1.8127E+02	-8.1850E+03	-2.6142E+01	0	0	298	374.15	< 10%	E
2-METHYLPYRIDINE	109-06-8	101	7.8877E+00	-1.2722E+03	0	0	0	298	313.15	< 5%	E
o-ETHYLANILINE	578-54-1	101	1.6012E+02	-7.0534E+03	-2.2745E+01	0	0	283.15	363.15	< 10%	P
2,6-DIMETHYLPYRIDINE	108-48-5	101	1.1003E+01	-1.9828E+03	0	0	0	298	313.15	< 25%	E
3-METHYLPYRIDINE	108-99-6	101	7.5009E+00	-1.0441E+03	0	0	0	298	313.15	< 25%	E
4-METHYLPYRIDINE	108-89-4	101	7.6057E+00	-1.1383E+03	0	0	0	298	313.15	< 5%	E
2,4-DIMETHYLANILINE	95-68-1	101	2.6650E+02	-1.2031E+04	-3.8444E+01	0	0	273.15	363.15	< 25%	P
N-ETHYLANILINE	103-69-5	101	1.8651E+02	-8.4052E+03	-2.6420E+01	0	0	273.15	363.15	< 10%	E
AROMATIC CHLORIDES											
BENZYL CHLORIDE	100-44-7	101	4.7978E+00	1.4249E+03	0	0	0	277.65	303.15	< 25%	E
MONOCHLOROBENZENE	108-90-7	101	3.0084E+02	-1.1883E+04	-4.4139E+01	0	0	274.15	353.16	< 10%	E
o-DICHLOROBENZENE	95-50-1	101	7.4286E+02	-3.1963E+04	-1.0964E+02	0	0	278.15	318.15	< 10%	E
m-DICHLOROBENZENE	541-73-1	101	7.3203E+00	1.1112E+03	0	0	0	278.15	318.15	< 10%	E
p-DICHLOROBENZENE	106-46-7	101	5.2227E+02	-2.3008E+04	-7.6204E+01	0	0	278.15	346.55	< 10%	E

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
HEXACHLOROBENZENE	118-74-1	101	9.4803E+00	2.8438E+03	0	0	0	278.15	318.15	< 25%	E
o-CHLOROTOLUENE	95-49-8	101	7.5502E+00	9.9128E+02	0	0	0	274.15	318.15	< 25%	E
p-CHLOROTOLUENE	106-43-4	101	7.8721E+00	9.0900E+02	0	0	0	278.15	318.15	< 25%	E
2,4-DICHLOROTOLUENE	95-73-8	101	8.0122E+00	1.4043E+03	0	0	0	278.15	323.1	< 10%	E
1,2,4-TRICHLOROBENZENE	120-82-1	101	1.0415E+01	6.0490E+02	0	0	0	278.15	318.15	< 25%	E
1,2,3-TRICHLOROBENZENE	87-61-6	101	6.1575E+00	1.9216E+03	0	0	0	277.15	323.15	< 25%	E
AROMATIC ESTERS											
BENZYL ACETATE	140-11-4	101	7.1499E+00	2.6685E+02	0	0	0	273.15	322.95	< 25%	E
DI-n-PROPYL PHTHALATE	131-16-8	100	1.5080E+05	0	0	0	0	298.15	298.15	< 25%	S
DI-n-HEXYL PHTHALATE	84-75-3	100	7.9098E+07	0	0	0	0	298.15	298.15	< 50%	S
METHYL BENZOATE	93-58-3	101	1.9660E+02	-8.1145E+03	-2.8291E+01	0	0	273.15	363.65	< 25%	E
ETHYL BENZOATE	93-89-0	101	1.4804E+02	-6.1565E+03	-2.0767E+01	0	0	273.15	363.45	< 25%	E
BENZYL FORMATE	104-57-4	101	7.6942E+00	-3.4741E+02	0	0	0	273.15	322.85	< 25%	E
DIETHYL PHTHALATE	84-66-2	101	2.9610E+02	-1.2896E+04	-4.2710E+01	0	0	278.15	374.15	< 10%	E
DI-n-BUTYL PHTHALATE	84-74-2	101	7.8577E+02	-3.4556E+04	-1.1507E+02	0	0	283.15	323.15	< 25%	E
DIMETHYL PHTHALATE	131-11-3	101	2.1362E+02	-8.7600E+03	-3.0969E+01	0	0	283.15	374.15	< 10%	EP
C, H, BR COMPOUNDS											
BROMOETHANE	74-96-4	101	7.9924E+00	-4.4577E+02	0	0	0	273.15	303.15	< 3%	E
1,1,2,2-TETRABROMOETHANE	79-27-6	101	-7.0038E+00	6.0714E+03	0	0	0	298.15	323.15	< 50%	E
1-BROMOPROPANE	106-94-5	101	9.9920E+00	-6.0960E+02	0	0	0	273.15	303.15	< 5%	E
2-BROMOPROPANE	75-26-3	101	3.4368E+02	-1.5159E+04	-5.0051E+01	0	0	273.15	303.15	< 5%	E
1-BROMOBUTANE	109-65-9	100	1.3502E+04	0	0	0	0	298.15	298.15	< 25%	S
1-BROMOHEPTANE	629-04-9	100	1.3438E+06	0	0	0	0	298.15	298.15	< 25%	S
1,2-DIBROMOETHANE	106-93-4	101	6.1031E+00	4.9221E+02	0	0	0	273.15	363.75	< 10%	E

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
m-DIBROMOBENZENE	108-36-1	100	1.3275E+05	0	0	0	0	298.15	298.15	< 25%	E
BROMOBENZENE	108-86-1	101	3.9706E+02	-1.6495E+04	-5.8226E+01	0	0	283.15	343.15	< 10%	E
TRIBROMOMETHANE	75-25-2	101	6.8448E+00	4.4238E+02	0	0	0	283	313.15	< 10%	E
DIBROMOMETHANE	74-95-3	101	6.9877E+01	-2.6491E+03	-9.5301E+00	0	0	273.15	363.25	< 5%	E
2-BROMOBUTANE	78-76-2	100	8.3200E+03	0	0	0	0	298.15	298.15	< 10%	E
p-BROMOTOLUENE	106-38-7	101	8.7771E+00	7.6479E+02	0	0	0	293.1	322.91	< 25%	E
C, H, F COMPOUNDS											
PERFLUORO-n-HEXANE	355-42-0	100	2.0497E+08	1.2453E+03	0	0	0	298.15	298.15	< 100%	E
PERFLUORO-n-OCTANE	307-34-6	100	1.4564E+10	-3.7208E+04	-1.0276E+02	0	0	298.15	298.15	< 100%	E
1,2-DIFLUOROBENZENE	367-11-3	101	6.6988E+00	5.9091E+02	0	0	0	298.15	322.89	< 25%	E
FLUOROBENZENE	462-06-6	101	1.4087E+02	-5.8074E+03	-1.9867E+01	0	0	273.15	353.15	< 10%	E
HEXAFLUOROBENZENE	392-56-3	101	2.7504E+02	-1.2435E+04	-3.9248E+01	0	0	281.46	339.76	< 3%	E
PERFLUORO-n-PENTANE	678-26-2	100	1.3835E+07	7.9863E+01	0	0	0	298.15	298.15	< 100%	E
PERFLUORO-n-HEPTANE	335-57-9	100	1.7852E+09	-6.8024E+03	0	0	0	298.15	298.15	< 100%	E
C, H, I COMPOUNDS											
METHYL IODIDE	74-88-4	101	3.2179E+02	-1.3816E+04	-4.7240E+01	0	0	273.15	322.91	< 5%	E
ETHYL IODIDE	75-03-6	101	2.9938E+02	-1.2802E+04	-4.3660E+01	0	0	273.15	303.15	< 3%	E
n-PROPYL IODIDE	107-08-4	101	9.8486E+00	-2.1844E+02	0	0	0	273.15	303.15	< 5%	E
ISOPROPYL IODIDE	75-30-9	101	1.0634E+01	-5.3563E+02	0	0	0	273.15	303.15	< 5%	E
IODOBENZENE	591-50-4	101	-1.6323E+01	8.1025E+03	0	0	0	298	303.15	< 25%	E
n-HEXYL IODIDE	638-45-9	100	8.3652E+05	0	0	0	0	298.15	298.15	< 25%	S
n-BUTYL IODIDE	542-69-8	100	4.3976E+04	-1.3108E+04	-5.1464E+01	4.2552E-02	1	298.15	298.15	< 25%	S
C, H, MULTIHALOGEN COMPOUNDS											
1,1,2-TRICHLOROTRIFLUORO-ETHANE	76-13-1	101	1.2459E+01	-5.4053E+02	0	0	0	275.15	303.15	< 25%	E

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
C, H, NO₂ COMPOUNDS											
NITROMETHANE	75-52-5	101	1.9786E+02	-8.0194E+03	-2.9403E+01	0	0	273.15	363.15	< 10%	E
NITROETHANE	79-24-3	101	1.1679E+02	-4.7515E+03	-1.6904E+01	0	0	273.15	363.25	< 5%	E
1-NITROPROPANE	108-03-2	101	1.4552E+02	-6.1723E+03	-2.0902E+01	0	0	273.15	363.65	< 10%	E
2-NITROPROPANE	79-46-9	101	1.1132E+02	-4.6173E+03	-1.5829E+01	0	0	292.85	363.35	< 3%	E
1-NITROBUTANE	627-05-4	100	1.5314E+03	0	0	0	0	298.15	298.15	< 25%	S
o-NITROTOLUENE	88-72-2	101	5.7392E+00	1.0851E+03	0	0	0	283.15	313.15	< 10%	E
p-NITROTOLUENE	99-99-0	101	7.6320E+00	5.3380E+02	0	0	0	281	338	< 10%	E
m-NITROTOLUENE	99-08-1	101	8.8637E+00	2.1248E+02	0	0	0	283.15	313.15	< 25%	E
NITROBENZENE	98-95-3	101	1.4633E+02	-5.4696E+03	-2.1029E+01	0	0	283.15	372.85	< 5%	E
2,4-DINITROTOLUENE	121-14-2	101	3.5728E+02	-1.4679E+04	-5.2362E+01	0	0	277.15	334.95	< 5%	E
2,6-DINITROTOLUENE	606-20-2	101	2.5130E-02	2.9932E+03	0	0	0	278	314	< 5%	E
3,4-DINITROTOLUENE	610-39-9	101	4.9257E+00	1.5634E+03	0	0	0	278.2	313.2	< 5%	E
2,4,6-TRINITROTOLUENE	118-96-7	101	4.9257E+02	-2.0991E+04	-7.2348E+01	0	0	275.55	334.15	< 10%	E
C1/C2 ALIPHATIC CHLORIDES											
CARBON TETRACHLORIDE	56-23-5	101	3.2892E+02	-1.4418E+04	-4.7608E+01	0	0	273.15	366.15	< 5%	E
DICHLOROMETHANE	75-09-2	101	7.7957E+01	-2.9989E+03	-1.0948E+01	0	0	273.15	366.15	< 10%	E
CHLOROFORM	67-66-3	101	1.7864E+02	-7.9325E+03	-2.5502E+01	0	0	273.15	366.15	< 3%	E
1,1-DICHLOROETHANE	75-34-3	101	2.1919E+02	-9.6308E+03	-3.1578E+01	0	0	273.15	366.15	< 5%	E
1,2-DICHLOROETHANE	107-06-2	101	2.2719E+02	-9.8351E+03	-3.2955E+01	0	0	273.15	366.15	< 10%	E
1,1,2-TRICHLOROETHANE	79-00-5	101	7.0028E+01	-2.5094E+03	-9.5188E+00	0	0	273.15	363.95	< 5%	E
HEXACHLOROETHANE	67-72-1	101	1.0718E+01	3.8043E+02	0	0	0	283.15	303.15	< 25%	E
1,1,1-TRICHLOROETHANE	71-55-6	101	2.9839E+01	-1.0774E+03	-3.0848E+00	0	0	273.15	348.15	< 10%	E
1,1,1,2- TETRACHLOROETHANE	630-20-6	101	8.5450E+00	1.3991E+02	0	0	0	273.15	323.15	< 10%	E

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
1,1,2,2-TETRACHLOROETHANE	79-34-5	101	7.0902E+00	2.8054E+02	0	0	0	283.15	328.65	< 10%	E
TRICHLOROETHYLENE	79-01-6	101	2.2817E+02	-9.8609E+03	-3.2730E+01	0	0	274.22	366.15	< 5%	E
TETRACHLORO-ETHYLENE	127-18-4	101	2.6382E+02	-1.1250E+04	-3.7821E+01	0	0	273.15	364.95	< 10%	E
cis-1,2-DICHLOROETHYLENE	156-59-2	101	1.7602E+02	-7.7385E+03	-2.5154E+01	0	0	281.15	366.15	< 10%	E
trans-1,2-DICHLOROETHYLENE	156-60-5	101	8.5045E+00	-4.1902E+02	0	0	0	283.15	323.15	< 10%	E
PENTACHLOROETHANE	76-01-7	101	1.7025E+01	-2.0327E+03	0	0	0	293.15	298	< 25%	E
1,1-DICHLOROETHYLENE	75-35-4	101	2.6075E+02	-1.1818E+04	-3.7462E+01	0	0	275.15	363.65	< 10%	E
C3 & HIGHER ALIPHATIC CHLORIDES											
1,4-DICHLOROBUTANE	110-56-5	101	2.1338E+00	1.9316E+03	0	0	0	286.65	298.05	< 10%	E
1,5-DICHLOROPENTANE	628-76-2	101	3.5460E+00	2.1016E+03	0	0	0	274.15	295.85	< 50%	E
1,2-DICHLOROPROPANE	78-87-5	101	2.1117E+02	-8.8441E+03	-3.0497E+01	0	0	273.15	353.35	< 10%	E
ISOPROPYL CHLORIDE	75-29-6	101	1.0546E+01	-9.8481E+02	0	0	0	273.15	303.15	< 10%	E
1,2,3-TRICHLOROPROPANE	96-18-4	101	-4.0021E+00	3.6230E+03	0	0	0	275.05	318.15	< 25%	E
HEXACHLORO-1,3-BUTADIENE	87-68-3	101	1.0004E+01	1.4831E+03	0	0	0	275.15	298.15	< 50%	E
PROPYL CHLORIDE	540-54-5	101	1.0409E+01	-9.0691E+02	0	0	0	273.15	310.15	< 10%	E
n-BUTYL CHLORIDE	109-69-3	101	1.0006E+01	-3.3086E+02	0	0	0	273.35	310.15	< 10%	E
sec-BUTYL CHLORIDE	78-86-4	101	1.1007E+01	-6.6744E+02	0	0	0	275.05	298.05	< 10%	E
1-CHLOROPENTANE	543-59-9	101	3.5428E+02	-1.4816E+04	-5.1650E+01	0	0	275.65	310.15	< 5%	E
1,3-DICHLOROPROPANE	142-28-9	101	2.5176E+00	1.6023E+03	0	0	0	275.05	303.15	< 25%	E
CYCLOALIPHATIC ALCOHOLS											
CYCLOHEXANOL	108-93-0	101	5.9244E+00	-2.8428E+02	0	0	0	298	333.15	< 25%	E
CYCLOALKANES											
CYCLOPENTANE	287-92-3	101	1.9711E+02	-8.3807E+03	-2.7898E+01	0	0	298	426.25	< 10%	E

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
CYCLOHEXANE	110-82-7	101	-1.5818E+03	4.1569E+04	2.7982E+02	-4.7222E-01	1	274.25	424.15	< 25%	E
CYCLOHEPTANE	291-64-5	100	2.8396E+05	-1.1604E+04	-3.5685E+01	0	0	298.15	298.15	< 10%	P
CYCLOOCTANE	292-64-8	101	1.4629E+02	-5.4727E+03	-2.0027E+01	0	0	273.15	313.15	< 10%	E
CYCLOALKENES											
CYCLOPENTENE	142-29-0	101	8.1479E+00	1.4497E+02	0	0	0	283.15	298.15	< 50%	E
CYCLOHEXENE	110-83-8	101	8.1071E+00	4.6844E+02	0	0	0	278.26	318.36	< 25%	E
CYCLOHEPTENE	628-92-2	100	8.0884E+04	0	0	0	0	298.15	298.15	< 25%	E
CYCLOOCTENE	931-88-4	101	1.4403E+02	-5.2275E+03	-2.0010E+01	0	0	273.15	313.15	< 5%	E
DIALKENES											
1,4-CYCLOHEXADIENE	628-41-1	101	7.5073E+00	2.9784E+02	0	0	0	278.26	318.36	< 25%	E
DIMETHYLALKANES											
2,2-DIMETHYLBUTANE	75-83-2	100	2.2886E+05	0	0	0	0	298.15	298.15	< 25%	S
2,3-DIMETHYLBUTANE	79-29-8	101	3.5524E+02	-1.5943E+04	-5.0796E+01	0	0	273.15	422.65	< 25%	E
2,4-DIMETHYLPENTANE	108-08-7	100	1.4765E+06	0	0	0	0	298.15	298.15	< 50%	S
3,3-DIMETHYLPENTANE	562-49-2	101	-1.9290E+03	5.3299E+04	3.3822E+02	-5.4678E-01	1	298.15	423.64	< 25%	E
2,2-DIMETHYLHEXANE	590-73-8	101	6.2123E+02	-2.6996E+04	-9.0435E+01	0	0	273.15	313.15	< 5%	E
2,5-DIMETHYLHEXANE	592-13-2	101	7.3845E+02	-3.1648E+04	-1.0825E+02	0	0	273.15	313.15	< 3%	E
DIPHENYL/POLYAROMATICS											
BIPHENYL	92-52-4	101	5.6749E+02	-2.3787E+04	-8.3311E+01	0	0	293.15	337.65	< 10%	E
EPOXIDES											
1,4-DIOXANE	123-91-1	101	8.2490E+01	-4.7383E+03	-1.1380E+01	0	0	273.35	374.15	< 25%	E
TETRAHYDROPYRAN	142-68-7	101	1.7525E+02	-9.0201E+03	-2.4698E+01	0	0	273.35	328.15	< 5%	E
TETRAHYDROFURAN	109-99-9	101	8.7722E+00	-1.7620E+03	0	0	0	273.35	328.15	< 10%	E
DIBENZOFURAN	132-64-9	101	7.8758E+00	1.5796E+03	0	0	0	278.15	318.15	< 25%	E
FORMATES											
METHYL FORMATE	107-31-3	101	3.9227E+00	-3.2472E+02	0	0	0	285.65	333.25	< 10%	E

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
ETHYL FORMATE	109-94-4	101	4.6479E+00	-2.0861E+02	0	0	0	293.15	333.25	< 25%	E
n-PROPYL FORMATE	110-74-7	101	6.5884E+00	-4.3744E+02	0	0	0	272.15	333.15	< 10%	E
n-BUTYL FORMATE	592-84-7	101	3.0690E+00	9.7352E+02	0	0	0	273.15	363.75	< 25%	E
ISOBUTYL FORMATE	542-55-2	101	8.8201E+00	-7.1323E+02	0	0	0	273.15	333.55	< 10%	E
n-PENTYL FORMATE	638-49-3	101	1.9856E+02	-8.9809E+03	-2.8183E+01	0	0	273.15	363.65	< 25%	E
tert-BUTYL FORMATE	762-75-4	101	3.8562E+00	5.9690E+02	0	0	0	278.15	308.15	< 5%	E
KETONES											
ACETONE	67-64-1	101	9.4580E+01	-5.3679E+03	-1.3112E+01	0	0	273	373.15	< 10%	E
METHYL ETHYL KETONE	78-93-3	101	1.8127E+02	-9.1056E+03	-2.5894E+01	0	0	273	373.15	< 10%	E
3-PENTANONE	96-22-0	101	1.9745E+02	-9.7433E+03	-2.8115E+01	0	0	273.15	310.2	< 10%	E
METHYL ISOBUTYL KETONE	108-10-1	101	1.6218E+02	-8.0097E+03	-2.2744E+01	0	0	273.15	363.55	< 10%	E
3-HEPTANONE	106-35-4	101	1.5436E+02	-7.6036E+03	-2.1344E+01	0	0	273.15	363.25	< 10%	E
4-HEPTANONE	123-19-3	101	2.0951E+02	-1.0110E+04	-2.9533E+01	0	0	273.15	363.35	< 10%	E
3-HEXANONE	589-38-8	101	1.2634E+01	-1.9968E+03	0	0	0	293.15	310.2	< 10%	E
2-PENTANONE	107-87-9	101	1.6360E+02	-8.2048E+03	-2.3100E+01	0	0	273.15	363.65	< 10%	E
METHYL ISOPROPYL KETONE	563-80-4	101	2.3714E+02	-1.1611E+04	-3.4030E+01	0	0	265.15	453.15	< 10%	P
2-HEXANONE	591-78-6	101	7.9859E+01	-4.3728E+03	-1.0415E+01	0	0	273.15	312.75	< 10%	E
2-HEPTANONE	110-43-0	101	2.2170E+02	-1.0473E+04	-3.1457E+01	0	0	273.15	363.65	< 5%	P
5-METHYL-2-HEXANONE	110-12-3	101	2.2974E+02	-1.0897E+04	-3.2652E+01	0	0	273.15	362.85	< 5%	P
3,3-DIMETHYL-2-BUTANONE	75-97-8	101	1.1488E+02	-5.8789E+03	-1.5707E+01	0	0	273.15	363.35	< 5%	E
DIISOBUTYL KETONE	108-83-8	101	2.7411E+02	-1.2906E+04	-3.8780E+01	0	0	273.15	363.85	< 10%	E
DIISOPROPYL KETONE	565-80-0	101	1.2060E+02	-6.0752E+03	-1.6336E+01	0	0	273.15	363.45	< 10%	E
5-NONANONE	502-56-7	101	2.1399E+02	-1.0276E+04	-2.9794E+01	0	0	273.15	363.75	< 25%	E
2-NONANONE	821-55-6	101	2.6336E+02	-1.2150E+04	-3.7302E+01	0	0	273.15	364.35	< 25%	E

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
ACETYLACETONE	123-54-6	101	3.0614E-01	1.0727E+03	0	0	0	303.15	373.15	< 25%	E
CYCLOPENTANONE	120-92-3	101	6.5404E+00	-9.4122E+02	0	0	0	293.2	353.15	< 100%	E
CYCLOHEXANONE	108-94-1	101	7.8156E+00	-1.0863E+03	0	0	0	298	353.15	< 10%	E
2-OCTANONE	111-13-7	101	2.7138E+02	-1.3009E+04	-3.8461E+01	0	0	273.15	364.15	< 25%	E
ACETOPHENONE	98-86-2	101	1.6261E+02	-7.9720E+03	-2.2643E+01	0	0	278	318.15	< 25%	E
3-OCTANONE	106-68-3	101	1.3269E+02	-6.2786E+03	-1.8085E+01	0	0	273.15	364.15	< 10%	E
3-NONANONE	925-78-0	101	9.4835E+00	5.2588E+01	0	0	0	273.15	353.35	< 25%	E
2,6,8-TRIMETHYL-4-NONANONE	123-18-2	100	4.5321E+05	0	0	0	0	298.15	298.15	< 25%	P
MERCAPTANS											
ISOPROPYL MERCAPTAN	75-33-2	101	7.2632E+00	-6.7794E+01	0	0	0	288.15	318.15	< 10%	E
n-BUTYL MERCAPTAN	109-79-5	101	3.4604E+02	-1.4901E+04	-5.0435E+01	0	0	288.15	333.15	< 10%	E
METHYLALKANES											
2-METHYLPENTANE	107-83-5	101	3.6405E+02	-1.6321E+04	-5.2063E+01	0	0	273.15	422.65	< 25%	E
3-METHYLPENTANE	96-14-0	100	3.6949E+05	0	0	0	0	298.15	298.15	< 3%	S
2-METHYLHEXANE	591-76-4	100	2.1898E+06	0	0	0	0	298.15	298.15	< 10%	E
3-METHYLHEXANE	589-34-4	100	2.0958E+06	0	0	0	0	298.15	298.15	< 10%	E
3-METHYLHEPTANE	589-81-1	100	8.0059E+06	0	0	0	0	298.15	298.15	< 10%	E
4-METHYLHEPTANE	589-53-7	100	7.5617E+06	0	0	0	0	298.15	298.15	< 5%	E
N-ALCOHOLS											
METHANOL	67-56-1	101	5.6961E+01	-3.1424E+03	-8.0557E+00	0	0	273	373.15	< 10%	E
ETHANOL	64-17-5	101	1.1923E+02	-6.2620E+03	-1.6999E+01	0	0	273	373.15	< 10%	E
1-PROPANOL	71-23-8	101	1.4622E+02	-7.4561E+03	-2.0812E+01	0	0	273	373.15	< 10%	E
1-BUTANOL	71-36-3	101	2.2282E+02	-1.0659E+04	-3.2139E+01	0	0	273	373.15	< 10%	E
1-PENTANOL	71-41-0	101	1.7028E+02	-7.8640E+03	-2.4335E+01	0	0	273.35	371.15	< 10%	E
1-HEXANOL	111-27-3	101	1.7476E+02	-7.6823E+03	-2.4982E+01	0	0	298	363.15	< 25%	E

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
1-HEPTANOL	111-70-6	101	2.0296E+02	-8.6151E+03	-2.9116E+01	0	0	273	375	< 10%	EP
1-OCTANOL	111-87-5	101	3.4096E+02	-1.5040E+04	-4.9310E+01	0	0	288.15	363.45	< 10%	EP
1-NONANOL	143-08-8	101	3.2991E+02	-1.4056E+04	-4.7688E+01	0	0	288.15	363.15	< 25%	E
1-DECANOL	112-30-1	101	3.9199E+02	-1.6451E+04	-5.6942E+01	0	0	288.15	363.15	< 10%	E
1-UNDECANOL	112-42-5	100	9.6801E+05	0	0	0	0	298.15	298.15	< 50%	P
1-DODECANOL	112-53-8	100	3.9184E+06	0	0	0	0	298.15	298.15	< 50%	P
N-ALIPHATIC PRIMARY AMINES											
ETHYLAMINE	75-04-7	100	4.7374E-01	-1.1698E+04	-3.3947E+01	-9.1695E-03	1	298.15	298.15	< 10%	S
n-PROPYLAMINE	107-10-8	100	1.9970E+00	-1.0633E+04	-2.8801E+01	0	0	298.15	298.15	< 10%	S
n-BUTYLAMINE	109-73-9	100	8.3162E+00	-1.2774E+04	-3.2890E+01	0	0	298.15	298.15	< 10%	S
n-PENTYLAMINE	110-58-7	100	3.2117E+01	-8.8828E+03	-2.1253E+01	0	0	298.15	298.15	< 10%	S
n-HEXYLAMINE	111-26-2	100	1.3205E+02	9.3895E+02	0	0	0	298.15	298.15	< 10%	S
n-HEPTYLAMINE	111-68-2	100	4.4889E+02	0	0	0	0	298.15	298.15	< 25%	S
n-OCTYLAMINE	111-86-4	100	1.8558E+03	0	0	0	0	298.15	298.15	< 25%	S
N-ALKANES											
n-PENTANE	109-66-0	101	-1.3467E+03	3.3236E+04	2.4123E+02	-4.2847E-01	1	273.15	422.65	< 25%	E
n-HEXANE	110-54-3	101	3.6592E+02	-1.6200E+04	-5.2402E+01	0	0	273.15	443.15	< 10%	E
n-HEPTANE	142-82-5	101	3.7426E+02	-1.6119E+04	-5.3641E+01	0	0	273.15	443.75	< 25%	E
n-OCTANE	111-65-9	101	4.5088E+02	-1.9596E+04	-6.4794E+01	0	0	273.15	456.15	< 25%	E
n-NONANE	111-84-2	101	6.6057E+02	-2.9453E+04	-9.5478E+01	0	0	287.95	409.75	< 50%	E
n-DECANE	124-18-5	100	2.3009E+08	-1.3625E+04	-3.5395E+01	0	0	298.15	298.15	< 50%	S
N-ALKYLBENZENES											
BENZENE	71-43-2	101	5.0455E+02	-1.6685E+04	-8.1666E+01	8.2232E-02	1	273.15	527.15	< 3%	E
TOLUENE	108-88-3	101	5.6897E+02	-1.8250E+04	-9.2692E+01	9.8716E-02	1	261.7	473.55	< 5%	E
ETHYLBENZENE	100-41-4	101	3.0277E+02	-1.2921E+04	-4.3705E+01	0	0	273.15	373.15	< 5%	E
n-PROPYLBENZENE	103-65-1	101	1.8059E+02	-6.8317E+03	-2.5610E+01	0	0	273.65	328.15	< 10%	E

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
n-BUTYLBENZENE	104-51-8	101	3.4172E+02	-1.3779E+04	-4.9545E+01	0	0	273.65	373.15	< 5%	E
n-HEPTYLBENZENE	1078-71-3	100	4.3293E+07	-2.7892E+03	0	6.6631E+05	-2	298.15	298.15	< 10%	P
n-DECYLBENZENE	104-72-3	100	4.8338E+09	-6.0768E+03	0	0	0	298.15	298.15	< 10%	S
n-PENTYLBENZENE	538-68-1	101	3.2086E+02	-1.2951E+04	-4.6121E+01	0	0	280.15	318.15	< 10%	E
n-HEXYLBENZENE	1077-16-3	101	2.8640E+02	-1.1376E+04	-4.0754E+01	0	0	278.15	318.15	< 10%	E
n-OCTYLBENZENE	2189-60-8	100	2.0136E+08	-7.0790E+03	-1.9442E+01	0	0	298.15	298.15	< 10%	P
n-NONYLBENZENE	1081-77-2	100	9.6961E+08	-8.8322E+02	0	0	0	298.15	298.15	< 25%	P
NAPHTHALENES											
NAPHTHALENE	91-20-3	101	2.7047E+02	-1.0551E+04	-3.9299E+01	0	0	275.05	346.55	< 3%	E
1-METHYLNAPHTHALENE	90-12-0	101	8.4227E+00	1.2191E+03	0	0	0	277.25	304.85	< 10%	E
2-METHYLNAPHTHALENE	91-57-6	101	7.1880E+00	1.6019E+03	0	0	0	277.25	304.15	< 25%	E
1-ETHYLNAPHTHALENE	1127-76-0	101	7.4549E+00	1.8432E+03	0	0	0	281.55	304.85	< 10%	E
NITRILES											
ACETONITRILE	75-05-8	101	-3.4249E+01	1.3791E+03	5.5893E+00	0	0	273	353.15	< 5%	E
PROPIONITRILE	107-12-0	101	4.1879E+01	-1.8363E+03	-5.6957E+00	0	0	278.15	298.15	< 25%	E
ACRYLONITRILE	107-13-1	101	2.7998E+00	2.7737E+02	0	0	0	273.15	343.15	< 3%	E
METHACRYLONITRILE	126-98-7	101	2.3474E+02	-1.0357E+04	-3.4203E+01	0	0	273.15	353.15	< 10%	P
n-BUTYRONITRILE	109-74-0	101	7.6172E+01	-3.3874E+03	-1.0547E+01	0	0	273.15	313.15	< 3%	E
VALERONITRILE	110-59-8	101	8.3220E+00	-7.2378E+02	0	0	0	273.15	303.15	< 25%	E
ISOBUTYRONITRILE	78-82-0	101	6.4645E+00	-5.1172E+02	0	0	0	273.15	303.15	< 25%	E
BENZONITRILE	100-47-0	101	2.6212E+02	-1.0995E+04	-3.8243E+01	0	0	273.15	363.15	< 25%	E
VINYLACETONITRILE	109-75-1	101	4.8136E+00	-3.2841E+01	0	0	0	273.15	303.15	< 5%	E
PHENYLACETONITRILE	140-29-4	101	7.9236E+00	1.1683E+02	0	0	0	273.15	303.15	< 100%	E
NITROAMINES											
o-NITROANILINE	88-74-4	101	-2.5696E+00	3.1847E+03	0	0	0	298.15	313.25	< 50%	E
p-NITROANILINE	100-01-6	101	-3.4304E+00	3.2308E+03	0	0	0	293.15	313.25	< 10%	E

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
ORGANIC SALTS											
DIETHYL CARBONATE	105-58-8	101	1.8930E+02	-8.8485E+03	-2.6978E+01	0	0	273.15	363.45	< 5%	E
DIMETHYL CARBONATE	616-38-6	101	1.7097E+02	-7.8524E+03	-2.4703E+01	0	0	273.35	333.15	< 3%	E
OTHER ALIPHATIC ALCOHOLS											
ISOPROPANOL	67-63-0	101	1.4959E+02	-7.7472E+03	-2.1323E+01	0	0	273	373.15	< 10%	E
2-METHYL-1-PROPANOL	78-83-1	101	2.7441E+02	-1.2923E+04	-3.9876E+01	0	0	273.35	374.15	< 10%	E
2-BUTANOL	78-92-2	101	1.8600E+02	-9.4446E+03	-2.6519E+01	0	0	273	371.15	< 5%	E
2-METHYL-2-PROPANOL	75-65-0	101	1.6764E+02	-9.0892E+03	-2.3636E+01	0	0	273	371.15	< 10%	E
2-PENTANOL	6032-29-7	101	2.1732E+02	-1.0616E+04	-3.1087E+01	0	0	273.35	371.15	< 5%	E
2-METHYL-2-BUTANOL	75-85-4	101	1.7560E+02	-9.3077E+03	-2.4714E+01	0	0	273.35	371.45	< 5%	E
2-METHYL-1-BUTANOL	137-32-6	101	2.5881E+02	-1.2308E+04	-3.7301E+01	0	0	273.35	363.15	< 5%	E
2,2-DIMETHYL-1-PROPANOL	75-84-3	101	2.2174E+02	-1.1079E+04	-3.1540E+01	0	0	285.15	363.15	< 25%	P
2-HEXANOL	626-93-7	101	2.3127E+02	-1.1094E+04	-3.2981E+01	0	0	273.15	363.35	< 25%	EP
3-HEXANOL	623-37-0	101	2.2426E+02	-1.1042E+04	-3.1824E+01	0	0	273.15	363.45	< 5%	EP
3-METHYL-1-PENTANOL	589-35-5	101	2.4864E+02	-1.1604E+04	-3.5685E+01	0	0	273.15	363.65	< 10%	P
3-METHYL-3-PENTANOL	77-74-7	101	7.3413E+01	-4.2766E+03	-9.4714E+00	0	0	282.95	363.55	< 10%	P
3-PENTANOL	584-02-1	101	2.1706E+02	-1.0682E+04	-3.1030E+01	0	0	273.35	371.15	< 5%	E
3-METHYL-1-BUTANOL	123-51-3	101	3.5250E+02	-1.6416E+04	-5.1307E+01	0	0	273.35	374.15	< 10%	E
3-METHYL-2-BUTANOL	598-75-4	101	2.1025E+02	-1.0387E+04	-3.0020E+01	0	0	273.35	371.15	< 3%	E
2-HEPTANOL	543-49-7	101	2.5987E+02	-1.2308E+04	-3.7057E+01	0	0	273.15	363.35	< 5%	P
3-HEPTANOL	589-82-2	101	2.2688E+02	-1.1106E+04	-3.2011E+01	0	0	273.15	363.35	< 10%	E
4-METHYL-2-PENTANOL	108-11-2	101	1.9584E+02	-9.5636E+03	-2.7711E+01	0	0	273.15	363.35	< 10%	EP
2-OCTANOL	123-96-6	101	3.7853E+02	-1.7786E+04	-5.4429E+01	0	0	273.15	363.65	< 10%	P
2-NONANOL	628-99-9	101	3.8316E+02	-1.8280E+04	-5.4709E+01	0	0	273.15	363.65	< 25%	P
2,6-DIMETHYL-4-HEPTANOL	108-82-7	101	1.6061E+02	-8.1347E+03	-2.1735E+01	0	0	273.15	363.65	< 25%	P

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
OTHER ALIPHATIC AMINES											
TRIETHYLAMINE	121-44-8	100	6.5469E+01	6.5156E+03	2.1156E+01	8.2308E-06	2	298.15	298.15	< 5%	S
DI-n-PROPYLAMINE	142-84-7	100	7.1297E+01	0	0	0	0	298.15	298.15	< 25%	S
DIETHYLAMINE	109-89-7	100	4.8211E+00	9.7708E+03	4.2873E+01	-3.6254E-02	1	298.15	298.15	< 25%	S
DI-n-BUTYLAMINE	111-92-2	100	1.2008E+03	1.3625E+04	3.5395E+01	0	0	298.15	298.15	< 50%	S
TRI-n-BUTYLAMINE	102-82-9	101	1.3366E+01	-1.1796E+03	0	0	0	273.15	353.15	< 25%	E
OTHER ALKANES											
2,3,4-TRIMETHYLPENTANE	565-75-3	100	4.7045E+06	1.7539E+03	0	0	0	298.15	298.15	< 5%	P
2,2,5-TRIMETHYLHEXANE	3522-94-9	101	2.0543E+01	-1.2384E+03	0	0	0	273.15	298.15	< 10%	E
OTHER ALKYL BENZENES											
o-XYLENE	95-47-6	101	2.6647E+02	-1.0911E+04	-3.8533E+01	0	0	273.15	353.15	< 10%	E
m-XYLENE	108-38-3	101	3.7934E+02	-1.2942E+04	4.4713E-02	-5.9466E+01	1	273.15	543.75	< 10%	E
p-XYLENE	106-42-3	101	7.1165E+02	-2.4322E+04	1.0910E-01	-1.1447E+02	1	273.15	568.05	< 10%	E
CUMENE	98-82-8	101	9.2874E+00	6.6679E+02	0	0	0	288.15	318.15	< 10%	E
1,2,3-TRIMETHYLBENZENE	526-73-8	101	7.5122E+00	1.2115E+03	0	0	0	288.15	318.15	< 25%	E
1,2,4-TRIMETHYLBENZENE	95-63-6	101	8.2919E+00	1.0180E+03	0	0	0	288.15	318.15	< 5%	E
MESITYLENE	108-67-8	101	1.6957E+02	-5.9219E+03	-2.4197E+01	0	0	283.15	373.15	< 25%	E
p-DIETHYLBENZENE	105-05-5	100	2.9628E+05	-5.1308E+03	0	0	0	293.15	293.15	< 100%	E
1,2,3,4-TETRAMETHYLBENZENE	488-23-3	101	1.0233E+01	8.1540E+02	0	0	0	288.15	308.15	< 100%	E
1,2,4,5-TETRAMETHYLBENZENE	95-93-2	100	6.0549E+05	0	0	0	0	298.15	298.15	< 10%	E
OTHER AMINES, IMINES											
CYCLOHEXYLAMINE	108-91-8	101	1.4299E+02	-8.3169E+03	-1.9689E+01	0	0	298.15	372.82	< 25%	E
TRIALLYLAMINE	102-70-5	101	3.2999E+02	-1.5808E+04	-4.7217E+01	0	0	273.15	363.15	< 25%	E

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
OTHER CONDENSED RINGS											
FLUORANTHENE	206-44-0	101	4.0885E+00	3.6092E+03	0	0	0	281.25	333.15	< 10%	E
FLUORENE	86-73-7	101	3.8046E+00	3.0197E+03	0	0	0	279.75	333.15	< 10%	E
ANTHRACENE	120-12-7	101	2.8549E+02	-2.0510E+04	-3.8180E+01	0	0	273.45	347.85	< 5%	E
PHENANTHRENE	85-01-8	101	2.4635E+02	-7.8027E+03	-3.6051E+01	0	0	277.15	346.55	< 5%	E
CHRYSENE	218-01-9	101	1.0112E+01	2.7117E+03	0	0	0	279.65	302.15	< 25%	E
PYRENE	129-00-0	101	2.8752E+02	-9.6725E+03	-4.1927E+01	0	0	277.85	347.85	< 5%	E
ACENAPHTHENE	83-32-9	101	4.2310E+00	2.6602E+03	0	0	0	277.25	347.85	< 25%	E
ACENAPHTHALENE	208-96-8	101	6.4639E+00	1.9187E+03	0	0	0	277.25	304.15	< 25%	E
BENZO[a]PYRENE	50-32-8	101	7.4109E+00	4.1135E+03	0	0	0	281.15	308.15	< 10%	E
BENZO[b]FLUORANTHENE	205-99-2	101	-2.0161E+00	6.3942E+03	0	0	0	283.15	328.15	< 10%	E
OTHER ETHERS/DIETHERS											
METHYLAL	109-87-5	101	6.6045E+00	-1.1224E+03	0	0	0	273.35	333.15	< 25%	E
1,2-DIMETHOXYETHANE	110-71-4	101	1.9979E+02	-1.1271E+04	-2.8177E+01	0	0	273.35	372.5	< 3%	E
DIETHYLENE GLYCOL DI-n-BUTYL ETHER	112-73-2	101	2.5849E+01	-5.0993E+03	0	0	0	273.15	293.15	< 100%	P
ANISOLE	100-66-3	101	3.7640E+00	1.3538E+03	0	0	0	275.15	363.85	< 5%	E
DIPHENYL ETHER	101-84-8	100	4.9071E+05	0	0	0	0	298.15	298.15	< 50%	S
OTHER INORGANICS											
HYDROGEN PEROXIDE	7722-84-1	101	1.1957E+00	-8.0509E+02	0	0	0	278.15	297.15	< 25%	E
OTHER MONOAROMATICS											
STYRENE	100-42-5	101	4.1793E+00	1.6920E+03	0	0	0	293.32	313.02	< 5%	E
OTHER POLYFUNCTIONAL C, H, O											
2-BUTOXYETHANOL	111-76-2	101	8.7199E+00	-1.5162E+03	0	0	0	293.15	303.15	< 25%	E
VANILLIN	121-33-5	101	-4.5897E+02	2.2705E+04	6.8209E+01	0	0	293.15	318.15	< 5%	E
OTHER POLYFUNCTIONAL C, H, N, O											

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
OTHER POLYFUNCTIONAL ORGANICS											
MALATHION	121-75-5	101	9.4811E+00	6.5738E+02	0	0	0	283.15	303.15	< 25%	E
OTHER SATURATED ALIPHATIC ESTERS											
ETHYL ISOVALERATE	108-64-5	101	1.9687E+02	-9.4132E+03	-2.7576E+01	0	0	273.15	363.65	< 10%	E
DIETHYL MALONATE	105-53-3	101	2.1777E+02	-1.0033E+04	-3.1264E+01	0	0	273.15	363.75	< 10%	P
ETHYL TRIMETHYL ACETATE	3938-95-2	101	1.7393E+02	-8.6446E+03	-2.3987E+01	0	0	273.15	363.55	< 25%	E
PEROXIDES											
METHYL HYDROPEROXIDE	3031-73-0	101	1.0534E+00	7.9863E+01	0	0	0	277.15	297.15	< 3%	E
ETHYL HYDROPEROXIDE	3031-74-1	101	2.7406E+00	-3.6257E+02	0	0	0	278.15	297.15	< 10%	E
POLYFUNCTIONAL AMIDES/AMINES											
N,N-DIMETHYLFORMAMIDE	68-12-2	101	1.4648E+02	-8.3423E+03	-2.0868E+01	0	0	289.4	372	< 25%	E
N-METHYLFORMAMIDE	123-39-7	101	2.0381E+00	-5.8045E+02	0	0	0	333.15	372.65	< 5%	E
N,N-DIMETHYLACETAMIDE	127-19-5	101	7.5170E+00	-2.4924E+03	0	0	0	343.15	372.28	< 10%	E
POLYFUNCTIONAL C, H, N, HALIDE, (O)											
o-CHLOROANILINE	95-51-2	101	1.2561E+03	-4.1122E+04	-2.0798E+02	2.4659E-01	1	298.15	431.15	< 25%	P
p-CHLORONITROBENZENE	100-00-5	101	2.1438E+00	2.3385E+03	0	0	0	283.15	313.15	< 10%	E
POLYFUNCTIONAL C, H, O, HALIDE											
DI(2-CHLOROETHYL)ETHER	111-44-4	101	1.5144E+02	-6.4595E+03	-2.1611E+01	0	0	273.15	364.85	< 5%	E
ETHYLCHLOROACETATE	105-39-5	101	1.9772E+02	-8.9873E+03	-2.8391E+01	0	0	273.15	333.25	< 10%	E
METHYL CHLOROACETATE	96-34-4	101	1.0528E+02	-4.4693E+03	-1.5019E+01	0	0	273.15	323.35	< 5%	E
POLYFUNCTIONAL C, H, O, N											
N-METHYL-2-PYRROLIDONE	872-50-4	101	3.9967E+02	-2.1700E+04	-5.7549E+01	0	0	298.8	372.85	< 5%	E

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
POLYFUNCTIONAL C, H, O, S											
DIMETHYL SULFOXIDE	67-68-5	101	2.9405E+00	-1.6085E+03	0	0	0	298.8	337.9	< 25%	E
POLYFUNCTIONAL ESTERS											
METHYL SALICYLATE	119-36-8	101	1.6195E+02	-6.2843E+03	-2.3185E+01	0	0	273.15	363.65	< 25%	E
2-ETHOXYETHYL ACETATE	111-15-9	100	4.7700E+01	0	0	0	0	303.15	303.15	< 10%	E
ETHYL ACETOACETATE	141-97-9	101	6.4902E+00	-6.8868E+02	0	0	0	273.15	303.15	< 25%	EP
ETHYLENE GLYCOL MONOBUTYL ETHER ACETATE	112-07-2	100	2.6459E+02	0	0	0	0	298.15	298.15	< 25%	E
PROPIONATES AND BUTYRATES											
METHYL PROPIONATE	554-12-1	101	8.7616E+00	-1.2878E+03	0	0	0	273.15	310.2	< 5%	E
ETHYL PROPIONATE	105-37-3	101	2.3017E+02	-1.0936E+04	-3.2963E+01	0	0	273.15	353.45	< 25%	E
n-PROPYL PROPIONATE	106-36-5	101	1.7662E+02	-8.7139E+03	-2.4641E+01	0	0	273.15	363.65	< 10%	E
n-BUTYL PROPIONATE	590-01-2	101	2.1594E+02	-1.0309E+04	-3.0373E+01	0	0	273.15	363.55	< 10%	E
n-PROPYL n-BUTYRATE	105-66-8	101	2.0811E+02	-9.7622E+03	-2.9327E+01	0	0	273.15	363.65	< 25%	E
METHYL n-BUTYRATE	623-42-7	101	1.0572E+02	-5.2463E+03	-1.4460E+01	0	0	273.15	363.65	< 25%	E
ETHYL n-BUTYRATE	105-54-4	101	1.4646E+02	-7.2997E+03	-2.0200E+01	0	0	273.15	363.65	< 25%	E
ISOBUTYL ISOBUTYRATE	97-85-8	101	2.2011E+02	-1.0408E+04	-3.0834E+01	0	0	273.15	363.45	< 25%	E
n-BUTYL n-BUTYRATE	109-21-7	101	2.9628E+02	-1.4154E+04	-4.2014E+01	0	0	273.15	363.45	< 10%	E
METHYL ISOBUTYRATE	547-63-7	101	1.4441E+02	-7.0836E+03	-2.0168E+01	0	0	273.15	353.45	< 10%	E
ETHYL ISOBUTYRATE	97-62-1	101	1.4146E+02	-7.0790E+03	-1.9442E+01	0	0	273.15	363.75	< 10%	P
n-PENTYL n-BUTYRATE	540-18-1	101	1.1097E+01	-2.1866E+02	0	0	0	273.15	363.65	< 25%	E
ISOPENTYL BUTYRATE	106-27-4	101	2.4293E+02	-1.1495E+04	-3.4028E+01	0	0	282.55	363.85	< 100%	E
SULFIDES/THIOPHENES											
DIMETHYL SULFIDE	75-18-3	101	1.6724E+02	-7.3055E+03	-2.4157E+01	0	0	272.35	353.45	< 10%	E
DIMETHYL DISULFIDE	624-92-0	101	4.5046E+00	8.8438E+02	0	0	0	273.2	343.15	< 10%	E

Table C.3: γ_i^∞ regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
2-METHYLTHIOPHENE	554-14-3	101	7.2541E+00	3.1691E+02	0	0	0	298.15	343.15	< 10%	E
DIISOPROPYL SULFIDE	625-80-9	101	9.7925E+00	-2.6053E+02	0	0	0	298.15	343.15	< 10%	E
UNSATURATED ALIPHATIC ESTERS											
ETHYL ACRYLATE	140-88-5	101	5.2318E+00	1.1910E+02	0	0	0	298.6	360.2	< 10%	P
METHYL METHACRYLATE	80-62-6	101	2.1314E+02	-9.7326E+03	-3.0636E+01	0	0	273.35	333.15	< 3%	E

Table C.4: x_i^{aq} regression summary

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
1-ALKENES											
1-PENTENE	109-67-1	100	4.5386E-05	0	0	0	0	298.15	298.15	< 25%	S
1-HEXENE	592-41-6	100	1.0983E-05	0	0	0	0	298.15	298.15	< 10%	S
1-HEPTENE	592-76-7	100	2.6578E-06	0	0	0	0	298.15	298.15	< 10%	S
1-OCTENE	111-66-0	100	6.4314E-07	-1.2631E+04	-3.2632E+01	0	0	298.15	298.15	< 10%	S
1-NONENE	124-11-8	100	1.5564E-07	0	0	0	0	298.15	298.15	< 25%	S
1-DECENE	872-05-9	100	3.7663E-08	-2.1023E+04	-4.4514E+01	0	0	298.15	298.15	< 25%	P
1-UNDECENE	872-05-9	100	9.1136E-09	-1.9152E+04	-3.3127E+01	0	0	298.15	298.15	< 25%	P
1-DODECENE	872-05-9	100	2.2055E-09	-8.1347E+03	-2.1735E+01	0	0	298.15	298.15	< 25%	P
1-TRIDECENE	592-41-6	100	5.3369E-10	1.6356E+03	0	0	0	298.15	298.15	< 25%	P
1-TETRADECENE	872-05-9	100	1.2916E-10	-1.8992E+04	-3.6878E+01	0	0	298.15	298.15	< 50%	P

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
1-PENTADECENE	13360-61-7	100	3.1253E-11	0	0	0	0	298.15	298.15	< 50%	S
2,3,4-ALKENES											
cis-2-PENTENE	627-20-3	100	5.2143E-05	0	0	0	0	298.15	298.15	< 5%	E
trans-2-PENTENE	646-04-8	100	5.2143E-05	0	0	0	0	298.15	298.15	< 5%	E
cis-2-HEPTENE	6443-92-1	100	2.7522E-06	0	0	0	0	298.15	298.15	< 10%	E
trans-2-HEPTENE	14686-13-6	100	2.7522E-06	0	0	0	0	298.15	298.15	< 10%	E
ACETATES											
METHYL ACETATE	79-20-9	101	-2.3084E+03	1.9554E+05	3.1910E+02	-1.4952E+07	-2	278.15	356.65	< 3%	E
ETHYL ACETATE	141-78-6	101	-1.3361E+03	1.0978E+05	1.8495E+02	-7.9991E+06	-2	273.15	343.65	< 3%	E
n-PROPYL ACETATE	109-60-4	101	4.8913E+00	-7.2626E+03	0	1.2351E+06	-2	273.15	363.35	< 5%	E
n-BUTYL ACETATE	123-86-4	101	3.1799E+00	-6.9848E+03	0	1.1870E+06	-2	273.15	363.65	< 10%	E
ISOBUTYL ACETATE	110-19-0	101	8.7151E+00	-1.0152E+04	0	1.6438E+06	-2	273.15	420.25	< 10%	E
ISOPENTYL ACETATE	123-92-2	101	6.9823E+00	-1.0058E+04	0	1.6541E+06	-2	273.15	363.85	< 10%	E
ISOPROPYL ACETATE	108-21-4	101	6.2087E+00	-8.1630E+03	0	1.4057E+06	-2	273.15	366.15	< 10%	E
sec-BUTYL ACETATE	105-46-4	101	3.8824E+00	-7.4666E+03	0	1.2842E+06	-2	273.15	363.65	< 10%	E
VINYL ACETATE	108-05-4	101	3.3355E+00	-5.4481E+03	0	8.6465E+05	-2	273.35	333.15	< 3%	E
n-PENTYL ACETATE	628-63-7	101	1.3007E+01	-1.4017E+04	0	2.2856E+06	-2	273.15	364.5	< 25%	E
n-HEXYL ACETATE	142-92-7	101	1.7463E+01	-1.7134E+04	0	2.6915E+06	-2	273.15	363.35	< 25%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
n-HEPTYL ACETATE	112-06-1	100	1.5184E-05	-1.5368E+04	-3.7557E+01	0	0	298.15	298.15	< 25%	P
n-OCTYL ACETATE	112-14-1	100	3.7799E-06	-1.1079E+04	-3.1540E+01	0	0	298.15	298.15	< 25%	P
ETHYLENE GLYCOL DIACETATE	111-55-7	101	6.4818E+00	-6.0277E+03	0	8.8752E+05	-2	273.15	363.25	< 3%	E
tert-BUTYL ACETATE	540-88-5	101	1.3823E+00	-6.1464E+03	0	1.1073E+06	-2	273.15	353.65	< 3%	E
CYCLOHEXYL ACETATE	622-45-7	101	4.5951E+00	-8.3650E+03	0	1.3772E+06	-2	273.15	363.45	< 10%	E
PHENYL ACETATE	122-79-2	101	6.0837E+00	-7.8765E+03	0	1.1694E+06	-2	273.15	363.95	< 10%	E
ALDEHYDES											
ACETALDEHYDE	75-07-0	101	-2.0000E+01	5.5703E+03	0	0	0	294.15	318.15	< 10%	P
PROPANAL	123-38-6	101	-1.0158E+01	2.4071E+03	0	0	0	285.45	323.15	< 25%	E
BUTANAL	123-72-8	101	-9.0064E+00	1.4733E+03	0	0	0	273.15	343.15	< 10%	E
2-METHYLPROPANAL	78-84-2	101	-8.9679E+00	1.3843E+03	0	0	0	273.15	333.15	< 5%	E
PENTANAL	110-62-3	101	-9.8645E+00	1.2720E+03	0	0	0	273.15	363.15	< 50%	E
HEPTANAL	111-71-7	101	1.5461E+01	-1.5603E+04	0	2.5355E+06	-2	273.15	363.15	< 10%	E
HEXANAL	66-25-1	101	1.7602E+01	-1.6591E+04	0	2.7535E+06	-2	273.15	373.15	< 25%	E
OCTANAL	124-13-0	101	-1.3858E+01	1.2453E+03	0	0	0	283.15	318.15	< 50%	E
NONANAL	124-19-6	101	-8.9336E+00	-8.6592E+02	0	0	0	283.15	318.15	< 50%	E
2-ETHYLHEXANAL	123-05-7	101	-8.2417E+00	-4.5642E+02	0	0	0	273.15	363.15	< 25%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
2-ETHYL-2-HEXENAL	645-62-5	101	-8.3157E+00	-1.0075E+02	0	0	0	273.15	363.15	< 25%	E
DECANAL	112-31-2	101	-1.6848E+01	1.0170E+03	0	0	0	298.15	318.15	< 50%	E
2-METHYL-BUTYRALDEHYDE	96-17-3	101	-9.8239E+00	1.2190E+03	0	0	0	273.15	363.15	< 25%	E
ACROLEIN	107-02-8	5000	-1.0495E+04	6.6687E+04	1.5256E+03	1.3239E+03	361.35	273.15	361.15	< 10%	E
trans-CROTONALDEHYDE	123-73-9	101	4.1230E+00	-4.8483E+03	0	8.0558E+05	-2	273.15	363.15	< 10%	E
METHACROLEIN	78-85-3	101	-6.5100E+00	7.1224E+02	0	0	0	273.15	333.15	< 10%	E
p-TOLUALDEHYDE	104-87-0	101	-5.9618E+00	-6.6380E+02	0	0	0	273.15	363.15	< 10%	E
BENZALDEHYDE	100-52-7	101	1.1413E+01	-1.0321E+04	0	1.4635E+06	-2	273.08	373.15	< 10%	E
ALIPHATIC ETHERS											
DIETHYL ETHER	60-29-7	101	-1.2269E+01	2.4068E+03	0	0	0	269.32	307.25	< 5%	E
DIISOPROPYL ETHER	108-20-3	101	-1.9779E+01	5.2627E+03	0	-3.8263E+05	-2	273.15	334.15	< 10%	E
DI-n-BUTYL ETHER	142-96-1	101	-1.5032E+01	1.3926E+03	0	0	0	273.15	363.65	< 25%	E
METHYL tert-BUTYL ETHER	1634-04-4	101	-8.9827E+00	2.3199E+02	0	3.0025E+05	-2	273.15	343.15	< 10%	E
METHYL ISOBUTYL ETHER	625-44-5	101	-1.5502E+01	2.8024E+03	0	0	0	273.15	298.15	< 5%	E
DI-n-HEXYL ETHER	112-58-3	100	6.8917E-08	-3.2876E+04	-7.4441E+01	0	0	298.15	298.15	< 25%	P
METHYL n-BUTYL ETHER	628-28-4	101	-1.8085E+01	3.4942E+03	0	0	0	273.15	298.15	< 25%	E
DI-n-PENTYL ETHER	693-65-2	100	2.2173E-06	-1.2187E+03	0	0	0	298.15	298.15	< 25%	P

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
METHYL tert-PENTYL ETHER	994-05-8	101	-1.4359E+00	-5.3257E+03	0	1.1530E+06	-2	273.15	352.35	< 10%	E
tert-BUTYL ETHYL ETHER	637-92-3	101	-1.5992E+01	2.9990E+03	0	0	0	278.15	318.15	< 25%	E
ETHYL tert-PENTYL ETHER	919-94-8	101	-1.7642E+01	2.9896E+03	0	0	0	288.15	318.15	< 25%	E
DI-n-PROPYL ETHER	111-43-3	101	1.4439E+01	-1.5271E+04	0	2.5850E+06	-2	273.15	333.25	< 10%	E
n-BUTYL ETHYL ETHER	628-81-9	101	5.5361E+00	-8.7602E+03	0	1.5059E+06	-2	273.15	363.85	< 25%	E
ALKYLCYCLOHEXANES											
METHYL-CYCLOHEXANE	108-87-2	101	2.2613E+01	-2.1770E+04	0	3.3465E+06	-2	293.15	443.95	< 25%	E
ETHYL-CYCLOHEXANE	1678-91-7	101	5.7107E+02	-6.9022E+04	-7.5554E+01	6.8089E+06	-2	274.15	443.95	< 10%	E
cis-1,2-DIMETHYL-CYCLOHEXANE	2207-01-4	101	9.2417E+00	-1.3497E+04	0	1.9708E+06	-2	274.15	313.15	< 3%	E
trans-1,2-DIMETHYL-CYCLOHEXANE	6876-23-9	101	4.0532E+02	-5.4401E+04	-5.2782E+01	5.6689E+06	-2	274.15	443.25	< 5%	E
ALKYL-CYCLOPENTANES											
METHYL-CYCLOPENTANE	96-37-7	100	9.1506E-06	-1.3517E+04	-3.1894E+01	0	0	298.15	298.15	< 3%	S
ALKYNES											
ETHYLACETYLENE	107-00-6	100	1.0280E-03	0	0	0	0	298.15	298.15	< 25%	S
1-PENTYNE	627-19-0	100	4.0450E-04	0	0	0	0	298.15	298.15	< 25%	S
1-HEXYNE	693-02-7	100	8.4660E-05	0	0	0	0	298.15	298.15	< 25%	S
1-OCTYNE	629-05-0	100	3.9470E-06	0	0	0	0	298.15	298.15	< 25%	S
1-NONYNE	3452-09-3	100	7.6320E-07	0	0	0	0	298.15	298.15	< 25%	S

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
1-HEPTYNE	628-71-7	100	1.8240E-05	0	0	0	0	298.15	298.15	< 25%	S
AROMATIC ALCOHOLS											
2,3-XYLENOL	526-75-0	6000	-1.5683E+01	1.6800E+03	2.0352E-02	-4.5266E+00	483.26	293.15	483.25	< 25%	E
2,4-XYLENOL	105-67-9	6000	-2.3457E+01	3.2524E+03	2.8887E-02	-3.8002E+00	486.06	298.15	486.05	< 25%	E
2,5-XYLENOL	95-87-4	6000	1.1654E+01	-3.1846E+03	-1.4892E-02	-5.5852E+00	492.16	293.15	492.15	< 25%	E
2,6-XYLENOL	576-26-1	6000	-2.9019E+01	4.1106E+03	3.6790E-02	-3.6459E+00	514.35	293.15	514.35	< 25%	E
3,4-XYLENOL	95-65-8	6000	-1.4145E+00	-4.9478E+02	1.1971E-04	-5.6703E+00	463.36	293.15	463.35	< 25%	E
3,5-XYLENOL	108-68-9	6000	3.1997E+00	-1.1427E+03	-6.1815E-03	-6.5122E+00	473.16	293.15	473.15	< 25%	E
BENZYL ALCOHOL	100-51-6	6000	-1.7550E+01	2.6630E+03	2.1755E-02	-4.1474E+00	421.66	273.15	421.65	< 10%	E
PHENOL	108-95-2	6000	-2.6083E+01	4.1611E+03	3.5914E-02	-5.3947E+00	341.46	273.15	341.45	< 10%	E
o-CRESOL	95-48-7	6000	-5.1769E+00	6.8531E+02	4.0436E-03	-5.4711E+00	439.66	273.15	439.65	< 25%	E
m-CRESOL	108-39-4	6000	-6.2483E+00	8.1278E+02	5.3851E-03	-5.4723E+00	421.86	272.95	421.85	< 10%	E
p-CRESOL	106-44-5	6000	1.3914E+01	-2.4840E+03	-2.4577E-02	-6.0623E+00	416.86	273.15	416.85	< 10%	E
p-ETHYLPHENOL	123-07-9	6000	-9.4585E+00	1.6912E+03	7.9368E-03	-7.7616E+00	460.26	298.15	460.25	< 50%	E
2-PHENYLETHANOL	60-12-8	101	1.0118E+01	-9.3144E+03	0	1.3785E+06	-2	283.15	373.15	< 10%	E
1-PHENYL-1-PROPANOL	93-54-9	101	8.8268E+00	-9.9590E+03	0	1.5984E+06	-2	293.15	363.65	< 5%	E
AROMATIC AMINES											
o-TOLUIDINE	95-53-4	101	-5.6696E+00	-4.1271E+01	0	0	0	298.15	323.15	< 25%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
m-TOLUIDINE	108-44-1	101	-5.6334E+00	-9.0608E+01	0	0	0	298.15	323.15	< 25%	E
p-TOLUIDINE	106-49-0	101	6.4902E+00	-3.8937E+03	0	0	0	293.15	313.15	< 5%	E
QUINOLINE	91-22-5	101	-6.5531E+02	5.0239E+04	9.1044E+01	-3.4520E+06	-2	273.15	498.2	< 25%	E
N,N-DIETHYLANILINE	91-66-7	101	-9.3555E+00	-4.0851E+02	0	0	0	273.15	333.15	< 10%	E
PYRIDINE	110-86-1	100	1	0	0	0	0	273.15	368.15	U	E
ANILINE	62-53-3	101	7.6638E+00	-7.1200E+03	0	1.0031E+06	-2	281.75	369.85	< 3%	E
2-METHYLPYRIDINE	109-06-8	100	1	0	0	0	0	273.15	368.15	U	E
o-ETHYLANILINE	578-54-1	101	5.2059E+00	-7.4833E+03	0	1.1569E+06	-2	283.15	363.15	< 10%	E
2,4,6-TRIMETHYL-PYRIDINE	108-75-8	6000	-2.4933E+01	3.0991E+03	3.6399E-02	-4.2595E+00	279.64	279.65	363.15	< 25%	E
2,6-DIMETHYL-PYRIDINE	108-48-5	6000	-5.0882E+01	7.5209E+03	7.6927E-02	-4.9826E+00	307.14	307.15	363.15	< 10%	E
3-METHYLPYRIDINE	108-99-6	100	1	0	0	0	0	273.15	368.15	U	E
4-METHYL-PYRIDINE	108-89-4	100	1	0	0	0	0	273.15	368.15	U	E
2,4-DIMETHYL-ANILINE	95-68-1	101	1.2321E+01	-1.2131E+04	0	1.8891E+06	-2	273.15	363.15	< 10%	E
N-ETHYLANILINE	103-69-5	101	7.1042E+00	-9.4288E+03	0	1.4841E+06	-2	273.15	363.15	< 5%	E
AROMATIC CARBOXYLIC ACIDS											
o-TOLUIC ACID	118-90-1	6000	1.5306E+00	-2.9748E+03	6.5706E-03	-3.3513E+00	434.36	278.15	434.35	< 25%	E
p-TOLUIC ACID	99-94-5	101	2.0163E+01	-1.4966E+04	0	1.7837E+06	-2	278.15	343.15	< 5%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
PHTHALIC ACID	88-99-3	101	-3.2559E+03	2.8099E+05	4.4935E+02	-2.2581E+07	-2	283.65	352.45	< 5%	E
m-TOLUIC ACID	99-04-7	101	2.4555E+01	-1.6880E+04	0	2.0431E+06	-2	278.15	348.15	< 3%	E
AROMATIC CHLORIDES											
BENZYL CHLORIDE	100-44-7	101	-4.8047E+00	-1.4232E+03	0	0	0	277.65	303.15	< 25%	E
MONO-CHLOROBENZENE	108-90-7	101	1.9378E+01	-1.6010E+04	0	2.2051E+06	-2	274.15	353.16	< 10%	E
o-DICHLOROBENZENE	95-50-1	101	4.6100E+01	-3.3265E+04	0	4.8466E+06	-2	278.15	318.15	< 10%	E
m-DICHLOROBENZENE	541-73-1	101	-7.2884E+00	-1.1202E+03	0	0	0	278.15	318.15	< 10%	E
p-DICHLOROBENZENE	106-46-7	101	1.9920E+01	-1.6897E+04	0	2.2424E+06	-2	278.15	346.55	< 10%	E
HEXACHLORO-BENZENE	118-74-1	101	-8.8149E+00	-3.8499E+03	0	0	0	278.15	318.15	< 25%	E
o-CHLOROTOLUENE	95-49-8	101	-7.5182E+00	-1.0005E+03	0	0	0	274.15	318.15	< 25%	E
p-CHLOROTOLUENE	106-43-4	101	-7.6410E+00	-9.8024E+02	0	0	0	278.15	318.15	< 25%	E
2,4-DICHLOROTOLUENE	95-73-8	101	-8.0197E+00	-1.4022E+03	0	0	0	278.15	323.1	< 10%	E
1,2,4-TRICHLORO-BENZENE	120-82-1	101	-7.9726E+00	-1.3232E+03	0	0	0	278.15	318.15	< 25%	E
1,2,3-TRICHLORO-BENZENE	87-61-6	101	-1.3063E+00	-3.5211E+03	0	0	0	277.15	323.15	< 25%	E
AROMATIC ESTERS											
BENZYL ACETATE	140-11-4	101	4.7216E+00	-7.2941E+03	0	1.0365E+06	-2	273.15	363.55	< 10%	E
DI-n-PROPYL PHTHALATE	131-16-8	100	6.6314E-06	0	0	0	0	298.15	298.15	< 25%	S
DI-n-HEXYL PHTHALATE	84-75-3	100	1.2643E-08	0	0	0	0	298.15	298.15	< 50%	S

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
METHYL BENZOATE	93-58-3	101	8.7933E+00	-9.7990E+03	0	1.4112E+06	-2	273.15	363.65	< 25%	E
ETHYL BENZOATE	93-89-0	101	2.7651E+00	-7.0158E+03	0	1.0392E+06	-2	273.15	363.45	< 25%	E
BENZYL FORMATE	104-57-4	101	2.3594E+01	-1.8379E+04	0	2.7903E+06	-2	273.15	363.65	< 25%	E
DIETHYL PHTHALATE	84-66-2	101	-3.2340E-01	-5.6461E+03	0	8.6853E+05	-2	278.15	323.15	< 10%	E
DI-n-BUTYL PHTHALATE	84-74-2	101	4.3454E+01	-3.4620E+04	0	5.1921E+06	-2	283.15	323.15	< 25%	E
DIMETHYL PHTHALATE	131-11-3	101	7.9570E+00	-8.8956E+03	0	1.2524E+06	-2	283.15	323.15	< 5%	E
C, H, BR COMPOUNDS											
BROMOMETHANE	74-83-9	101	-1.7476E+01	3.4457E+03	0	0	0	273.15	305.15	< 25%	E
BROMOETHANE	74-96-4	101	-8.0017E+00	4.4804E+02	0	0	0	273.15	303.15	< 3%	E
1,1,2,2-TETRABROMOETHANE	79-27-6	101	6.9918E+00	-6.0683E+03	0	0	0	298.15	323.15	< 50%	E
1-BROMOPROPANE	106-94-5	101	-1.0015E+01	6.1560E+02	0	0	0	273.15	303.15	< 5%	E
2-BROMOPROPANE	75-26-3	101	1.4812E+01	-1.3646E+04	0	2.0705E+06	-2	273.15	303.15	< 5%	E
1-BROMOBUTANE	109-65-9	100	7.4063E-05	0	0	0	0	298.15	298.15	< 25%	S
1-BROMOHEPTANE	629-04-9	100	7.4417E-07	0	0	0	0	298.15	298.15	< 25%	S
1,2-DIBROMOETHANE	106-93-4	101	-5.9500E+00	-5.3923E+02	0	0	0	273.15	363.75	< 10%	E
m-DIBROMOBENZENE	108-36-1	100	7.5328E-06	0	0	0	0	298.15	298.15	< 25%	E
BROMOBENZENE	108-86-1	101	2.4459E+01	-1.9666E+04	0	2.8011E+06	-2	283.15	343.15	< 10%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
TRIBROMO-METHANE	75-25-2	101	-6.7179E+00	-4.8067E+02	0	0	0	283	313.15	< 10%	E
DIBROMO-METHANE	74-95-3	101	-9.0240E-01	-3.2490E+03	0	4.5394E+05	-2	273.15	363.25	< 5%	E
2-BROMOBUTANE	78-76-2	100	1.2019E-04	0	0	0	0	298.15	298.15	< 10%	E
p-BROMOTOLUENE	106-38-7	101	-7.8116E+00	-1.0704E+03	0	0	0	293.1	322.91	< 25%	E
C, H, F COMPOUNDS											
METHYL FLUORIDE	593-53-3	101	-4.3912E+00	-3.6301E+03	0.0000E+00	8.6381E+05	-2	273.21	353.4	< 1%	E
PERFLUORO-n-HEXANE	355-42-0	100	4.8787E-09	2.9932E+03	0	0	0	298.15	298.15	< 100%	E
PERFLUORO-n-OCTANE	307-34-6	100	6.8663E-11	-3.6046E+04	0	5.5730E+06	-2	298.15	298.15	< 100%	E
1,2-DIFLUORO-BENZENE	367-11-3	101	-6.6918E+00	-5.9345E+02	0	0	0	298.15	322.89	< 25%	E
FLUOROBENZENE	462-06-6	101	2.9097E+00	-6.4889E+03	0	9.4766E+05	-2	273.15	353.15	< 10%	E
HEXAFLUORO-BENZENE	392-56-3	101	8.8895E+00	-1.1813E+04	0	1.8684E+06	-2	281.46	339.76	< 3%	E
PERFLUORO-n-PENTANE	678-26-2	100	7.2276E-08	3.2308E+03	0	0	0	298.15	298.15	< 100%	E
PERFLUORO-n-HEPTANE	335-57-9	100	5.6014E-10	0	0	0	0	298.15	298.15	< 100%	E
C, H, I COMPOUNDS											
METHYL IODIDE	74-88-4	101	1.7790E+01	-1.4071E+04	0	2.0544E+06	-2	273.15	322.91	< 5%	E
ETHYL IODIDE	75-03-6	101	1.3260E+01	-1.2282E+04	0	1.8002E+06	-2	273.15	303.15	< 3%	E
n-PROPYL IODIDE	107-08-4	101	-9.8622E+00	2.2204E+02	0	0	0	273.15	303.15	< 5%	E
ISOPROPYL IODIDE	75-30-9	101	-1.0652E+01	5.4025E+02	0	0	0	273.15	303.15	< 10%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
IODOBENZENE	591-50-4	101	1.7033E+01	-8.3179E+03	0	0	0	298	303.15	< 10%	E
n-HEXYL IODIDE	638-45-9	100	1.1954E-06	0	0	0	0	298.15	298.15	< 25%	S
n-BUTYL IODIDE	542-69-8	100	2.2740E-05	1.2631E+04	3.2632E+01	0	0	298.15	298.15	< 10%	S
C, H, MULTIHALOGEN COMPOUNDS											
DICHLORODIFLUORO-METHANE	75-71-8	101	-4.7872E+02	3.0993E+04	6.6398E+01	-1.1878E+06	-2	274	313.9	< 3%	E
TRICHLOROFLUORO-METHANE	75-69-4										
1,1,2-TRICHLORO-TRIFLUOROETHANE	76-13-1	101	-1.2463E+01	5.4155E+02	0	0	0	275.15	303.15	< 25%	E
C, H, NO2 COMPOUNDS											
NITROMETHANE	75-52-5	101	1.5480E+01	-1.0523E+04	0	1.4556E+06	-2	273.15	363.15	< 10%	E
NITROETHANE	79-24-3	101	5.9340E+00	-5.9512E+03	0	8.4315E+05	-2	273.15	363.25	< 5%	E
1-NITROPROPANE	108-03-2	101	6.2988E+00	-7.1087E+03	0	1.0499E+06	-2	273.15	363.65	< 10%	E
2-NITROPROPANE	79-46-9	101	4.1019E+00	-5.7427E+03	0	8.4502E+05	-2	292.85	363.35	< 5%	E
1-NITROBUTANE	627-05-4	100	6.5301E-04	0	0	0	0	298.15	298.15	< 25%	S
o-NITROTOLUENE	88-72-2	101	-5.7342E+00	-1.0869E+03	0	0	0	283.15	313.15	< 10%	E
p-NITROTOLUENE	99-99-0	101	-2.6968E+00	-2.1673E+03	0	0	0	281	338	< 10%	E
m-NITROTOLUENE	99-08-1	101	-7.4215E+00	-6.5552E+02	0	0	0	283.15	313.15	< 10%	E
NITROBENZENE	98-95-3	101	6.8466E+00	-8.1755E+03	0	1.1026E+06	-2	283.15	372.85	< 5%	E
m-DINITROBENZENE	99-65-0	101	1.3019E+00	-3.2988E+03	0	0	0	277.65	310.15	< 10%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
p-DINITROBENZENE	100-25-4	101	-2.1454E+00	-2.9164E+03	0	0	0	277.65	310.15	< 5%	E
2,4-DINITROTOLUENE	121-14-2	101	2.8417E+01	-1.9872E+04	0	2.4358E+06	-2	277.15	334.95	< 5%	E
2,6-DINITROTOLUENE	606-20-2	101	3.3784E+00	-4.2272E+03	0	0	0	278	314	< 5%	E
3,4-DINITROTOLUENE	610-39-9	101	2.0447E+00	-3.8841E+03	0	0	0	278.2	313.2	< 5%	E
2,4,6-TRINITROTOLUENE	118-96-7	101	3.5464E+01	-2.4251E+04	0	3.0504E+06	-2	275.55	334.15	< 10%	E
RDX	121-82-4	101	5.2818E+00	-5.7566E+03	0	1.4123E+05	-2	275.25	307.35	< 5%	E
HMX	2691-41-0	101	7.0734E+00	-6.6105E+03	0	0	0	275.25	305.45	< 10%	E
C1/C2 ALIPHATIC CHLORIDES											
CARBON TETRACHLORIDE	56-23-5	101	1.5476E+01	-1.4953E+04	0	2.2550E+06	-2	273.15	366.15	< 5%	E
METHYL CHLORIDE	74-87-3	101	-2.8687E+00	-4.8446E+03	0	1.1429E+06	-2	273.15	366.15	< 10%	E
ETHYL CHLORIDE	75-00-3	101	-1.8893E+02	1.0561E+04	2.5805E+01	0	0	285.45	366.15	< 25%	E
VINYL CHLORIDE	75-01-4	101	-1.5752E+01	2.1198E+03	0	1.3139E+05	-2	273.35	366.15	< 25%	E
DICHLORO-METHANE	75-09-2	101	1.4073E+00	-3.8511E+03	0	5.3262E+05	-2	273.15	366.15	< 10%	E
CHLOROFORM	67-66-3	101	6.1301E+00	-7.9847E+03	0	1.2365E+06	-2	273.15	366.15	< 3%	E
1,1-DICHLOROETHANE	75-34-3	101	9.8021E+00	-1.0195E+04	0	1.5486E+06	-2	273.15	366.15	< 5%	E
1,2-DICHLOROETHANE	107-06-2	101	1.1248E+01	-1.0523E+04	0	1.5653E+06	-2	273.15	366.15	< 10%	E
1,1,2-TRICHLOROETHANE	79-00-5	101	-1.4213E+00	-3.2057E+03	0	4.2655E+05	-2	273.15	363.95	< 5%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
HEXA-CHLOROETHANE	67-72-1	101	-1.0570E+01	-6.4713E+02	0	0	0	283.15	303.15	< 25%	E
1,1,1-TRICHLORO-ETHANE	71-55-6	101	-7.0243E+00	-1.1282E+03	0	1.9183E+05	-2	273.15	348.15	< 10%	E
1,1,1,2-TETRACHLORO-ETHANE	630-20-6	101	1.5281E+01	-1.4390E+04	0	2.1225E+06	-2	273.15	323.15	< 10%	E
1,1,2,2-TETRACHLORO-ETHANE	79-34-5	101	-7.3648E+00	-2.1019E+02	0	0	0	283.15	328.65	< 10%	E
TRICHLORO-ETHYLENE	79-01-6	101	9.3855E+00	-1.0828E+04	0	1.6277E+06	-2	274.22	366.15	< 5%	E
TETRACHLORO-ETHYLENE	127-18-4	101	1.0289E+01	-1.2404E+04	0	1.8415E+06	-2	273.15	364.95	< 10%	E
cis-1,2-DICHLORO-ETHYLENE	156-59-2	101	6.5856E+00	-8.1835E+03	0	1.2548E+06	-2	281.15	366.15	< 10%	E
trans-1,2-DICHLORO-ETHYLENE	156-60-5	101	-8.4863E+00	4.1422E+02	0	0	0	283.15	323.15	< 10%	E
PENTACHLORO-ETHANE	76-01-7	101	-1.7027E+01	2.0331E+03	0	0	0	293.15	298	< 25%	E
1,1-DICHLORO-ETHYLENE	75-35-4	101	1.0285E+01	-1.1316E+04	0	1.7781E+06	-2	275.15	363.65	< 10%	E
C3 & HIGHER ALIPHATIC CHLORIDES											
1,4-DICHLOROBUTANE	110-56-5	101	-2.1428E+00	-1.9291E+03	0	0	0	286.65	298.05	< 25%	E
1,5-DICHLOROPENTANE	628-76-2	101	-3.5394E+00	-2.1037E+03	0	0	0	274.15	295.85	< 50%	E
1,2-DICHLOROPROPANE	78-87-5	101	9.0358E+00	-9.7318E+03	0	1.4095E+06	-2	273.15	353.35	< 10%	E
ISOPROPYL CHLORIDE	75-29-6	101	-1.0554E+01	9.8685E+02	0	0	0	273.15	303.15	< 10%	E
1,2,3-TRICHLOROPROPANE	96-18-4	101	3.9845E+00	-3.6177E+03	0	0	0	275.05	318.15	< 25%	E
HEXACHLORO-1,3-BUTADIENE	87-68-3	101	-1.0024E+01	-1.4778E+03	0	0	0	275.15	298.15	< 50%	E

Table C.4: x_i^{aq} regression summary (cont.)[illegible]

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
DIALKENES											
1,4-CYCLOHEXADIENE	628-41-1	101	-7.5088E+00	-2.9759E+02	0	0	0	278.26	318.36	< 25%	E
DICARBOXYLIC ACIDS											
ADIPIC ACID	124-04-9	101	3.5900E+01	-2.0360E+04	0	2.3624E+06	-2	278.15	338.15	< 3%	E
SUCCINIC ACID	110-15-6	101	1.1306E+01	-5.6087E+03	0	2.8154E+05	-2	278.15	338.15	< 3%	E
GLUTARIC ACID	110-94-1	101	-2.4823E+03	2.1516E+05	3.4284E+02	-1.7292E+07	-2	276.55	342.15	< 3%	E
DIMETHYLALKANES											
2,2-DIMETHYLBUTANE	75-83-2	100	4.4181E-06	0	0	0	0	298.15	298.15	< 25%	S
2,3-DIMETHYLBUTANE	79-29-8	101	1.6456E+01	-1.7985E+04	0	2.7997E+06	-2	273.15	422.65	< 25%	E
2,4-DIMETHYL-PENTANE	108-08-7	100	7.3837E-07	0	0	0	0	298.15	298.15	< 50%	S
3,3-DIMETHYL-PENTANE	562-49-2	101	-1.7960E+03	1.5381E+05	2.4627E+02	-1.2163E+07	-2	298.15	423.64	< 25%	E
2,2-DIMETHYLHEXANE	590-73-8	101	2.9049E+01	-2.6493E+04	0	3.9458E+06	-2	274.35	313.15	< 3%	E
2,5-DIMETHYLHEXANE	592-13-2	101	3.8363E+01	-3.1483E+04	0	4.5955E+06	-2	275.15	313.05	< 3%	E
DIPHENYL/POLYAROMATICS											
BIPHENYL	92-52-4	101	4.2986E+01	-3.0707E+04	0	4.0885E+06	-2	293.15	337.65	< 10%	E
EPOXIDES											
2-METHYL-TETRAHYDROFURAN	96-47-9	101	-1.0213E+01	2.0161E+03	0	-4.0757E+03	-2	273.15	343.75	< 10%	E
TRIOXANE	110-88-3	6000	-5.1321E+02	7.6276E+04	8.5628E-01	-2.5488E+00	321.65	293.15	321.65	< 25%	E
TETRAHYDROPYRAN	142-68-7	101	-1.1151E+03	9.0997E+04	1.5415E+02	-6.4427E+06	-2	273.15	354.45	< 5%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
DIBENZOFURAN	132-64-9	101	-1.3449E+00	-3.9001E+03	0	0	0	278.15	318.15	< 25%	E
FORMATES											
ETHYL FORMATE	109-94-4	101	8.8137E+00	-7.2082E+03	0	1.0206E+06	-2	278.15	348.65	< 3%	E
n-PROPYL FORMATE	110-74-7	101	6.2828E+00	-7.2734E+03	0	1.1493E+06	-2	272.15	333.15	< 10%	E
n-BUTYL FORMATE	592-84-7	101	-3.0776E+00	-9.7124E+02	0	0	0	273.15	363.75	< 25%	E
ISOBUTYL FORMATE	542-55-2	101	1.6896E+00	-5.6214E+03	0	9.5112E+05	-2	273.15	333.55	< 10%	E
n-PENTYL FORMATE	638-49-3	101	6.3123E+00	-9.0250E+03	0	1.4307E+06	-2	273.15	363.65	< 25%	E
tert-BUTYL FORMATE	762-75-4	101	-3.8596E+00	-5.9611E+02	0	0	0	278.15	308.15	< 5%	E
KETONES											
METHYL ETHYL KETONE	78-93-3	101	-1.4231E+03	1.1758E+05	1.9698E+02	-8.5370E+06	-2	254.64	415.75	< 5%	E
3-PENTANONE	96-22-0	101	6.6297E+00	-7.9913E+03	0	1.3864E+06	-2	273.15	353.35	< 10%	E
METHYL ISOBUTYL KETONE	108-10-1	101	2.6363E+00	-6.2055E+03	0	1.1064E+06	-2	273.15	363.55	< 10%	E
3-HEPTANONE	106-35-4	101	4.8630E-01	-5.8367E+03	0	1.0527E+06	-2	273.15	363.25	< 10%	E
4-HEPTANONE	123-19-3	101	4.6691E+00	-8.4432E+03	0	1.4500E+06	-2	273.15	363.35	< 10%	E
3-HEXANONE	589-38-8	101	-1.2639E+01	1.9983E+03	0	0	0	293.15	310.2	< 25%	E
2-PENTANONE	107-87-9	101	3.8928E+00	-6.2893E+03	0	1.1324E+06	-2	273.15	363.65	< 10%	E
METHYL ISOPROPYL KETONE	563-80-4	101	-4.6269E+02	3.2695E+04	6.4733E+01	-1.7848E+06	-2	265.15	453.15	< 10%	E
2-HEXANONE	591-78-6	101	2.9923E+00	-6.5018E+03	0	1.1514E+06	-2	273.15	364.65	< 5%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
2-HEPTANONE	110-43-0	101	6.4375E+00	-9.3023E+03	0	1.5481E+06	-2	273.15	363.65	< 5%	E
5-METHYL-2-HEXANONE	110-12-3	101	7.1688E+00	-9.6863E+03	0	1.6139E+06	-2	273.15	362.85	< 5%	E
3,3-DIMETHYL-2-BUTANONE	75-97-8	101	-8.1543E-01	-4.0863E+03	0	7.8661E+05	-2	273.15	363.35	< 5%	E
DIISOBUTYL KETONE	108-83-8	101	7.1063E+00	-1.1438E+04	0	1.9006E+06	-2	273.15	363.85	< 10%	E
DIISOPROPYL KETONE	565-80-0	101	-2.1253E+00	-4.1921E+03	0	8.0333E+05	-2	273.15	363.45	< 10%	E
5-NONANONE	502-56-7	101	1.7303E+00	-8.2199E+03	0	1.4280E+06	-2	273.15	363.75	< 25%	E
2-NONANONE	821-55-6	101	5.8740E+00	-1.0490E+04	0	1.7112E+06	-2	273.15	364.35	< 25%	E
ACETYLACETONE	123-54-6	101	1.6717E+01	-1.1122E+04	0	1.5290E+06	-2	273.15	363.65	< 10%	E
ISOPHORONE	78-59-1	101	9.5687E+00	-1.0617E+04	0	1.7565E+06	-2	273.15	364.25	< 5%	E
CYCLOPENTANONE	120-92-3	6000	-1.6081E+01	3.1202E+03	1.6210E-02	-2.9219E+00	378.15	267.45	378.15	< 3%	E
CYCLOHEXANONE	108-94-1	101	6.6031E+00	-7.5035E+03	0	1.2904E+06	-2	273.15	363.85	< 5%	E
2-OCTANONE	111-13-7	101	7.7020E+00	-1.1250E+04	0	1.9033E+06	-2	273.15	364.15	< 25%	E
ACETOPHENONE	98-86-2	101	2.1398E+00	-4.5194E+03	0	5.3558E+05	-2	293.05	353.35	< 25%	E
3-OCTANONE	106-68-3	101	-1.6485E+00	-5.0117E+03	0	8.7634E+05	-2	273.15	364.15	< 10%	E
3-NONANONE	925-78-0	101	-9.4462E+00	-6.6198E+01	0	0	0	273.15	353.35	< 25%	E
2,6,8-TRIMETHYL-4-NONANONE	123-18-2	100	2.2065E-06	0	0	0	0	298.15	298.15	< 25%	P

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
MERCAPTANS											
ISOPROPYL MERCAPTAN	75-33-2	101	-7.3515E+00	9.3654E+01	0	0	0	288.15	318.15	< 10%	E
n-BUTYL MERCAPTAN	109-79-5	101	1.8888E+01	-1.6328E+04	0	2.4137E+06	-2	288.15	333.15	< 5%	E
METHYLALKANES											
2-METHYLPENTANE	107-83-5	101	1.6772E+01	-1.8364E+04	0	2.8566E+06	-2	273.15	422.65	< 25%	E
3-METHYLPENTANE	96-14-0	100	2.7067E-06	0	0	0	0	298.15	298.15	< 3%	S
2-METHYLHEXANE	591-76-4	100	4.5667E-07	0	0	0	0	298.15	298.15	< 5%	E
3-METHYLHEXANE	589-34-4	100	4.7716E-07	0	0	0	0	298.15	298.15	< 5%	E
3-METHYLHEPTANE	589-81-1	100	1.2491E-07	0	0	0	0	298.15	298.15	< 5%	E
4-METHYLHEPTANE	589-53-7	100	1.3224E-07	0	0	0	0	298.15	298.15	< 5%	E
N-ALCOHOLS											
1-BUTANOL	71-36-3	6000	-1.6135E+01	3.2105E+03	1.5570E-02	-5.1405E+00	398.31	273.15	398.3	< 3%	E
1-PENTANOL	71-41-0	101	7.8962E+00	-8.7669E+03	0	1.4281E+06	-2	273.15	368.95	< 3%	E
1-HEXANOL	111-27-3	101	8.2658E+00	-9.8737E+03	0	1.5999E+06	-2	273.15	363.45	< 10%	E
1-HEPTANOL	111-70-6	101	1.8876E+01	-1.7053E+04	0	2.6686E+06	-2	273.15	363.65	< 10%	E
1-OCTANOL	111-87-5	101	1.8301E+01	-1.7025E+04	0	2.5985E+06	-2	288.15	363.45	< 10%	EP
1-NONANOL	143-08-8	101	1.8177E+01	-1.7366E+04	0	2.5783E+06	-2	288.15	363.15	< 25%	E
1-DECANOL	112-30-1	101	2.2457E+01	-2.0308E+04	0	2.9573E+06	-2	288.15	363.15	< 10%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
1-UNDECANOL	112-42-5	100	1.0330E-06	0	0	0	0	298.15	298.15	< 50%	P
1-DODECANOL	112-53-8	100	2.5520E-07	0	0	0	0	298.15	298.15	< 50%	P
1-TRIDECANOL	112-70-9	100	6.3040E-08	0	0	0	0	298.15	298.15	< 50%	P
1-TETRADECANOL	112-72-1	100	1.5573E-08	-2.1975E+04	-4.5599E+01	0	0	298.15	298.15	< 50%	P
1-PENTADECANOL	629-76-5	100	3.8469E-09	-1.1094E+04	-3.2981E+01	0	0	298.15	298.15	< 100%	P
1-HEXADECANOL	36653-82-4	100	9.5034E-10	-7.2665E+03	0	1.0944E+06	-2	298.15	298.15	< 100%	P
1-HEPTADECANOL	1454-85-9	100	2.3477E-10	-1.2866E+04	-2.5451E+01	0	0	298.15	298.15	< 100%	P
1-OCTADECANOL	112-92-5	100	5.7993E-11	-5.9638E+03	-1.8057E+01	0	0	298.15	298.15	< 100%	P
N-ALIPHATIC PRIMARY AMINES											
METHYLAMINE	74-89-5	100	1	0	0	0	0	298.15	298.15	U	E
ETHYLAMINE	75-04-7	100	1	0	0	0	0	298.15	298.15	U	E
n-PROPYLAMINE	107-10-8	100	1	0	0	0	0	273.15	321.15	U	E
n-BUTYLAMINE	109-73-9	100	1	0	0	0	0	273.15	351.15	U	E
n-PENTYLAMINE	110-58-7	100	1	0	0	0	0	273.15	368.15	U	E
N-ALKANES											
n-PENTANE	109-66-0	101	-1.3911E+03	1.1170E+05	1.9221E+02	-8.0108E+06	-2	273.15	422.65	< 25%	E
n-HEXANE	110-54-3	101	-5.9005E+02	3.7354E+04	8.2875E+01	-1.8163E+06	-2	273.15	443.15	< 10%	E
n-HEPTANE	142-82-5	101	-6.1790E+02	3.9030E+04	8.6780E+01	-1.9566E+06	-2	273.15	443.75	< 25%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
n-OCTANE	111-65-9	101	2.6143E+01	-2.5670E+04	0	3.9079E+06	-2	273.15	456.15	< 25%	E
n-NONANE	111-84-2	101	4.0403E+01	-3.6046E+04	0	5.5730E+06	-2	287.95	409.75	< 50%	E
n-DECANE	124-18-5	100	4.3460E-09	0	0	0	0	298.15	298.15	< 50%	S
N-ALKYLBENZENES											
BENZENE	71-43-2	101	8.8569E+00	-9.5142E+03	0	1.3553E+06	-2	273.15	527.15	< 3%	E
TOLUENE	108-88-3	101	8.9793E+00	-1.0140E+04	0	1.4189E+06	-2	261.7	473.55	< 5%	E
ETHYLBENZENE	100-41-4	101	1.4054E+01	-1.4454E+04	0	2.1340E+06	-2	273.15	373.15	< 5%	E
n-PROPYLBENZENE	103-65-1	101	4.5679E+00	-8.9194E+03	0	1.2076E+06	-2	273.65	328.15	< 10%	E
n-BUTYLBENZENE	104-51-8	101	1.7850E+01	-1.7506E+04	0	2.4576E+06	-2	273.65	373.15	< 5%	E
n-HEPTYLBENZENE	1078-71-3	100	2.3098E-08	-1.0897E+04	-3.2652E+01	0	0	298.15	298.15	< 5%	P
n-DECYLBENZENE	104-72-3	100	2.0688E-10	-9.7708E+03	-4.2873E+01	3.6254E-02	1	298.15	298.15	< 10%	S
n-PENTYLBENZENE	538-68-1	101	1.1262E+01	-1.4633E+04	0	2.0603E+06	-2	280.15	318.15	< 10%	E
n-HEXYLBENZENE	1077-16-3	101	6.7333E+00	-1.2793E+04	0	1.7900E+06	-2	278.15	318.15	< 3%	E
n-OCTYLBENZENE	2189-60-8	100	4.9663E-09	0	0	0	0	298.15	298.15	< 10%	P
n-NONYLBENZENE	1081-77-2	100	1.0313E-09	0	0	0	0	298.15	298.15	< 10%	P
NAPHTHALENES											
NAPHTHALENE	91-20-3	101	2.0659E+01	-1.6308E+04	0	1.9288E+06	-2	275.05	346.55	< 3%	E
1-METHYL-NAPHTHALENE	90-12-0	101	-8.4391E+00	-1.2142E+03	0	0	0	277.25	304.85	< 10%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
2-METHYL-NAPHTHALENE	91-57-6	101	-2.6452E+00	-3.0013E+03	0	0	0	277.25	304.15	< 25%	E
1-ETHYL-NAPHTHALENE	1127-76-0	101	-7.4377E+00	-1.8483E+03	0	0	0	281.55	304.85	< 10%	E
NITRILES											
PROPIONITRILE	107-12-0	101	1.5998E+01	-1.1438E+04	0	1.6904E+06	-2	298.15	353.15	< 5%	E
ACRYLONITRILE	107-13-1	101	5.6317E+00	-5.4279E+03	0	7.8429E+05	-2	273.15	343.15	< 5%	E
METHACRYLO-NITRILE	126-98-7	101	1.2848E+01	-1.0849E+04	0	1.6366E+06	-2	273.15	353.15	< 10%	E
SUCCINONITRILE	110-61-2	6000	1.4744E+02	-2.4452E+04	-2.2705E-01	-3.3471E+00	329.01	305	329	< 3%	E
GLUTARONITRILE	544-13-8	5000	-1.3892E+04	8.9362E+04	2.0360E+03	1.7487E+03	342.00	290	342	< 10%	E
n-BUTYRONITRILE	109-74-0	101	1.1169E+00	-3.7470E+03	0	5.9779E+05	-2	273.15	363.15	< 3%	E
VALERONITRILE	110-59-8	101	1.5015E+01	-1.2632E+04	0	1.9077E+06	-2	273.15	363.15	< 10%	E
ISOBUTYRO-NITRILE	78-82-0	101	-4.5050E-01	-2.6707E+03	0	4.2005E+05	-2	273.15	363.15	< 5%	E
BENZONITRILE	100-47-0	101	-3.8294E+00	-1.0182E+03	0	0	0	273.15	363.15	< 25%	E
VINYL-ACETONITRILE	109-75-1	101	9.3859E+00	-8.2424E+03	0	1.2031E+06	-2	273.15	363.15	< 5%	E
PHENYL-ACETONITRILE	140-29-4	101	1.5843E+01	-1.3292E+04	0	1.8234E+06	-2	273.15	363.15	< 25%	E
NITROAMINES											
o-NITROANILINE	88-74-4	101	7.2633E+00	-4.8220E+03	0	0	0	298.15	313.25	< 50%	E
p-NITROANILINE	100-01-6	101	7.2258E+00	-4.9899E+03	0	0	0	293.15	313.25	< 10%	E
m-NITROANILINE	99-09-2	6000	2.2651E+00	-3.5004E+03	5.9660E-03	-1.8641E+00	458.36	273.15	458.35	< 10%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
ORGANIC SALTS											
DIETHYL CARBONATE	105-58-8	101	6.3888E+00	-8.1262E+03	0	1.3289E+06	-2	273.15	363.45	< 5%	E
DIMETHYL CARBONATE	616-38-6	101	4.6151E+00	-4.7448E+03	0	6.8616E+05	-2	282.95	333.55	< 3%	E
OTHER ALIPHATIC ALCOHOLS											
2-METHYL-1-PROPANOL	78-83-1	101	-2.3310E+03	1.9509E+05	3.2240E+02	-1.4587E+07	-2	273.15	407.25	< 10%	E
2-BUTANOL	78-92-2	6000	-2.3732E+01	4.4883E+03	2.6387E-02	-3.5525E+00	386.65	273.15	386.65	< 3%	E
2-PENTANOL	6032-29-7	101	6.9003E+00	-8.2230E+03	0	1.4184E+06	-2	273.15	363.45	< 5%	E
2-METHYL-2-BUTANOL	75-85-4	101	-1.8723E+03	1.5768E+05	2.5837E+02	-1.1765E+07	-2	273.65	401.75	< 10%	E
2-METHYL-1-BUTANOL	137-32-6	101	8.0268E+00	-8.9236E+03	0	1.4838E+06	-2	273.65	363.95	< 5%	E
2,2-DIMETHYL-1-PROPANOL	75-84-3	101	4.5624E+00	-6.9514E+03	0	1.2234E+06	-2	285.15	363.15	< 10%	E
2-HEXANOL	626-93-7	101	7.9408E+00	-9.6532E+03	0	1.6249E+06	-2	273.15	363.35	< 10%	E
3-HEXANOL	623-37-0	101	6.2630E+00	-8.7820E+03	0	1.5363E+06	-2	273.15	363.45	< 5%	E
3-METHYL-1-PENTANOL	589-35-5	101	1.0298E+01	-1.0896E+04	0	1.7641E+06	-2	273.15	363.65	< 5%	E
3-METHYL-3-PENTANOL	77-74-7	101	-9.2854E+00	1.2486E+03	0	0	0	282.95	363.55	< 10%	E
3-PENTANOL	584-02-1	101	7.0880E+00	-8.4508E+03	0	1.4859E+06	-2	273.15	363.15	< 5%	E
3-METHYL-1-BUTANOL	123-51-3	101	5.1443E+00	-7.1281E+03	0	1.2022E+06	-2	273.15	363.15	< 5%	E
3-METHYL-2-BUTANOL	598-75-4	101	-1.3435E+03	1.1083E+05	1.8565E+02	-8.0438E+06	-2	273.15	363.15	< 5%	E
2-HEPTANOL	543-49-7	101	8.9387E+00	-1.1008E+04	0	1.8246E+06	-2	273.15	363.35	< 10%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
3-HEPTANOL	589-82-2	101	5.5168E+00	-9.1575E+03	0	1.5956E+06	-2	273.15	363.35	< 10%	E
4-METHYL-2-PENTANOL	108-11-2	101	5.0667E+00	-7.8098E+03	0	1.3557E+06	-2	273.15	363.35	< 10%	E
2-OCTANOL	123-96-6	101	1.6068E+01	-1.6328E+04	0	2.6604E+06	-2	273.15	363.65	< 10%	E
2-NONANOL	628-99-9	101	1.3272E+01	-1.5882E+04	0	2.6529E+06	-2	273.15	363.65	< 25%	E
2,6-DIMETHYL-4-HEPTANOL	108-82-7	101	-3.2973E+00	-5.3209E+03	0	1.0358E+06	-2	273.15	363.65	< 25%	E
OTHER ALIPHATIC AMINES											
TRIETHYLAMINE	121-44-8	6000	2.7364E+01	-5.2885E+03	-3.9158E-02	-7.5913E+00	291.14	291.15	353.15	< 25%	E
DI-n-PROPYLAMINE	142-84-7	6000	1.1999E+01	-2.2185E+03	-2.2788E-02	-5.6220E+00	268.34	268.35	363.15	< 25%	E
N,N-DIMETHYL-n-BUTYLAMINE	927-62-8	6000	-2.4897E+01	4.2843E+03	1.9112E-02	-1.2491E+00	276.94	276.95	363.15	< 25%	E
N,N-DIETHYL-METHYLAMINE	616-39-7	6000	3.7296E+02	-6.6421E+04	-5.2238E-01	-1.0773E+01	322.05	322.06	351.43	< 25%	E
DIETHYLAMINE	109-89-7	100	1	0	0	0	0	273.15	328.15	U	E
DIISOPROPYLAMINE	108-18-9	6000	1.2988E+02	-2.2447E+04	-1.9105E-01	-8.2043E+00	300.17	300.18	357.15	< 25%	E
DI-n-BUTYLAMINE	111-92-2	101	4.2595E+00	-9.3265E+03	0	1.7549E+06	-2	273.15	363.15	< 10%	E
TRI-n-BUTYLAMINE	102-82-9	101	-1.3391E+01	1.1860E+03	0	0	0	273.15	353.15	< 25%	E
OTHER ALKANES											
2,3,4-TRIMETHYL-PENTANE	565-75-3	100	2.1260E-07	1.1974E+04	3.1657E+01	0	0	298.15	298.15	< 5%	S
2,2,5-TRIMETHYL-HEXANE	3522-94-9	101	-2.0551E+01	1.2394E+03	0	0	0	273.15	298.15	< 5%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
OTHER ALKYL BENZENES											
o-XYLENE	95-47-6	101	1.2005E+01	-1.2703E+04	0	1.8021E+06	-2	273.15	353.15	< 10%	E
m-XYLENE	108-38-3	101	1.1687E+01	-1.2479E+04	0	1.7520E+06	-2	273.15	543.75	< 10%	E
p-XYLENE	106-42-3	101	1.2851E+01	-1.3397E+04	0	1.9251E+06	-2	273.15	568.05	< 10%	E
CUMENE	98-82-8	101	-9.0761E+00	-7.3125E+02	0	0	0	288.15	318.15	< 10%	E
1,2,3-TRIMETHYL-BENZENE	526-73-8	101	-7.7694E+00	-1.1242E+03	0	0	0	288.15	318.15	< 25%	E
1,2,4-TRIMETHYL-BENZENE	95-63-6	101	-8.3822E+00	-9.8904E+02	0	0	0	288.15	318.15	< 5%	E
MESITYLENE	108-67-8	101	6.7148E+00	-9.8463E+03	0	1.2888E+06	-2	283.15	373.15	< 25%	E
p-DIETHYLBENZENE	105-05-5	100	3.3752E-06	-2.0510E+04	-3.8180E+01	0	0	293.15	293.15	< 100%	E
1,2,3,4-TETRAMETHYL-BENZENE	488-23-3	101	-1.0237E+01	-8.1465E+02	0	0	0	288.15	308.15	< 100%	E
1,2,4,5-TETRAMETHYL-BENZENE	95-93-2	100	4.6710E-07	0	0	0	0	298.15	298.15	< 10%	E
OTHER AMINES, IMINES											
CYCLOHEXYL-AMINE	108-91-8	100	1	0	0	0	0	273.15	363.15	U	E
HEXAMETHYLENE-TETRAMINE	100-97-0	101	-2.8632E+00	1.4809E+02	0	0	0	293.15	353.15	< 3%	E
N-ETHYL-2-METHYLALLYL-AMINE	18328-90-0	6000	-5.2494E+01	9.2838E+03	5.8177E-02	-4.3737E-02	291.76	291.75	363.15	< 25%	E
TRIALLYLAMINE	102-70-5	101	1.2461E+01	-1.3869E+04	0	2.3202E+06	-2	273.15	363.15	< 25%	E
N-METHYL-CYCLOHEXYLAMINE	100-60-7	6000	-2.1877E+01	2.8736E+03	2.7524E-02	-2.7938E+00	284.74	284.75	363.15	< 3%	E
DIALLYLAMINE	124-02-7	101	1.2788E+01	-1.3282E+04	0	2.4774E+06	-2	279.15	363.15	< 25%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
N-METHYLPIPERIDINE	626-67-5	6000	-2.6040E+01	3.9739E+03	3.2166E-02	-3.2579E+00	317.14	317.15	363.15	< 10%	E
OTHER CONDENSED RINGS											
FLUORANTHENE	206-44-0	101	3.6246E-01	-5.3775E+03	0	0	0	281.25	333.15	< 10%	E
FLUORENE	86-73-7	101	-4.1946E-01	-4.4885E+03	0	0	0	279.75	333.15	< 10%	E
ANTHRACENE	120-12-7	101	2.3500E+01	-1.9563E+04	0	2.0384E+06	-2	273.45	347.85	< 5%	E
PHENANTHRENE	85-01-8	101	1.9291E+01	-1.6237E+04	0	1.6996E+06	-2	277.15	346.55	< 5%	E
CHRYSENE	218-01-9	101	-8.7436E+00	-4.1710E+03	0	0	0	279.65	302.15	< 25%	E
PYRENE	129-00-0	101	2.0676E+01	-1.8288E+04	0	1.9905E+06	-2	277.85	347.85	< 5%	E
ACENAPHTHENE	83-32-9	101	2.2202E+00	-5.0460E+03	0	0	0	277.25	347.85	< 25%	E
ACENAPHTHALENE	208-96-8	101	-4.9130E+00	-2.5248E+03	0	0	0	277.25	304.15	< 25%	E
BENZO[a]PYRENE	50-32-8	101	-6.0895E+00	-5.0166E+03	0	0	0	281.15	308.15	< 10%	E
BENZANTHRACENE	56-55-3	101	1.1722E+02	-7.6026E+04	0	1.0373E+07	-2	276.85	308.15	< 10%	E
BENZO[b]FLUOR-ANTHENE	205-99-2	101	6.9573E+00	-8.9149E+03	0	0	0	283.15	328.15	< 10%	E
OTHER ETHERS/DIETHERS											
METHYLAL	109-87-5	101	-3.7046E+00	3.8855E+02	0	0	0	273.15	313.15	< 10%	E
1,2-DIMETHOXYETHANE	110-71-4	101	-5.6795E+00	8.8936E+02	0	0	0	273.15	313.15	< 25%	E
DIETHYLENE GLYCOL DIETHYL ETHER	112-36-7	6000	-6.1762E+01	1.1670E+04	6.9790E-02	4.4908E-02	301.14	301.15	363.25	< 10%	E
DIETHYLENE GLYCOL DI-n-BUTYL ETHER	112-73-2	101	1.4893E+01	-1.7993E+04	0	3.2683E+06	-2	273.15	363.15	< 10%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
ANISOLE	100-66-3	101	-3.7251E+00	-1.3661E+03	0	0	0	275.15	363.85	< 5%	E
DIPHENYL ETHER	101-84-8	100	1.9486E-06	-1.1616E+04	-2.9743E+01	0	0	298.15	298.15	< 5%	S
1,2-DIETHOXYETHANE	629-14-1	101	-1.6857E+01	5.9191E+03	0	-5.6886E+05	-2	273.15	363.15	< 5%	E
OTHER MONOAROMATICS											
STYRENE	100-42-5	101	-4.2008E+00	-1.6858E+03	0	0	0	293.32	313.02	< 5%	E
OTHER POLYFUNCTIONAL C, H, O											
FURFURAL	98-01-1	6000	-1.8103E+01	2.6482E+03	2.4606E-02	-3.5623E+00	394.05	273.15	394.05	< 5%	E
2-BUTOXYETHANOL	111-76-2	101	6.8405E+01	-5.1749E+04	0	9.2209E+06	-2	322.15	401.15	< 10%	E
2-HEXOXYETHANOL	112-25-4	101	1.1508E+01	-1.3242E+04	0	2.3543E+06	-2	273.15	363.15	< 25%	E
2-(2-HEXOXYETHOXY)-ETHANOL	112-59-4	101	4.5719E+01	-3.4663E+04	0	5.7078E+06	-2	285.15	363.15	< 25%	E
VANILLIN	121-33-5	101	-2.5626E+01	1.5128E+04	0	-2.8283E+06	-2	293.15	318.15	< 3%	E
PROPYLENE GLYCOL n-BUTYL ETHER	5131-66-8	101	-1.8709E+00	-3.8554E+03	0	8.8426E+05	-2	268.15	363.15	< 10%	E
OTHER POLYFUNCTIONAL ORGANICS											
MALATHION	121-75-5	101	-9.4825E+00	-6.5710E+02	0	0	0	283.15	303.15	< 25%	E
THIOUREA	62-56-6	101	-1.2285E+00	2.9718E+03	0	-1.0679E+06	-2	290.75	358.15	< 1%	E
OTHER SATURATED ALIPHATIC ESTERS											
ETHYL ISOVALERATE	108-64-5	101	3.2211E+00	-7.9778E+03	0	1.3642E+06	-2	273.15	363.65	< 10%	E
DIETHYL MALONATE	105-53-3	101	9.1270E+00	-9.7095E+03	0	1.5509E+06	-2	273.15	363.75	< 10%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
DIETHYL SUCCINATE	123-25-1	101	6.8326E+00	-8.6091E+03	0	1.4125E+06	-2	273.15	363.85	< 10%	E
ETHYL TRIMETHYL ACETATE	3938-95-2	101	1.5042E-01	-6.4989E+03	0	1.1890E+06	-2	273.15	363.55	< 25%	E
DIMETHYL SUCCINATE	106-65-0	101	7.0324E+00	-6.6630E+03	0	1.0004E+06	-2	293.75	364.75	< 3%	E
DIMETHYL-MALONATE	108-59-8	101	1.2864E+01	-9.3645E+03	0	1.3133E+06	-2	273.15	363.35	< 10%	E
POLYFUNCTIONAL ACIDS											
SALICYLIC ACID	69-72-7	101	2.8389E+01	-1.8578E+04	0	2.2896E+06	-2	273.08	348.15	< 5%	E
POLYFUNCTIONAL AMIDES/AMINES											
UREA	57-13-6	101	3.6465E+00	-1.5491E+03	0	1.7577E+04	-2	273.15	373.15	< 3%	E
ACETAMIDE	60-35-5	101	5.4067E+00	-2.3153E+03	0	1.3645E+05	-2	273.15	336.15	< 3%	E
N,N-DIETHYL-HYDROXYLAMINE	3710-84-7	101	-6.6884E+02	5.7896E+04	9.1700E+01	-4.4550E+06	-2	273.15	363.15	< 3%	E
POLYFUNCTIONAL C, H, N, HALIDE, (O)											
o-CHLOROANILINE	95-51-2	101	1.3102E+01	-1.2484E+04	0	1.9616E+06	-2	298.15	431.15	< 10%	E
p-CHLORONITRO-BENZENE	100-00-5	101	-1.8812E+00	-2.5635E+03	0	0	0	283.15	313.15	< 10%	E
POLYFUNCTIONAL C, H, O, HALIDE											
DI(2-CHLOROETHYL)-ETHER	111-44-4	101	5.3292E+00	-7.1469E+03	0	1.0658E+06	-2	273.15	364.85	< 5%	E
ETHYL-CHLOROACETATE	105-39-5	101	6.9181E+00	-8.0820E+03	0	1.2777E+06	-2	273.15	363.65	< 10%	E
o-CHLOROPHENOL	95-57-8	6000	-1.1011E+01	1.0640E+03	1.3317E-02	-3.2801E+00	446.15	288.55	446.15	< 5%	E
m-CHLOROPHENOL	108-43-0	6000	-5.3005E+00	2.3222E+02	5.1282E-03	-4.2920E+00	403.96	298.15	403.95	< 5%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
p-CHLOROPHENOL	106-48-9	6000	-1.6519E+01	2.3389E+03	2.0971E-02	-5.1532E+00	402.41	272.95	402.4	< 25%	E
METHYL CHLOROACETATE	96-34-4	101	5.2524E+00	-5.9225E+03	0	8.7840E+05	-2	273.15	363.25	< 5%	E
POLYFUNCTIONAL C, H, O, N											
ACETAMINOPHEN	103-90-2	101	3.0067E+01	-1.9166E+04	0	2.4749E+06	-2	278.15	343.15	< 5%	E
ACETANILIDE	103-84-4	101	2.3705E+00	-2.8256E+03	0	0	0	293.15	303.15	< 25%	E
POLYFUNCTIONAL ESTERS											
METHYL SALICYLATE	119-36-8	101	6.3814E+00	-8.3884E+03	0	1.1532E+06	-2	273.15	363.65	< 25%	E
2-ETHOXYETHYL ACETATE	111-15-9	101	-7.8319E+00	1.3675E+03	0	0	0	273.15	363.15	< 5%	E
ETHYL ACETOACETATE	141-97-9	101	-6.1053E+00	5.8199E+02	0	0	0	273.15	303.15	< 25%	E
ETHYLENE GLYCOL MONOBUTYL ETHER ACETATE	112-07-2	101	1.0526E+01	-1.1809E+04	0	2.0227E+06	-2	273.15	363.35	< 25%	E
POLYOLS											
p-HYDROQUINONE	123-31-9	101	3.4838E+01	-2.0077E+04	0	2.4975E+06	-2	280.5	342.1	< 3%	E
XYLITOL	87-99-0	101	6.3503E+00	-2.4082E+03	0	0	0	293.28	328.21	< 5%	E
1,3-BENZENEDIOL	108-46-3	101	5.3815E+00	-2.0610E+03	0	0	0	298.15	382.55	< 3%	E
PROPIONATES AND BUTYRATES											
METHYL PROPIONATE	554-12-1	101	-5.4523E-01	-3.0228E+03	0	5.6240E+05	-2	273.15	343.35	< 3%	E
ETHYL PROPIONATE	105-37-3	101	8.6870E+00	-9.6576E+03	0	1.6020E+06	-2	273.15	353.45	< 25%	E
n-PROPYL PROPIONATE	106-36-5	101	2.3221E+00	-6.9160E+03	0	1.2331E+06	-2	273.15	363.65	< 10%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
n-BUTYL PROPIONATE	590-01-2	101	4.6182E+00	-8.9538E+03	0	1.5193E+06	-2	273.15	363.55	< 5%	E
n-PROPYL n-BUTYRATE	105-66-8	101	4.9548E+00	-8.9029E+03	0	1.4776E+06	-2	273.15	363.65	< 25%	E
METHYL n-BUTYRATE	623-42-7	101	2.4057E+00	-5.8594E+03	0	1.0181E+06	-2	273.15	363.65	< 25%	E
ETHYL n-BUTYRATE	105-54-4	101	-1.0853E-01	-5.2994E+03	0	9.7757E+05	-2	273.15	363.65	< 25%	E
ISOBUTYL ISOBUTYRATE	97-85-8	101	3.3149E+00	-8.8385E+03	0	1.4943E+06	-2	273.15	363.45	< 25%	E
n-BUTYL n-BUTYRATE	109-21-7	101	8.5669E+00	-1.2335E+04	0	2.0772E+06	-2	273.15	363.45	< 10%	E
METHYL ISOBUTYRATE	547-63-7	101	1.6346E+00	-5.4561E+03	0	9.7055E+05	-2	273.15	353.45	< 3%	E
ETHYL ISOBUTYRATE	97-62-1	101	-2.9548E-01	-5.2398E+03	0	9.7076E+05	-2	273.15	363.75	< 10%	E
n-PENTYL n-BUTYRATE	540-18-1	101	-1.1103E+01	2.1889E+02	0	0	0	273.15	363.65	< 25%	E
ISOPENTYL BUTYRATE	106-27-4	101	4.3536E+00	-1.0210E+04	0	1.7238E+06	-2	282.55	363.85	< 100%	E
SULFIDES/THIOPHENES											
DIMETHYL SULFIDE	75-18-3	101	6.9652E+00	-7.2665E+03	0	1.0944E+06	-2	272.35	353.45	< 25%	EP
DIMETHYL DISULFIDE	624-92-0	101	-4.5077E+00	-8.8322E+02	0	0	0	273.2	343.15	< 10%	P
2-METHYLTHIOPHENE	554-14-3	101	-7.2777E+00	-3.1058E+02	0	0	0	298.15	343.15	< 10%	E
DIISOPROPYL SULFIDE	625-80-9	101	-1.2690E+01	2.1122E+03	0	-2.9537E+05	-2	298.15	343.15	< 25%	E
UNSATURATED ALIPHATIC ESTERS											
ETHYL ACRYLATE	140-88-5	101	1.0372E+00	-4.1664E+03	0	6.4977E+05	-2	298.6	360.2	< 10%	E
METHYL METHACRYLATE	80-62-6	101	7.2435E+00	-8.2103E+03	0	1.2803E+06	-2	273.15	353.15	< 5%	E

Table C.4: x_i^{aq} regression summary (cont.)

Name	CAS	Eq	A	B	C	D	E	Tmin	Tmax	s	DT
DIETHYL MALEATE	141-05-9	101	7.0429E+00	-8.5650E+03	0	1.3541E+06	-2	273.15	364.05	< 5%	E
DIMETHYL MALEATE	624-48-6	101	3.5518E+00	-4.5854E+03	0	6.4474E+05	-2	273.15	363.65	< 3%	E

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm)

		x_i^{aq} (mol frac.)			k_H (kPa)			γ_i^∞ (unitless)		
Name	CAS	Value	s	DT	Value	s	DT	Value	s	DT
1-ALKENES										
1-PENTENE	109-67-1	4.5386E-05	< 25%	S	1.8667E+06	< 25%	S	2.2033E+04	< 25%	S
1-HEXENE	592-41-6	1.0983E-05	< 10%	S	2.1902E+06	< 25%	S	9.1053E+04	< 10%	S
1-HEXENE	592-41-6	1.0983E-05	< 10%	S	2.1902E+06	< 25%	S	9.1053E+04	< 10%	S
1-HEPTENE	592-76-7	2.6578E-06	< 10%	S	2.8281E+06	< 10%	S	3.7625E+05	< 10%	S
1-OCTENE	111-66-0	6.4314E-07	< 10%	S	3.5550E+06	< 10%	S	1.5549E+06	< 10%	S
1-NONENE	124-11-8	1.5564E-07	< 25%	S	4.8537E+06	< 25%	S	6.4250E+06	< 25%	S
1-DECENE	872-05-9	3.7663E-08	< 25%	P	6.4586E+06	< 25%	P	2.6552E+07	< 25%	P
1-DECENE	872-05-9	3.7663E-08	< 25%	P	6.4586E+06	< 25%	P	2.6552E+07	< 25%	P
1-DECENE	872-05-9	3.7663E-08	< 25%	P	6.4586E+06	< 25%	P	2.6552E+07	< 25%	P
1-DECENE	872-05-9	3.7663E-08	< 25%	P	6.4586E+06	< 25%	P	2.6552E+07	< 25%	P
1-PENTADECENE	13360-61-7	3.1253E-11	< 50%	S	2.4928E+07	< 50%	S	3.1997E+10	< 50%	S
2,3,4-ALKENES										
cis-2-PENTENE	627-20-3	5.2143E-05	< 5%	E	1.2624E+06	< 5%	D	1.9178E+04	< 5%	D
trans-2-PENTENE	646-04-8	5.2143E-05	< 5%	E	1.2932E+06	< 5%	D	1.9178E+04	< 5%	D
cis-2-HEPTENE	6443-92-1	2.7522E-06	< 10%	E	2.3412E+06	< 10%	D	3.6334E+05	< 10%	D
trans-2-HEPTENE	14686-13-6	2.7522E-06	< 10%	E	2.2918E+06	< 10%	D	3.6334E+05	< 10%	D
ACETATES										

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

		x_i^{aq} (mol frac.)			k_H (kPa)			γ_i^∞ (unitless)		
Name	CAS	Value	s	DT	Value	s	DT	Value	s	DT
METHYL ACETATE	79-20-9	6.9553E-02	< 3%	S	6.7019E+02	< 10%	S	2.3439E+01	< 10%	S
ETHYL ACETATE	141-78-6	1.6041E-02	< 3%	S	8.4064E+02	< 10%	S	6.7660E+01	< 10%	S
n-PROPYL ACETATE	109-60-4	3.7965E-03	< 5%	S	1.2278E+03	< 10%	S	2.7549E+02	< 10%	S
n-BUTYL ACETATE	123-86-4	1.0141E-03	< 10%	S	1.4682E+03	< 10%	S	9.8461E+02	< 10%	S
ISOBUTYL ACETATE	110-19-0	1.0669E-03	< 10%	S	2.1658E+03	< 10%	S	9.1034E+02	< 10%	S
ISOPENTYL ACETATE	123-92-2	2.9009E-04	< 10%	S	2.7926E+03	< 10%	S	3.4340E+03	< 10%	S
ISOPROPYL ACETATE	108-21-4	4.7173E-03	< 10%	S	1.6585E+03	< 25%	S	2.0593E+02	< 25%	S
sec-BUTYL ACETATE	105-46-4	1.2134E-03	< 10%	S	2.4256E+03	< 10%	S	8.1913E+02	< 10%	S
VINYL ACETATE	108-05-4	5.4579E-03	< 3%	S	2.7687E+03	< 3%	S	1.8316E+02	< 10%	S
n-PENTYL ACETATE	628-63-7	2.5034E-04	< 10%	S	2.2402E+03	< 10%	S	3.9828E+03	< 10%	S
n-HEXYL ACETATE	142-92-7	5.9728E-05	< 25%	S	2.9135E+03	< 25%	S	1.6595E+04	< 25%	S
n-HEPTYL ACETATE	112-06-1	1.5184E-05	< 25%	P	3.8109E+03	< 25%	P	6.5861E+04	< 25%	P
n-OCTYL ACETATE	112-14-1	3.7799E-06	< 25%	P	5.4713E+03	< 25%	P	2.6455E+05	< 25%	P
ETHYLENE GLYCOL DIACETATE	111-55-7	2.3492E-02	< 3%	S	-	-	-	-	-	-
tert-BUTYL ACETATE	540-88-5	1.1407E-03	< 3%	S	4.7259E+03	< 3%	S	8.7411E+02	< 10%	S
CYCLOHEXYL ACETATE	622-45-7	3.4601E-04	< 10%	S	5.0496E+02	< 10%	S	2.8802E+03	< 25%	S
PHENYL ACETATE	122-79-2	7.6266E-04	< 5%	S	4.2332E+01	< 5%	S	1.3057E+03	< 25%	S
ALDEHYDES										
ACETALDEHYDE	75-07-0	2.6796E-01	< 10%	P	3.7813E+02	< 10%	S	-	-	-
PROPANAL	123-38-6	1.2435E-01	< 25%	S	4.3628E+02	< 5%	S	1.0275E+01	< 5%	S
BUTANAL	123-72-8	1.7165E-02	< 10%	S	6.0597E+02	< 5%	S	4.0709E+01	< 10%	S
2-METHYLPROPANAL	78-84-2	1.3236E-02	< 3%	S	1.0934E+03	< 5%	S	5.1912E+01	< 5%	S
PENTANAL	110-62-3	3.7043E-03	< 25%	S	8.3509E+02	< 5%	S	1.8344E+02	< 10%	S
HEPTANAL	111-71-7	2.3651E-04	< 10%	S	1.6308E+03	< 10%	S	3.1958E+03	< 25%	S

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

Name	CAS	x_i^{aq} (mol frac.)			k_H (kPa)	s	DT	γ_i^∞ (unitless)		
		Value	s	DT				Value	s	DT
HEXANAL	66-25-1	8.5071E-04	< 25%	S	1.2229E+03	< 25%	S	8.4624E+02	< 25%	S
OCTANAL	124-13-0	6.2433E-05	< 50%	S	2.7045E+03	< 10%	S	1.7662E+04	< 25%	S
NONANAL	124-19-6	7.2257E-06	< 50%	S	4.9990E+03	< 25%	S	1.3820E+05	< 25%	S
2-ETHYLHEXANAL	123-05-7	5.6992E-05	< 25%	S	4.7357E+03	< 25%	S	1.7457E+04	< 25%	S
2-ETHYL-2-HEXENAL	645-62-5	1.7449E-04	< 25%	S	1.0617E+03	< 50%	S	5.6988E+03	< 50%	S
DECANAL	112-31-2	1.4607E-06	< 50%	S	8.7393E+03	< 100%	S	6.8367E+05	< 100%	S
2-METHYLBUTYRALDEHYDE	96-17-3	3.2304E-03	< 10%	S	2.2328E+03	< 25%	P	3.0863E+02	< 25%	P
ACROLEIN	107-02-8	8.6576E-02	< 10%	S	7.6218E+02	< 100%	E	2.0967E+01	> 100%	S
trans-CROTONALDEHYDE	123-73-9	4.6146E-02	< 10%	S	1.0285E+02	< 25%	S	2.1898E+01	< 25%	S
METHACROLEIN	78-85-3	1.6226E-02	< 10%	S	1.2347E+03	< 10%	S	6.1659E+01	< 10%	S
p-TOLUALDEHYDE	104-87-0	2.7792E-04	< 10%	S	1.0440E+02	< 50%	S	2.4875E+03	< 50%	S
BENZALDEHYDE	100-52-7	1.1842E-03	< 10%	S	1.4523E+02	< 10%	S	8.5631E+02	< 10%	S
m-TOLUALDEHYDE	620-23-5	-	-	-	1.6029E+02	< 25%	S	3.5526E+03	< 50%	S
ALIPHATIC ETHERS										
DIETHYL ETHER	60-29-7	1.5055E-02	< 3%	S	5.7579E+03	< 3%	S	8.0282E+01	< 5%	S
DIISOPROPYL ETHER	108-20-3	1.6091E-03	< 10%	S	1.2343E+04	< 10%	S	6.2256E+02	< 10%	S
DI-n-BUTYL ETHER	142-96-1	3.1637E-05	< 25%	S	2.6272E+04	< 25%	S	3.1840E+04	< 25%	S
METHYL tert-BUTYL ETHER	1634-04-4	8.0111E-03	< 5%	S	4.1832E+03	< 5%	S	1.2555E+02	< 10%	S
METHYL ISOBUTYL ETHER	625-44-5	2.2379E-03	< 5%	S	1.2742E+04	< 10%	S	4.4655E+02	< 10%	S
DI-n-HEXYL ETHER	112-58-3	6.8917E-08	< 25%	P	8.9153E+04	< 25%	P	1.4510E+07	< 25%	P
METHYL n-BUTYL ETHER	628-28-4	1.7201E-03	< 25%	S	1.0077E+04	< 25%	S	5.5095E+02	< 25%	S
DI-n-PENTYL ETHER	693-65-2	2.2173E-06	< 25%	P	5.1508E+04	< 25%	P	4.5100E+05	< 25%	P
METHYL tert-PENTYL ETHER	994-05-8	1.7852E-03	< 10%	S	5.6708E+03	< 10%	S	5.6458E+02	< 10%	S
tert-BUTYL ETHYL ETHER	637-92-3	2.6488E-03	< 25%	S	7.7653E+03	< 10%	S	4.6299E+02	< 25%	S
ETHYL tert-PENTYL ETHER	919-94-8	4.9313E-04	< 25%	S	1.1017E+04	< 25%	S	2.0257E+03	< 25%	S

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

		x_i^{aq} (mol frac.)			k_H (kPa)			γ_i^∞ (unitless)		
Name	CAS	Value	s	DT	Value	s	DT	Value	s	DT
DI-n-PROPYL ETHER	111-43-3	4.5240E-04	< 5%	S	1.8369E+04	< 5%	S	2.2134E+03	< 10%	S
n-BUTYL ETHYL ETHER	628-81-9	1.0019E-03	< 25%	S	7.2904E+03	< 25%	S	9.9042E+02	< 25%	S
ALKYLCYCLOHEXANES										
METHYLCYCLOHEXANE	108-87-2	2.8767E-06	< 25%	S	2.1421E+06	< 25%	S	3.4764E+05	< 25%	S
ETHYLCYCLOHEXANE	1678-91-7	6.1015E-07	< 5%	S	2.7811E+06	< 10%	S	1.6267E+06	< 10%	S
cis-1,2-DIMETHYLCYCLOHEXANE	2207-01-4	9.5855E-07	< 3%	S	2.0121E+06	< 10%	S	1.0422E+06	< 25%	S
trans-1,2-DIMETHYLCYCLOHEXANE	6876-23-9	7.4961E-07	< 3%	S	3.4285E+06	< 5%	S	1.3281E+06	< 5%	S
ALKYLCYCLOPENTANES										
METHYLCYCLOPENTANE	96-37-7	9.1506E-06	< 3%	S	2.0047E+06	< 3%	S	1.0935E+05	< 3%	S
ALKYNES										
ETHYLACETYLENE	107-00-6	1.0280E-03	< 25%	S	9.8530E+04	< 25%	S	-	-	-
1-PENTYNE	627-19-0	4.0450E-04	< 25%	S	1.4360E+05	< 25%	S	2.4720E+03	< 25%	S
1-HEXYNE	693-02-7	8.4660E-05	< 25%	S	2.0930E+05	< 25%	S	1.1810E+04	< 25%	S
1-OCTYNE	629-05-0	3.9470E-06	< 25%	S	4.4450E+05	< 25%	S	2.5340E+05	< 25%	S
1-NONYNE	3452-09-3	7.6320E-07	< 25%	S	6.4770E+05	< 25%	S	1.3100E+06	< 25%	S
1-HEPTYNE	628-71-7	1.8240E-05	< 25%	S	3.0500E+05	< 25%	S	5.4810E+04	< 25%	S
AROMATIC ALCOHOLS										
2,3-XYLENOL	526-75-0	6.9753E-04	< 50%	S	-	-	-	-	-	-
2,4-XYLENOL	105-67-9	1.2252E-03	< 25%	S	1.5966E+01	> 100%	S	9.0061E+02	> 100%	S
2,5-XYLENOL	95-87-4	5.1962E-04	< 25%	S	7.0922E+00	< 25%	S	4.7001E+02	< 25%	S
2,6-XYLENOL	576-26-1	9.1715E-04	< 50%	S	3.4833E+01	< 10%	S	8.9965E+02	< 25%	S
3,4-XYLENOL	95-65-8	8.5953E-04	< 25%	S	2.1606E+00	< 25%	S	4.9566E+02	< 50%	S
3,5-XYLENOL	108-68-9	7.8450E-04	< 25%	S	3.5323E+00	< 25%	S	6.0576E+02	< 25%	S
BENZYL ALCOHOL	100-51-6	7.5540E-03	< 5%	S	-	-	-	-	-	-

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

		x_i^{aq} (mol frac.)			k_H (kPa)			γ_i^∞ (unitless)		
Name	CAS	Value	s	DT	Value	s	DT	Value	s	DT
PHENOL	108-95-2	1.6095E-02	< 10%	S	2.5082E+00	< 25%	S	4.3116E+01	< 25%	S
o-CRESOL	95-48-7	4.4173E-03	< 25%	S	8.2552E+00	< 5%	S	1.9993E+02	< 10%	S
m-CRESOL	108-39-4	3.8787E-03	< 10%	S	4.8030E+00	< 25%	S	2.5942E+02	< 25%	S
p-CRESOL	106-44-5	3.2357E-03	< 10%	S	4.3850E+00	< 3%	S	2.4356E+02	< 10%	S
p-ETHYLPHENOL	123-07-9	1.0065E-03	< 100%	S	-	-	-	-	-	-
2-PHENYLETHANOL	60-12-8	3.6435E-03	< 10%	S	3.1574E+00	< 10%	P	2.7319E+02	< 25%	P
1-PHENYL-1-PROPANOL	93-54-9	1.3678E-03	< 5%	S	-	-	-	-	-	-
AROMATIC AMINES										
o-TOLUIDINE	95-53-4	3.0033E-03	< 25%	S	1.1565E+01	< 25%	S	3.3124E+02	< 25%	S
m-TOLUIDINE	108-44-1	2.6391E-03	< 25%	S	9.8018E+00	< 25%	S	3.7689E+02	< 25%	S
p-TOLUIDINE	106-49-0	1.4028E-03	< 3%	S	1.9157E+01	< 3%	S	4.5491E+02	< 50%	S
QUINOLINE	91-22-5	1.0010E-03	< 25%	S	-	-	-	-	-	-
N,N-DIETHYLANILINE	91-66-7	2.1975E-05	< 5%	S	8.3101E+02	< 10%	S	4.4708E+04	< 25%	S
PYRIDINE	110-86-1	1.0000E+00	Unknown	E	6.1142E+01	< 25%	S	2.1558E+01	< 25%	S
ANILINE	62-53-3	7.2058E-03	< 3%	S	1.1727E+01	< 10%	S	1.3007E+02	< 10%	S
2-METHYLPYRIDINE	109-06-8	1.0000E+00	Unknown	E	5.6453E+01	< 5%	S	3.7361E+01	< 5%	S
o-ETHYLANILINE	578-54-1	1.0298E-03	< 10%	S	2.2524E+01	< 10%	P	9.6769E+02	< 25%	P
DIBENZOPYRROLE	86-74-8	-	-	-	5.9192E-01	< 50%	S	-	-	-
2,4,6-TRIMETHYLPYRIDINE	108-75-8	4.4755E-03	< 25%	S	-	-	-	-	-	-
2,6-DIMETHYLPYRIDINE	108-48-5	1.0000E+00	Unknown	E	6.0121E+01	< 25%	S	7.7723E+01	< 25%	S
3-METHYLPYRIDINE	108-99-6	1.0000E+00	Unknown	E	4.4215E+01	< 25%	S	5.4547E+01	< 25%	S
4-METHYLPYRIDINE	108-89-4	1.0000E+00	Unknown	E	3.3943E+01	< 5%	S	4.4159E+01	< 5%	S
2,4-DIMETHYLANILINE	95-68-1	8.1102E-04	< 10%	S	2.0392E+01	< 25%	P	1.2231E+03	< 50%	P
N-ETHYLANILINE	103-69-5	3.9949E-04	< 3%	S	6.5222E+01	< 10%	S	2.4020E+03	< 25%	S
AROMATIC CARBOXYLIC ACIDS										

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

		x_i^{aq} (mol frac.)			k_H (kPa)			γ_i^∞ (unitless)		
Name	CAS	Value	s	DT	Value	s	DT	Value	s	DT
o-TOLUIC ACID	118-90-1	1.5612E-04	< 25%	S	-	-	-	-	-	-
p-TOLUIC ACID	99-94-5	4.6846E-05	< 5%	S	-	-	-	-	-	-
PHTHALIC ACID	88-99-3	6.9289E-04	< 5%	S	-	-	-	-	-	-
m-TOLUIC ACID	99-04-7	1.1411E-04	< 3%	S	-	-	-	-	-	-
AROMATIC CHLORIDES										
BENZYL CHLORIDE	100-44-7	6.9223E-05	< 25%	S	2.5098E+03	< 25%	S	1.4426E+04	< 25%	S
MONOCHLOROBENZENE	108-90-7	7.3850E-05	< 5%	S	2.1129E+04	< 5%	S	1.3325E+04	< 10%	S
o-DICHLOROBENZENE	95-50-1	1.7571E-05	< 10%	S	1.0267E+04	< 10%	S	5.6832E+04	< 10%	S
m-DICHLOROBENZENE	541-73-1	1.5956E-05	< 10%	S	1.7908E+04	< 10%	S	6.2774E+04	< 10%	S
p-DICHLOROBENZENE	106-46-7	9.8562E-06	< 10%	S	1.3908E+04	< 10%	S	5.5428E+04	< 10%	S
HEXACHLOROBENZENE	118-74-1	3.6635E-10	< 25%	S	8.4566E+03	< 25%	S	1.8179E+08	< 100%	S
o-CHLOROTOLUENE	95-49-8	1.8949E-05	< 25%	S	2.4693E+04	< 25%	S	5.2836E+04	< 25%	S
p-CHLOROTOLUENE	106-43-4	1.7936E-05	< 25%	S	2.0445E+04	< 25%	S	5.5321E+04	< 25%	S
2,4-DICHLOROTOLUENE	95-73-8	2.9825E-06	< 5%	S	2.0337E+04	< 5%	S	3.3512E+05	< 10%	S
1,2,4-TRICHLOROBENZENE	120-82-1	4.0753E-06	< 25%	S	1.3112E+04	< 25%	S	2.5357E+05	< 25%	S
1,2,3-TRICHLOROBENZENE	87-61-6	2.0122E-06	< 10%	S	8.9256E+03	< 10%	S	2.9730E+05	< 25%	S
AROMATIC ESTERS										
BENZYL ACETATE	140-11-4	3.0871E-04	< 5%	S	6.5097E+01	< 10%	S	3.1178E+03	< 25%	S
DI-n-PROPYL PHTHALATE	131-16-8	6.6314E-06	< 25%	S	2.5555E+00	< 25%	S	1.5080E+05	< 25%	S
DI-n-HEXYL PHTHALATE	84-75-3	1.2643E-08	< 50%	S	6.6872E+00	< 50%	S	7.9098E+07	< 50%	S
METHYL BENZOATE	93-58-3	2.7549E-04	< 25%	S	1.8175E+02	< 25%	S	3.6117E+03	< 25%	S
ETHYL BENZOATE	93-89-0	1.1434E-04	< 25%	S	3.3110E+02	< 25%	S	8.7005E+03	< 25%	S
BENZYL FORMATE	104-57-4	1.2812E-03	< 25%	S	2.8015E+01	< 10%	S	6.8470E+02	< 25%	S
DIETHYL PHTHALATE	84-66-2	7.5610E-05	< 10%	S	8.9347E-01	< 5%	S	1.3457E+04	< 25%	S
DI-n-BUTYL PHTHALATE	84-74-2	6.4443E-07	< 25%	S	5.2831E+00	< 25%	S	1.5514E+06	< 25%	S

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

		x_i^{aq} (mol frac.)			k_H (kPa)			γ_i^∞ (unitless)		
Name	CAS	Value	s	DT	Value	s	DT	Value	s	DT
DIMETHYL PHTHALATE	131-11-3	4.1370E-04	< 5%	S	1.1053E+00	< 5%	P	2.4220E+03	< 25%	P
C, H, BR COMPOUNDS										
BROMOMETHANE	74-83-9	2.6893E-03	< 25%	S	3.7821E+04	< 25%	S	-	-	-
BROMOETHANE	74-96-4	1.5050E-03	< 3%	S	4.1374E+04	< 3%	S	6.6333E+02	< 10%	S
1,1,2,2-TETRABROMOETHANE	79-27-6	1.5750E-06	< 50%	S	3.8812E+03	< 100%	S	6.3403E+05	< 100%	S
1-BROMOPROPANE	106-94-5	3.5257E-04	< 3%	S	5.2435E+04	< 3%	S	2.8283E+03	< 5%	S
2-BROMOPROPANE	75-26-3	4.6986E-04	< 3%	S	6.1437E+04	< 3%	S	2.1294E+03	< 5%	S
1-BROMOBUTANE	109-65-9	7.4063E-05	< 25%	S	7.4474E+04	< 25%	S	1.3502E+04	< 25%	S
1-BROMOHEPTANE	629-04-9	7.4417E-07	< 25%	S	2.4825E+05	< 25%	S	1.3438E+06	< 25%	S
1,2-DIBROMOETHANE	106-93-4	4.2706E-04	< 10%	S	4.0636E+03	< 10%	S	2.3308E+03	< 10%	S
m-DIBROMOBENZENE	108-36-1	7.5328E-06	< 25%	E	4.7500E+03	< 25%	D	1.3275E+05	< 25%	D
BROMOBENZENE	108-86-1	4.5894E-05	< 5%	S	1.2337E+04	< 5%	S	2.1731E+04	< 10%	S
TRIBROMOMETHANE	75-25-2	2.4116E-04	< 5%	S	3.0319E+03	< 10%	S	4.1402E+03	< 10%	S
DIBROMOMETHANE	74-95-3	1.2397E-03	< 3%	S	4.8569E+03	< 3%	S	8.0641E+02	< 5%	S
2-BROMOBUTANE	78-76-2	1.2019E-04	< 10%	D	6.3195E+04	< 10%	D	8.3200E+03	< 10%	E
p-BROMOTOLUENE	106-38-7	1.1174E-05	< 25%	S	1.5321E+04	< 25%	S	8.4309E+04	< 50%	S
C, H, F COMPOUNDS										
METHYL FLUORIDE	593-53-3	1.0604E-03	< 1%	S	9.5359E+04	< 3%	S	-	-	-
PERFLUORO-n-HEXANE	355-42-0	4.8787E-09	< 100%	E	5.9661E+09	< 100%	D	2.0497E+08	< 100%	D
PERFLUORO-n-OCTANE	307-34-6	6.8663E-11	< 100%	E	6.8797E+10	< 100%	D	1.4564E+10	< 100%	D
1,2-DIFLUOROBENZENE	367-11-3	1.6957E-04	< 25%	S	4.2141E+04	< 25%	S	5.8883E+03	< 25%	S
FLUOROBENZENE	462-06-6	2.7647E-04	< 10%	S	3.7174E+04	< 10%	S	3.6155E+03	< 10%	S
HEXAFLUOROBENZENE	392-56-3	6.0415E-05	< 3%	S	1.8597E+05	< 3%	S	1.6515E+04	< 5%	S
PERFLUORO-n-PENTANE	678-26-2	7.2276E-08	< 100%	E	1.2114E+09	< 100%	D	1.3835E+07	< 100%	D
PERFLUORO-n-HEPTANE	335-57-9	5.6014E-10	< 100%	E	1.8284E+10	< 100%	D	1.7852E+09	< 100%	D

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

		x_i^{aq} (mol frac.)			k_H (kPa)			γ_i^∞ (unitless)		
Name	CAS	Value	s	DT	Value	s	DT	Value	s	DT
C, H, I COMPOUNDS										
METHYL IODIDE	74-88-4	1.8484E-03	< 5%	S	2.9424E+04	< 5%	S	5.4150E+02	< 10%	S
ETHYL IODIDE	75-03-6	4.6009E-04	< 3%	S	3.8621E+04	< 3%	S	2.1744E+03	< 10%	S
n-PROPYL IODIDE	107-08-4	1.0973E-04	< 5%	S	5.2297E+04	< 5%	S	9.0990E+03	< 10%	S
ISOPROPYL IODIDE	75-30-9	1.4480E-04	< 10%	S	6.5461E+04	< 5%	S	6.8884E+03	< 10%	S
IODOBENZENE	591-50-4	1.9110E-05	< 10%	S	7.3212E+03	< 25%	S	5.1681E+04	< 25%	S
n-HEXYL IODIDE	638-45-9	1.1954E-06	< 25%	S	1.6347E+05	< 25%	S	8.3652E+05	< 25%	S
n-BUTYL IODIDE	542-69-8	2.2740E-05	< 10%	S	8.0572E+04	< 25%	S	4.3976E+04	< 10%	S
C, H, MULTIHALOGEN COMPOUNDS										
DICHLORODIFLUOROMETHANE	75-71-8	5.4251E-05	< 3%	S	1.8669E+06	< 10%	S	-	-	-
TRICHLOROFLUOROMETHANE	75-69-4	-	-	-	5.2851E+05	< 10%	S	-	-	-
1,1,2- TRICHLOROTRIFLUOROETHAN E	76-13-1	2.3783E-05	< 25%	S	1.8719E+06	< 25%	S	4.2032E+04	< 25%	S
C, H, NO2 COMPOUNDS										
NITROMETHANE	75-52-5	3.2090E-02	< 10%	S	1.4923E+02	< 10%	S	3.1049E+01	< 10%	S
NITROETHANE	79-24-3	1.0659E-02	< 5%	S	2.5784E+02	< 5%	S	9.3542E+01	< 10%	S
1-NITROPROPANE	108-03-2	3.2369E-03	< 5%	S	4.1378E+02	< 5%	S	3.0767E+02	< 10%	S
2-NITROPROPANE	79-46-9	3.5066E-03	< 5%	S	6.5009E+02	< 3%	S	2.8516E+02	< 10%	S
1-NITROBUTANE	627-05-4	6.5301E-04	< 25%	S	7.2451E+02	< 25%	S	1.5314E+03	< 25%	S
o-NITROTOLUENE	88-72-2	8.4414E-05	< 5%	S	2.3476E+02	< 10%	S	1.1833E+04	< 10%	S
p-NITROTOLUENE	99-99-0	4.6972E-05	< 10%	S	6.2304E+01	< 10%	S	1.2362E+04	< 50%	S
m-NITROTOLUENE	99-08-1	6.6378E-05	< 5%	S	2.0274E+02	< 10%	S	1.4420E+04	< 25%	S
NITROBENZENE	98-95-3	2.8289E-04	< 5%	S	1.1835E+02	< 5%	S	3.5294E+03	< 10%	S
m-DINITROBENZENE	99-65-0	5.7572E-05	< 10%	S	-	-	-	-	-	-
p-DINITROBENZENE	100-25-4	6.6100E-06	< 3%	S	-	-	-	-	-	-

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

		x_i^{aq} (mol frac.)			k_H (kPa)			γ_i^∞ (unitless)		
Name	CAS	Value	s	DT	Value	s	DT	Value	s	DT
2,4-DINITROTOLUENE	121-14-2	1.9749E-05	< 5%	S	1.4084E+00	< 5%	S	1.6382E+04	< 25%	S
2,6-DINITROTOLUENE	606-20-2	2.0406E-05	< 5%	S	1.1571E+01	< 3%	S	2.3491E+04	< 50%	S
3,4-DINITROTOLUENE	610-39-9	1.6997E-05	< 5%	S	1.6741E+00	< 5%	S	2.6088E+04	< 50%	S
2,4,6-TRINITROTOLUENE	118-96-7	9.5510E-06	< 5%	S	1.2083E-01	< 5%	S	2.1202E+04	< 50%	S
RDX	121-82-4	3.9683E-06	< 5%	S	-	-	-	-	-	-
HMX	2691-41-0	2.7726E-07	< 10%	S	-	-	-	-	-	-
C1/C2 ALIPHATIC CHLORIDES										
CARBON TETRACHLORIDE	56-23-5	9.0660E-05	< 5%	S	1.6765E+05	< 5%	S	1.1029E+04	< 5%	S
METHYL CHLORIDE	74-87-3	1.9096E-03	< 10%	S	5.2937E+04	< 10%	S	-	-	-
ETHYL CHLORIDE	75-00-3	1.5332E-03	< 25%	S	6.6090E+04	< 10%	S	-	-	-
VINYL CHLORIDE	75-01-4	7.7389E-04	< 25%	S	1.3062E+05	< 25%	S	-	-	-
DICHLOROMETHANE	75-09-2	4.0143E-03	< 10%	S	1.4546E+04	< 10%	S	2.4938E+02	< 10%	S
CHLOROFORM	67-66-3	1.1812E-03	< 3%	S	2.2108E+04	< 3%	S	8.4620E+02	< 5%	S
1,1-DICHLOROETHANE	75-34-3	9.3878E-04	< 5%	S	3.2146E+04	< 5%	S	1.0626E+03	< 5%	S
1,2-DICHLOROETHANE	107-06-2	1.5990E-03	< 5%	S	6.6406E+03	< 5%	S	6.2532E+02	< 5%	S
1,1,2-TRICHLOROETHANE	79-00-5	6.2690E-04	< 5%	S	4.9329E+03	< 5%	S	1.5984E+03	< 5%	S
HEXACHLOROETHANE	67-72-1	2.9288E-06	< 25%	S	2.0130E+04	< 25%	S	1.6173E+05	> 100%	S
1,1,1-TRICHLOROETHANE	71-55-6	1.7505E-04	< 5%	S	9.4916E+04	< 5%	S	5.7096E+03	< 10%	S
1,1,1,2-TETRACHLOROETHANE	630-20-6	1.1099E-04	< 10%	S	1.3082E+04	< 10%	S	8.2197E+03	< 10%	S
1,1,2,2-TETRACHLOROETHANE	79-34-5	3.1284E-04	< 10%	S	2.3451E+03	< 10%	S	3.0753E+03	< 10%	S
TRICHLOROETHYLENE	79-01-6	1.8045E-04	< 3%	S	5.4319E+04	< 3%	S	5.5283E+03	< 5%	S
TETRACHLOROETHYLENE	127-18-4	2.4977E-05	< 5%	S	9.8707E+04	< 5%	S	3.9944E+04	< 10%	S
cis-1,2-DICHLOROETHYLENE	156-59-2	1.1760E-03	< 10%	S	2.3105E+04	< 10%	S	8.4974E+02	< 10%	S
trans-1,2-DICHLOROETHYLENE	156-60-5	8.2754E-04	< 10%	S	5.3633E+04	< 10%	S	1.2109E+03	< 10%	S
PENTACHLOROETHANE	76-01-7	3.6884E-05	< 25%	S	1.3266E+04	< 25%	S	2.7105E+04	< 25%	S

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

		x_i^{aq} (mol frac.)			k_H (kPa)			γ_i^∞ (unitless)		
Name	CAS	Value	s	DT	Value	s	DT	Value	s	DT
1,1-DICHLOROETHYLENE	75-35-4	4.6806E-04	< 10%	S	1.7069E+05	< 10%	S	2.1390E+03	< 10%	S
C3 & HIGHER ALIPHATIC CHLORIDES										
1,4-DICHLOROBUTANE	110-56-5	1.8176E-04	< 25%	S	2.7961E+03	< 10%	S	5.4994E+03	< 25%	S
1,5-DICHLOROPENTANE	628-76-2	2.5038E-05	< 100%	S	5.3788E+03	< 100%	S	3.9925E+04	< 100%	S
1,2-DICHLOROPROPANE	78-87-5	4.3120E-04	< 5%	S	1.6001E+04	< 5%	S	2.3227E+03	< 10%	S
ISOPROPYL CHLORIDE	75-29-6	7.1457E-04	< 10%	S	9.6129E+04	< 5%	S	1.3976E+03	< 10%	S
1,2,3-TRICHLOROPROPANE	96-18-4	2.8896E-04	< 25%	S	1.6851E+03	< 25%	S	3.4614E+03	< 25%	S
HEXACHLORO-1,3-BUTADIENE	87-68-3	3.1189E-07	< 50%	S	9.8863E+04	< 50%	S	3.1995E+06	< 50%	S
PROPYL CHLORIDE	540-54-5	6.3068E-04	< 10%	S	7.2709E+04	< 10%	S	1.5838E+03	< 10%	S
n-BUTYL CHLORIDE	109-69-3	1.3685E-04	< 10%	S	9.8517E+04	< 10%	S	7.3035E+03	< 10%	S
sec-BUTYL CHLORIDE	78-86-4	1.5548E-04	< 10%	S	1.3477E+05	< 10%	S	6.4263E+03	< 10%	S
1-CHLOROPENTANE	543-59-9	3.3579E-05	< 5%	S	1.2764E+05	< 5%	S	2.9802E+04	< 5%	S
1,3-DICHLOROPROPANE	142-28-9	3.7373E-04	< 25%	S	4.7562E+03	< 25%	S	2.6752E+03	< 25%	S
CYCLOALIPHATIC ALCOHOLS										
CYCLOHEXANOL	108-93-0	6.9303E-03	< 5%	S	1.2847E+01	< 50%	S	1.4416E+02	< 50%	S
CYCLOALKANES										
CYCLOPENTANE	287-92-3	4.3311E-05	< 10%	S	9.8100E+05	< 10%	S	2.3099E+04	< 10%	S
CYCLOHEXANE	110-82-7	1.3004E-05	< 10%	S	9.4972E+05	< 10%	S	7.2858E+04	< 10%	S
CYCLOHEPTANE	291-64-5	3.5216E-06	< 10%	P	8.2667E+05	< 10%	P	2.8396E+05	< 10%	P
CYCLOOCTANE	292-64-8	9.7862E-07	< 10%	S	7.6345E+05	< 10%	S	1.0159E+06	< 10%	S
CYCLOALKENES										
CYCLOPENTENE	142-29-0	1.7784E-04	< 50%	S	2.8372E+05	< 50%	S	5.6203E+03	< 50%	S
CYCLOHEXENE	110-83-8	6.2535E-05	< 10%	S	1.8948E+05	< 10%	S	1.5966E+04	< 10%	S
CYCLOHEPTENE	628-92-2	1.2363E-05	< 10%	E	2.7050E+05	< 25%	D	8.0884E+04	< 10%	D
CYCLOOCTENE	931-88-4	3.8290E-06	< 3%	S	2.6909E+05	< 3%	S	2.6479E+05	< 3%	S

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

		x_i^{aq} (mol frac.)			k_H (kPa)			γ_i^∞ (unitless)		
Name	CAS	Value	s	DT	Value	s	DT	Value	s	DT
DIALKENES										
1,4-CYCLOHEXADIENE	628-41-1	2.0206E-04	< 25%	S	4.3001E+04	< 25%	S	4.9455E+03	< 25%	S
DICARBOXYLIC ACIDS										
ADIPIC ACID	124-04-9	2.9884E-03	< 3%	S	-	-	-	-	-	-
SUCCINIC ACID	110-15-6	1.3056E-02	< 3%	S	-	-	-	-	-	-
GLUTARIC ACID	110-94-1	1.6229E-01	< 3%	S	-	-	-	-	-	-
DIMETHYLALKANES										
2,2-DIMETHYLBUTANE	75-83-2	4.4181E-06	< 25%	S	9.7812E+06	< 25%	S	2.2886E+05	< 25%	S
2,3-DIMETHYLBUTANE	79-29-8	4.2380E-06	< 25%	S	7.1887E+06	< 25%	S	2.3026E+05	< 25%	S
2,4-DIMETHYLPENTANE	108-08-7	7.3837E-07	< 50%	S	1.9367E+07	< 50%	S	1.4765E+06	< 50%	S
3,3-DIMETHYLPENTANE	562-49-2	1.0612E-06	< 25%	S	1.0373E+07	< 25%	S	9.4075E+05	< 25%	S
2,2-DIMETHYLHEXANE	590-73-8	2.0105E-07	< 3%	S	2.2630E+07	< 5%	S	4.9870E+06	< 5%	S
2,5-DIMETHYLHEXANE	592-13-2	1.7933E-07	< 3%	S	2.2712E+07	< 3%	S	5.6221E+06	< 5%	S
DIPHENYL/POLYAROMATICS										
BIPHENYL	92-52-4	8.2180E-07	< 10%	S	1.5402E+03	< 10%	S	4.5922E+05	< 10%	S
EPOXIDES										
2-METHYLTETRAHYDROFURAN	96-47-9	3.0305E-02	< 5%	S	-	-	-	-	-	-
1,4-DIOXANE	123-91-1	-	-	-	2.9564E+01	< 10%	S	5.8161E+00	< 10%	S
TRIOXANE	110-88-3	4.3058E-02	< 50%	S	-	-	-	-	-	-
TETRAHYDROPYRAN	142-68-7	1.7361E-02	< 3%	S	6.7975E+02	< 5%	S	7.1786E+01	< 5%	S
TETRAHYDROFURAN	109-99-9	-	-	-	3.7471E+02	< 10%	S	1.7501E+01	< 10%	S
DIBENZOFURAN	132-64-9	5.4315E-07	< 10%	S	7.4383E+02	< 10%	S	5.2639E+05	< 25%	S
FORMATES										
METHYL FORMATE	107-31-3	-	-	-	1.3237E+03	< 10%	D	1.7005E+01	< 10%	S
ETHYL FORMATE	109-94-4	2.0624E-02	< 3%	S	1.6926E+03	< 25%	S	5.1845E+01	< 25%	S

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

Name	CAS	x_i^{aq} (mol frac.)			k_H (kPa)	s	DT	γ_i^∞ (unitless)		
		Value	s	DT				Value	s	DT
n-PROPYL FORMATE	110-74-7	5.6073E-03	< 5%	S	1.8180E+03	< 10%	S	1.6754E+02	< 10%	S
n-BUTYL FORMATE	592-84-7	1.7728E-03	< 25%	S	2.1415E+03	< 25%	P	5.6349E+02	< 25%	P
ISOBUTYL FORMATE	542-55-2	1.5568E-03	< 10%	S	3.3686E+03	< 10%	S	6.1884E+02	< 10%	P
n-PENTYL FORMATE	638-49-3	3.8458E-04	< 25%	S	3.0716E+03	< 25%	S	2.5812E+03	< 25%	S
tert-BUTYL FORMATE	762-75-4	2.8541E-03	< 5%	S	3.8948E+03	< 5%	S	3.5011E+02	< 25%	S
KETONES										
ACETONE	67-64-1	-	-	-	1.9892E+02	< 5%	S	6.4743E+00	< 10%	S
METHYL ETHYL KETONE	78-93-3	8.0636E-02	< 3%	S	3.0126E+02	< 10%	S	2.4485E+01	< 10%	S
3-PENTANONE	96-22-0	1.0287E-02	< 10%	S	5.8894E+02	< 5%	S	9.7081E+01	< 10%	S
METHYL ISOBUTYL KETONE	108-10-1	3.2450E-03	< 10%	S	8.1503E+02	< 10%	S	3.0755E+02	< 25%	S
3-HEPTANONE	106-35-4	7.1140E-04	< 10%	S	7.8587E+02	< 10%	S	1.3989E+03	< 25%	S
4-HEPTANONE	123-19-3	6.5059E-04	< 10%	S	1.1765E+03	< 10%	S	1.5279E+03	< 25%	S
3-HEXANONE	589-38-8	2.6395E-03	< 25%	S	7.0108E+02	< 10%	S	3.7866E+02	< 10%	S
2-PENTANONE	107-87-9	1.1528E-02	< 5%	S	4.0969E+02	< 5%	S	8.6500E+01	< 10%	S
METHYL ISOPROPYL KETONE	563-80-4	1.3715E-02	< 5%	S	5.1455E+02	< 5%	P	7.4588E+01	< 10%	P
2-HEXANONE	591-78-6	2.8457E-03	< 5%	S	5.4080E+02	< 5%	S	3.4847E+02	< 10%	S
2-HEPTANONE	110-43-0	6.4445E-04	< 3%	S	8.0639E+02	< 3%	S	1.5469E+03	< 5%	P
5-METHYL-2-HEXANONE	110-12-3	7.7380E-04	< 5%	S	8.8878E+02	< 5%	S	1.2828E+03	< 5%	S
3,3-DIMETHYL-2-BUTANONE	75-97-8	3.4404E-03	< 5%	S	1.2372E+03	< 5%	S	2.8979E+02	< 25%	S
DIISOBUTYL KETONE	108-83-8	5.1396E-05	< 10%	S	4.3295E+03	< 10%	S	1.9301E+04	< 10%	S
DIISOPROPYL KETONE	565-80-0	7.8573E-04	< 10%	S	2.4265E+03	< 10%	S	1.2694E+03	< 25%	S
5-NONANONE	502-56-7	5.6855E-05	< 25%	S	1.3894E+03	< 25%	S	1.7526E+04	< 25%	S
2-NONANONE	821-55-6	4.2752E-05	< 25%	S	1.5963E+03	< 25%	S	2.3810E+04	< 25%	S
ACETYLACETONE	123-54-6	3.3792E-02	< 5%	S	6.4870E+01	< 25%	S	4.9594E+01	< 50%	S
ISOPHORONE	78-59-1	1.8712E-03	< 3%	S	-	-	-	-	-	-

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

		x_i^{aq} (mol frac.)				k_H (kPa)			γ_i^∞ (unitless)		
Name	CAS	Value	s	DT	Value	s	DT	Value	s	DT	
CYCLOPENTANONE	120-92-3	8.0153E-02	< 3%	S	4.4727E+01	< 100%	S	2.9473E+01	< 100%	S	
CYCLOHEXANONE	108-94-1	1.7464E-02	< 5%	S	3.6312E+01	< 25%	S	6.4858E+01	< 25%	S	
2-OCTANONE	111-13-7	1.8040E-04	< 25%	S	1.0319E+03	< 25%	S	5.5108E+03	< 25%	S	
ACETOPHENONE	98-86-2	9.1777E-04	< 25%	S	5.0333E+01	< 25%	S	9.5225E+02	< 25%	S	
3-OCTANONE	106-68-3	1.8418E-04	< 10%	S	1.3047E+03	< 10%	S	5.4089E+03	< 25%	S	
3-NONANONE	925-78-0	6.3260E-05	< 25%	S	1.0827E+03	< 25%	S	1.5675E+04	< 25%	S	
2,6,8-TRIMETHYL-4-NONANONE	123-18-2	2.2065E-06	< 25%	P	5.2094E+03	< 25%	P	4.5321E+05	< 25%	P	
MERCAPTANS											
ISOPROPYL MERCAPTAN	75-33-2	8.7844E-04	< 10%	S	4.1922E+04	< 10%	S	1.1366E+03	< 10%	S	
n-BUTYL MERCAPTAN	109-79-5	1.6284E-04	< 3%	S	3.7378E+04	< 5%	S	6.0435E+03	< 10%	S	
METHYLALKANES											
2-METHYLPENTANE	107-83-5	3.0912E-06	< 25%	S	8.9514E+06	< 10%	S	3.1860E+05	< 25%	S	
3-METHYLPENTANE	96-14-0	2.7067E-06	< 3%	S	9.3257E+06	< 3%	S	3.6949E+05	< 3%	S	
2-METHYLHEXANE	591-76-4	4.5667E-07	< 5%	E	1.9204E+07	< 10%	D	2.1898E+06	< 5%	D	
3-METHYLHEXANE	589-34-4	4.7716E-07	< 5%	S	1.7157E+07	< 10%	S	2.0958E+06	< 5%	S	
3-METHYLHEPTANE	589-81-1	1.2491E-07	< 5%	E	2.0935E+07	< 10%	S	8.0059E+06	< 10%	S	
4-METHYLHEPTANE	589-53-7	1.3224E-07	< 3%	E	2.0657E+07	< 5%	D	7.5617E+06	< 5%	D	
METHYLALKENES											
2-METHYL-2-BUTENE	513-35-9	-	-	-	9.6127E+05	< 3%	S	-	-	-	
N-ALCOHOLS											
METHANOL	67-56-1	-	-	-	2.8376E+01	< 5%	S	1.6882E+00	< 5%	S	
ETHANOL	64-17-5	-	-	-	3.1518E+01	< 5%	S	3.9792E+00	< 5%	S	
1-PROPANOL	71-23-8	-	-	-	3.9058E+01	< 10%	S	1.3915E+01	< 10%	S	
1-BUTANOL	71-36-3	1.8884E-02	< 3%	S	4.7161E+01	< 5%	S	5.2370E+01	< 10%	S	
1-PENTANOL	71-41-0	4.3266E-03	< 3%	S	6.3569E+01	< 10%	S	1.9216E+02	< 10%	S	

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

Name	CAS	x_i^{aq} (mol frac.)			k_H (kPa)	s	DT	γ_i^∞ (unitless)		
		Value	s	DT				Value	s	DT
1-HEXANOL	111-27-3	1.0560E-03	< 5%	S	7.6111E+01	< 25%	S	7.7971E+02	< 25%	S
1-HEPTANOL	111-70-6	2.4859E-04	< 5%	S	1.0435E+02	< 10%	S	3.5458E+03	< 10%	S
1-OCTANOL	111-87-5	6.9824E-05	< 10%	S	1.5895E+02	< 10%	S	1.4289E+04	< 10%	S
1-NONANOL	143-08-8	1.5673E-05	< 25%	S	2.0923E+02	< 25%	S	6.3464E+04	< 25%	S
1-DECANOL	112-30-1	4.1701E-06	< 10%	S	3.0679E+02	< 10%	S	2.3876E+05	< 10%	S
1-UNDECANOL	112-42-5	1.0330E-06	< 50%	P	4.0302E+02	< 50%	P	9.6801E+05	< 50%	P
1-DODECANOL	112-53-8	2.5520E-07	< 50%	P	4.6684E+02	< 50%	P	3.9184E+06	< 50%	P
1-TRIDECANOL	112-70-9	6.3040E-08	< 50%	P	-	-	-	-	-	-
1-TETRADECANOL	112-72-1	1.5573E-08	< 50%	P	-	-	-	-	-	-
1-PENTADECANOL	629-76-5	3.8469E-09	< 100%	P	-	-	-	-	-	-
1-HEXADECANOL	36653-82-4	9.5034E-10	< 100%	P	-	-	-	-	-	-
1-HEPTADECANOL	1454-85-9	2.3477E-10	< 100%	P	-	-	-	-	-	-
1-OCTADECANOL	112-92-5	5.7993E-11	< 100%	P	-	-	-	-	-	-
N-ALIPHATIC PRIMARY AMINES										
METHYLAMINE	74-89-5	1.0000E+00	Unknown	E	-	-	-	-	-	-
ETHYLAMINE	75-04-7	1.0000E+00	Unknown	E	6.6181E+01	< 10%	S	4.7374E-01	< 25%	S
n-PROPYLAMINE	107-10-8	1.0000E+00	Unknown	E	8.2550E+01	< 10%	S	1.9970E+00	< 10%	S
n-BUTYLAMINE	109-73-9	1.0000E+00	Unknown	E	1.0298E+02	< 10%	S	8.3162E+00	< 10%	S
n-PENTYLAMINE	110-58-7	1.0000E+00	Unknown	E	1.2844E+02	< 10%	S	3.2117E+01	< 10%	S
n-HEXYLAMINE	111-26-2	-	-	-	1.6021E+02	< 10%	S	1.3205E+02	< 10%	S
n-HEPTYLAMINE	111-68-2	-	-	-	1.9986E+02	< 25%	S	4.4889E+02	< 25%	S
n-OCTYLAMINE	111-86-4	-	-	-	2.4929E+02	< 25%	S	1.8558E+03	< 25%	S
N-ALKANES										
n-PENTANE	109-66-0	1.0456E-05	< 10%	S	6.5485E+06	< 10%	S	9.5728E+04	< 10%	S
n-HEXANE	110-54-3	2.2390E-06	< 10%	S	9.0978E+06	< 10%	S	4.5143E+05	< 10%	S

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

		x_i^{aq} (mol frac.)			k_H (kPa)			γ_i^∞ (unitless)		
Name	CAS	Value	s	DT	Value	s	DT	Value	s	DT
n-HEPTANE	142-82-5	4.7371E-07	< 10%	S	1.2916E+07	< 10%	S	2.1361E+06	< 10%	S
n-OCTANE	111-65-9	1.1329E-07	< 25%	S	1.6258E+07	< 25%	S	8.7203E+06	< 25%	S
n-NONANE	111-84-2	1.8576E-08	< 50%	S	3.0731E+07	< 50%	S	5.2941E+07	< 50%	S
n-DECANE	124-18-5	4.3460E-09	< 50%	S	4.1648E+07	< 50%	S	2.3009E+08	< 50%	S
N-ALKYLBENZENES										
BENZENE	71-43-2	4.0652E-04	< 3%	S	3.1114E+04	< 3%	S	2.4561E+03	< 3%	S
TOLUENE	108-88-3	1.1535E-04	< 3%	S	3.3054E+04	< 3%	S	8.6874E+03	< 5%	S
ETHYLBENZENE	100-41-4	2.9892E-05	< 3%	S	4.2613E+04	< 3%	S	3.3370E+04	< 5%	S
n-PROPYLBENZENE	103-65-1	7.7818E-06	< 10%	S	5.9359E+04	< 10%	S	1.2796E+05	< 10%	S
n-BUTYLBENZENE	104-51-8	1.8154E-06	< 5%	S	7.8555E+04	< 5%	S	5.4877E+05	< 5%	S
n-HEPTYLBENZENE	1078-71-3	2.3098E-08	< 5%	P	2.0792E+05	< 10%	P	4.3293E+07	< 5%	P
n-DECYLBENZENE	104-72-3	2.0688E-10	< 10%	S	7.2846E+05	< 10%	S	4.8338E+09	< 10%	S
n-PENTYLBENZENE	538-68-1	4.3812E-07	< 10%	S	1.0026E+05	< 10%	S	2.2824E+06	< 10%	S
n-HEXYLBENZENE	1077-16-3	1.0832E-07	< 3%	S	1.3470E+05	< 5%	S	9.2359E+06	< 10%	S
n-OCTYLBENZENE	2189-60-8	4.9663E-09	< 10%	P	3.0050E+05	< 10%	P	2.0136E+08	< 10%	P
n-NONYLBENZENE	1081-77-2	1.0313E-09	< 10%	P	3.8660E+05	< 25%	P	9.6961E+08	< 10%	P
NAPHTHALENES										
NAPHTHALENE	91-20-3	4.3668E-06	< 3%	S	2.5234E+03	< 3%	S	7.0849E+04	< 5%	S
1-METHYLNAPHTHALENE	90-12-0	3.6833E-06	< 10%	S	2.7244E+03	< 10%	S	2.7144E+05	< 10%	S
2-METHYLNAPHTHALENE	91-57-6	3.0163E-06	< 25%	S	2.7289E+03	< 25%	S	2.8517E+05	< 25%	S
1-ETHYLNAPHTHALENE	1127-76-0	1.1956E-06	< 10%	S	2.7757E+03	< 10%	S	8.3651E+05	< 10%	S
NITRILES										
ACETONITRILE	75-05-8	-	-	-	1.1210E+02	< 5%	S	9.2319E+00	< 5%	S
PROPIONITRILE	107-12-0	3.5087E-02	< 10%	S	1.6565E+02	< 25%	S	2.6265E+01	< 25%	S
ACRYLONITRILE	107-13-1	2.3502E-02	< 3%	S	5.4807E+02	< 3%	S	4.1682E+01	< 25%	S

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

Name	CAS	x_i^{aq} (mol frac.)			k_H (kPa)	s	DT	γ_i^∞ (unitless)		
		Value	s	DT				Value	s	DT
METHACRYLONITRILE	126-98-7	5.9311E-03	< 5%	S	1.5916E+03	< 5%	P	1.6780E+02	< 10%	P
SUCCINONITRILE	110-61-2	2.2642E-02	< 10%	S	-	-	-	-	-	-
GLUTARONITRILE	544-13-8	1.9661E-02	< 10%	S	-	-	-	-	-	-
n-BUTYRONITRILE	109-74-0	8.8648E-03	< 3%	S	2.9080E+02	< 3%	S	1.1185E+02	< 5%	S
VALERONITRILE	110-59-8	2.7594E-03	< 10%	S	3.5359E+02	< 25%	S	3.6302E+02	< 25%	S
ISOBUTYRONITRILE	78-82-0	9.2527E-03	< 5%	S	5.0410E+02	< 25%	S	1.1538E+02	< 25%	S
BENZONITRILE	100-47-0	7.1412E-04	< 25%	S	1.6637E+02	< 25%	S	1.5605E+03	< 25%	S
VINYLACETONITRILE	109-75-1	8.8683E-03	< 5%	S	2.7177E+02	< 1%	S	1.1032E+02	< 5%	S
PHENYLACETONITRILE	140-29-4	2.6710E-04	< 25%	S	5.0250E+01	< 100%	S	4.0865E+03	< 100%	S
NITROAMINES										
o-NITROANILINE	88-74-4	1.3507E-04	< 50%	S	7.7514E-01	< 50%	S	3.3338E+03	< 100%	S
p-NITROANILINE	100-01-6	7.4078E-05	< 10%	S	6.3686E-03	< 10%	S	1.6454E+03	< 50%	S
m-NITROANILINE	99-09-2	1.2221E-04	< 10%	S	-	-	-	-	-	-
ORGANIC SALTS										
DIETHYL CARBONATE	105-58-8	2.6923E-03	< 3%	S	5.3166E+02	< 3%	S	3.6990E+02	< 10%	S
DIMETHYL CARBONATE	616-38-6	2.7871E-02	< 3%	S	3.5902E+02	< 3%	S	4.8661E+01	< 5%	S
OTHER ALIPHATIC ALCOHOLS										
ISOPROPANOL	67-63-0	-	-	-	4.8094E+01	< 10%	S	8.2933E+00	< 10%	S
2-METHYL-1-PROPANOL	78-83-1	2.1995E-02	< 10%	S	6.7391E+01	< 10%	S	4.8123E+01	< 10%	S
2-BUTANOL	78-92-2	5.0591E-02	< 3%	S	5.9124E+01	< 5%	S	2.5291E+01	< 5%	S
2-METHYL-2-PROPANOL	75-65-0	-	-	-	6.8926E+01	< 5%	S	1.2064E+01	< 10%	S
2-PENTANOL	6032-29-7	8.8835E-03	< 5%	S	7.9763E+01	< 3%	S	9.8222E+01	< 5%	S
2-METHYL-2-BUTANOL	75-85-4	2.4227E-02	< 10%	S	8.1409E+01	< 3%	S	3.5808E+01	< 5%	S
2-METHYL-1-BUTANOL	137-32-6	5.4588E-03	< 5%	S	7.2382E+01	< 5%	S	1.4816E+02	< 5%	S
2,2-DIMETHYL-1-PROPANOL	75-84-3	6.8035E-03	< 10%	S	1.7596E+02	< 25%	P	1.3088E+02	< 25%	P

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

Name	CAS	x_i^{aq} (mol frac.)			k_H (kPa)	s	DT	γ_i^∞ (unitless)		
		Value	s	DT				Value	s	DT
2-HEXANOL	626-93-7	2.1193E-03	< 5%	S	1.4331E+02	< 5%	S	4.6989E+02	< 10%	S
3-HEXANOL	623-37-0	2.7146E-03	< 5%	S	1.5227E+02	< 3%	S	3.6732E+02	< 25%	S
3-METHYL-1-PENTANOL	589-35-5	1.6550E-03	< 3%	S	7.6763E+01	< 10%	P	5.9895E+02	< 10%	P
3-METHYL-3-PENTANOL	77-74-7	6.1117E-03	< 10%	S	2.1990E+02	< 10%	P	1.6495E+02	< 25%	P
3-PENTANOL	584-02-1	1.0665E-02	< 3%	S	9.2713E+01	< 3%	S	8.3949E+01	< 5%	S
3-METHYL-1-BUTANOL	123-51-3	5.3033E-03	< 5%	S	6.9731E+01	< 10%	S	1.6643E+02	< 25%	S
3-METHYL-2-BUTANOL	598-75-4	1.1849E-02	< 5%	S	1.1070E+02	< 3%	S	7.9081E+01	< 10%	S
2-HEPTANOL	543-49-7	5.7725E-04	< 5%	S	1.7236E+02	< 5%	P	1.7183E+03	< 10%	P
3-HEPTANOL	589-82-2	7.1191E-04	< 10%	S	1.7651E+02	< 10%	S	1.3947E+03	< 10%	S
4-METHYL-2-PENTANOL	108-11-2	2.8026E-03	< 10%	S	2.4972E+02	< 5%	S	3.5580E+02	< 10%	S
2-OCTANOL	123-96-6	1.5562E-04	< 10%	S	2.1047E+02	< 10%	P	6.3825E+03	< 10%	P
2-NONANOL	628-99-9	3.8944E-05	< 25%	S	2.7628E+02	< 25%	P	2.5433E+04	< 25%	P
2,6-DIMETHYL-4-HEPTANOL	108-82-7	7.5500E-05	< 25%	S	5.3226E+02	< 25%	P	1.3210E+04	< 25%	P
OTHER ALIPHATIC AMINES										
DIMETHYLAMINE	124-40-3	-	-	-	8.9023E+01	< 50%	S	-	-	-
TRIETHYLAMINE	121-44-8	1.4394E-02	< 10%	S	5.9606E+02	< 5%	S	6.5469E+01	< 5%	S
DI-n-PROPYLAMINE	142-84-7	7.1637E-03	< 25%	S	2.5765E+02	< 25%	S	7.1297E+01	< 25%	S
N,N-DIMETHYL-n-BUTYLAMINE	927-62-8	4.7045E-03	< 25%	S	-	-	-	-	-	-
N,N-DIETHYLMETHYLAMINE	616-39-7	1.0000E+00	Unknown	E	-	-	-	-	-	-
DIETHYLAMINE	109-89-7	1.0000E+00	Unknown	E	1.5144E+02	< 25%	S	4.8211E+00	< 25%	S
DIISOPROPYLAMINE	108-18-9	1.0000E+00	Unknown	E	-	-	-	-	-	-
DI-n-BUTYLAMINE	111-92-2	6.8942E-04	< 10%	S	4.3834E+02	< 50%	S	1.2008E+03	< 50%	S
TRI-n-BUTYLAMINE	102-82-9	8.1648E-05	< 25%	S	1.4890E+02	< 25%	S	1.2208E+04	< 25%	S
OTHER ALKANES										
2,3,4-TRIMETHYLPENTANE	565-75-3	2.1260E-07	< 5%	S	1.7009E+07	< 5%	S	4.7045E+06	< 5%	S

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

		x_i^{aq} (mol frac.)			k_H (kPa)			γ_i^∞ (unitless)		
Name	CAS	Value	s	DT	Value	s	DT	Value	s	DT
2,2,5-TRIMETHYLHEXANE	3522-94-9	7.5851E-08	< 5%	E	2.9316E+07	< 10%	D	1.3113E+07	< 10%	D
OTHER ALKYL BENZENES										
o-XYLENE	95-47-6	3.2745E-05	< 10%	S	2.7047E+04	< 10%	S	3.0533E+04	< 10%	S
m-XYLENE	108-38-3	2.8685E-05	< 5%	S	3.8659E+04	< 5%	S	3.4498E+04	< 10%	S
p-XYLENE	106-42-3	2.9633E-05	< 10%	S	3.8088E+04	< 10%	S	3.2575E+04	< 10%	S
CUMENE	98-82-8	9.8434E-06	< 5%	S	6.1375E+04	< 5%	S	1.0110E+05	< 10%	S
1,2,3-TRIMETHYLBENZENE	526-73-8	9.7333E-06	< 25%	S	2.3000E+04	< 25%	S	1.0645E+05	< 25%	S
1,2,4-TRIMETHYLBENZENE	95-63-6	8.2984E-06	< 5%	S	3.4730E+04	< 3%	S	1.2134E+05	< 5%	S
MESITYLENE	108-67-8	7.4151E-06	< 25%	S	4.6913E+04	< 25%	S	1.3931E+05	< 25%	S
p-DIETHYLBENZENE	105-05-5	3.3790E-06	< 100%	E	2.8409E+04	< 100%	D	2.9628E+05	< 100%	D
1,2,3,4-TETRAMETHYLBENZENE	488-23-3	2.3311E-06	< 100%	S	1.8900E+04	< 100%	S	4.2831E+05	< 100%	S
1,2,4,5-TETRAMETHYLBENZENE	95-93-2	4.6710E-07	< 10%	E	4.0791E+04	< 10%	D	6.0549E+05	< 25%	D
OTHER AMINES, IMINES										
CYCLOHEXYLAMINE	108-91-8	5.1435E-02	Unknown	P	2.4813E+01	< 25%	S	1.8486E+01	< 25%	S
HEXAMETHYLENETETRAMINE	100-97-0	9.3809E-02	< 3%	S	-	-	-	-	-	-
N-ETHYL-2-METHYLALLYLAMINE	18328-90-0	1.7908E-02	< 25%	S	-	-	-	-	-	-
TRIALLYLAMINE	102-70-5	3.5083E-04	< 25%	S	1.3674E+03	< 10%	S	2.8223E+03	< 25%	S
N-METHYLCYCLOHEXYLAMINE	100-60-7	6.4662E-03	< 3%	S	-	-	-	-	-	-
DIALLYLAMINE	124-02-7	2.0405E-02	< 10%	S	-	-	-	-	-	-
N-METHYLPYPERIDINE	626-67-5	1.0000E+00	< 10%	E	-	-	-	-	-	-
OTHER CONDENSED RINGS										
FLUORANTHENE	206-44-0	2.1103E-08	< 10%	S	7.9065E+01	< 10%	S	1.0787E+07	< 100%	S
FLUORENE	86-73-7	1.9043E-07	< 10%	S	4.1834E+02	< 10%	S	1.1242E+06	< 25%	S
ANTHRACENE	120-12-7	4.6580E-09	< 5%	S	1.7268E+02	< 5%	S	4.3246E-01	< 100%	S

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

Name	CAS	x_i^{aq} (mol frac.)			k_H (kPa)	s	DT	γ_i^∞ (unitless)		
		Value	s	DT				Value	s	DT
PHENANTHRENE	85-01-8	1.0729E-07	< 5%	S	1.6014E+02	< 5%	S	2.6089E+06	< 10%	P
CHRYSENE	218-01-9	1.3398E-10	< 10%	S	3.7646E+00	< 10%	S	2.1955E+08	> 100%	P
PYRENE	129-00-0	1.1600E-08	< 5%	S	4.9163E+01	< 5%	S	1.0798E+07	< 10%	S
ACENAPHTHENE	83-32-9	4.1124E-07	< 25%	S	7.7124E+02	< 25%	S	5.1573E+05	< 25%	S
ACENAPHTHALENE	208-96-8	1.5440E-06	< 25%	S	6.4945E+02	< 25%	S	3.9997E+05	Unknown	S
BENZO[a]PYRENE	50-32-8	1.1168E-10	< 10%	S	1.2412E+01	< 5%	S	1.6231E+09	< 50%	S
BENZANTHRACENE	56-55-3	7.0358E-10	< 5%	S	-	-	-	-	-	-
BENZO[b]FLUORANTHENE	205-99-2	1.0860E-10	< 10%	S	3.6764E+00	< 10%	S	2.7444E+08	> 100%	S
OTHER ETHERS/DIETHERS										
METHYLAL	109-87-5	9.0588E-02	< 5%	S	9.0067E+02	< 25%	S	1.7117E+01	< 25%	S
1,2-DIMETHOXYETHANE	110-71-4	6.7434E-02	< 25%	S	3.9438E+01	< 3%	S	4.2545E+00	< 10%	S
DIETHYLENE GLYCOL DIETHYL ETHER	112-36-7	1.0000E+00	Unknown	E	-	-	-	-	-	-
DIETHYLENE GLYCOL DI-n-BUTYL ETHER	112-73-2	1.6847E-04	< 10%	S	2.2058E+01	> 100%	P	6.2862E+03	> 100%	P
ANISOLE	100-66-3	2.4678E-04	< 5%	S	1.9271E+03	< 5%	S	4.0420E+03	< 5%	S
DIPHENYL ETHER	101-84-8	1.9486E-06	< 5%	S	2.9031E+03	< 50%	S	4.9071E+05	< 50%	S
1,2-DIETHOXYETHANE	629-14-1	3.3244E-02	< 5%	S	-	-	-	-	-	-
OTHER INORGANICS										
HYDROGEN PEROXIDE	7722-84-1	-	-	-	5.7967E-02	< 25%	S	2.2211E-01	< 25%	S
OTHER MONOAROMATICS										
STYRENE	100-42-5	5.2493E-05	< 5%	S	1.5542E+04	< 5%	S	1.9043E+04	< 5%	S
OTHER POLYFUNCTIONAL C, H, O										
FURFURAL	98-01-1	1.6437E-02	< 5%	S	-	-	-	-	-	-
2-BUTOXYETHANOL	111-76-2	1.0000E+00	< 3%	E	4.3241E+00	< 25%	S	3.7889E+01	< 25%	S
2-HEXOXYETHANOL	112-25-4	1.6257E-03	< 25%	S	-	-	-	-	-	-

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

Name	CAS	x_i^{aq} (mol frac.)			k_H (kPa)	s	DT	γ_i^∞ (unitless)		
		Value	s	DT				Value	s	DT
2-(2-HEXOXYETHOXY)ETHANOL	112-59-4	1.7793E-03	< 25%	S	-	-	-	-	-	-
VANILLIN	121-33-5	1.2265E-03	< 3%	S	3.1102E-02	< 3%	S	3.3360E+02	< 50%	S
PROPYLENE GLYCOL n-BUTYL ETHER	5131-66-8	7.7931E-03	< 10%	S	-	-	-	-	-	-
OTHER POLYFUNCTIONAL ORGANICS										
MALATHION	121-75-5	8.4076E-06	< 25%	S	2.8250E+00	< 3%	S	1.1889E+05	< 25%	S
THIOUREA	62-56-6	3.7861E-02	< 3%	S	-	-	-	-	-	-
OTHER SATURATED ALIPHATIC ESTERS										
ETHYL ISOVALERATE	108-64-5	2.7722E-04	< 10%	S	3.8440E+03	< 10%	S	3.5841E+03	< 10%	S
DIETHYL MALONATE	105-53-3	2.4991E-03	< 5%	S	1.4010E+01	< 10%	P	3.9755E+02	< 25%	P
DIETHYL SUCCINATE	123-25-1	2.1269E-03	< 10%	S	-	-	-	-	-	-
ETHYL TRIMETHYL ACETATE	3938-95-2	2.5561E-04	< 25%	S	8.7394E+03	< 25%	S	3.8894E+03	< 25%	S
DIMETHYL SUCCINATE	106-65-0	1.7223E-02	< 3%	S	-	-	-	-	-	-
DIMETHYLMALONATE	108-59-8	2.3027E-02	< 10%	S	-	-	-	-	-	-
PEROXIDES										
METHYL HYDROPEROXIDE	3031-73-0	-	-	-	1.8200E+01	< 5%	S	3.7482E+00	< 50%	S
ETHYL HYDROPEROXIDE	3031-74-1	-	-	-	1.6528E+01	< 10%	S	4.5931E+00	< 50%	S
POLYFUNCTIONAL ACIDS										
SALICYLIC ACID	69-72-7	2.8433E-04	< 3%	S	-	-	-	-	-	-
POLYFUNCTIONAL AMIDES/AMINES										
N,N-DIMETHYLFORMAMIDE	68-12-2	-	-	-	3.6460E-01	< 25%	S	6.7237E-01	< 50%	S
UREA	57-13-6	2.5890E-01	< 3%	S	-	-	-	-	-	-
N-METHYLFORMAMIDE	123-39-7	-	-	-	4.0404E-02	< 25%	S	1.0955E+00	< 25%	S
ACETAMIDE	60-35-5	4.3874E-01	< 1%	S	-	-	-	-	-	-
N,N-DIMETHYLACETAMIDE	127-19-5	-	-	-	1.2519E-01	< 50%	P	4.3060E-01	< 50%	P

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

		x_i^{aq} (mol frac.)			k_H (kPa)			γ_i^∞ (unitless)		
Name	CAS	Value	s	DT	Value	s	DT	Value	s	DT
N,N-DIETHYLHYDROXYLAMINE	3710-84-7	1.0021E-01	< 3%	S	-	-	-	-	-	-
POLYFUNCTIONAL C, H, N, HALIDE, (O)										
o-CHLOROANILINE	95-51-2	1.2271E-03	< 25%	S	2.6749E+01	< 25%	P	8.0958E+02	< 25%	P
p-CHLORONITROBENZENE	100-00-5	2.8118E-05	< 10%	S	8.6341E+01	< 10%	S	2.1744E+04	< 50%	S
POLYFUNCTIONAL C, H, O, HALIDE										
DI(2-CHLOROETHYL)ETHER	111-44-4	1.2915E-03	< 3%	S	1.1134E+02	< 3%	S	7.7156E+02	< 10%	S
ETHYLCHLOROACETATE	105-39-5	2.9808E-03	< 5%	S	2.1209E+02	< 10%	S	3.3591E+02	< 25%	S
o-CHLOROPHENOL	95-57-8	3.2052E-03	< 3%	S	-	-	-	-	-	-
m-CHLOROPHENOL	108-43-0	3.2188E-03	< 5%	S	-	-	-	-	-	-
p-CHLOROPHENOL	106-48-9	3.3222E-03	< 25%	S	-	-	-	-	-	-
METHYL CHLOROACETATE	96-34-4	8.8232E-03	< 3%	S	1.1352E+02	< 3%	S	1.1243E+02	< 5%	S
POLYFUNCTIONAL C, H, O, N										
N-METHYL-2-PYRROLIDONE	872-50-4	-	-	-	1.6824E-02	< 10%	S	3.6610E-01	< 10%	S
ACETAMINOPHEN	103-90-2	1.7020E-03	< 3%	S	-	-	-	-	-	-
ACETANILIDE	103-84-4	8.1951E-04	< 10%	S	-	-	-	-	-	-
POLYFUNCTIONAL C, H, O, S										
DIMETHYL SULFOXIDE	67-68-5	-	-	-	6.9311E-03	< 25%	S	8.5921E-02	< 25%	S
POLYFUNCTIONAL ESTERS										
METHYL SALICYLATE	119-36-8	1.5371E-04	< 10%	S	3.3064E+01	< 25%	S	6.4658E+03	< 25%	S
2-ETHOXYETHYL ACETATE	111-15-9	3.8954E-02	< 5%	S	1.8500E+01	< 10%	E			
ETHYL ACETOACETATE	141-97-9	1.5712E-02	< 25%	S	6.1986E+00	< 25%	S	6.5389E+01	< 25%	S
ETHYLENE GLYCOL MONOBUTYL ETHER ACETATE	112-07-2	1.7864E-03	< 25%	S	3.0618E+01	< 25%	E	2.6500E+02	< 25%	D
POLYOLS										
p-HYDROQUINONE	123-31-9	1.2199E-02	< 3%	S	-	-	-	-	-	-

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

Name	CAS	x_i^{aq} (mol frac.)			k_H (kPa)	s	DT	γ_i^∞ (unitless)		
		Value	s	DT				Value	s	DT
XYLITOL	87-99-0	1.7785E-01	< 5%	S	-	-	-	-	-	-
1,3-BENZENEDIOL	108-46-3	2.1628E-01	< 3%	S	-	-	-	-	-	-
PROPIONATES AND BUTYRATES										
METHYL PROPIONATE	554-12-1	1.2818E-02	< 3%	S	9.6262E+02	< 5%	S	8.4969E+01	< 5%	S
ETHYL PROPIONATE	105-37-3	3.4032E-03	< 25%	S	1.4323E+03	< 25%	S	2.9291E+02	< 25%	S
n-PROPYL PROPIONATE	106-36-5	9.0986E-04	< 10%	S	2.0359E+03	< 10%	S	1.0930E+03	< 10%	S
n-BUTYL PROPIONATE	590-01-2	2.4314E-04	< 5%	S	2.4355E+03	< 5%	S	4.0803E+03	< 10%	S
n-PROPYL n-BUTYRATE	105-66-8	2.5274E-04	< 25%	S	3.0510E+03	< 25%	S	3.9278E+03	< 25%	S
METHYL n-BUTYRATE	623-42-7	3.0473E-03	< 25%	S	1.3329E+03	< 25%	S	3.1059E+02	< 25%	E
ETHYL n-BUTYRATE	105-54-4	1.0223E-03	< 10%	S	2.1767E+03	< 10%	S	9.7580E+02	< 10%	S
ISOBUTYL ISOBUTYRATE	97-85-8	7.3381E-05	< 25%	S	7.9493E+03	< 25%	S	1.3565E+04	< 25%	S
n-BUTYL n-BUTYRATE	109-21-7	7.9584E-05	< 10%	S	3.4019E+03	< 10%	S	1.2443E+04	< 10%	S
METHYL ISOBUTYRATE	547-63-7	3.1917E-03	< 3%	S	2.0910E+03	< 10%	S	3.1234E+02	< 10%	S
ETHYL ISOBUTYRATE	97-62-1	9.5920E-04	< 10%	S	3.5044E+03	< 10%	P	1.0373E+03	< 10%	P
n-PENTYL n-BUTYRATE	540-18-1	3.1389E-05	< 25%	S	3.3166E+03	< 25%	S	3.1692E+04	< 25%	S
ISOPENTYL BUTYRATE	106-27-4	2.7546E-05	< 100%	S	5.2634E+03	< 100%	S	3.6102E+04	< 100%	S
SULFIDES/THIOPHENES										
DIMETHYL SULFIDE	75-18-3	6.1242E-03	< 10%	S	1.0548E+04	< 10%	S	1.6343E+02	< 10%	S
DIMETHYL DISULFIDE	624-92-0	5.6992E-04	< 10%	P	6.6448E+03	< 10%	S	1.7561E+03	< 10%	S
2-METHYLTHIOPHENE	554-14-3	2.4374E-04	< 10%	S	1.3663E+04	< 10%	S	4.0928E+03	< 10%	S
DIISOPROPYL SULFIDE	625-80-9	1.3264E-04	< 25%	S	1.9586E+04	< 10%	S	7.4705E+03	< 25%	S
UNSATURATED ALIPHATIC ESTERS										
ETHYL ACRYLATE	140-88-5	3.5978E-03	< 10%	S	1.4139E+03	< 10%	P	2.7901E+02	< 10%	P
METHYL METHACRYLATE	80-62-6	2.7645E-03	< 3%	S	1.8891E+03	< 3%	S	3.8100E+02	< 10%	S
DIETHYL MALEATE	141-05-9	1.5787E-03	< 3%	S	-	-	-	-	-	-

Table C.5: Summary of constant value recommendations (298.15 K and 1 atm) (cont.)

Name	CAS	x_i^{aq} (mol frac.)			k_H (kPa)			γ_i^∞ (unitless)		
		Value	s	DT	Value	s	DT	Value	s	DT
DIMETHYL MALEATE	624-48-6	1.0309E-02	< 3%	S	-	-	-	-	-	-

Appendix D. Experimental Data

Details of each experimental run are included in this appendix. The abbreviations of the headings are given in Table D.1. Any variables not included in the table are described in Section 5.2.

Table D.1: Abbreviations used in tables with experimental run information

ID	Experimental run number
ChemID	502 = toluene, 1105 = 1-butanol, 1461 = anisole, 1637 = 1,2-difluorobenzene, 2592 = 1,2,3-trichlorobenzene, 2661 = 4-bromotoluene, 1579 = 2,4-dichlorotoluene
Date	Date of experiment
Cell Order	Experimental number with a given cell
Cell T	Cell temperature
Room T	Room temperature
Cell P	Cell pressure
Flow Rate	N ₂ flow rate at room temperature
n	Number of GC measurements used to calculate the slope, a
a	Slope (Equation 5.23)
s_k	Error of variable k
R ²	R ² of plot used to obtain a
H ₂ O (g)	Water in stripping cell
Solute	Initial amount of solute; values only included for the first cell run
y_i max	The maximum headspace concentration possible due to the initial amount of solute
use	Y = run was included when determining final value; N = adsorption effects were significant so run was not included

Table D.2: Cell data for each experimental run

ID	ChemID	Date	Cell Order	Cell T (K)	Room T (K)	Cell P (atm)	H ₂ O (g)	Solute	y_i max
116	502	1/4/2012	1	298.15	296.99	0.887	356.518	0.156 g	3.09E-02
133	502	3/29/2012	8	298.15	296.99	0.859	394.566		
135	502	3/30/2012	2	298.15	296.99	0.872	395.055		
123	502	3/21/2012	2	298.15	296.99	0.852	393.915		
126	502	3/23/2012	2	298.15	296.99	0.870	394.566		
117	502	1/4/2012	2	298.15	296.99	0.871	356.518		
116	502	12/15/2012	1	298.15	296.99	0.904	352.976	0.03 g	
118	502	1/20/2012	1	298.15	296.99	0.851	362.157	0.112 g	2.07E-02
119	502	1/20/2012	2	298.15	296.99	0.872	362.157		
122	502	3/21/2012	1	298.15	296.99	0.906	393.915	0.12 mL	2.07E-02
124	502	3/22/2012	3	298.15	296.99	0.901	393.915		
125	502	3/23/2012	1	298.15	296.99	0.901	394.566	0.133 g	1.89E-02
128	502	3/26/2012	3	298.15	296.99	0.845	394.566		
129	502	3/26/2012	4	298.15	296.99	0.900	394.566		
130	502	3/26/2012	5	298.15	296.99	0.872	394.566		
131	502	3/29/2012	6	298.15	296.99	0.909	394.566		
132	502	3/29/2012	7	298.15	296.99	0.866	394.566		
134	502	3/30/2012	1	298.15	296.99	0.859	395.055		
120	502	1/30/2012	1	308.15	296.99	0.850	387.738	0.125 g	3.27E-02
121	502	1/30/2012	2	308.15	296.99	0.871	387.738		
5	1105	4/11/2012	4	298.15	296.99	1.031	375.291		
2	1105	4/10/2012	1	308.15	296.99	1.043	375.291	4.265 g	2.86E-03
4	1105	4/11/2012	3	313.15	296.99	1.042	375.291		
1	1105	4/6/2012	1	323.15	297.02	1.063	396.165	0.9246 g	1.56E-03
3	1105	4/10/2012	2	323.15	296.99	1.058	375.291		
103	1461	5/30/2012	1	281.05	296.85	1.046	388.706	0.42 g	1.37E-03
107	1461	5/30/2012	5	281.06	296.35	1.039	388.706		
92	1461	5/25/2012	3	293.12	297.05	1.027	390.332		
89	1461	5/24/2012	3	293.14	298.35	1.034	394.685		
88	1461	5/24/2012	2	293.18	298.75	1.036	394.685		
101	1461	5/29/2012	6	281.04	296.65	1.037	389.472		
102	1461	5/29/2012	7	281.04	296.65	1.031	389.472		
108	1461	5/31/2012	6	281.04	296.95	1.044	388.706		
100	1461	5/29/2012	5	281.05	297.25	1.040	389.472		
111	1461	6/1/2012	2	281.07	296.45	1.045	390.316		
113	1461	6/6/2012	2	281.07	298.15	1.036	387.785		
115	1461	6/7/2012	4	281.07	298.45	1.039	387.785		

Table D.2: Cell data for each experimental run (cont.)

ID	ChemID	Date	Cell Order	Cell T (K)	Room T (K)	Cell P (atm)	H ₂ O (g)	Solute	y_i max
109	1461	5/31/2012	7	281.09	296.75	1.041	388.706		
90	1461	5/25/2012	1	293.1	297.55	0.940	390.332	0.458 g	2.74E-03
106	1461	5/30/2012	4	293.12	296.35	0.996	388.706		
93	1461	5/26/2012	4	293.16	297.75	0.986	390.332		
91	1461	5/25/2012	2	293.18	297.05	0.978	390.332		
112	1461	6/6/2012	1	293.19	298.45	0.998	387.785	0.343 g	2.15E-03
114	1461	6/7/2012	3	293.2	298.65	0.995	387.785		
99	1461	5/28/2012	4	308.11	296.75	0.998	389.472		
105	1461	5/30/2012	3	308.14	296.25	1.001	388.706		
97	1461	5/28/2012	2	308.17	296.95	0.975	389.472		
104	1461	5/30/2012	2	308.19	296.35	1.001	388.706		
96	1461	5/28/2012	1	322.87	296.65	0.975	389.472	0.46 g	1.06E-02
98	1461	5/28/2012	3	322.9	296.65	0.971	389.472		
94	1461	5/26/2012	5	322.91	297.25	1.007	390.332		
110	1461	6/1/2012	1	322.91	296.65	1.017	390.316	0.43 g	1.00E-02
95	1461	5/26/2012	6	322.92	297.75	0.965	390.332		
35	1637	4/13/2012	4	298.15	296.99	0.901	394.062		
40	1637	4/16/2012	5	298.15	296.99	0.911	394.542		
38	1637	4/16/2012	3	298.15	296.99	0.855	394.542		
39	1637	4/16/2012	4	298.15	296.99	0.895	394.542		
59	1637	5/2/2012	3	298.15	297.15	0.885	394.516		
57	1637	5/2/2012	1	298.15	297.15	0.905	394.516	0.25 g	3.43E-02
61	1637	5/3/2012	5	298.18	296.85	0.904	394.516		
62	1637	5/3/2012	6	298.18	296.65	0.886	394.516		
33	1637	4/13/2012	2	298.35	296.99	0.844	394.062		
41	1637	4/17/2012	1	298.35	296.99	0.855	392.354	0.201 g	2.46E-02
34	1637	4/13/2012	3	298.35	296.99	0.884	394.062		
36	1637	4/16/2012	1	298.35	296.99	0.913	394.542	0.21 g	3.25E-02
50	1637	4/24/2012	3	308.15	295.95	0.874	396.527		
47	1637	4/23/2012	2	317.9	296.95	0.879	394.874		
44	1637	4/17/2012	4	323.15	296.99	0.882	392.354		
45	1637	4/17/2012	5	323.15	296.99	0.883	392.354		
58	1637	5/2/2012	2	298.13	297.15	0.854	394.516		
60	1637	5/3/2012	4	298.14	296.95	0.854	394.516		
37	1637	4/16/2012	2	298.15	296.99	0.880	394.542		
32	1637	4/13/2012	1	298.35	296.99	0.871	394.062	0.198 g	3.44E-02
51	1637	4/26/2012	1	308.05	296.85	0.870	396.177	0.264 g	7.17E-02

Table D.2: Cell data for each experimental run (cont.)

ID	ChemID	Date	Cell Order	Cell T (K)	Room T (K)	Cell P (atm)	H ₂ O (g)	Solute	y_i max
42	1637	4/17/2012	2	308.15	296.99	0.881	392.354		
43	1637	4/17/2012	3	308.15	296.99	0.881	392.354		
52	1637	4/26/2012	2	317.89	296.65	0.870	396.177		
48	1637	4/24/2012	1	317.9	296.15	0.878	396.527	0.243 g	9.01E-02
53	1637	4/27/2012	1	317.92	297.65	0.881	393.315	0.24 g	8.60E-02
54	1637	4/27/2012	2	322.79	297.75	0.881	393.315		2.12E-03
55	1637	4/27/2012	1	322.82	297.65	0.881	393.308	0.25 g	1.08E-01
56	1637	4/27/2012	2	322.83	298.15	0.880	393.308		3.50E-03
46	1637	4/23/2012	1	322.95	296.15	0.882	394.874	0.186 g	8.10E-02
49	1637	4/24/2012	2	323.05	295.65	0.878	396.527		
62.5	2592	5/14/2012		293.1	297.15	0.881	394.228	0.002 g	
82	2592	6/4/2012	1	281.04	296.45	0.891	394.028		
83	2592	6/4/2012	2	281.06	296.45	0.907	394.028		
84	2592	6/4/2012	3	281.07	297.05	0.904	394.028		
86	2592	6/5/2012	5	281.07	297.15	0.902	394.028		
85	2592	6/5/2012	4	281.08	296.25	0.899	394.028		
68	2592	5/16/2012	6	292.8	296.35	0.948	395.873		
67	2592	5/16/2012	5	292.95	296.25	0.873	395.873		
65	2592	5/15/2012	3	293.1	296.65	0.911	395.873		
66	2592	5/15/2012	4	293.11	297.35	0.948	395.873		
69	2592	5/16/2012	7	293.11	296.15	0.907	395.873		
70	2592	5/16/2012	8	293.11	296.25	0.889	395.873		
63	2592	5/15/2012	1	293.15	297.15	0.894	395.873	0.002 g	2.85E-05
79	2592	5/23/2012	4	293.21	297.65	0.888	392.05		
71	2592	5/16/2012	1	308.1	296.99	0.891	395.47	0.002 g	7.31E-05
72	2592	5/17/2012	2	308.1	296.99	0.837	395.47		
73	2592	5/17/2012	3	308.1	296.99	0.837	395.47		
81	2592	5/23/2012	6	308.18	297.15	0.887	392.05		
80	2592	5/23/2012	5	308.22	297.35	0.889	392.05		
74	2592	5/18/2012	4	322.85	296.99	0.898	395.47		
75	2592	5/18/2012	5	322.87	296.99	0.898	395.47		
76	2592	5/22/2012	1	322.97	296.35	0.896	392.05	0.002 g	1.46E-04
77	2592	5/22/2012	2	322.97	296.35	0.896	392.05		
78	2592	5/22/2012	3	322.97	296.05	0.896	392.05		
31	2661	5/12/2012	5	293.12	298.15	0.881	395.451		
10	2661	5/8/2012	1	308.07	297.15	0.864	394.770	0.020 g	1.04E-03
20	2661	5/11/2012	3	322.9	297.25	0.880	395.451		

Table D.2: Cell data for each experimental run (cont.)

ID	ChemID	Date	Cell Order	Cell T (K)	Room T (K)	Cell P (atm)	H ₂ O (g)	Solute	y_i max
26	2661	5/10/2012	5	293.07	297.25	0.872	394.844		
27	2661	5/10/2012	6	293.09	297.15	0.872	394.844		
30	2661	5/11/2012	1	293.15	297.45	0.876	395.451	0.020 g	6.10E-04
29	2661	5/10/2012	9	298.13	296.95	0.872	394.844		
28	2661	5/10/2012	7	298.14	297.05	0.873	394.844		
25	2661	5/4/2012	1	298.2	297.85	0.874	393.956	0.0005 g	2.41E-05
12	2661	5/8/2012	3	308.05	297.15	0.897	394.770		
17	2661	5/9/2012	4	308.07	295.65	0.892	394.844		
16	2661	5/9/2012	3	308.08	295.65	0.911	394.844		
8	2661	5/7/2012	4	308.093	297.35	0.876	393.956		
6	2661	5/7/2012	2	308.1	297.65	0.895	393.956	0.030 g	2.18E-03
9	2661	5/7/2012	5	308.1	295.65	0.912	393.956		
13	2661	5/8/2012	4	308.1	296.65	0.878	394.770		
15	2661	5/9/2012	2	308.1	296.95	0.877	394.844		
21	2661	5/11/2012	4	308.1	297.35	0.876	395.451		
7	2661	5/7/2012	3	308.15	297.65	0.861	393.956		
11	2661	5/8/2012	2	308.15	297.25	0.916	394.770		
14	2661	5/9/2012	1	308.15	297.25	0.862	394.844	0.020 g	1.47E-03
24	2661	5/12/2012	8	317.95	296.75	0.881	395.451		
18	2661	5/10/2012	8	318.1	295.75	0.875	394.844		
22	2661	5/12/2012	6	322.89	297.05	0.885	395.451		
19	2661	5/11/2012	2	322.91	297.65	0.881	395.451		
23	2661	5/12/2012	7	322.92	297.05	0.884	395.451		
136	1461	8/28/2012	1	280.96	296.35	0.992	386.149	0.420 g	
137	1461	8/28/2012	2	280.97	297.15	0.990	386.149		
138	1461	8/28/2012	3	280.95	297.15	0.988	386.149		
139	1461	8/28/2012	4	280.94	297.35	0.987	386.149		
140	1461	8/29/2012	5	280.95	296.85	0.992	386.149		
141	2592	8/30/2012	1	293.1	297.35	0.878	394.989	0.002 g	
142	2592	8/30/2012	2	293.09	297.15	0.862	394.989		
143	2592	8/30/2012	3	323.08	297.15	0.882	394.989		
144	2592	8/30/2012	4	323.14	296.95	0.880	394.989		
145	2592	8/31/2012	5	323.09	296.95	0.886	394.989		
146	2592	8/31/2012	6	308.12	296.95	0.879	394.989		
147	2592	8/31/2012	7	280.95	297.05	0.876	394.989		
148	2592	9/3/2012	1	293.09	296.95	0.881	393.47	0.002 g	
149	2592	9/3/2012	2	280.94	297.15	0.862	393.47		

Table D.2: Cell data for each experimental run (cont.)

ID	ChemID	Date	Cell Order	Cell T (K)	Room T (K)	Cell P (atm)	H ₂ O (g)	Solute	y_i max
150	2592	9/3/2012	3	280.95	297.25	0.860	393.47		
151	2592	9/3/2012	4	280.96	297.25	0.862	393.47		
177	1579	9/24/2012	2	323.09	297.25	0.886	395.214	0.013 g	5.38E-03
165	1579	9/10/2012	5	280.91	297.05	0.896	395.521		
176	1579	9/24/2012	1	280.95	297.45	0.896	395.214	0.0077 g	1.72E-04
178	1579	9/24/2012	4	280.93	298.05	0.892	395.214		
174	1579	9/14/2012	3	280.96	297.95	0.900	394.095		
158	1579	9/6/2012	7	293.09	297.25	0.861	395.495		
154	1579	9/5/2012	3	293.1	297.15	0.864	395.495		
163	1579	9/7/2012	3	293.11	297.55	0.867	395.521		
160	1579	9/6/2012	9	293.07	297.15	0.876	395.495		
157	1579	9/6/2012	6	293.09	297.15	0.877	395.495		
159	1579	9/6/2012	8	293.09	297.15	0.877	395.495		
161	1579	9/7/2012	1	293.11	297.25	0.884	395.521	0.0092 g	4.08E-04
173	1579	9/14/2012	2	293.07	298.55	0.903	394.095		
155	1579	9/5/2012	4	293.08	297.05	0.896	395.495		
153	1579	9/5/2012	2	293.09	297.25	0.898	395.495		
162	1579	9/7/2012	2	293.09	297.65	0.901	395.521		
156	1579	9/6/2012	5	293.07	296.95	0.912	395.495		
152	1579	9/5/2012	1	293.1	297.25	0.916	395.495	0.0092 g	4.95E-04
164	1579	9/7/2012	4	293.12	297.25	0.917	395.521		
172	1579	9/14/2012	1	308.14	298.15	0.888	394.095	0.01 g	9.04E-04
168	1579	9/11/2012	1	308.15	297.05	0.898	395.133		
169	1579	9/11/2012	2	308.15	297.05	0.897	395.133		
166	1579	9/10/2012	6	308.16	296.85	0.896	395.521		
171	1579	9/13/2012	4	308.16	298.05	0.906	395.133		
178	1579	9/24/2012	3	323.09	297.25	0.867	395.214		
175	1579	9/15/2012	4	323.09	296.85	0.890	394.095	0.0123 g	2.98E-03
167	1579	9/10/2012	7	323.09	296.45	0.903	395.521		
170	1579	9/13/2012	3	323.14	297.85	0.913	395.133	0.0123 g	3.04E-03

Table D.3: Experimental run information—additional parameters

ID	Flow Rate (mL/min)	n	a (1/min)	s_a	R^2	use	P_i^{sat} (Pa)	B_{ii} (L/mol)	ϕ_i^{sat}
116	40	8	-2.976E-02	2.47E-04	1.000	N	3803.90	-2.821	0.996
133	12	12	-4.408E-03	8.43E-04	0.646	N	3803.90	-2.821	0.996
135	40	24	-2.204E-02	5.79E-04	0.985	N	3803.90	-2.821	0.996
123	12	18	-3.290E-03	3.49E-04	0.847	N	3803.90	-2.821	0.996
126	40	18	-1.275E-02	2.12E-04	0.996	N	3803.90	-2.821	0.996
117	20.6	10	-1.152E-02	3.25E-04	0.994	Y	3803.90	-2.821	0.996
116	75	5	-4.713E-02	2.66E-04	1.000	Y	3803.90	-2.821	0.996
118	12	24	-8.416E-03	7.31E-05	0.998	Y	3803.90	-2.821	0.996
119	40	21	-3.023E-02	2.39E-04	0.999	Y	3803.90	-2.821	0.996
122	75	19	-4.995E-02	4.31E-04	0.999	Y	3803.90	-2.821	0.996
124	75	12	-5.176E-02	2.93E-04	1.000	Y	3803.90	-2.821	0.996
125	75	12	-4.140E-02	3.75E-04	0.999	Y	3803.90	-2.821	0.996
128	12	25	-7.072E-03	1.43E-04	0.991	Y	3803.90	-2.821	0.996
129	75	12	-5.308E-02	2.28E-04	1.000	Y	3803.90	-2.821	0.996
130	40	9	-2.786E-02	4.30E-04	0.998	Y	3803.90	-2.821	0.996
131	75	14	-5.389E-02	1.77E-04	1.000	Y	3803.90	-2.821	0.996
132	40	8	-2.583E-02	7.31E-04	0.995	Y	3803.90	-2.821	0.996
134	12	31	-6.215E-03	2.57E-04	0.953	Y	3803.90	-2.821	0.996
120	12	21	-1.243E-02	8.79E-05	0.999	Y	6239.14	-2.509	0.994
121	40	8	-4.443E-02	1.04E-03	0.997	Y	6239.14	-2.509	0.994
5	200	21	-2.784E-04	2.18E-05	0.896	Y	900.84	-10.036	0.996
2	200	27	-4.328E-04	1.64E-05	0.965	Y	1797.89	-7.800	0.995
4	200	21	-5.840E-04	2.35E-05	0.970	Y	2487.26	-6.906	0.993
1	200	29	-1.163E-03	4.84E-05	0.955	Y	4580.99	-5.459	0.991
3	200	21	-1.063E-03	2.03E-05	0.993	Y	4580.99	-5.459	0.991
103	200	19	-2.935E-03	4.40E-04	0.723	N	150.11	0.000	1.000
107	200	14	-2.095E-03	3.85E-04	0.712	N	150.22	0.000	1.000
92	200	19	-4.867E-03	1.43E-04	0.986	N	348.35	0.000	1.000
89	200	24	-4.429E-03	1.73E-04	0.967	N	348.81	0.000	1.000
88	200	20	-4.693E-03	1.54E-04	0.981	N	349.74	0.000	1.000
101	200	14	-3.084E-03	3.63E-05	0.998	N	150.00	0.000	1.000
102	200	13	-2.576E-03	1.63E-04	0.958	N	150.00	0.000	1.000
108	200	21	-2.468E-03	1.32E-04	0.949	N	150.00	0.000	1.000
100	200	16	-3.128E-03	3.74E-04	0.833	N	150.11	0.000	1.000
111	200	44	-2.334E-03	6.83E-05	0.965	N	150.33	0.000	1.000
113	200	29	-2.378E-03	4.36E-05	0.991	N	150.33	0.000	1.000
115	200	24	-2.350E-03	3.33E-05	0.996	N	150.33	0.000	1.000

Table D.3: Experimental run information—additional parameters (cont.)

ID	Flow Rate (mL/min)	n	a (1/min)	s_a	R^2	use	P_i^{sat} (Pa)	B_{ii} (L/mol)	ϕ_i^{sat}
109	200	29	-2.655E-03	1.02E-04	0.962	N	150.55	0.000	1.000
90	125	20	-3.397E-03	2.04E-04	0.939	Y	347.89	0.000	1.000
106	160	22	-4.455E-03	1.58E-04	0.975	Y	348.35	0.000	1.000
93	160	21	-4.422E-03	1.61E-04	0.975	Y	349.28	0.000	1.000
91	160	15	-4.456E-03	2.13E-04	0.971	Y	349.74	0.000	1.000
112	160	28	-4.538E-03	8.35E-05	0.991	Y	349.97	0.000	1.000
114	160	14	-4.907E-03	6.49E-05	0.998	Y	350.20	0.000	1.000
99	160	19	-8.650E-03	2.49E-04	0.986	Y	888.17	0.000	1.000
105	160	13	-9.745E-03	2.68E-04	0.992	Y	889.74	0.000	1.000
97	160	20	-9.316E-03	2.09E-04	0.991	Y	891.31	0.000	1.000
104	160	13	-9.818E-03	2.94E-04	0.990	Y	892.35	0.000	1.000
96	125	16	-1.460E-02	4.03E-04	0.989	Y	2016.78	0.000	1.000
98	125	15	-1.583E-02	3.49E-04	0.994	Y	2019.96	0.000	1.000
94	160	13	-1.955E-02	5.63E-04	0.991	Y	2021.02	0.000	1.000
110	160	15	-1.878E-02	2.57E-04	0.998	Y	2021.02	0.000	1.000
95	125	12	-1.568E-02	4.11E-04	0.993	Y	2022.08	0.000	1.000
35	80	11	-3.950E-02	1.24E-03	0.991	N	7140.67	-3.173	0.991
40	80	6	-5.569E-02	8.70E-04	0.999	N	7140.67	-3.173	0.991
38	12	16	-7.035E-03	1.35E-04	0.995	N	7140.67	-3.173	0.991
39	60	11	-4.440E-02	1.69E-04	1.000	N	7140.67	-3.173	0.991
59	60	12	-4.460E-02	1.47E-04	1.000	N	7140.67	-3.173	0.991
57	80	10	-5.260E-02	4.99E-04	0.999	N	7140.67	-3.173	0.991
61	80	7	-6.091E-02	3.93E-04	1.000	N	7151.17	-3.172	0.991
62	60	11	-4.696E-02	1.30E-04	1.000	N	7151.17	-3.172	0.991
33	12	14	-7.577E-03	2.81E-04	0.984	N	7210.90	-3.164	0.991
41	12	14	-7.064E-03	1.78E-04	0.992	N	7210.90	-3.164	0.991
34	60	16	-4.461E-02	4.44E-04	0.999	N	7210.90	-3.164	0.991
36	80	10	-5.924E-02	4.28E-04	1.000	N	7210.90	-3.164	0.991
50	40	13	-3.719E-02	1.18E-03	0.989	N	11441.31	-2.762	0.988
47	40	13	-6.564E-02	1.35E-03	0.995	N	17530.56	-2.434	0.984
44	40	10	-7.455E-02	2.09E-03	0.994	N	21787.89	-2.282	0.982
45	40	11	-3.209E-02	3.85E-03	0.886	N	21787.89	-2.282	0.982
58	20	12	-1.674E-02	1.21E-04	0.999	Y	7133.68	-3.174	0.991
60	20	9	-1.655E-02	3.02E-04	0.997	Y	7137.18	-3.174	0.991
37	40	10	-3.179E-02	2.55E-04	0.999	Y	7140.67	-3.173	0.991
32	40	20	-3.322E-02	1.83E-04	0.999	Y	7210.90	-3.164	0.991
51	40	10	-5.333E-02	4.79E-04	0.999	Y	11389.49	-2.765	0.988

Table D.3: Experimental run information—additional parameters (cont.)

ID	Flow Rate (mL/min)	n	a (1/min)	s_a	R^2	use	P_i^{sat} (Pa)	B_{ii} (L/mol)	ϕ_i^{sat}
42	40	11	-5.131E-02	3.25E-04	1.000	Y	11441.31	-2.762	0.988
43	40	11	-4.992E-02	4.92E-04	0.999	Y	11441.31	-2.762	0.988
52	40	10	-7.685E-02	5.57E-04	1.000	Y	17523.17	-2.434	0.984
48	40	10	-7.620E-02	6.40E-04	0.999	Y	17530.56	-2.434	0.984
53	40	8	-7.298E-02	6.26E-04	1.000	Y	17545.37	-2.433	0.984
54	40	7	-7.877E-02	1.20E-03	0.999	Y	21471.13	-2.292	0.982
55	40	10	-9.078E-02	7.59E-04	0.999	Y	21497.39	-2.291	0.982
56	40	10	-8.442E-02	1.29E-03	0.998	Y	21506.14	-2.291	0.982
46	40	11	-9.225E-02	7.88E-04	0.999	Y	21611.45	-2.288	0.982
49	40	7	-9.020E-02	1.10E-03	0.999	Y	21699.52	-2.285	0.982
62.5	40	12	-4.826E-03	6.46E-04	0.848	N	11.25	0.000	1.000
82	60	30	-2.853E-03	9.86E-05	0.968	Y	3.43	0.000	1.000
83	80	29	-3.674E-03	1.01E-04	0.980	Y	3.44	0.000	1.000
84	80	27	-3.503E-03	2.07E-04	0.919	Y	3.44	0.000	1.000
86	80	29	-3.857E-03	6.77E-05	0.992	Y	3.44	0.000	1.000
85	80	27	-3.613E-03	1.13E-04	0.976	Y	3.45	0.000	1.000
68	120	13	-1.082E-02	2.77E-04	0.993	Y	10.93	0.000	1.000
67	40	16	-5.050E-03	1.72E-04	0.984	Y	11.09	0.000	1.000
65	80	16	-8.354E-03	1.99E-04	0.992	Y	11.25	0.000	1.000
66	120	13	-1.215E-02	3.03E-04	0.993	Y	11.26	0.000	1.000
69	80	16	-8.347E-03	2.47E-04	0.988	Y	11.26	0.000	1.000
70	60	8	-5.120E-03	1.48E-04	0.995	Y	11.26	0.000	1.000
63	60	13	-6.519E-03	2.01E-04	0.990	Y	11.30	0.000	1.000
79	60	17	-6.887E-03	2.21E-04	0.985	Y	11.36	0.000	1.000
71	60	27	-1.734E-02	7.29E-05	1.000	Y	43.19	0.000	1.000
72	60	16	-1.714E-02	1.85E-04	0.998	Y	43.19	0.000	1.000
73	60	12	-1.528E-02	3.97E-04	0.993	Y	43.19	0.000	1.000
81	60	12	-1.692E-02	3.14E-04	0.997	Y	43.49	0.000	1.000
80	60	14	-1.826E-02	3.21E-04	0.996	Y	43.64	0.000	1.000
74	60	20	-3.492E-02	1.84E-04	1.000	Y	143.60	0.000	1.000
75	60	11	-3.028E-02	3.34E-04	0.999	Y	143.83	0.000	1.000
76	60	10	-3.748E-02	5.15E-04	0.998	Y	144.95	0.000	1.000
77	60	10	-3.525E-02	1.31E-03	0.989	Y	144.95	0.000	1.000
78	60	10	-3.076E-02	8.36E-04	0.994	Y	144.95	0.000	1.000
31	40	9	-8.788E-03	3.25E-04	0.990	N	113.95	0.000	1.000
10	20	13	-7.729E-03	3.94E-04	0.972	N	331.94	0.000	1.000
20	40	12	-4.311E-02	6.95E-04	0.997	N	755.72	0.000	1.000

Table D.3: Experimental run information—additional parameters (cont.)

ID	Flow Rate (mL/min)	n	a (1/min)	s_a	R^2	use	P_i^{sat} (Pa)	B_{ii} (L/mol)	ϕ_i^{sat}
26	40	10	-8.981E-03	1.44E-04	0.998	Y	113.45	0.000	1.000
27	40	13	-9.041E-03	2.80E-04	0.990	Y	113.65	0.000	1.000
30	40	11	-8.746E-03	2.31E-04	0.994	Y	114.24	0.000	1.000
29	40	12	-1.248E-02	2.33E-04	0.997	Y	175.09	0.000	1.000
28	40	13	-1.272E-02	3.23E-04	0.993	Y	175.24	0.000	1.000
25	40	20	-1.392E-02	2.13E-04	0.996	Y	176.13	0.000	1.000
12	60	10	-3.227E-02	3.12E-04	0.999	Y	331.56	0.000	1.000
17	60	9	-3.249E-02	5.99E-04	0.998	Y	331.94	0.000	1.000
16	80	10	-4.126E-02	3.83E-04	0.999	Y	332.14	0.000	1.000
8	40	10	-2.260E-02	5.24E-04	0.996	Y	332.39	0.000	1.000
6	60	13	-3.234E-02	2.20E-04	0.999	Y	332.52	0.000	1.000
9	80	8	-4.429E-02	9.81E-04	0.997	Y	332.52	0.000	1.000
13	40	10	-2.230E-02	4.79E-04	0.996	Y	332.52	0.000	1.000
15	40	11	-2.152E-02	3.08E-04	0.998	Y	332.52	0.000	1.000
21	40	13	-1.944E-02	2.94E-04	0.997	Y	332.52	0.000	1.000
7	20	11	-1.182E-02	3.43E-04	0.992	Y	333.49	0.000	1.000
11	80	10	-4.034E-02	2.68E-04	1.000	Y	333.49	0.000	1.000
14	20	10	-1.095E-02	2.58E-04	0.996	Y	333.49	0.000	1.000
24	40	9	-3.049E-02	5.47E-04	0.998	Y	579.48	0.000	1.000
18	40	12	-3.559E-02	6.52E-04	0.997	Y	584.24	0.000	1.000
22	40	14	-4.404E-02	4.81E-04	0.999	Y	755.32	0.000	1.000
19	40	12	-4.385E-02	8.40E-04	0.996	Y	756.11	0.000	1.000
23	40	9	-4.572E-02	1.06E-03	0.996	Y	756.51	0.000	1.000
136	160	28	-2.983E-03	2.06E-04	0.890	Y	149.12	0.000	1.000
137	160	22	-2.533E-03	2.91E-04	0.791	Y	149.23	0.000	1.000
138	160	29	-2.308E-03	1.78E-04	0.862	Y	149.01	0.000	1.000
139	160	25	-2.492E-03	1.76E-04	0.897	Y	148.90	0.000	1.000
140	160	21	-2.431E-03	1.55E-04	0.928	Y	149.01	0.000	1.000
141	40	17	-5.297E-03	1.42E-04	0.989	Y	11.25	0.000	1.000
142	20	19	-2.734E-03	5.46E-05	0.993	Y	11.23	0.000	1.000
143	40	11	-2.529E-02	6.12E-04	0.995	Y	146.19	0.000	1.000
144	40	13	-2.454E-02	2.35E-04	0.999	Y	146.87	0.000	1.000
145	40	6	-2.922E-02	1.41E-03	0.991	Y	146.31	0.000	1.000
146	40	15	-1.025E-02	1.55E-04	0.997	Y	43.27	0.000	1.000
147	40	27	-1.862E-03	7.83E-05	0.958	Y	3.40	0.000	1.000
148	40	19	-4.912E-03	1.59E-04	0.982	Y	11.23	0.000	1.000
149	20	23	-1.325E-03	4.75E-05	0.974	Y	3.40	0.000	1.000

Table D.3: Experimental run information—additional parameters (cont.)

ID	Flow Rate (mL/min)	n	a (1/min)	s_a	R^2	use	P_i^{sat} (Pa)	B_{ii} (L/mol)	ϕ_i^{sat}
150	20	22	-1.319E-03	5.88E-05	0.962	Y	3.40	0.000	1.000
151	20	21	-1.127E-03	6.26E-05	0.947	Y	3.41	0.000	1.000
177	40	5	-1.223E-01	1.29E-02	0.968	N	312.47	0.000	1.000
165	60	17	-9.249E-03	1.87E-04	0.994	Y	16.21	0.000	1.000
176	60	18	-8.918E-03	2.70E-04	0.986	Y	16.27	0.000	1.000
178	60	23	-8.111E-03	1.89E-04	0.989	Y	16.24	0.000	1.000
174	60	29	-5.586E-03	1.27E-04	0.986	Y	16.28	0.000	1.000
158	20	21	-6.145E-03	1.79E-04	0.984	Y	42.14	0.000	1.000
154	20	19	-7.578E-03	1.83E-04	0.990	Y	42.17	0.000	1.000
163	20	17	-4.762E-03	2.01E-04	0.974	Y	42.21	0.000	1.000
160	40	12	-1.257E-02	4.26E-04	0.989	Y	42.08	0.000	1.000
157	40	15	-1.141E-02	2.50E-04	0.994	Y	42.14	0.000	1.000
159	40	9	-1.426E-02	3.49E-04	0.996	Y	42.14	0.000	1.000
161	40	11	-1.197E-02	2.98E-04	0.994	Y	42.21	0.000	1.000
173	60	8	-1.530E-02	3.84E-04	0.996	Y	42.08	0.000	1.000
155	60	9	-2.187E-02	1.09E-03	0.983	Y	42.11	0.000	1.000
153	60	12	-2.201E-02	2.30E-04	0.999	Y	42.14	0.000	1.000
162	60	11	-1.915E-02	4.05E-04	0.996	Y	42.14	0.000	1.000
156	80	13	-2.657E-02	2.49E-04	0.999	Y	42.08	0.000	1.000
152	80	12	-2.897E-02	3.54E-04	0.999	Y	42.17	0.000	1.000
164	80	12	-1.974E-02	2.16E-04	0.999	Y	42.24	0.000	1.000
172	40	12	-2.513E-02	2.43E-04	0.999	Y	121.81	0.000	1.000
168	60	9	-2.597E-02	2.21E-04	0.999	Y	121.89	0.000	1.000
169	60	6	-3.236E-02	8.08E-04	0.998	Y	121.89	0.000	1.000
166	60	7	-4.541E-02	6.52E-04	0.999	Y	121.97	0.000	1.000
171	60	6	-4.801E-02	1.18E-03	0.998	Y	121.97	0.000	1.000
178	20	7	-2.992E-02	1.08E-03	0.994	Y	312.47	0.000	1.000
175	40	7	-7.261E-02	9.32E-04	0.999	Y	312.47	0.000	1.000
167	60	6	-7.037E-02	3.04E-03	0.993	Y	312.47	0.000	1.000
170	60	5	-1.106E-01	3.08E-03	0.998	Y	313.40	0.000	1.000

Table D.4: Experimental run calculated values

ID	Ψ	s_{Ψ}	γ_i^{∞}	$s_{\gamma\infty} (\%)$	k_H (kPa·mol/mol)	$s_{kH} (\%)$
116	1	0	9.65E+03	5.2%	3.66E+04	7.9%
133	1	0	5.09E+03	5.2%	1.93E+04	19.3%
135	1	0	7.67E+03	5.2%	2.90E+04	8.0%
123	1	0	3.78E+03	5.2%	1.43E+04	10.8%
126	1	0	4.41E+03	5.2%	1.67E+04	7.7%
117	1	0	7.18E+03	5.2%	2.72E+04	15.3%
116	1	0	8.04E+03	5.2%	3.05E+04	4.2%
118	1	0	9.17E+03	5.2%	3.47E+04	2.0%
119	1	0	9.92E+03	5.2%	3.76E+04	7.8%
122	1	0	9.28E+03	5.2%	3.52E+04	4.2%
124	1	0	9.62E+03	5.2%	3.64E+04	4.1%
125	1	0	7.69E+03	5.2%	2.91E+04	4.2%
128	1	0	8.19E+03	5.2%	3.10E+04	2.6%
129	1	0	9.88E+03	5.2%	3.74E+04	4.1%
130	1	0	9.71E+03	5.2%	3.68E+04	7.8%
131	1	0	1.00E+04	5.2%	3.80E+04	4.1%
132	1	0	8.99E+03	5.2%	3.41E+04	8.1%
134	1	0	7.20E+03	5.2%	2.73E+04	4.5%
120	1	0	8.49E+03	5.2%	5.26E+04	2.0%
121	1	0	9.13E+03	5.2%	5.66E+04	7.9%
5	1	0	7.74E+01	25.3%	6.94E+01	8.0%
2	1	0	5.89E+01	25.2%	1.05E+02	4.1%
4	1	0	5.65E+01	25.2%	1.40E+02	4.3%
1	1	0	6.16E+01	25.1%	2.80E+02	4.4%
3	1	0	5.33E+01	25.1%	2.42E+02	2.4%
103	1	0	5.16E+03	25.4%	7.74E+02	15.1%
107	1	0	3.67E+03	25.4%	5.51E+02	18.4%
92	1	0	3.66E+03	25.4%	1.27E+03	3.3%
89	1	0	3.37E+03	25.4%	1.18E+03	4.2%
88	1	0	3.57E+03	25.4%	1.25E+03	3.6%
101	1	0	5.43E+03	25.4%	8.14E+02	2.0%
102	1	0	4.53E+03	25.4%	6.80E+02	6.5%
108	1	0	4.34E+03	25.4%	6.51E+02	5.6%
100	1	0	5.51E+03	25.4%	8.28E+02	12.1%
111	1	0	4.10E+03	25.4%	6.17E+02	3.3%
113	1	0	4.18E+03	25.4%	6.28E+02	2.4%

Table D.4: Experimental run calculated values (cont.)

ID	Ψ	s_{Ψ}	γ_i^{∞}	$s_{\gamma\infty} (\%)$	k_H (kPa·mol/mol)	$s_{kH} (\%)$
115	1	0	4.13E+03	25.4%	6.21E+02	2.1%
109	1	0	4.65E+03	25.4%	7.00E+02	4.1%
90	1	0	4.09E+03	25.4%	1.42E+03	6.5%
106	1	0	4.15E+03	25.4%	1.45E+03	4.0%
93	1	0	4.15E+03	25.4%	1.45E+03	4.1%
91	1	0	4.16E+03	25.4%	1.46E+03	5.1%
112	1	0	4.23E+03	25.4%	1.48E+03	2.7%
114	1	0	4.58E+03	25.4%	1.60E+03	2.3%
99	1	0	3.07E+03	25.4%	2.73E+03	3.4%
105	1	0	3.44E+03	25.4%	3.06E+03	3.3%
97	1	0	3.29E+03	25.4%	2.93E+03	2.9%
104	1	0	3.46E+03	25.4%	3.08E+03	3.5%
96	1	0	2.71E+03	25.4%	5.47E+03	3.5%
98	1	0	2.93E+03	25.4%	5.93E+03	3.1%
94	1	0	2.86E+03	25.4%	5.77E+03	3.4%
110	1	0	2.74E+03	25.4%	5.54E+03	2.2%
95	1	0	2.92E+03	25.4%	5.90E+03	3.4%
35	1	0	3.67E+03	10.0%	2.60E+04	5.0%
40	1	0	5.20E+03	10.0%	3.68E+04	4.2%
38	1	0	4.36E+03	10.0%	3.09E+04	2.5%
39	1	0	5.53E+03	10.0%	3.91E+04	5.1%
59	1	0	5.56E+03	10.0%	3.93E+04	5.1%
57	1	0	4.91E+03	10.0%	3.48E+04	4.0%
61	1	0	5.68E+03	10.0%	4.03E+04	3.9%
62	1	0	5.84E+03	10.0%	4.14E+04	5.1%
33	1	0	4.65E+03	10.0%	3.32E+04	4.1%
41	1	0	4.32E+03	10.0%	3.08E+04	3.0%
34	1	0	5.49E+03	10.0%	3.93E+04	5.2%
36	1	0	5.48E+03	10.0%	3.92E+04	3.9%
50	1	0	4.23E+03	10.0%	4.78E+04	8.0%
47	1	0	4.70E+03	9.9%	8.11E+04	7.6%
44	1	0	4.14E+03	9.9%	8.86E+04	7.8%
45	1	0	1.76E+03	9.9%	3.76E+04	14.0%
58	1	0	6.26E+03	10.0%	4.43E+04	15.2%
60	1	0	6.19E+03	10.0%	4.38E+04	15.3%
37	1	0	5.94E+03	10.0%	4.20E+04	7.7%

Table D.4: Experimental run calculated values (cont.)

ID	Ψ	s_Ψ	γ_i^∞	$s_{\gamma\infty}$ (%)	k_H (kPa·mol/mol)	s_{kH} (%)
32	1	0	6.14E+03	10.0%	4.39E+04	7.7%
51	1	0	6.14E+03	10.0%	6.91E+04	7.6%
42	1	0	5.85E+03	10.0%	6.61E+04	7.5%
43	1	0	5.69E+03	10.0%	6.43E+04	7.6%
52	1	0	5.52E+03	9.9%	9.52E+04	7.4%
48	1	0	5.47E+03	9.9%	9.43E+04	7.4%
53	1	0	5.24E+03	9.9%	9.05E+04	7.5%
54	1	0	4.48E+03	9.9%	9.44E+04	7.5%
55	1	0	5.17E+03	9.9%	1.09E+05	7.5%
56	1	0	4.80E+03	9.9%	1.01E+05	7.6%
46	1	0	5.20E+03	9.9%	1.10E+05	7.4%
49	1	0	5.05E+03	9.9%	1.08E+05	7.4%
62.5	5.35E-01	4.49E-02	3.03E+02	6.7%	6.37E+03	15.4%
82	4.41E-01	6.23E-02	3.26E+02	8.0%	2.54E+03	6.3%
83	4.41E-01	6.23E-02	3.14E+02	8.0%	2.45E+03	4.8%
84	4.41E-01	6.23E-02	3.00E+02	8.0%	2.34E+03	7.1%
86	4.41E-01	6.23E-02	3.30E+02	8.0%	2.58E+03	4.3%
85	4.41E-01	6.22E-02	3.08E+02	8.0%	2.41E+03	5.0%
68	5.33E-01	4.54E-02	2.33E+02	6.8%	4.78E+03	3.6%
67	5.34E-01	4.52E-02	3.21E+02	6.7%	6.67E+03	8.3%
65	5.35E-01	4.49E-02	2.63E+02	6.7%	5.53E+03	4.5%
66	5.36E-01	4.49E-02	2.56E+02	6.7%	5.38E+03	3.6%
69	5.36E-01	4.49E-02	2.62E+02	6.7%	5.51E+03	4.8%
70	5.36E-01	4.49E-02	2.14E+02	6.7%	4.51E+03	5.8%
63	5.36E-01	4.49E-02	2.73E+02	6.7%	5.76E+03	5.9%
79	5.36E-01	4.48E-02	2.85E+02	6.7%	6.04E+03	6.0%
71	7.03E-01	2.51E-02	2.40E+02	5.6%	1.48E+04	4.9%
72	7.03E-01	2.51E-02	2.36E+02	5.6%	1.45E+04	5.0%
73	7.03E-01	2.51E-02	2.11E+02	5.6%	1.29E+04	5.5%
81	7.04E-01	2.50E-02	2.31E+02	5.6%	1.43E+04	5.2%
80	7.05E-01	2.50E-02	2.49E+02	5.6%	1.54E+04	5.2%
74	9.43E-01	4.66E-03	1.81E+02	5.0%	2.75E+04	4.7%
75	9.44E-01	4.63E-03	1.56E+02	5.0%	2.38E+04	4.8%
76	9.46E-01	4.47E-03	1.91E+02	5.0%	2.92E+04	4.9%
77	9.46E-01	4.47E-03	1.79E+02	5.0%	2.75E+04	6.0%
78	9.46E-01	4.47E-03	1.56E+02	5.0%	2.39E+04	5.4%

Table D.4: Experimental run calculated values (cont.)

ID	Ψ	s_{Ψ}	γ_i^{∞}	$s_{\gamma\infty}(\%)$	k_H (kPa·mol/mol)	$s_{kH}(\%)$
31	8.70E-01	0.05172	8.93E+01	25.5%	1.17E+04	8.5%
10	1	0	5.93E+04	10.9%	1.97E+04	15.4%
20	1	0	6.76E+04	10.9%	5.11E+04	7.2%
26	8.69E-01	0.052071	9.11E+01	25.5%	1.19E+04	7.8%
27	8.70E-01	0.051931	9.16E+01	25.5%	1.20E+04	8.2%
30	8.71E-01	0.051509	8.84E+01	25.5%	1.16E+04	8.1%
29	9.64E-01	0.014676	9.02E+01	25.0%	1.64E+04	7.8%
28	9.64E-01	0.014598	9.19E+01	25.0%	1.67E+04	7.9%
25	9.66E-01	0.014126	1.00E+02	25.0%	1.83E+04	7.7%
12	1	0	8.31E+04	10.9%	2.76E+04	5.0%
17	1	0	8.31E+04	10.9%	2.76E+04	5.2%
16	1	0	7.92E+04	10.9%	2.63E+04	3.8%
8	1	0	8.69E+04	10.9%	2.89E+04	7.7%
6	1	0	8.30E+04	10.9%	2.76E+04	5.0%
9	1	0	8.48E+04	10.9%	2.82E+04	4.3%
13	1	0	8.56E+04	10.9%	2.85E+04	7.6%
15	1	0	8.27E+04	10.9%	2.75E+04	7.5%
21	1	0	7.48E+04	10.9%	2.49E+04	7.5%
7	1	0	9.06E+04	10.9%	3.02E+04	14.9%
11	1	0	7.75E+04	10.9%	2.59E+04	3.7%
14	1	0	8.39E+04	10.9%	2.80E+04	14.8%
24	1	0	6.43E+04	10.9%	3.73E+04	7.3%
18	1	0	7.41E+04	10.9%	4.33E+04	7.3%
22	1	0	6.91E+04	10.9%	5.22E+04	7.1%
19	1	0	6.88E+04	10.9%	5.21E+04	7.3%
23	1	0	7.17E+04	10.9%	5.42E+04	7.5%
136	1	0	6.54E+03	10.9%	9.75E+02	7.2%
137	1	0	5.56E+03	10.9%	8.30E+02	11.7%
138	1	0	5.07E+03	10.9%	7.56E+02	8.0%
139	1	0	5.49E+03	10.9%	8.17E+02	7.3%
140	1	0	5.34E+03	10.9%	7.96E+02	6.7%
141	5.35E-01	4.49E-02	3.34E+02	6.7%	7.01E+03	8.1%
142	5.35E-01	4.50E-02	3.44E+02	6.7%	7.23E+03	15.4%
143	9.48E-01	4.29E-03	1.93E+02	5.0%	2.97E+04	7.4%
144	9.49E-01	4.20E-03	1.86E+02	5.0%	2.88E+04	7.0%
145	9.48E-01	4.28E-03	2.23E+02	5.0%	3.44E+04	8.5%

Table D.4: Experimental run calculated values (cont.)

ID	Ψ	s_{Ψ}	γ_i^{∞}	$s_{\gamma\infty} (\%)$	k_H (kPa·mol/mol)	$s_{kH} (\%)$
146	7.03E-01	2.51E-02	2.12E+02	5.6%	1.30E+04	7.4%
147	4.40E-01	6.24E-02	3.22E+02	8.0%	2.49E+03	9.0%
148	5.35E-01	4.49E-02	3.08E+02	6.7%	6.46E+03	8.3%
149	4.40E-01	6.25E-02	4.58E+02	8.0%	3.54E+03	16.3%
150	4.40E-01	6.24E-02	4.56E+02	8.0%	3.52E+03	16.5%
151	4.40E-01	6.24E-02	3.89E+02	8.0%	3.01E+03	16.8%
177	1	0	4.74E+05	10.9%	1.48E+05	13.3%
165	1	0	5.11E+05	10.9%	8.29E+03	5.7%
176	1	0	4.91E+05	10.9%	7.99E+03	6.1%
178	1	0	4.49E+05	10.9%	7.28E+03	5.8%
174	1	0	3.07E+05	10.9%	5.00E+03	5.8%
158	1	0	3.87E+05	10.9%	1.63E+04	15.6%
154	1	0	4.78E+05	10.9%	2.01E+04	15.5%
163	1	0	3.00E+05	10.9%	1.26E+04	15.9%
160	1	0	3.97E+05	10.9%	1.67E+04	8.4%
157	1	0	3.59E+05	10.9%	1.51E+04	8.0%
159	1	0	4.50E+05	10.9%	1.89E+04	8.0%
161	1	0	3.77E+05	10.9%	1.59E+04	8.1%
173	1	0	3.22E+05	10.9%	1.36E+04	5.7%
155	1	0	4.60E+05	10.9%	1.94E+04	7.2%
153	1	0	4.63E+05	10.9%	1.95E+04	5.2%
162	1	0	4.03E+05	10.9%	1.70E+04	5.5%
156	1	0	4.20E+05	10.9%	1.77E+04	4.0%
152	1	0	4.57E+05	10.9%	1.93E+04	4.0%
164	1	0	3.10E+05	10.9%	1.31E+04	4.0%
172	1	0	2.65E+05	10.9%	3.23E+04	7.4%
168	1	0	1.82E+05	10.9%	2.21E+04	4.9%
169	1	0	2.27E+05	10.9%	2.76E+04	5.5%
166	1	0	3.19E+05	10.9%	3.89E+04	5.1%
171	1	0	3.39E+05	10.9%	4.13E+04	5.6%
178	1	0	2.27E+05	10.9%	7.09E+04	14.6%
175	1	0	2.77E+05	10.9%	8.66E+04	7.4%
167	1	0	1.78E+05	10.9%	5.57E+04	6.5%
170	1	0	2.83E+05	10.9%	8.86E+04	5.8%

Appendix E. Prediction Method Sample Calculations

E.1 First-Order Group Contribution Method

E.1.1. 1,2,4,5-Tetramethylbenzene

Structural information and group contributions:

Group	$n_i \cdot \Delta G_{hyd,i}^\infty$ (kJ/mol)	$n_i \cdot B_i$ (K)	$n_i \cdot C_i$
CH ₃	4·(3.67)	4·3683	4·(-8.73)
CH _{ar}	2·(-0.65)	2·1849	2·(-4.16)
C _{ar}	4·(-3.85)	4·(-102)	4·(0.05)
I(C-C)	2·(-1.01)	2·6047	2·(-19.2)
Y ⁰	7.95	0	0
Total	3.91	30116	-81.44

Equation 7.2 is rearranged to solve for $k_{H,ref}$:

$$k_{H,ref} = 101.325 \cdot \text{kPa} \cdot 55.5084 \frac{\text{mol}}{\text{mol}} \cdot \exp \left(\frac{3.91 \cdot 1000 \frac{\text{J}}{\text{mol}}}{8.3145 \frac{\text{J}}{\text{mol} \cdot \text{K}} \cdot 298.15 \cdot \text{K}} \right) = 2.723 \cdot 10^4 \cdot \text{kPa} \frac{\text{mol}}{\text{mol}}.$$

The overall temperature-dependent equation for k_H is predicted by:

$$\ln(k_H) = (30116 \cdot \text{K}) \left(\frac{1}{298.15 \cdot \text{K}} - \frac{1}{T} \right) + (-81.44) \left(\ln \left(\frac{T}{298.15 \cdot \text{K}} \right) \right) + \ln \left(2.723 \cdot 10^4 \cdot \text{kPa} \frac{\text{mol}}{\text{mol}} \right).$$

E.1.2. 2,2-Dimethylhexane

Structural information and group contributions:

Group	$n_i \cdot \Delta G_{hyd,i}^{\infty}$ (kJ/mol)	$n_i \cdot B_i$ (K)	$n_i \cdot C_i$
CH ₃	4·(3.67)	4·3683	4·(-8.73)
CH ₂	3·(0.70)	3·2962	3·(-8.43)
C	1·(-4.51)	1·1203	1·(-8.1)
Y ⁰	7.95	0	0
Total	20.22	24821	-68.31

Equation 7.2 is rearranged to solve for $k_{H,ref}$:

$$k_{H,ref} = 101.325 \cdot \text{kPa} \cdot 55.5084 \frac{\text{mol}}{\text{mol}} \cdot \exp \left(\frac{20.22 \cdot 1000 \frac{\text{J}}{\text{mol}}}{8.3145 \frac{\text{J}}{\text{mol} \cdot \text{K}} \cdot 298.15 \cdot \text{K}} \right) = 1.961 \cdot 10^7 \cdot \text{kPa} \frac{\text{mol}}{\text{mol}}.$$

The overall temperature-dependent equation for k_H is predicted by:

$$\ln(k_H) = (24821 \cdot \text{K}) \left(\frac{1}{298.15 \cdot \text{K}} - \frac{1}{T} \right) + (-68.31) \left(\ln \left(\frac{T}{298.15 \cdot \text{K}} \right) \right) + \ln \left(1.961 \cdot 10^7 \cdot \text{kPa} \frac{\text{mol}}{\text{mol}} \right).$$

E.1.3. Methyl ethyl ketone

Structural information and group contributions:

Group	$n_i \cdot \Delta G_{hyd,i}^{\infty}$ (kJ/mol)	$n_i \cdot B_i$ (K)	$n_i \cdot C_i$
CH ₃	2·(3.67)	2·3683	2·(-8.73)
CH ₂	1·(0.70)	1·2962	1·(-8.43)
C=O (ketone)	1·(-22.74)	1·4520	1·(-4)
Y ⁰	7.95	0	0
Total	-6.75	14848	-29.89

Equation 7.2 is rearranged to solve for $k_{H,ref}$:

$$k_{H,ref} = 101.325 \cdot \text{kPa} \cdot 55.5084 \frac{\text{mol}}{\text{mol}} \cdot \exp \left(\frac{-6.75 \cdot 1000 \frac{\text{J}}{\text{mol}}}{8.3145 \frac{\text{J}}{\text{mol} \cdot \text{K}} \cdot 298.15 \cdot \text{K}} \right) = 3.694 \cdot 10^2 \cdot \text{kPa} \frac{\text{mol}}{\text{mol}}.$$

The overall temperature-dependent equation for k_H is predicted by:

$$\ln(k_H) = (14848 \cdot \text{K}) \left(\frac{1}{298.15 \cdot \text{K}} - \frac{1}{T} \right) + (-68.31) \left(\ln \left(\frac{T}{298.15 \cdot \text{K}} \right) \right) + \ln \left(3.694 \cdot 10^2 \cdot \text{kPa} \frac{\text{mol}}{\text{mol}} \right).$$

E.2 Second-Order Group Contribution Method

E.2.1. 3-Methyl-1-butanol

Structural information and group contributions:

Group	$n_i \cdot \Delta G_{hyd,i}^\infty$ (kJ/mol)	$n_i \cdot B_i$ (K)	$n_i \cdot C_i$	D
O-(C)(H)	1·(-25.46)	1·(0)	1·(0)	-
C-(C)(H) ₂ (O)	1·(0.77)	1·22467	1·(-49.8)	-
C-(H)(C) ₃	1·(-1.93)	1·(-4267)	1·(11.8)	-
C-(H) ₂ (C) ₂	1·(0.68)	1·868	1·(-1.89)	-
C-(H) ₃ (C)	2·3.72	2·6820	2·(-18.6)	-
Y ⁰	7.95	0	0	-
Total	-10.55	32708	-77.09	0.7774

Equation 7.2 is rearranged to solve for $k_{H,ref}$:

$$k_{H,ref} = 101.325 \cdot \text{kPa} \cdot 55.5084 \frac{\text{mol}}{\text{mol}} \cdot \exp \left(\frac{-10.55 \cdot 1000 \frac{\text{J}}{\text{mol}}}{8.3145 \frac{\text{J}}{\text{mol} \cdot \text{K}} \cdot 298.15 \cdot \text{K}} \right) = 79.76 \cdot \text{kPa} \frac{\text{mol}}{\text{mol}}.$$

The overall temperature-dependent equation for k_H is predicted by:

$$\ln(k_H) = 0.7774 \cdot \left[(32708 \cdot \text{K}) \left(\frac{1}{298.15 \cdot \text{K}} - \frac{1}{T} \right) + (-77.09) \left(\ln \left(\frac{T}{298.15 \cdot \text{K}} \right) \right) \right] + \ln \left(79.76 \cdot \text{kPa} \frac{\text{mol}}{\text{mol}} \right)$$

E.2.2. Ethylbenzene

Structural information and group contributions:

Group	$n_i \cdot \Delta G_{hyd,i}^{\infty}$ (kJ/mol)	$n_i \cdot B_i$ (K)	$n_i \cdot C_i$	D
C-(H) ₂ (C)(Cb)	1·(1.02)	1·19962	1·(-65)	-
Cb-(C)(Cb) ₂	1·(-4.53)	1·(-13358)	1·45	-
Cb-(H)(Cb) ₂	5·(-0.6)	5·1870	5·(-4.28)	-
C-(H) ₃ (C)	1·3.72	1·6820	1·(-18.6)	-
Y ⁰	7.95	0	0	-
Total	5.16	22774	-60	1

Equation 7.2 is rearranged to solve for $k_{H,ref}$:

$$k_{H,ref} = 101.325 \cdot \text{kPa} \cdot 55.5084 \frac{\text{mol}}{\text{mol}} \cdot \exp \left(\frac{5.16 \cdot 1000 \frac{\text{J}}{\text{mol}}}{8.3145 \frac{\text{J}}{\text{mol} \cdot \text{K}} \cdot 298.15 \cdot \text{K}} \right) = 4.509 \cdot 10^4 \cdot \text{kPa} \frac{\text{mol}}{\text{mol}}.$$

The overall temperature-dependent equation for k_H is predicted by:

$$\ln(k_H) = 1 \cdot \left[(22774 \cdot \text{K}) \left(\frac{1}{298.15 \cdot \text{K}} - \frac{1}{T} \right) + (-60) \left(\ln \left(\frac{T}{298.15 \cdot \text{K}} \right) \right) \right] + \ln \left(4.509 \cdot 10^4 \cdot \text{kPa} \frac{\text{mol}}{\text{mol}} \right).$$

E.2.3. *n*-Butyl ethyl ether

Structural information and group contributions:

Group	$n_i \cdot \Delta G_{hyd,i}^{\infty}$ (kJ/mol)	$n_i \cdot B_i$ (K)	$n_i \cdot C_i$	D
O-(C) ₂	1·(-15.52)	1·(-29379)	1·(68.9)	-
C-(C)(H) ₂ (O)	2·(0.77)	2·22467	2·(-49.8)	-
C-(H) ₂ (C) ₂	2·(0.68)	2·868	2·(-1.89)	-
C-(H) ₃ (C)	2·3.72	2·6820	2·(-18.6)	-
[CH ₃ -CH ₂ -O-CH ₂] corr	1·(-1.42)	1·1106	1·(-6)	-
Y ⁰	7.95	0	0	-
Total	1.35	32037	-77.68	0.7774

Equation 7.2 is rearranged to solve for $k_{H,ref}$:

$$k_{H,ref} = 101.325 \cdot \text{kPa} \cdot 55.5084 \frac{\text{mol}}{\text{mol}} \cdot \exp \left(\frac{1.35 \cdot 1000 \frac{\text{J}}{\text{mol}}}{8.3145 \frac{\text{J}}{\text{mol} \cdot \text{K}} \cdot 298.15 \cdot \text{K}} \right) = 9.696 \cdot 10^3 \cdot \text{kPa} \frac{\text{mol}}{\text{mol}}.$$

The overall temperature-dependent equation for k_H is predicted by:

$$\ln(k_H) = 0.7774 \cdot \left[(32037 \cdot \text{K}) \left(\frac{1}{298.15 \cdot \text{K}} - \frac{1}{T} \right) + (-77.68) \left(\ln \left(\frac{T}{298.15 \cdot \text{K}} \right) \right) \right] + \ln \left(9.696 \cdot 10^3 \cdot \text{kPa} \frac{\text{mol}}{\text{mol}} \right)$$